BRENNAN DAVIS | HUNTER GLANZ

DATA SCIENCE FOR ALL

Brennan Davis
California Polytechnic State University

Hunter Glanz
California Polytechnic State University

DATA SCIENCE FOR ALL

Pearson

Content Development: David Dietz, Thomas Atwater, Amit Verma, Ravi Indiver
Content Management: Laura Briskman, Monique Bettencourt
Content Production: Mahalakshmi Usha, Nupur Anand, Audra Walsh, Robert Carroll, Dominick Frank, Sandra Rodriguez, Scott Disanno, Rachel Reeve, Guadalupe Edgar
Product Management: Karen Montgomery
Product Marketing: Demetrius Hall
Rights and Permissions: Ben Ferrini, Anjali Singh, Vignesh Murugamoorthy

Please contact www.AskPearsonSupport.com with any queries on this content.
Pearson puts care into the creation of our products. Our learning materials undergo content reviews to minimize bias and ensure the content is inclusive, accurate, and relevant. Please use this link to report instances of bias and inaccuracies: https://www.pearson.com/report-bias.html
You can learn more about Pearson's commitment to accessibility at https://www.pearson.com/us/accessibility.html

Cover images from Shutterstock, D.R.3D/Shutterstock, AAVAA / Shutterstock, and NasonovVasiliy/Shutterstock.

Acknowledgments of third-party content appear on the appropriate page within the text.

Library of Congress Cataloging-in-Publication Data

Data Science For All
Library of Congress Cataloging in Publication Control Number: 2024044768

$PrintCode

Pearson

Rental Edition:

ISBN 10: 0-13-832314-3
ISBN 13: 978-0-13-832314-1

Print Offer:

ISBN 10: 0-13-531141-1
ISBN 13: 978-0-13-531141-7

DEDICATION

To my beautiful wife, Anna, for her unwavering support and culinary delights; to my sons, Canyon, Levi, and Tobin, who were my pillars of encouragement; and to my son, Noe, and his wife, Dani, for their unwavering backing—this book is a testament to your love and support.

—Brennan

To my family, Maddie and Jake, for their unconditional support throughout all of this. To all of my friends and other family for always being there, no matter what.

—Hunter

ABOUT THE AUTHORS

Brennan Davis is the Richard and Julie Hood Professor and director of graduate analytics programs at the Orfalea College of Business at California Polytechnic State University (Cal Poly, San Luis Obispo). He received a BS in mathematics from the University of California, Los Angeles, an MBA from the Wharton School of Business at the University of Pennsylvania, and a PhD from the University of California, Irvine. Brennan currently teaches undergraduate and graduate analytics courses. In 2019, Brennan received the Emeritus Faculty Award for significant and meritorious achievement in contributing to student welfare.

Hunter Glanz is a Professor of Statistics and Data Science at California Polytechnic State University (Cal Poly, San Luis Obispo). He received a BS in mathematics and a BS in statistics from Cal Poly, followed by an MA and PhD in statistics from Boston University. He maintains a passion for data science, machine learning, and statistical computing and enjoys teaching courses in those areas. Hunter serves on numerous committees and organizations dedicated to delivering cutting-edge statistical and data science content to students and professionals alike, including being a founding board member of the California Alliance for Data Science Education. In 2019, Hunter received the Terrance Harris Excellence in Mentorship Award, and in 2020, he received the Outstanding Faculty Award in the Master's in Business Analytics program at Cal Poly.

CONTENTS

PREFACE

Data Science: Why Now?

In writing this book, we aim to democratize data literacy and empower every student with the skills to navigate our data-rich world. We recognize the need for this growing field of data science to be a part of everyone's life, both in their consumption of data day-to-day and in their working with data professionally or personally.

For decades, we have witnessed data's involvement in coursework and scholarly projects across campus. We have observed the necessity for data-related skills and comprehension in the diverse professions into which our students have graduated. The ubiquity of data has the potential to unite us in the process of ingesting information, thinking critically to draw conclusions, and communicating with those around us.

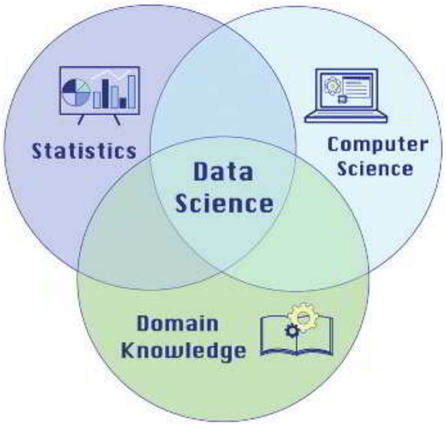
As an exciting and rich blend of statistics, computer science, and domain application, the field of data science has much to offer everyone in their pursuits. This book aims to take the most applicable ideas from these component fields and put them in the hands of students so they can apply them immediately to their everyday lives. By giving them hands-on experience with every concept, this book will provide a *lingua franca*, enabling them to fluently converse with data no matter their background, major, or professional goals.

The time is right for a textbook that *distinguishes* itself by offering unparalleled support through meticulously designed content that caters to true beginners, ensuring quality, clarity, and accessibility without sacrificing depth. With a commitment to fostering an inclusive learning environment, we provide a comprehensive, up-to-date curriculum, complemented by innovative, evolving tools for assessment and engagement. This holistic approach guarantees a learning experience that is both enriching and enduring, setting a new standard for academic resources in the data science domain.

Approach

As California Polytechnic State University instructors, we work daily to help our students "Learn by Doing," a motto that is also a core part of our professional identity. We, therefore, make it a core part of this text. The firm belief that data work requires technology is closely linked with this motto, but software should not be a barrier to engaging with data science. To this end, we have filled our text with opportunities to get students working with data early and often in the software package of the instructor's choice (R, Python, Excel, or StatCrunch).

A foundation in data science is not exclusive to those with a quantitative background. We have built our text to be accessible to students with no prior statistics or computer science knowledge and minimal mathematical experience. Instead, this text provides students of all backgrounds with an entry point to the world of data and any future coursework that involves data. It is not merely a modernization of a traditional first course in statistics or computer science. Instead, it distills these component fields into an overview of what data science means and what it involves in today's world.



Prepare, Analyze, and Communicate

Today's students already are consumers of data and will likely be directly engaged with data work in their future careers. We believe it is essential that they understand what the complete life cycle of a data science project can entail. Chapter 1 introduces the field of data science and our three overarching components of a data science project: data preparation, data analysis, and data storytelling.

One of the most exciting facets of data work is its iterative nature. Working with data is a labor of learning, analyzing, and communicating. We then revisit these steps as new pieces of data are introduced. Analysis can result in findings that necessitate revisiting or using a different analysis technique. Completing a data science project also requires communicating these analyses in different formats.

Data Preparation

Data preparation can include:

- Data collection
- Data organization
- Understanding the data
- Wrangling (cleaning, transforming, and integrating) the data.

Data often require preparation before they're suitable for visualization and analysis. It's essential to first grasp the data's structure and source. As we delve into data exploration, it becomes clear that further organization or manipulation might be necessary for practical analysis and storytelling. We'll explore several prevalent techniques for transforming data into formats ready for analytical and narrative purposes.

Data Analysis

Data analysis can include:

- Describing data with numerical summaries
- Testing hypotheses
- Predictive modeling
- Understand the various relationships in a dataset.

Data analysis spans a spectrum of methods, each serving distinct objectives and questions. Initially, we describe data using summary statistics and other essential metrics, condensing key characteristics into concise numerical or verbal forms. This is where data science starts to branch away from introductory statistics, which typically focuses on making inferences about various population parameters. While hypothesis testing remains vital for deriving conclusions, data science emphasizes examining proportions, means, and correlations using single samples, extending to two-sample inference in contexts like A/B testing. Furthermore, data science expands beyond traditional linear regression, encompassing a more comprehensive range of supervised and unsupervised modeling techniques. This includes, but is not limited to, methods like decision trees, random forests, *k*-nearest neighbors, logistic regression, and *k*-means clustering, providing a more comprehensive toolkit for analyzing complex datasets.

Data Storytelling

Data storytelling can include:

- Visualizations
- Written reports
- Oral presentations
- Recommendations to a client.

Students must master compelling data storytelling, enabling them to uncover and articulate the narratives hidden within data. This skill goes beyond merely presenting findings. It involves probing what the data reveal and identifying what information might still be missing.

A pivotal aspect of data storytelling is creating and interpreting compelling data visualizations with software tools. Students can effectively convey complex information by weaving a narrative that links data preparation, analysis, and visual storytelling, making it accessible and actionable for decision-makers. Our text integrates a data storytelling framework called STAR, encouraging students to apply this approach at each stage of their data-related tasks.

Topical Coverage

We designed this text and the accompanying MyLab Statistics course for a first-semester course in introductory data science. The structure of our Table of Contents mirrors this approach, allocating additional time for certain sections at the instructor's discretion. However, this text can be used flexibly. Chapters 1–4 introduce data preparation, storytelling, visualization, and foundational analysis. Chapters 5–7 delve into more advanced data analysis techniques, and Chapters 8–10 cover contemporary machine learning techniques. Recognizing the dynamic nature of data science, we've designed the curriculum with the flexibility to incorporate specific topics or projects that align with your unique emphasis.

Chapter 1 introduces data science as an academic discipline, an increasingly critical component of every business, and an inevitable piece of our everyday lives. It also introduces the anatomy of data as consisting of observations, variables, and values. We distill the data science workflow stages into three overarching categories: data preparation, data analysis, and data storytelling.

Students *apply* this knowledge, including:

- distinguishing between different data types,
- practicing storytelling, and
- recognizing analysis types.

Chapter 2 introduces data wrangling as the first stage in the data science workflow to preprocess data into a cleaner, more friendly form for analysis and storytelling. At the end of data wrangling efforts, data should be consistent and in the structure needed for analysis and storytelling.

Students *apply* this knowledge, including:

- applying strategies for addressing missing data,
- finding and addressing anomalous values, and
- transforming data into a more usable structure.

Chapter 3 introduces data visualization as a primary way to communicate data using graphical techniques. We introduce the grammar of graphics and continue the STAR data storytelling framework as ways to unite the creation of all data visualizations in insightful ways. By the end of the chapter, students will have seen standard techniques for visualizing one, two, three, and four variables simultaneously. We emphasize the dangers of misrepresentation and provide guidelines.

Students *apply* this knowledge, including creating and interpreting:

- histograms and density plots;
- box plots and violin plots;
- bar charts; and
- scatterplots, line charts, and bubble charts.

Chapter 4 introduces exploratory data analysis via numerical metrics that capture key characteristics in a dataset, such as central tendency and spread. Students are reminded of the different shapes distributions have, but then shown how these shapes and the resistance of specific metrics to extreme values can affect the values of summary metrics. We finish with the characterization and identification of outliers.

Students *apply* this knowledge, including:

- calculating and evaluating summary statistics,
- identifying distribution shapes,
- computing correlations, and
- determining outliers.

Chapter 5 introduces the process of formulating questions we can ask of our data and how to make those requests using a software tool. We show students how to perform standard data queries (in generic, tool-agnostic ways) by subsetting a dataset by variables or observations and sorting, aggregating, and combining datasets. For this chapter only, we introduce structured query language (SQL) as a tool option alongside Excel, Python, R, and StatCrunch.

Students *apply* this knowledge, including:

- selecting data,
- filtering data,

- ordering data,
- summarizing data, and
- merging data.

Chapter 6 revisits the concept of variability, first introduced in Chapter 4, and explores its relationship to uncertainty and predictability in our data and the target population. To clarify the role of uncertainty in data analysis, we introduce fundamental ideas in probability. Subsequently, we focus on diagnosing and assessing variability in data visualizations. Students also leave Chapter 6 with a better understanding of sampling methods.

Students *apply* this knowledge, including:

- assessing variability,
- estimating likelihoods of events using data, and
- performing simulations.

Chapter 7 broadly introduces statistical inference and then walks students through simulation-based inference for a population proportion and correlation. We cover inference for a population mean and the difference between two population means via t-tests. In particular, the two-sample t-test is introduced in the context of an A/B Test. Students leave Chapter 7 with a better understanding of statistical inference and the difference between experiments and observational studies.

Students *apply* the knowledge, including:

- choosing appropriate methods for statistical inference,
- making statistical inferences in observational studies and experiments, and
- understanding false positives and false negatives.

Chapter 8 provides an overview of machine learning and artificial intelligence, with particular attention to the choices and considerations required in a data science problem involving machine learning. In particular, students are introduced to differentiating model characteristics, with the goal of guiding the students on identifying which models to explore. Then, they see the metrics that inform final model evaluation and selection.

Students *apply* the knowledge, including:

- distinguishing between regression and classification methods,
- comparing methods with respect to interpretability and complexity, and
- evaluating machine learning methods using training and test data.

Chapter 9 dives deeper into supervised machine learning as the class of methods appropriate for a problem involving a particular response variable of interest, such as regression or classification. We build intuition and understanding about traditional linear regression models with different types of explanatory variables and introduce logistic regression for classification. We also present decision trees, random forests, and k-nearest neighbors models for regression and classification.

Students *apply* this knowledge, including:

- distinguishing different types of supervised learning methods,
- evaluating and comparing regression methods on test data, and
- evaluating and comparing classification methods on test data.

Chapter 10 expands on unsupervised machine learning as the class of methods appropriate for when our problem does not involve a response variable of interest. We outline three of the most common categories of unsupervised learning: dimension reduction, anomaly detection, and clustering. Students are introduced to k-means and hierarchical clustering, emphasizing the former in evaluating clustering results.

Students *apply* this knowledge, including:

- detecting anomalies,
- calculating similarity using distance, and
- comparing cluster analysis results.

Alternative Pathways through the Content

We have arranged the 10 chapters of this book to align with much of the data science work we are familiar with and our recommended order. However, data science work is iterative and can proceed in a different order than we present. For example, many professors feel more comfortable teaching data visualization earlier than it is currently placed.

Chapters 2–5 can be taught in any order. To ensure the richest possible experience for students, they should come before Chapters 6–10. Then, we recommend the following:

- Chapter 6 should come first after Chapters 1–5.
- Chapter 7 should come immediately after Chapter 6.
- Chapter 8 should come immediately after Chapter 7.
- Chapters 9 and 10 could be covered in either order after completing Chapter 8.

While the table of contents has 10 chapters, we recommend teaching at a slower pace for Chapters 8–10, especially Chapter 9. While students might have been exposed to graphing concepts, foundational statistics, and coding, machine learning will likely be new to most, and these chapters benefit from a slower pace. We promise that it is worthwhile!

Integrating Software

In this text, we have elected to support R, Python, Microsoft Excel, and StatCrunch (with support for SQL as topics allow). Screenshots from different tools are used throughout the text. While we recognize that instructors will likely select one of these, we believe exposure to multiple tools is helpful to students who will encounter different technologies in their careers.

Tool Time

Excel

Python

R

StatCrunch

The use of technology should not be a barrier to success in this course. We recommend cloud-based tools (e.g., Posit Cloud and Google Colab) for running the code in this course. However, we support any tool, cloud-based or device-installed, that runs R and Python code. We have created numerous resources (available in MyLab Statistics) to support students and instructors as they get started:

- Video guides in setting up cloud-based tools, Posit Cloud and Google Colab, for running R and Python code (recommended)
- Video guides in setting up R and Python on students' devices
- Video support for software functions in R, Python, Excel, and StatCrunch
- R and Python code for every "Try It Yourself" feature, where applicable
- Excel and StatCrunch click paths for every "Try It Yourself" feature, where available and applicable

All datasets are available in MyLab Statistics and the Pearson Math and Statistics Resource site (www.pearsonhighered.com/mathstatsresources).

Join our GitHub for code, data, and other digital assets: https://github.com/datasci4all

R

If you're using R for any portion of the journey through our text, please install the correct packages so that your output matches ours. Each chapter involves a different set of packages. Check the corresponding Tech Center resource for the packages needed in that chapter.

If the Tech Center includes a line such as

library(tidyverse)

then you need to make sure you have the tidyverse package installed. This only needs to be done once per machine by running the following line of code:

install.packages("tidyverse")

Python

If you're using Python for any portion of the journey through our text, please install the correct libraries so that your output matches ours. Each chapter involves a different set of libraries. Check the corresponding Tech Center resource for the libraries needed in that chapter.

If the Tech Center includes a line such as

import pandas

then you need to make sure you have the pandas library installed. This only needs to be done once per machine by running the following line of code:

pip install pandas

Excel and StatCrunch

If you're using Excel and/or StatCrunch for any portion of the journey through our text, make sure that you are working with a fresh copy of supplied data files for any exercise. It can be easy to overlook work from previous exercises or activities that will give you incorrect results. Note that not every activity is currently possible with standard Excel and StatCrunch functionality; we have noted these where applicable and recommend exploring the activities in R or Python if desired.

Video Support

As instructors, we know that students are always asking for video support. They want to watch videos if they miss lectures, for test preparation, for software support, or just to remind themselves how to do something that might have looked easy in class. We also know students don't want to watch 15-minute lecture videos. So, we have created a suite of video support for students, ensuring that their textbook authors are essentially "on call" to help them when needed. We have recorded over 500 videos:

- Lecture videos that introduce each section;
- Example videos that walk students through selected exercises in each chapter, and
- Software support videos.

Guiding Students

All chapters (except Chapter 1) open with a **story** centering on a young person encountering a data science investigation in their school, place of work, or personal life. Lessons presented in a given chapter tie back to the opening narrative, applying concepts to real life.

Case Studies are integrated throughout each chapter, exploring chapter-wise concepts in business, technology, environmental sciences, health, sports, and more.

In the margins, we supply **Quick Tips** as additional guidance on challenging material, warnings about common mistakes, and further details expounding on pedagogy.

We periodically answer the ubiquitous question, "Why does this all matter?" via **Big Picture Breaks**, which take a step back from minutiae to offer additional context.

Each chapter concludes with a **Putting It Together** feature, with broken-down bullets so that students can reinforce what they have learned.

Storytelling is a fundamental aspect of data science, and to emphasize its importance, we have developed a unique framework specific to this title. The **STAR Framework** invites students to communicate an engaging narrative that not only showcases data but also provides context and prompts action. We introduce this framework in Chapter 1 and extensively apply it throughout. Here is an overview of the STAR Framework's components:

1. *Scope the Data's Context*: We begin by detailing the data's background, including the observational units, variables, their types, and the data source.
2. *Track the Question Behindthe Exploration*: We identify the crucial questions and underlying tensions the data aim to resolve.
3. *Articulate the Results*: We focus on clearly presenting the findings derived from the data.
4. *Respond with Next Steps*: We conclude by suggesting further actions or investigations guided by the results.

We designed the STAR Framework as a teaching tool, guiding students through a structured process of thinking about and communicating with data.

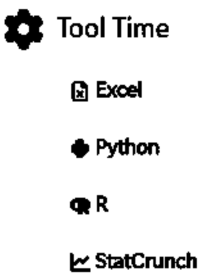
Guiding Instructors

Teaching a new course is always a challenge. We have included a robust instructor resource package, including the following:

- An instructor resource manual written by Hunter Glanz, with notes from his own introduction to data science course. This provides an overview of each chapter. It includes notes, on what to emphasize in each class and suggestions for presenting the material.
- PowerPoint lecture slides that include essential graphics and follow the sequence and philosophy of the text. They can be downloaded from MyLab Statistics or www.pearson.com.
- A complete instructor solutions manual containing worked-out solutions to all text exercises and case study answers. It can be downloaded from MyLab Statistics or www.pearson.com.
- TestGen® (www.pearson.com/testgen), which enables instructors to build, edit, print, and administer tests using a computerized bank of questions developed to cover all the text's objectives. TestGen is algorithmically based, allowing instructors to create multiple but equivalent versions of the same question or test with the click of a button. Instructors can also modify test bank questions or add new questions. The software and test bank are available for download from Pearson's online catalog, www.pearson.com. The questions are also assignable in MyLab Statistics.
- Accessibility: Pearson works continuously to ensure our products are as accessible as possible to all students. We are working toward achieving WCAG 2.0 Level AA and Section 508 standards, as expressed in the Pearson Guidelines for Accessible Educational Web Media, www.pearson.com/mylab/statistics/accessibility.

Active Learning

Because these lessons are buildable, we heavily incorporate student engagement as they learn. Concepts are interspersed with **Try It Yourself** (TiY) exercises, where students can practice what they have just learned. Where applicable, the TiY exercises are accompanied by our Tool Time feature, which offers technology-specific solutions and tips (e.g., Excel, Python, R, and StatCrunch). Answers to all TiYs are available to students at the end of the text.



For each TiY exercise, we also provide an accompanying **Applying the Concepts** (AtC) activity at the end of the chapter. The goal is a scaffolding approach, wherein students practice with their in-chapter TiYs and then complete AtCs as assigned with less guidance. Comprehensive answers are available to instructors in MyLab Statistics.

AtCs are also available in MyLab Statistics, which can be assigned as MediaShareproject-based assignments for group or individual work. They come with customizable rubrics designed to offer grading support to instructors.

Designed to provoke further thought and investigation, we have also included **Explaining the Concept** questions at the end of each chapter, and available in MyLab Statistics for assignment. These open-ended questions are great for formative or summative assessment or just as an in-class exercise.

Integrated Review

Data Science for All was created with no prerequisites necessary for success beyond the completion of high school math. However, we realize that students may need extra support with some mathematical or statistical topics. This content has been added to the MyLab Statistics course with just-in-time support via auto-graded exercises, worksheets, and videos. This allows for targeted support for students in their time of need. This also fits a corequisite model.

Putting It Together

Data Science for All is a comprehensive yet accessible journey into the world of data science designed for students of all majors and backgrounds. Our book demystifies data science, covering its entire lifecycle from preparation and analysis to storytelling, all through a hands-on approach. We emphasize "Learning By Doing," integrating the STAR framework and various software tools and making the experience engaging and practical.

We warmly invite you to explore *Data Science for All*. Whether you're a student, educator, or curious learner, this book is a gateway to the vital skills of data science, applicable in every facet of modern life. We invite you to connect, joining us on this enlightening journey to uncover the power of data.

Sincerely,
Brennan Davis and Hunter Glanz
bdavis39@calpoly.edu
hglanz@calpoly.edu

ACKNOWLEDGMENTS

We express our profound gratitude to those closest to us. Hunter thanks Maddie Schroth-Glanz and Jake for their invaluable support and encouragement. Brennan thanks Anna, Canyon, Levi, Tobin, Noe, and Dani Davis for their immeasurable advocacy and inspiration.

We also extend our heartfelt thanks to those we feel professionally and personally close through this process. Laura Briskman, Karen Montgomery, Demetrius Hall, and Monique Bettencourt believed in our idea to launch a new book from the very beginning. They gave us their time, introduced us to their networks, and skillfully guided us, knowing exactly when to rein us in and when to encourage us forward. Through this process, they have become more than colleagues to us; they are friends. We love our tradition of enjoying fun and unusual desserts together, a testament to the sweet journey we've shared.

We are also grateful to Melissa Melton, Sean Pepper Leader, and Brogan Pietrocini for their dedicated efforts in bringing this project to fruition.

We sincerely thank the content production team led by Nupur Anand, Sandra Rodriguez, and Scott Disanno for their meticulous work. We are also thankful for the exceptional media production overseen by Audra Walsh and Stephanie Woodward.

We acknowledge the valuable contributions of the MathXL development team, particularly Bob Carroll and Phil Oslin, for their technical expertise. Lastly, we express our gratitude to our developmental editor, David Dietz, whose insights and guidance have been instrumental in shaping this book.

REVIEWERS

We are indebted to the many reviewers, class testers, and students who provided us with guidance and feedback as we developed this product. We apologize for any omissions.

We are also indebted to our many corporate reviewers for their support and guidance through this process, including Abe Weston at Google, John Frith at Urban Science, Rex Briggs at Claritas, and Lucas Pierce at Character.AI.

Bagui, Sikha—University of West Florida

Ballion, Tatiana—Midland University

Beagley, Jonathan—Valparaiso University

Beckman, Matthew—Pennsylvania State University

Bergeler, Elmar—University of Texas at San Antonio

Brown, Austin—Kennesaw State University

Brown, Robert—Carroll Community College

Burgin, Joe—College of Southern Maryland

Burns, Jared W.—Seton Hill University

Carter, Perry—Texas Tech University

Cheng, Fuxia—Illinois State University

Coe, Jacquelyn—Central Oregon Community College

Combs, Adam—Robert Morris University

Dalkilic, Mehmet—Indiana University

Davis, Amanda—Forsyth Technical Community College

Ding, Junhua—University of North Texas

Do, Thuy—Luther College

Drummey, Kevin—Loyola University Maryland

Fernando, Harshini—Purdue University Northwest

Finotti, Luis—University of Tennessee, Knoxville

Firouzian, Shawn—MiraCosta College

Fitzpatrick, Kelly—County College of Morris

Ghriga, Mohammed—Long Island University

Gold, Kevin—Boston University

Habimana, Jean Remy—Southwestern University

Hardin, Jo—Pomona College

Hashemi, Navid—College of Charleston

Hendricks, Todd—GSU Perimeter College

Holay, Sandeep—Southeast Community College

Hong, Sungbum—Jackson State University

Houser, Shannon—University of Alabama at Birmingham

Kader, Idrissa—Northern Virginia Community College

Karunaratne, Susitha—Purdue University

Kinney, David—Bellevue University

Kirmani, Syed—University of Northern Iowa

Kitchings, N. Clayton—University of North Georgia

Kox, Erin—Fanshawe College

Kristin, Lilly—Columbus State University

Kurban, Hasan—Texas A&M University

Kuyper, Arend—Northwestern University

Lewis, Mark—Alfred University

Long, Michael—Polk State College

Lyne, Heidi—Joliet Junior College

Magnum, Amanda—Converse College

Malachowski, James—Northwest Arkansas Community College

McLean, Jeffrey—University of North Carolina at Chapel Hill

Mogull, Michael—Cuesta College

Montgomery, Aaron—Baldwin Wallace University

Neal, Matthew—Denison University

Nicolae, Dan—University of Chicago

Onunwor, Enyinda—Saint Paul College

Pearsall, Sam—Los Angeles Pierce College

Poliak, Adam—Bryn Mawr College

Rampure, Suraj—University of California at San Diego

Harsy Ramsey, Amanda—Lewis University

Rathor, Shekhar—University of Central Oklahoma

Ritter, Carrie—Randolph Community College
Roberts, David—University of Calgary
Rosson, Holly—Warren Wilson College
Schell, William—Montana State University
Scoville, Nicholas—Ursinus College
Seefield, Kim—Nashua Community College
Shamir, Lior—Kansas State University
Sharma, Sharad—University of North Texas
Singh, Vivek—Rutgers University
Smith, Alice E.—Auburn University
Sourile, Daniel—Joliet Junior College
Spence, Dianna—University of North Georgia
Starin, Karen—Columbus State Community College
Szczurek, Piotr—Lewis University
Thurman, Alexis—County College of Morris
Williams, Therese—University of Central Oklahoma
Xu, Jie—George Mason University
Xue, Yajiong (Lucky)—East Carolina University
Zhou, Yi—Columbus State University

INDEX OF ACTIVITIES

These activities are designed for students to learn how to interact with data in different ways. They take a scaffolded approach, where students start with software support and then are asked to apply what they've learned to the same objective.

Chapter 5: Data Management

Chapter 6: Understanding Uncertainty, Probability, and Variability

Chapter 7: Drawing Conclusions from Data

Chapter 8: Machine Learning

Chapter 9: Supervised Learning

Software Libraries to Install

For those using Python or R, the following libraries are required to complete activities in each chapter. In Chapter 7 only, an extra tool pack will be required for those using Excel. No libraries need to be installed for those using StatCrunch.

Python

Chapters 2–10
pandas

Chapter 3
plotnine
wordcloud
matplotlib
squarify

Chapter 4
numpy
scipy.stats
plotnine

Chapter 5
numpy

Chapter 6
numpy
plotnine
random

Chapter 7
numpy
scipy
sklearn.utils
plotnine
random

Chapter 8
sklearn.model_selection
sklearn.tree
sklearn.metrics
sklearn.pipeline
sklearn.compose
sklearn.preprocessing

Chapter 9
numpy
scipy
sklearn.linear_model
sklearn.preprocessing
plotnine
sklearn.model_selection
sklearn.tree
sklearn.ensemble
sklearn.neighbors
sklearn.linear_model
sklearn.metrics
sklearn.compose
sklearn.pipeline
statsmodels.api

Chapter 10
numpy
sklearn.cluster
plotnine
scipy.spatial

R

Chapters 2–10
tidyverse

Chapter 2
lubridate

Chapter 3
wordcloud
RColorBrewer

Chapter 7
broom

Chapter 8
rpart
caret
tidymodels

Chapter 9
kknn
vip
tidymodels
broom
caret
ranger
rsample
rpart
randomForest
Metrics

Excel

Activities in Microsoft Excel are all able to be completed in standard Excel except for Chapter 7, which requires the (free) Analysis ToolPak. To enable the Analysis ToolPak, follow these steps:

1. File → Options → Add-Ins.
2. In the Manage box.
   a. Select Excel Add-Ins.
   b. Click Go.
3. Check the Analysis ToolPak checkbox.
   a. Click OK.

CHAPTER 1

WHAT IS DATA SCIENCE?

GaudiLab/Shutterstock

LEARNING OBJECTIVES

- Interpret the definition and scope of data science, including its interdisciplinary nature and lifecycle.
- Recognize the benefits of presenting data as a tidy table.
- Distinguish between different data types.
- Describe a coherent data preparation pipeline, considering the various formats and types of data that may be encountered.
- Explore the various types of data analysis for compelling storytelling.
- Evaluate the growing impact of data science in society and industry, including ethical considerations.

Introduction

Data science impacts many aspects of our world. Streaming media services use viewers' past entertainment choices to customize recommended content. Doctors use human genome data to prescribe medicines tailored to patients' unique gene profiles. Farmers use soil and weather data to determine suitable watering and harvest times, thereby improving crop production and reducing waste. Historians mine thousands of declassified U.S. State Department memos to learn how diplomacy affected the Cold War in the 20th century. University administrators quickly filter data on tens of thousands of students to match them with donors' scholarship criteria. Coaches use real-time wearable technology data to help their athletes win championships. These are just a few examples of how data science revolutionizes our lives.

Shutterstock

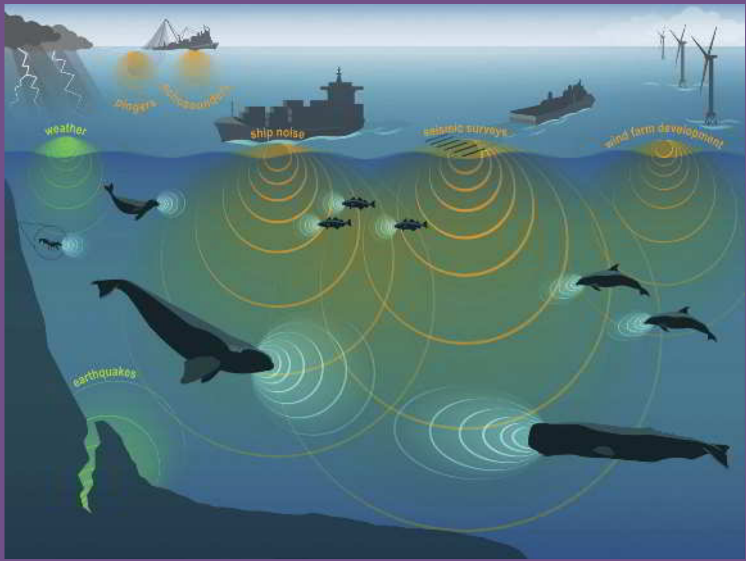
A whale surfaces near offshore wind energy technologies in the Pacific Ocean. From Ocean News & Technology.

Data science also helps us do good on a more global scale. Consider the tensions between energy progress and marine life preservation. While energy providers aim to increase renewable energy through offshore wind technologies, marine scientists are concerned about how new offshore wind energy sites might affect nearby marine mammals. Offshore wind technologies have the potential to benefit civilization, but their noise might disrupt marine mammals' foraging, orientation, hunting, and communication, which are already threatened by human-made disturbances (Figure 1.1).

California marine scientists use data science to monitor the deep-water ocean ecosystem near the state's offshore wind energy technologies. Specifically, they combine computer science and statistics using multiple data sources: acoustic recordings, underwater camera images, and Monterey Bay Aquarium Research Institute classifications. They use computer algorithms and statistics to recognize marine mammals and track their behavior. This automated data science helps the California Energy Commission promote clean energy while simultaneously protecting marine life.

Using data sources as diverse as photographic images, acoustic recordings, animal tagging data, and animal classifications to make important planning decisions would have been far less feasible just a few years ago. New approaches provided by data science make such advances possible.

FIGURE 1.1
Sound waves overlap in the ocean soundscape. From Ocean News & Technology.



NOAA Fisheries

SECTION 1.1: INTRODUCTION TO DATA SCIENCE

Before defining *data science*, we should define *data*. For this textbook, we will use the following definition:

DEFINITION

Data are information from a source or sources that can be useful for insights and decision-making after being processed in various ways.

As we will see, data can be more than just numbers. We'll use the word as a plural noun (e.g., "these data are recorded . . ."). We can be less formal about the *science* part of data science; think of science as a way of obtaining knowledge.

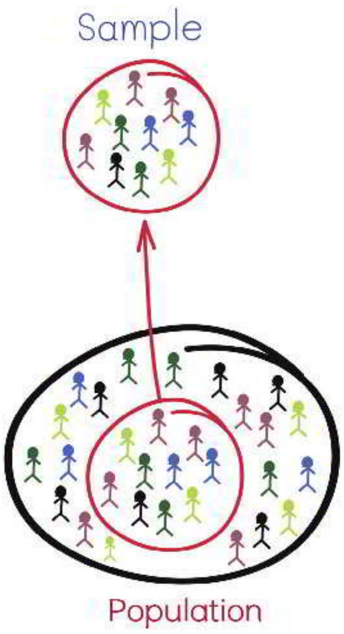
DEFINITION

Data science is an interdisciplinary field that uses scientific methods, processes, and systems to extract knowledge and insights from data across various domains.

The main reason data and their analyses are important is because we are constantly interested in describing or drawing conclusions about a *population* of interest based on a more feasibly accessed subset of that population, called a *sample*.

DEFINITION

A **population** is the set of every item or individual that is of interest for a particular question. For example, all used car owners in the United States.



lamnee/123RF

Because it's not practical to collect data on an entire population of interest, we are forced to take a sample from this population and work with data from this sample.

DEFINITION

A **sample** is a subset of the population of interest.

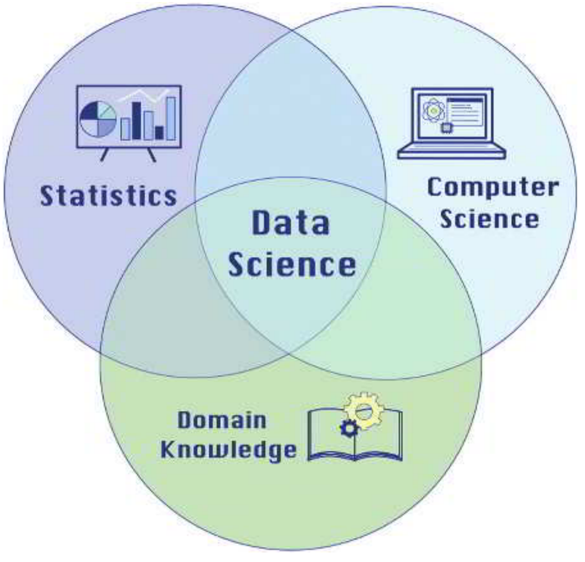
For example, if our population of interest were all used car owners in Europe, one possible sample would be the used car owners responding to a survey sent to a randomly generated list of 1000 people in Europe.

The ability to understand how people work with data, from a sample, to communicate insights about populations of interest is crucial for living an informed life in today's society of news and other media.

Combining Statistics, Computing, and Expert Knowledge

Data science combines statistics, computer science, and expertise from various domains to extract value from data (Figure 1.2). For example, when we ask Spotify to play our favorite songs, when Netflix recommends shows we enjoy binge-watching, or when we see products on Amazon's homepage tailored to our preferences, we experience the power of data science. These online suggestions are made using algorithms based on user-provided data, running on lightning-fast servers.

FIGURE 1.2
Data science overlaps with computer science, statistics, and domain knowledge.



Sometimes, the distinctions between data science and data analytics need to be clarified. *Data analytics* is the applicationof data science to a domain-specific problem. The term *data science* has been traced back to 1974 when Peter Naur proposed it as an alternative name for computer science. In these ways, it also resembles the field of statistics, which is the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data. More modern challenges, such as the largeness, complexity, and accessibility of datasets, and the increased complexity of many analysis methods, have necessitated the incorporation of ideas and tools from computer science and statistics to handle such data and extract insights successfully.

Modern data science recognizes that data are everywhere and can be incredibly useful through collaboration with statistics, computer science, and every imaginable domain. While the field of data science is still evolving, there is a consensus that it involves all three pieces of the Venn diagram in Figure 1.2.

It may be helpful to distinguish "data science" as a field and "data scientist" as a job title. Different companies and organizations will likely have distinct responsibilities for those working as data scientists. However, most data scientists are expected to be able to gain actionable insights from data.

As data science has evolved alongside technology, it has become one of the most in-demand professions. Data scientists and data science skills are necessary assets for almost all organizations.

Data Science Lifecycle

As data scientists, we generally follow a process that consists of *iterative* rounds of (1) data preparation, (2) data analysis, and (3) data storytelling for better insights and decision-making. By *iterative*, we mean that the steps often repeat and build on each other, improving over time. Figure 1.3 illustrates these components as useful exercises at multiple points in the workflow to help people make better decisions. We typically start with preliminary data preparation, which involves cleaning and organizing the data. After preliminary data preparation, data scientists may repeat the other components of the process to refine their understanding, which could include additional data preparation. For example, after preparing visualization data, we may dive into data analysis with our cleaned data to find patterns and relationships between variables. Alternatively, we may jump to data storytelling with a visualization that tells us how to prepare or analyze the data better.

FIGURE 1.3

The data science lifecycle comprises three iterative data science components that are repeated and improved as necessary: preparation, analysis, and storytelling.

We often revisit data preparation after data analysis and storytelling. For instance, analyzing or visualizing data can reveal new insights that prompt a return to preparing the data in a different way. At any point in the cycle, a discovery can spark further questions and analyses, feeding back into the cycle. Finally, we are ready for publicized data storytelling (e.g., posting it online, or distributing it privately to a decision maker), wherein we share a summary of our findings, often accompanied by compelling visual narratives.

Sometimes, this process also includes a data collection stage before the first data preparation step. Data collection may involve conducting surveys to gather customer feedback, using sensors to collect environmental data, or scraping websites for social media trends. Data scientists may collect more data as needs arise at any point in the cycle.

We call this the *data science lifecycle*, which occurs when all stages are completed at least once. Restarting the process based on findings from the previous cycle is essential because data science often introduces as many new questions as it answers. Figure 1.3 illustrates the data science lifecycle by showing data preparation, analysis, and storytelling as a rotation.

Quick Tip

Remember, before diving into data preparation, it's crucial to identify the question, problem or conjecture driving our data analysis. This initial step guides the purpose and direction of our data preparation, ensuring that the process is aligned with our objectives.

DEFINITION

The **data science lifecycle** is a system of three recurring steps: preparation, analysis, and storytelling. It starts with preliminary preparation and ends with publicized storytelling.

Data preparation is the process of collecting, cleaning, transforming, integrating, and managing data to structure it for effective analysis, storytelling, and decision-making.

Data analysis is the process of examining prepared data to uncover valuable insights and guide decision-making.

Data storytelling is the communication of data insights through summaries, visualizations, and narratives.


Evgeny Tomeev/Shutterstock

Data Preparation

Data preparation is the first component in which we collect (e.g., access) and wrangle (e.g., clean, transform, and integrate) data. As the name suggests, data preparation involves working with the data to get it into an increasingly structured format that is helpful for analysis and communication. We will spend the first few chapters of *Data Science for All* learning how to prepare data. Specifically, we will learn about representing data in tabular form (Chapter 1) and wrangling data to make it more useful (Chapter 2). Data preparation is often the most time-consuming part of a data scientist's job. It is like preparing ingredients to make a salad. In data preparation, much like meticulously washing, chopping, and mixing various ingredients to create a well-balanced salad, we methodically clean, transform, and integrate data to build a solid foundation for insightful analysis and storytelling.


Christian Jung/Shutterstock

Data Analysis

Data analysis is the extraction of meaning from prepared data to answer questions such as "What has happened?", "What might happen in the future?", and "What should we do?". We can perform data analysis by exploring (Chapter 4), querying (Chapter 5), understanding uncertainty (the study of how likely an event is to occur, covered in Chapter 6), learning how to draw conclusions from data (Chapter 7), and modeling data (e.g., supervised and unsupervised *machine learning* covered in Chapters 8, 9, and 10). We will learn how to conduct and interpret typical data science analyses. It is akin to creating your salad from the prepared ingredients.


Evgeny Karandaev/Shutterstock

Data Storytelling

Data storytelling is a component of data science aimed at maximizing decision-making utility. The goal is to ensure that findings from data analysis are suitable for making informed decisions. Success in communicating our discoveries depends on how well we prepare and analyze data. Chapter 3 covers the topic of *visualization in storytelling.* We use storytelling to assist data scientists in making decisions about data preparation and analysis. Storytelling can help both technical and non-technical stakeholders understand the results of data analysis.Consider this as presenting the salad in a way that makes people want to eat it.

Data preparation, analysis and storytelling are interrelated, so you will find them in all chapters. One reason these components can reoccur

Case Study: Netflix Uses Data Science for a Better Customer Experience

Urbazon/E+/Getty Images

Netflix uses data science to create a state-of-the-art customer service streaming site. It collects all kinds of customer data: when and how often subscribers watch a show, whether they binge-watch it or take some time to complete a series, and whether they pause an episode and resume it later. These data are used to create personalized account attributes for each subscriber, helping Netflix make content recommendations. Netflix's data scientists take advantage of visualization by structuring raw data into consumable dashboards, such as graphs and charts, for analysts to explore and translate into new features for the service.

Netflix uses the data science practices outlined in this textbook for preparing its data, analyzing its traffic and operations, maintaining its information security, and managing experimentation using A/B tests. These tests are trials in which, for example, the company sends one of two subscription offers to randomly selected customers to see which offer is accepted more often and by whom.

Overall, Netflix has wisely put all the data to good use. By incorporating concepts such as data analysis, machine learning, and statistics, data science has helped Netflix grow exponentially and become one of the top streaming sites in the world.

throughout the data science workflow is that the audience of the work can change. For example, the first iterations of data preparation and data storytelling are often most useful for the data scientist who is actually working with and analyzing the data. These rounds give them the understanding of the data needed to process, analyze, and communicate it in the best ways for downstream consumers like their professor, boss, colleagues, or customers. To this end, the final step in the data science lifecycle is to publicize the data storytelling component with visualizations and communication built for these consumers.

Data Science Tools

Data tools support the implementation of data science. These tools are software applications that facilitate preparation, analysis, and storytelling. They include coding tools (e.g., R and Python) as well as spreadsheet tools (e.g., Excel). Imagine trying to analyze online reviews without these technologies. Let's say you're an analyst at Amazon, aiming to understand if and how comments listing pros and cons influence customers' purchase decisions. If you or your team had to read and categorize every review for every product on Amazon, then compare those results to product sales histories, it could take a lifetime. However, tools like R, Python, or Excel could expedite the process, scanning through comments and searching for pros and cons, lists, and keywords in mere seconds.

We will include information and resources about data technologies in every chapter throughout this text. We will also cover scalable and robust tools for data preparation, analysis, and storytelling. *Data ethics*, also included in every chapter throughout this text, is the application of social justice and moral and ethical principles to data science.

DEFINITION

The **cloud** is a system of computer servers accessible over the internet, allowing us to deal with large amounts of data.

Case Study: NASA Uses Cloud Services to Stream Real-Time Mars Footage

Triff/Shutterstock

The National Aeronautics and Space Administration (NASA) is an agency of the United States government focused on the civil space program and space research. NASA's Curiosity Mission involved a meticulous and complex landing procedure for its interplanetary rover, Curiosity. NASA wanted to share real-time footage of the launch with people worldwide, so it turned to *cloud* services to achieve this. NASA's Jet Propulsion Laboratory uses cloud services to stream images and videos from Curiosity's Mars landing. The use of cloud services enables NASA to ensure the availability of its content around the globe. Thus, in real time, NASA can provide hundreds of gigabits per second of groundbreaking images of Mars for hundreds of thousands of viewers.

SECTION 1.2: DATA IN TABLES

As we will see, data come from a wide variety of sources in a wide variety of forms. Both the sources and the forms will help dictate the specific steps we need to take to extract meaningful insight from the data. Although the exact steps will almost never be the same, the general stages of this process will always include preparation, analysis, and storytelling. Before we delve into the ways in which we prepare data, we need to understand the anatomy of data.

Observations and Variables

Tables are a way to organize our data, structuring them in rows of *observational units* and columns of *variables*.

DEFINITION

An entity about which data are recorded is called an **observational unit**.

The recorded characteristic of an observational unit is the **variable** of interest.

The data for a single observational unit is sometimes called an **observation**.

Data in **tabular form** are organized into a table of rows and columns for clarity and ease of use.

A **dataset** is a collection of data.

A variable's **value** is the attribute or quantity given to a single observational unit within a dataset.

Many datasets exist in a *tabular form***,** meaning a table with rows and columns (Table 1.1). In the dataset below, the observational units in the rows are students, and the variables in the columns are first name, age, GPA, favorite pet species, and whether each person (observational unit) voted in a recent election.

For the tabular data we will discuss in this chapter, a dataset is a collection of variables with multiple observations. Therefore, a tabular dataset is like a table with many rows and columns. A dataset can also be represented in tabular form as a sheet in spreadsheets (e.g., Excel and Google Sheets), which we will discuss throughout this text.

TABLE 1.1
Example data from student election in tabular form.

variable

observational unit

first_name	age	gpa	favorite_pet	voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

Quick Tip

In practice, the data scientist collecting these data would use a student ID number or some other confidential identifier in place of names to protect subjects' personal information.

A variable's value is the information stored about each observational unit. In Table 1.1, each student is an observational unit in a particular row of our dataset. For example, Selma is someone in our dataset whose age is 18 and whose GPA is 3.7. Her age (18) and GPA (3.7) serve as values.

The values of the variables `age` and `gpa` would be stored on the computer as 18 and 3.7, respectively, for Selma. We would be able to find the correct age and GPA for any of our students, assuming the information stored in each student's `age` and `gpa` variables is accurate.

Pavel Bobrovskiy/Shutterstock

Variables and Data Types

Tables with defined variables (e.g., spreadsheets) are a very common form of structured data. Unstructured data, such as documents or images, do not directly fit into analysis tools like spreadsheets. Structured data versus unstructured data is like comparing organized belongings to unorganized ones. Structured data are often ordered in rows and columns.

DEFINITION

Structured data are information that is organized in a standardized format for efficient access, often in tables or databases.

A **database** is an organized collection of data, typically stored electronically, that allows for easy storage, retrieval, and manipulation of information.

Unstructured data are information that lacks predefined organization, such as text, images, or videos.

For data that can be represented as variables of observational units in something as neat as a spreadsheet, there are two critical ways to think about variables and their values:

1. The variable's use. Throughout the rest of this book, we will discuss numerous ways to use data, including creating summaries, visualizations, and analyses.
   a. *Quantitative* variables take numeric values on which mathematical operations make sense. In Table 1.1, `gpa` is an example of a quantitative variable.
      i. *Discrete variables* are quantitative variables that take on values that can be written down in a list. In Table 1.1, `age` (in whole numbers of years) is an example of a discrete variable.
      ii. *Continuous variables* are quantitative variables that can take on any value within an interval of values. The body temperature of the students is an example of a continuous variable.

b. *Categorical (qualitative) variables* take values that are category designations. In Table 1.1, `favorite_pet` is an example of a categorical variable.

2. The variable's type. The type of data the variable contains determines how the variable's values are stored in a computer. These variable types include:

a. *Text*: The variable contains words or characters without order or numerical utility (e.g., a `first_name` value of "Hector" or `favorite_pet` value of "Dog"). A text variable is sometimes referred to as a string variable. Categorical variables are often stored as text variables.

b. *Integer*: The variable contains numbers with no decimal points (e.g., an `age` value of 18).

c. *Float*: The variable holds numbers with decimal points (e.g., a `gpa` value of 3.1). Because a number may have many decimal places, the number can be quite long. This variable is sometimes referred to as a decimal variable.

d. *Boolean*: The variable contains only two possible values: TRUE or FALSE. Booleans are also referred to as *indicator variables*, *logical variables*, *binary variables*, *dichotomous variables*, or *dummy variables*. Sometimes, the values are stored as 1 or 0, representing TRUE or FALSE, respectively (e.g., the `voted` value of 1 or 0, meaning it is TRUE or FALSE that the student indicated they voted).

Try It Yourself: Identify Variable Uses and Types

Consider the following ten variables and a sample of their values:

`height`: 5.7, 6.0, 5.8, 5.9, 6.1
`age`: 21, 22, 23, 24, 25
`name`: 'Alex', 'Beth', 'Charlie', 'Dana', 'Eli'
`gpa`: 3.5, 3.6, 3.7, 3.8, 3.9
`graduated`: True, False, True, True, False
`pets`: 0, 1, 2, 0, 1
`birth_month`: 'January', 'February', 'March', 'April', 'May'
`siblings`: 2, 1, 0, 3, 2
`fulltime_student`: False, True, True, False, True
`shoe_size`: 8, 9, 10, 11, 12

Your task is to identify each variable's use and type. Remember, the possible uses are quantitative and categorical, and the possible types are text, integer, float, and Boolean. This exercise should help you better understand and identify different variable types, which is an essential skill when working with data.

Data scientists name their variables. Sometimes, the names are helpful to others, and sometimes they are not. Consider a variable named `var_1`. This variable name is unhelpful. Without a data key or codebook, we have no idea what information is stored in this variable. A *data key*, or *codebook,* is a reference document that defines the variables and the types of information they contain. Table 1.2 shows an example data key.

Creating meaningful names related to the contents of their variables is a thoughtful process. For example, suppose we create a Boolean variable where people with a cell value of 1 indicate that their favorite pet is a fish and 0 if they indicate some other animal. Table 1.3 has a poorly named variable, `var_1`, which does not describe the Boolean variable well. A better name might be `favorite_pet_is_fish`.

Fesus/123RF

It is best practice—often required by software—to name variables using a single word without symbols like %, ^, &, *, $, #, ~, or @ that often have other coding purposes. Spaces are also usually prohibited in variable names when coding. In Table 1.3, the variables are named to follow these conventions. We use the snake_case convention in this textbook because it is the most common naming method used in Python and R. However, other conventions exist like UpperCamelCase, which separates words without spaces using changes in case. To prepare you for the real world we may use UpperCamelCase occasionally in this textbook, too.

TABLE 1.2
Data key with both uninformative and meaningful variable names.

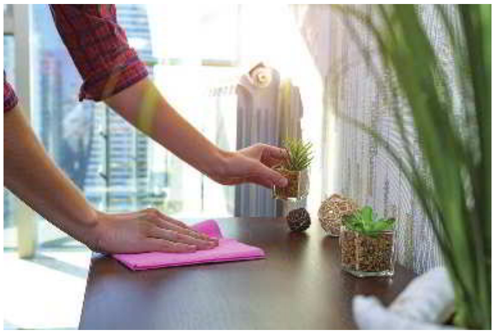
Uninformative Variable Name	Meaningful Variable Name	Description	Type of Data
ar_n	artist_name	Name of artist	Text
al_t	album_title	Album title	Text
p_r	pitch_score	Review score from online music magazine *Pitchfork*	Float
g_n	grammy_nominated	Was the album nominated for a Grammy Award?	Boolean
b_p	peak_chart	Peak charting position on the U.S. *Billboard* 200	Integer
s	sales_us	Number of certified sales in the U.S.	Integer

TABLE 1.3
Data with an uninformatively named variable, `var_1`.

first_name	favorite_pet	var_1
Selma	Dog	0
Jerry	Cat	0
Sanjiv	Dog	0
Olivia	Other	0
Hector	Fish	1

Tidy Data

Tabular datasets are often delivered to us in a format that is not yet ready to work with. While a tabular shape is considered structured, it's important to remember that data can be further structured in preparation for analysis. The creation of summaries, data visualizations, and analyses will require that our data be organized in a consistent, well-defined structure. If our dataset is not originally organized in such a structure, it's referred to as messy data. While unstructured data are unorganized, messy data have been poorly organized. Table 1.4 provides an example of messy data, where the first two rows appear to contain a mixture of column headers and data values, and the third column contains data on more than one variable.



Goffkein.pro/Shutterstock

TABLE 1.4
Messy data.

Class	Sporting	Hound	NonSporting	Herding	
	Age	Name_and_ID	Breed		Class
	Ten	Otis18654	BORDERCOLLIE		he
	Five	Tito20862	DACHSHUND		ho
	Fifteen	Chloe12368	POODLE		ns
	One	Olive56731	BULLDOG		ns
	One	Olive56731	BULLDOG		ns
	Six	Georgie90052	LABRADOR		sp

The process of converting messy data into a form ready for summarizing, visualizing, and analyzing is a large part of data preparation and usually involves both cleaning (discussed later in this chapter) and structuring. At the data analysis stage of a data science project, it's helpful to be working with *tidy data*. With tidy data:

1. Every column is a variable.
2. Every row is an observation.
3. Every cell is a single value.

DEFINITION

Data structuring is the process of organizing data, often into tables.

Tidy data is a data structure where each column represents a distinct variable, each row corresponds to a unique observation, and each cell contains only one value.

Table 1.4 is an example of data that are not tidy. Further preparation may need to be done for us to implement specific visualizations or analyses, but a tidy dataset is well understood and straightforward to work from. Table 1.5 displays a clean, tidy version of the data in Table 1.4, where we now have a column for every variable, a row for every observation, and a cell for every value. Whether data are considered tidy or messy may depend on how a unit is defined. For example, tidying data depends on knowing that `dog_id` and `age` are numerical variables, and `class`, `name`, and `breed` are text variables.

TABLE 1.5

The messy data are now tidy.

dog_id	class	name	age	breed
18654	Herding	Otis	10	Collie
20862	Hound	Tito	5	Dachshund
12368	NonSporting	Chloe	15	Poodle
56731	NonSporting	Olive	1	Bulldog
90052	Sporting	Georgie	6	Labrador

The previous messy data we reviewed obviously needed cleaning. However, some data are messy because they do not fit a purpose for analysis. For example, suppose we observed multiple student performances across different subjects: Math, English, and History. We want to analyze the data in a way that considers each subject a different observation. Table 1.7 is considered messy because it does not adhere to the second principle of tidy data: every row should be a single observation. In other words, each row in Table 1.7 represents multiple observations for the stated purpose: to analyze the data across multiple subjects. It certainly looks tidier than Table 1.6. However, we still want to tidy it more to adhere to the purpose.

Table 1.8 is a tidy version of Table 1.7, with separate columns for `subject`, `score`, and `hours_studied`. Each row in Table 1.8 represents a single observation, i.e., a student's score and hours studied for a specific subject. This structure might make the table easier to analyze and visualize.

Try It Yourself: Tidy Messy Data

Africa Studio/Shutterstock

Take a look at Table 1.6, which shows student performance across different subjects. You might notice that the table is a bit messy! It has displaced column headers and confused values in the rows. However, each observational unit is fortunately within one row, making your task easier. Please tidy this table!

TABLE 1.6
A messy table example.

student_id		Score
	Subject	|
1	Math	85
1	90	Chemistry
2	Physics	88
2	Physics	88
88	Math	2
Match	3	88
3	Chemistry	85

TABLE 1.7
The data are messy because each row is a series of observations.

student_id	math_score	math_hours_studied	english_score	english_hours_studied	history_score	history_hours_studied
1	85	10	90	12	88	9
2	90	11	88	11	90	10
3	88	10	89	12	85	8
4	92	12	85	11	90	9

TABLE 1.8
The messy data are now tidy.

student_id	subject	score	hours_studied
1	Math	85	10
1	English	90	12
1	History	88	9
2	Math	90	11
2	English	88	11
2	History	90	10
3	Math	88	10
3	English	89	12
3	History	85	8
4	Math	92	12
4	English	85	11
4	History	90	9

Try It Yourself: Tidy Data for Better Analysis

Over three days, Coach John rated the athletic performance of camp participants from 0 (low performance) to 100 (high performance) in three different sports. He also recorded the number of minutes they took to run a mile in the morning of the observation, as a measure of their endurance. Specifically, he rated their performance in volleyball on Monday, basketball on Wednesday, and water polo on Friday, each time after recording their endurance. Coach John wants each observation to be a distinct sports performance. A tidier version of Table 1.9 should be created for Coach John's purpose.

TABLE 1.9
The original data are considered messy for Coach John's purpose.

student_jersey	volleyball_rating	volleyball_endurance	basketball_rating	basketball_endurance	water_polo_rating	water_polo_endurance
1	87	7.11	90	7.09	88	7.10
2	97	6.75	88	6.57	90	6.66
3	81	8.11	89	7.75	85	8.01
4	97	5.98	85	6.03	90	6.12

Big Picture Break

We just described the usefulness of data organized in a *tidy* way, and what characteristics make a dataset tidy. Many of the data preparation techniques discussed in the next section, and in future chapters, are especially useful for getting *messy* data into a *tidy* format. However, it should be noted that there are many formats that a structured dataset can be in that are consistent, well-defined, and *not technically tidy*. In the end, it is often the case that the intended uses of a dataset help drive the formats it needs to take. For example, one structure may be especially useful for a particular visualization, while another structure may be better suited for certain modeling or analysis methods. People doing data science work with a dataset should be familiar with multiple techniques for wrangling a dataset into the shapes needed for different types of visualization, analysis, and storytelling.

Metadata

It is extremely common that the data you are working with will require their own description and documentation. These descriptions are not about the values or observations within your dataset, but about the dataset itself. These data about your dataset are appropriately called *metadata*.

Quick Tip

Metadata are almost always the first things we want to know about a dataset before actually looking at any of the data itself. Think of metadata as the information you would want someone to tell you when they hand you a new dataset, so that you can start work quickly and easily.

DEFINITION

Metadata are information that describe and provide context about a dataset.

The codebook, or data key, mentioned in the previous section, is a common place to store important metadata. To this end, the following characteristics are important pieces of metadata:

- Variable names in the dataset.
- Description of the variables in the dataset.

- The type of data stored in each variable in the dataset.
- The coding scheme (if the variables are coded).
    - For example, in Table 1.3, we need to understand what a value of 0 signifies and what a value of 1 represents in the `var_1` variable.
- The units of measurement for each of the variables in the dataset.
- The date the dataset was created.
- The author and/or sources of the dataset.
- The number of missing values.

This information should be stored in a file accompanying the dataset, wherever it resides, and not in the dataset file itself.

Try It Yourself: Investigate Metadata

Dataset: Titanic-Dataset

Consider Table 1.10 with the metadata for a dataset about survivors of the Titanic's historic shipwreck on April 15, 1912.

TABLE 1.10

From a dataset provided by M. Yasser H., accessed at Kaggle.[1]

variable_name	description	data_type	coding_scheme	unit_of_ measurement	missing_values
passenger_id	Unique identifier for each passenger	integer	-	-	-
survived	Survival status	boolean	0 = Did not survive, 1 = Survived	-	-
p_class	Ticket class	integer	1 = 1st class, 2 = 2nd class, 3 = 3rd class	-	-
name	Name of the passenger	text	-	-	-
sex	Sex of the passenger	text	-	-	-
age	Age of the passenger	float	-	Years	177
sib_sp	Number of siblings/spouses aboard	integer	-	Count	-
par_ch	Number of parents/children aboard	integer	-	Count	-
ticket	Ticket number	text	-	-	-
fare	Passenger fare	float	-	Currency (likely Great British Pound or GBP)	-
cabin	Cabin number	text	-	-	687
embarked	Port of embarkation	text	C = Cherbourg, Q = Queenstown, S = Southampton	-	2

[1] Yasser H., M. (2022). Titanic Dataset. Kaggle. https://www.kaggle.com/datasets/yasserh/titanic-dataset/

(Continued) ▶

(*Continued*)

Question

Fill in the blank: Why is metadata important when analyzing a dataset, like the Titanic passenger list?

Metadata helps us understand what the data mean. For example, we understand that 'Survived' coded as '1' means ________________, which is crucial for analyzing survival patterns in the Titanic dataset.

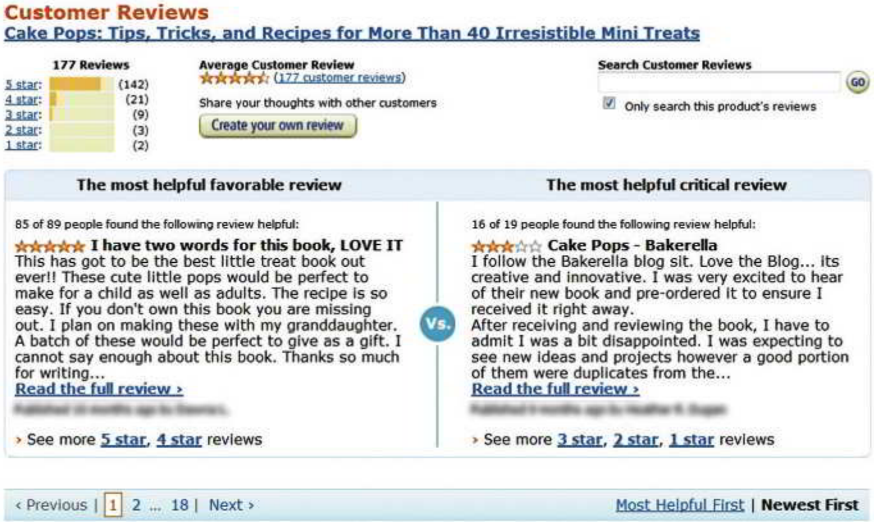
SECTION 1.3: DATA PREPARATION

Data come from many sources in different formats. As we just saw, they may be quantitative—using numbers—or qualitative—using words, photos, or graphs. They may be stored on a computer, such as in a database of demographic data organized by zip code, downloadable from a government server, or accessed "live," such as real-time data from a stock exchange. They may come ready for analysis, like a dataset of media consumption patterns purchased from a data provider like Nielsen, or they may need to be cleaned before use, as with the extraction of social media posts.

We will see in the discussion of data wrangling below that *data cleaning* is different from *data tidying*, as discussed earlier in the chapter. Data can be structured, such as those found in spreadsheets, or unstructured, like the text found in Amazon reviews (Figure 1.4).

FIGURE 1.4

A sample Amazon product review.



Amazon.com, Inc.

In any form, recall that unstructured datameans the data are not organized in predefined data formats and, therefore, are not yet useful for analysis. Note that structured dataare the opposite: they are organized in predefined formats and fit nicely into relatable frameworks like tables or spreadsheets, many of which are ready for analysis. Extracting meaning from unstructured data is more difficult. For example, grouping Amazon reviews by the same geographic region would be challenging before the data are structured. In addition to structuring data, a data scientist must prepare it by *cleaning* it to integrate the data with other data sources.

Data preparation has three parts:

- data collection
- data wrangling
- data management.

DEFINITION

Raw data are data in the form originally collected that have not been processed for use in analysis, storytelling or other purposes.

Data collection is the systematic gathering of relevant data from various sources.

Data wrangling, also known as **data preprocessing**, is the process of converting raw data into a more usable format. This is achieved by cleaning, transforming, and integrating the data to facilitate effective analysis and storytelling.

Data management is the storage of data to make it available for analysis.

Data Collection

Abhishekspadmanabhan/123RF

Data collection is the systematic process of gathering relevant data from various sources such as databases, websites, surveys, or logs recorded in scientific experiments or field research. The quality and relevance of the data collected form the bedrock of our analyses and subsequent insights. Even the most advanced data analysis cannot compensate for low-quality or irrelevant data. Therefore, it's important that we contemplate the specifics of what data are needed, the sources from which they can be procured, and the method of collection to be employed.

We can collect data from a variety of sources, and these sources often depend on the field of study. For example, suppose a wildlife conservation group wants to study the movement patterns and behaviors of birds in a particular wetland. They decide to collect data by:

- Tagging a sample of birds with small tracking devices to record GPS locations over time.
- Installing motion-sensor cameras throughout the wetland to capture images of the birds.
- Making visual observations from viewing blinds and documenting behaviors such as feeding, nesting, and interactions.
- Recording audio of bird calls and analyzing their characteristics and frequency.
- Collecting samples of plants and insects in the habitat, and noting the birds associating with each food source.
- Acquiring regional weather and seasonal data to associate environmental factors with activity.
- Partnering with other researchers to obtain existing observational datasets on similar species in nearby wetlands.

Creativity is often required to gather relevant data when not working with traditional transactions and metrics. We discuss data collection further in Chapters 6 and 7.

After data collection, our data don't just sit there—they undergo a makeover. In the data wrangling stage, we clean and transform the data, preparing it for its big moment: the analysis. We might need to clean the data, deal with any errors, or address missing values, to ensure our analysis can accurately reflect what the data are trying to tell us.

Data Wrangling: Clean, Transform, and Integrate

Data wrangling, also known as data preprocessing, is the process of converting raw data into forms necessary for analysis and storytelling. Specifically, wrangling data involves cleaning, transforming, and integrating data to make them useful for analysis and storytelling.

DEFINITION

Cleaning data involves fixing or removing incomplete, abnormal, or inconsistent data to enhance their quality for accurate analysis and storytelling.

Transforming data involves organizing and formatting data into a structure that is useful for efficient analysis.

Integrating data is the process of merging multiple data sources to provide comprehensive insights that surpass what each individual source could offer alone.

Quick Tip

Even though data cleaning is distinct from data tidying, they are both critical pieces of data preparation. Many forms of visualization and analysis require data to be in a tidy format.

Step 1: Clean

Cleaning deals with data that are incomplete, abnormal, or inconsistent. We clean data to ensure that they are of the highest quality possible before analyzing them and presenting our findings. Cleaning data is distinct from tidying data: cleaning improves data quality, while tidying optimizes their structure.

Cleaning data can be a complicated process that requires judgment calls. The best strategy for a given dataset that needs cleaning may depend on the industry or the situation. However, it's always important to report any adjustments that were made to your data during the cleaning process. We discuss some best practices of data cleaning in Chapter 2.

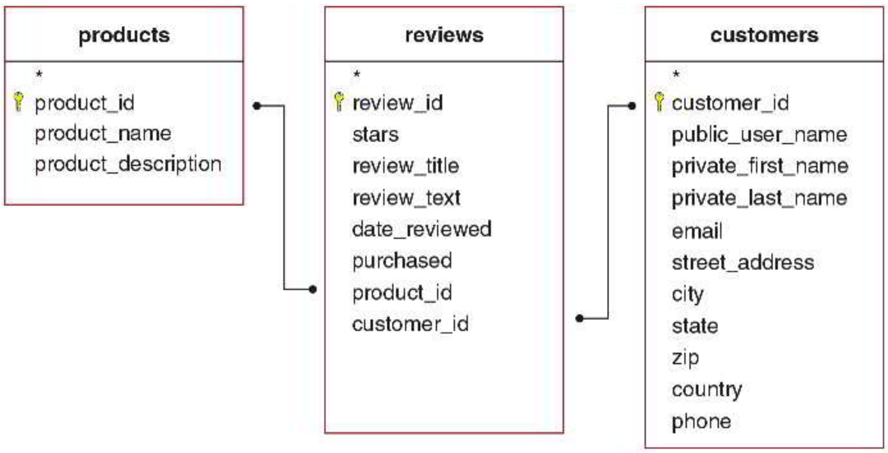
Step 2: Transform

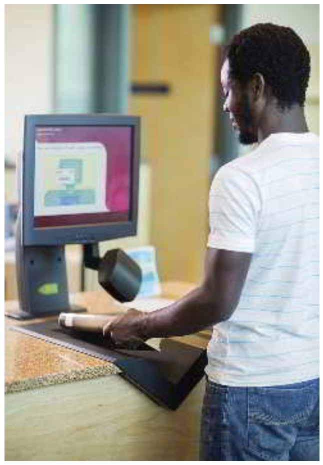
MK photograp55/Shutterstock

As mentioned, we often need to structure data when it comes to us in ways that would otherwise be difficult to analyze. *Data transformation* is the process of organizing and formatting data to facilitate efficient analysis. For example, consider a database of blog entries from the past ten years. How would we analyze those millions of unstructured blog entries? Would we only include text written by the blog owner? What about responses? How would we explore everything? It would be difficult because there needs to be a structure relating responses to original posts.

Unstructured data do not reside in traditional, organized databases (such as a collection of related spreadsheets). Instead, they have an internal structure (e.g., the name of a blog's author with her product review). Figure 1.5 is an example of transforming the unstructured data from Figure 1.4 into structured data. The *Cake Pops* book, the reviews of which are shown in Figure 1.4, would be one entry in the Products table in Figure 1.5, along with all the other millions of products sold by Amazon.

FIGURE 1.5
Transforming unstructured Amazon reviews into structured tables, connected by identification variables such as `product_id` and `customer_id`.





Tyler Olson/Shutterstock

Similarly, the two reviews we see in Figure 1.4 would be two entries in the Reviews table in Figure 1.5, which could contain tens or hundreds of millions of entries. The two customers who wrote these reviews would be two entries in the Customers table in Figure 1.5, which could also contain millions of entries. We acknowledge that Amazon's data management methods are more sophisticated, handling billions of daily transactions in the cloud.

Unstructured data may be generated by humans interacting with machines. A turnstile counter at an amusement park is an example of humans interacting with a machine. There can also be machine-to-machine communication, sometimes called M2M. For example, automated weather monitors inform an energy company's power plants about projected electricity demand and productivity.

Unstructured data are diverse. They may include personal messaging such as email, instant messages, and chats; business documents like reports, presentations, and survey responses; web content such as site pages, blogs, wikis, audio files, photos, and videos; and sensor output like scanner transactions.

Sometimes, there is a trade-off between efficiency and organization. In such cases, the data scientist may structure the data by organizing them in whatever convenient format works for the analysis needed. For example, large amounts of big data accessed in real time may need to be processed quickly. It may be impractical to force the data into a structure because of the time and storage space limitations. Instead, the data scientist would analyze the data *without* structuring it. On the other hand, structuring even large amounts of data may be worthwhile if the data are to be deeply analyzed and integrated with other data sources to discover patterns.

It's often the case that a structured dataset can be further structured, or prepared for analysis, by transforming some of the existing data. For example, a spreadsheet showing reported crimes with a single column for the date of each report is already well-structured. However, it may be of interest to separate the day, month, and year into their own columns for visualization and analysis. Such transformations fall under the umbrella of data wrangling.

Step 3: Integrate

Still AB/Shutterstock

Once a dataset has been cleaned and structured, it is ready for analysis. However, integrating it with other datasets often offers essential insights. *Data integration* involves combining two or more datasets to provide more insights than each dataset would yield individually. Integration is crucial for publicly available data, where analysts have likely already harvested insights. When data are easily obtained from a single, publicly available source, the competitive advantage of the information by itself is low.

We can typically find unique insights by integrating one data source with another, especially if the second data source is proprietary (i.e., not public). Suppose an umbrella company considers a list of keyword search frequencies for "umbrella" within each U.S. state over particular time periods, from a data source like Google Trends. A competitor could easily find these data too. The value of the keyword search frequencies for competitive insights is limited. However, after integrating this dataset with internal sales information and external weather

Try It Yourself: Illustrate Unstructured Data Using Structured Tables

Consider the related tables in Figure 1.5, based on the Amazon reviews in Figure 1.4. Please fill in the blanks.

1. "Cake Pops" from Figure 1.4 is likely stored in the _________ field of Figure 1.5's _________ table.
2. "Cake Pops - Bakerella" from Figure 1.4 is likely a record in Figure 1.5's _________ field in the _________ table.
3. To find all reviews for a specific product on Amazon, the _________ field in Figure 1.5's **`reviews table`** is used to link to the corresponding product in the **`products table`**.
4. To identify the customer who wrote a review on Amazon, the _________ field in Figure 1.5's **`reviews table`** is used to link to the customer's record in the **`customers table`**.

data, it would be possible to model the predictive value of keyword searches, and to use that data to estimate expected umbrella sales across states at different times of the year.

We will delve more into data wrangling in Chapter 2 and data management in Chapter 5. However, next, we will briefly discuss the place of data management within data preparation, alongside data collection and data wrangling.

Data Management

Data management is the process of storing data and facilitating its access. After collecting and wrangling data, we often want to store them in a tabular form within a secure database. When data are integrated from multiple sources, we may connect multiple tables with each other. We also want people to access our data efficiently, ensuring the right information is available when needed. When executed effectively, data management allows us to store raw data as a meaningful resource for in-depth analyses in various academic and industry contexts.

SECTION 1.4: DATA ANALYSIS AND STORYTELLING

Data Analysis

Data analysis is the process of examining datasets to uncover useful information, which can inform conclusions and support decision-making. It involves applying statistical, logical, and mathematical techniques to sift through data, identify patterns, trends, and relationships. This phase is critical in data science because it translates data into insights that can drive informed actions and strategies.

Exploratory Data Analysis (EDA) is a preliminary step in the data analysis process to explore and examine datasets. Data analysis can be broken into four main categories: descriptive, diagnostic, predictive, and prescriptive analysis. As the first step, exploratory data analysis often focuses on descriptive and diagnostic statistics but can sometimes adopt predictive and prescriptive analysis as well.

DEFINITION

Descriptive analysis systematically examines current and historical data, utilizing various benchmarks, to provide a detailed view of past and present outcomes.

Diagnostic analysis investigates data to determine the possible causes and reasons behind past events.

Predictive analysis uses statistical methods to predict future events based on historical data.

Prescriptive analysis provides recommendations on actions to take.

Ruth Black/Shutterstock

Descriptive analysis examines what has happened in the past or what is happening at the present moment, based on current data. For instance, we could use the data represented in Figure 1.4 to determine the average number of stars the *Cake Pops* book receives compared to the average ratings of products in the "dessert books" category. We might find that our *Cake Pops* book has an average rating of 4.68, while the average rating for all dessert books is higher, at 4.81.

Diagnostic analysis goes beyond asking *what* happened to *why* it happened. For example, we could ask whether our *Cake Pops* book's average ratings are lower than all dessert books because of the reviewers' geography. We might discover that half of the *Cake Pops* book ratings come primarily from a single region that is particularly tough on dessert books in general. Within this single region, the *Cake Pops* book's average rating is 4.58, while the average rating for all dessert books is also lower than the global average at 4.61. Outside that region, the *Cake Pops* book's ratings are much higher at 4.81, equal to the category's global average rating. These data would help the *Cake Pops* authors understand why their book has lower average ratings. In other words, diagnostic analysis reveals that they may be focusing too much on selling to customers from a region that does not appreciate dessert books, and they can adjust their marketing strategy based on that information.

Predictive analysis seeks to determine what is likely to happen in the future by estimating future behavior based on past patterns. For instance, we could predict next month's average customer review ratings based on past seasonal data. These data show that reviews for the product category of dessert books tend to worsen from January to July and improve from August to December. Data scientists could also create a mathematical formula, known as a "predictive model." This model uses machine learning to identify which regions are most likely to appreciate the *Cake Pops* book, based on other facts about the customers and product.

Prescriptive analysis asks what actions should be taken, building on predictive analysis. For example, marketers could use predictive models based on seasonality and geographical region to recommend the action of a promotional campaign in the more advantageous months of August to December, in the particular areas more favorable to dessert books during those months. In other words, prescriptive analysis would optimize seasonality and location data for good reviews.

Try It Yourself: Identify Analysis Types

Consider four simple analyses related to the *Cake Pops* book ratings, and identify each as either descriptive, diagnostic, predictive, or prescriptive:

- A targeted marketing campaign in regions with a higher appreciation for bakery books should increase *Cake Pops* book ratings by 4%.
- Customers who bought at least two other bakery books rated the *Cake Pops* book 4.71 on average, which is higher than the 4.55 average rating from customers who only bought the *Cake Pops* book. This indicates that a general interest in bakery played a role in its ratings.
- Based on current bakery book review trends, the average rating of the *Cake Pops* book is projected to decrease by 0.5 stars (half a star) in the next year.
- The *Cake Pops* book has an average rating of 4.68, whereas the overall average rating for general bakery books is 4.79.

Data Storytelling: Visualization, Narrative, and the STAR Framework

Data storytelling is the translation of complex data preparation and data analysis into a narrative that can be easily understood by an audience not necessarily trained in data science. Many data scientists focus on gaining data and analysis skill sets, but eventually, they must learn how best to communicate their findings. However, we approach the concept of storytelling cautiously. While it is certainly not the creation of a fictitious story, it is also not sufficient to tell a true story without presenting evidence based on the hard work of preparation and analysis. It is also insufficient to omit the complex assumptions and decisions that data scientists must make to obtain their results. It is crucial that any data scientist who hears the story can replicate it themselves based on the description of the process. Storytelling supports decision-making by succinctly communicating the data scientist's highly sophisticated mathematical, statistical, and computer science methods. We tell stories with data through visualization and narrative. We offer the STAR Framework as a helpful aid when communicating data results.

DEFINITION

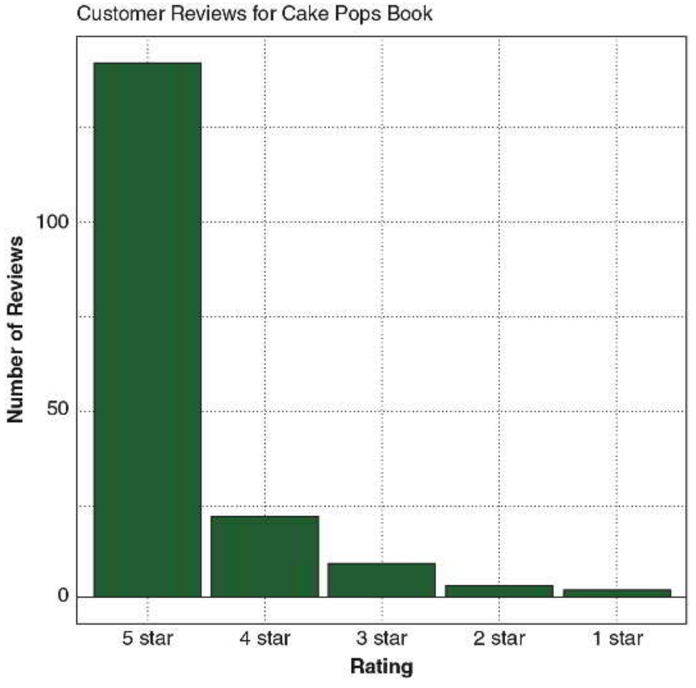
Visualization is the presentation of data in a graphical format.

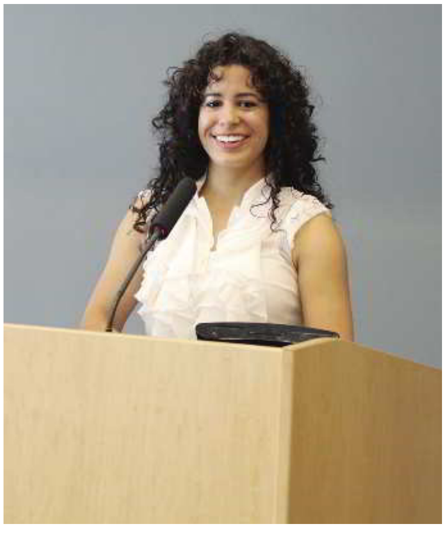
A **storytelling narrative** in data science is the art of presenting data in an accessible and engaging manner.

Visualization

Visualizationis a meaningful way to represent information concisely in a graphical format, facilitating storytelling. For example, Figure 1.6 shows the distribution of customer reviews for the *Cake Pops* book from Figure 1.4 in a visual format. This communicates that the book has received an overwhelming number of five-star reviews.

FIGURE 1.6
Visualization shows the number of customer reviews for the *Cake Pops* book.





Val lawless/Shutterstock

Learning data science requires practice in choosing chart types, making charts, and interpreting charts. In addition to the focus on data visualization in Chapter 3, we will see visualization examples throughout many chapters. This is because visualization is central to effective data preparation, analysis, and storytelling.

Narrative

The best data preparation and analysis are only helpful if communicated effectively to the right people. A well-crafted narrative makes information more accessible to a diverse audience, including scientists, managers, farmers, artists, scholars, and policymakers. Therefore, we also aim to enhance both spoken and written presentations. The ultimate goal of data science is to tell a story that aids people in making better decisions in any field. For instance, geneticists can make more informed decisions about medicine, historians can interpret digitized documents more accurately, farmers can identify which crops will maximize sustainable cultivation, and ecologists can better manage wildlife populations. The art of storytelling requires presenting multiple facets of data in a manner that most people can understand.

STAR Framework

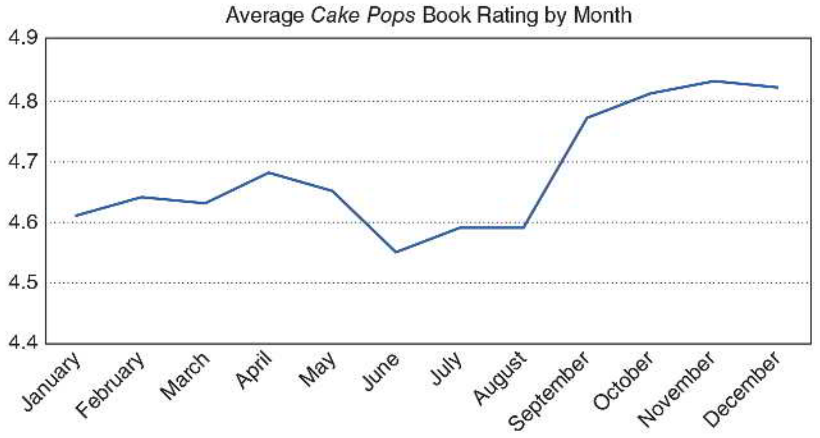
We introduce the STAR Framework as a structured approach to communication in data storytelling. Each component of STAR serves as a step towards crafting a compelling narrative that not only presents data but also contextualizes it and drives action. Here's how each part of the STAR Framework functions:

1. **Scope the Data's Context**: This initial step characterizes the data and their context, including the observational units, variables and their types, and data source.
2. **Track the Question Behind the Exploration**: Here, we identify the questions and core tensions we hope the data will address.
3. **Articulate the Results**: In this phase, we clearly explain the results.
4. **Respond with Next Steps**: The final component encourages potential actions or investigations.

Try It Yourself: Optimize the Narrative

Consider the following visualization that draws a line between the average *Cake Pops* book rating for each month of the year. Answer questions about the visualization based on Figure 1.7.

FIGURE 1.7
Average Amazon ratings of the *Cake Pops* book by month over the past three years.



1. Which statement best describes Figure 1.7 in terms of months and seasons (e.g., fall, winter, spring, and summer)?

   The average *Cake Pops* book rating . . .

   A. is unrelated to the time of year.
   B. changes with the seasons.
   C. increases each month.
   D. changes each month unrelated to the seasons.

2. Which statement about the *Cake Pops* book is most likely, given the data in Figure 1.7?

   The *Cake Pops* book features . . .

   A. month-themed cake pop recipes.
   B. a new cake pop recipe each week.
   C. cake pop recipes for each season of the year.

(*Continued*) ▶

(*Continued*)

D. cake pop recipes for each season emphasizing holidays in September, October, November, and December.
E. cake pop recipes for each season emphasizing December holidays.
F. cake pop recipes for each season emphasizing spring holidays.

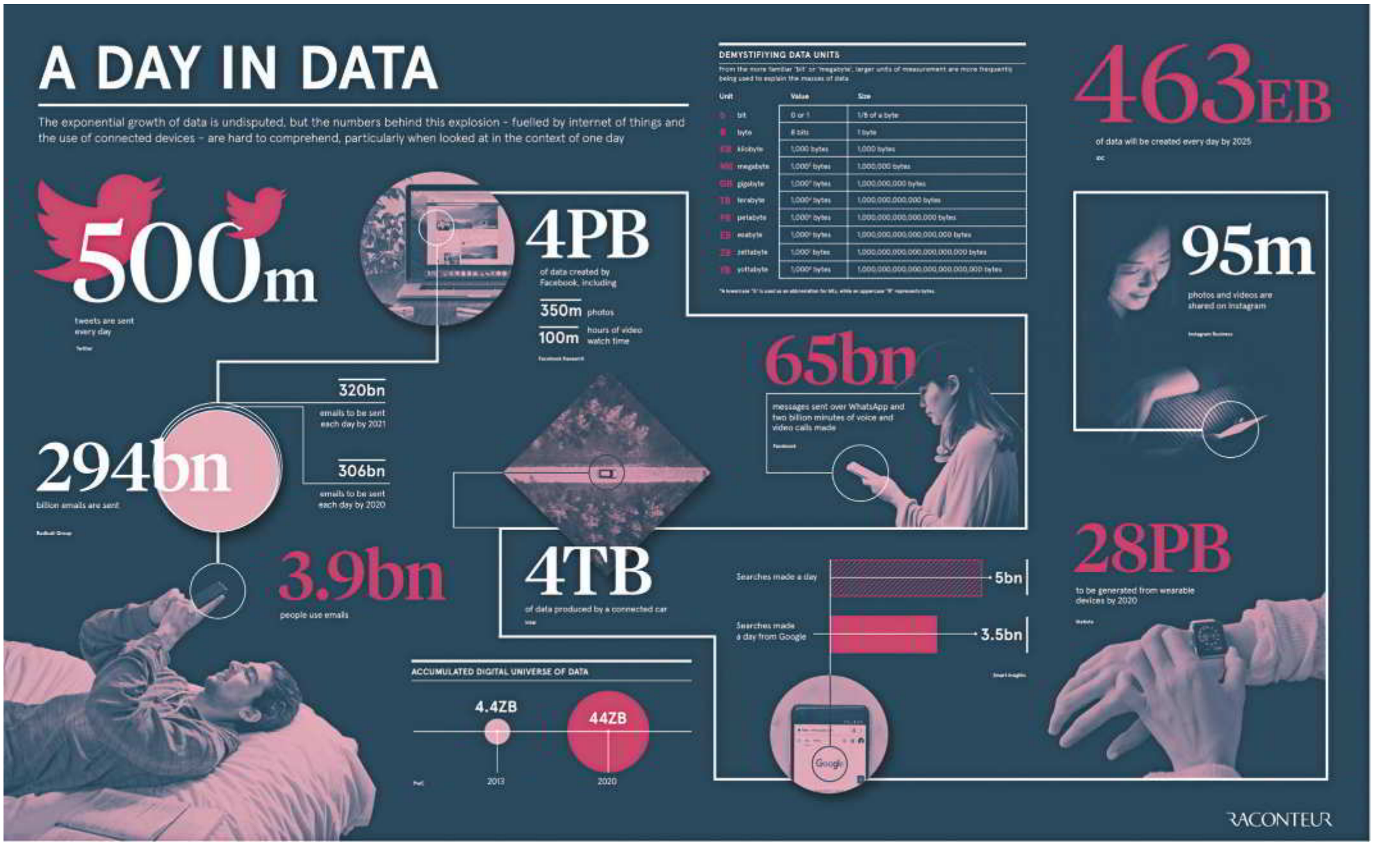
3. Open-Response Question: Use the STAR Framework to tell a story about Figure 1.7.

SECTION 1.5: DATA SCIENCE IN SOCIETY AND INDUSTRY

The Growth of Data Science

Data science is a growing field due to previously unimaginable amounts of available data. The number of people engaged in exploring and learning from these data is not keeping pace with this growth. The amount of global data is staggering (Figure 1.8). Living in the digital age, we can access new kinds of data that were rarely seen in previous decades.

FIGURE 1.8
Global data are increasing exponentially. From Jeff Desjardins, Visual Capitalist.



Jeff Desjardins, How Much Data Is Generated Each Day?, Visual Capitalist, April 15, 2019, https://www.visualcapitalist.com/how-much-data-is-generated-each-day/

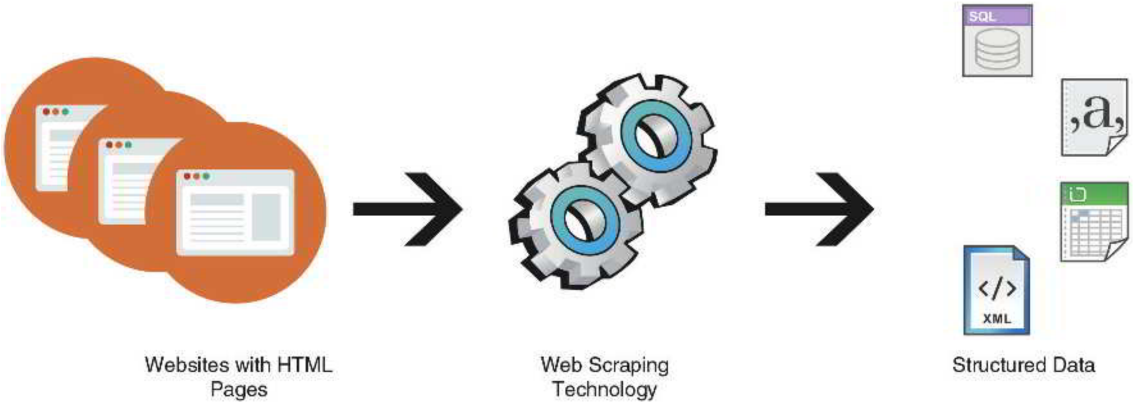
Consider how the following data sources could benefit people:

- Amazon reviews assist customers in making more informed purchase decisions. For instance, review data aids firms in improving their products and services for the benefit of customers.

Olegdudko/123RF

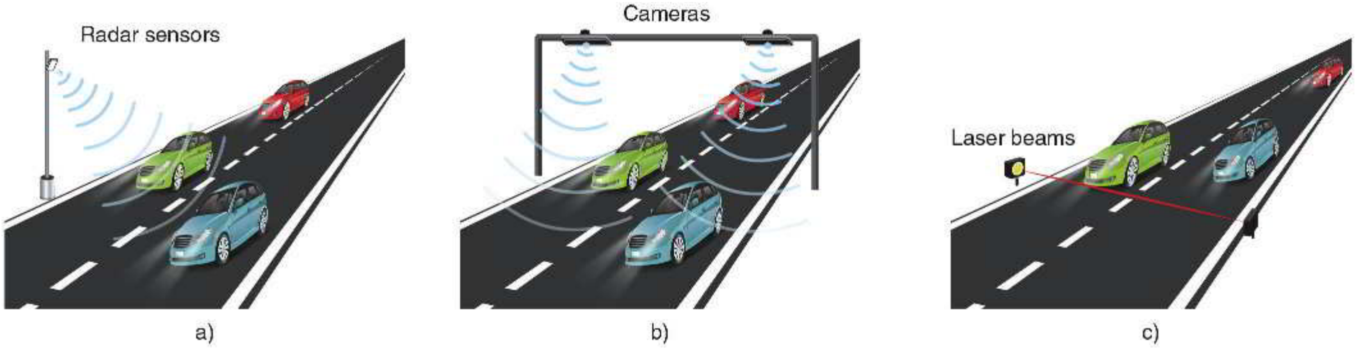
- Social scientists monitor videos featuring cultural trends found on social media. For example, a consulting firm performs video analysis of people wearing yellow for World Mental Health Day in videos posted on TikTok.
- Impactful technological innovations from online forums provide business insights. For example, a start-up company searches online forums devoted to new uses of artificial intelligence in household products, like Amazon Alexa.
- Web-scraping, the automated extraction of information from public websites, supports the study of technological trends. For example, a professor monitors technology review websites to teach students about the latest product offerings (Figure 1.9).

FIGURE 1.9
Web-scraping translates HTML from thousands of websites into useful data tables.



- Machine-generated data influence public-sector administration. For example, a city planner analyzes changing traffic patterns from road sensors (Figure 1.10).

FIGURE 1.10
Radar sensors, cameras, and laser beams generate data to estimate traffic problems.



Data Science Everywhere

Irabel8/Shutterstock

We don't have to be in the workplace to experience data science. Data science affects nearly every part of our lives.

Do you know anyone who looks at their Fitbit or smartwatch for anything other than discovering the number of steps they've taken that day? These wearable devices can generate vast amounts of data. Data scientists and other public health researchers are figuring out how to use these large but simple accelerometer data to predict a user's activity at any given moment. These health professionals may be able to apply this new source of data to improve health by monitoring the activity levels of Fitbit wearers. It turns out that people are not very good at self-reporting how much time they spend each day doing physical activities. Therefore, if medical professionals can take advantage of wearable monitor data volunteered by at-risk individuals, there is hope that health outcomes can be improved.

Have you ever been distracted by the sounds of cars or trains traveling near your home? Now, imagine this, but many times worse, and in the ocean! Sound travels faster and more easily in the sea than in the air. Therefore, locating noise-generating coastal roads and mapping shipping routes to minimize the negative impacts on ocean wildlife present challenging tasks. The ocean is vast, though. So, how do we know where creatures are in order to avoid them? Because various shapes and sizes of sound vibrations traveling through water are detectable at great distances, many tools can be used to collect and analyze acoustic data. Researchers use special microphones to listen for sounds made by wildlife, and other tools like sonar can be used to assess what's under the water in real time. Regardless of the technology, these acoustic data must be prepared, analyzed, and communicated to inform what we do in the ocean.

The *scientific method* involves posing a question, making a hypothesis, conducting a test with an experiment, and analyzing the resulting data to draw conclusions. Therefore, by definition, anyone conducting science or testing hypotheses will be doing some form of data science through their work with data to investigate their question of interest, as we saw in the previous two situations.

Today's unprecedented plethora of data has often been dubbed the "data deluge," and for a good reason. While the number of people doing data science grows daily, everyone is a consumer of data science through news, social media, pop culture, etc. To this end, *Data Science for All* aims to equip everyone with the literacy to be a competent consumer of data science and a few tools to start doing data science.

Data's Importance to Business and Industry

Drazen Zigic/Shutterstock

Data science is in high demand. In a recent global survey by Gartner, over 400 leaders revealed their highest-priority technology opportunities for growth. Data science is the primary tool for addressing these opportunities. In particular, data science identifies areas ripe for disruption.

For example, data show marketers which product categories have responded well to past disruptions and reveal which related categories still need to be disrupted. Similarly, data science helps firms understand how best to engage their customers. For instance, data may immediately show which strategies—such as contests, news, or giveaways—engage more customers for fewer dollars. In summary, data science applies technology to provide a deeper understanding of issues critical to organizations' missions.

Job expectations are increasingly integrated with data science. *Data Science for All* reveals the importance of data science for professional roles that may seem to have little use for quantitative analysis. For example, graphic designers must understand the power of A/B testing to validate their designs. Copywriters are interested in ways to use social listening to inform their content strategies. Journalists should understand how a machine learning model might help them design targeted narratives for various audiences. Customer service representatives need analytics to assess the success of the interactive strategies they practice. Professional sellers may focus on predicting sales leads. Public relations managers may analyze social and news media to participate in and exchange information with the public.

Likewise, traditionally, quantitative jobs are also more integrated across data science domains. Rather than hiring analysts with deep expertise in a narrow area, companies seek data professionals with a broad understanding of how data interact with fields such as agriculture, architecture, biology, business, history, journalism, etc. Today's data scientists need to know how their analyses and storytelling affect the qualitative aspects of the workplace. Equipped with these competencies, they can design enduring employee experiences, create engaging media content, craft effective public communication, produce fantastic new goods and services, and make the world a better place.

Case Study: Amazon Uses Data for Customers, Ads, and Fraud Prevention


Elpisterra/Shutterstock

Amazon uses data science to analyze customer purchase patterns, tailor its ads, and audit fraudulent activity. Sometimes, it feels like Amazon knows what we need even before we do. Amazon manages millions of purchases daily without direct personal contact. It does this with three specific strategies: dynamic pricing, product recommendations, and fraud protection.

Dynamic pricing is a strategy in which companies continually change the price of a product based on the data they collect from customers, estimating how much each customer is willing to pay. Amazon, for example, changes product prices approximately 2.5 million times daily. This means the cost of a typical product shifts every ten minutes. Thanks to the company's massive amounts of data, it can analyze factors ranging from competitor pricing to available inventory. It then uses sophisticated algorithms to make informed decisions about pricing.

Customers benefit when Amazon configures their product recommendations based on items already in their carts, products they've purchased before, and products they have browsed. Any action a customer takes on their website, any click—Amazon will use that data. They can learn what each customer wants and likes and can recommend appropriate products. Product recommendations are attributed to 35% of Amazon's annual sales.

Amazon protects itself and its customers from retail fraud. It collects more than 2000 historical and real-time data points on every order and then uses machine learning algorithms to identify transactions with an elevated likelihood of being fraudulent. Additionally, the company uses data science to tweak algorithms, enabling it to scrutinize suspicious return requests. Therefore, if the data shows that a person has returned too many items or reveals other unusual patterns, the company is notified to investigate further.

"Data Science for All" implies that everyone should understand how data science impacts their areas of interest. These skills can be applied to almost every job and academic field. Moreover, some readers of *Data Science for All* may discover that they are interested in pursuing a career in data science. Companies appreciate employees who can serve as intermediaries between the qualitative and quantitative aspects of business, fostering understanding between high-level business managers and information systems specialists. Data science values curiosity, creativity, strategy, and insight in all careers. While data science does not necessarily require the full expertise of a computer scientist or statistician, we aim to impart the ability to formulate meaningful data queries, interpret data science results, critique its methods, and make sound recommendations based on data science output.

Chapter 1: Putting It Together

Data science is a new world made possible by the digital age, where data are rich and plentiful. We explored data science's multidisciplinary applications and overall lifecycle. We also navigated through different data types, preparation pipelines, methods of analysis, and the societal impact of data science.

Interpret the Definition and Scope of Data Science

- Discussed the interdisciplinary nature of data science.
- Emphasized its applicability across various fields, such as healthcare, finance, and social sciences.
- Outlined the typical lifecycle of a data science project, from data collection to insights.

Recognize the Need for Data to Be Organized as a Tidy Table

- Explained how the source and form of data influence the subsequent analysis.
- Recognized tidy data resides in tables where every column is a variable, every row is an observation, and every cell is a single value.

Distinguish between Different Data Types

- Examined the different uses of data, such as quantitative and categorical.
- Highlighted the importance of understanding data types for effective data preparation and analysis.

Describe a Coherent Data Preparation Pipeline

- Explored various data preparation techniques and their sequence in a typical pipeline.
- Provided insights on assembling an effective pipeline tailored to specific data types and formats.

Evaluate Various Types of Data Analysis

- Covered the four major types of data analysis: descriptive, diagnostic, predictive, and prescriptive.
- Explained how these methods contribute to compelling storytelling and decision-making.
- Illustrated the use of these methods in a real-world scenario (e.g., *Cake Pops* customer reviews on Amazon), to emphasize their practical importance.

Evaluate the Impact and Growth of Data Science

- Discussed the transformative role of data science in society and industry.
- Highlighted its rapid growth and the increasing demand for data science skills.

Now that we appreciate the depth and breadth of data science, we understand the importance of careful data preparation, the diversity of analytical methods available, and the significant impact of data science in the modern world. We are ready to wrangle with data to get it ready for storytelling and analysis.

Chapter 1: Ethics in Practice: Some Risks in Data Science

While data science provides valuable insights, it also carries ethical risks around bias, transparency, and consent that should be carefully considered. A nuanced, conscientious approach to data ethics is just as important as the data themselves. If we can record or measure something, we can collect data about it. And if we can collect data about an entity or process, we can try to optimize it. By keeping ethics front and center, data collectors can gather meaningful insights while respecting human dignity.

Consider the following examples that have likely affected us all:

- Email spam filters are designed to divert unwanted emails from our inboxes.
- University planners predict the number of sections offered in a particular course based on historical and present data.
- Policymakers use aerial pictures of wildlife to estimate population sizes in state and national parks, and other protected areas.
- Medical researchers use the human genome and health-related data to determine factors that decrease disease risk.
- Demographers and political scientists use polls and surveys to collect demographic and public opinion data, then summarize and report on them.

Because it's nearly impossible for us as individuals to perceive the world around us at all possible scales, we rely on other individuals or organizations to summarize the information and data we lack. A fundamental understanding of data science is paramount to trusting what these other authorities provide. Each of us must ask the following critical questions:

- What are the sources of the data I have?
- What are the sources of the data summarized for me?
- Were the data collected ethically?
- Are the data being misrepresented?

As useful as data science can be for understanding the world around us and enhancing our lives, it is not without its shortcomings. Some of these are related to issues of ethics. Two examples are: (1) issues of privacy, and (2) issues of fairness/bias in decision-making algorithms.

A massive amount of the technology and software that we use on a daily basis collects a plethora of data about us; particularly our behavior and usage of these tools (e.g., web browsers, websites, and mobile phone apps). With enough of this information, people and companies can explicitly identify users and uncover their personal information. Some companies may want to use this information to customize a user's web browsing experience, while other entities may have more malicious goals. Even with the former, having your entire web browsing experience potentially controlled by advertisers may not be desirable. Your private life may become much less private.

Machine learning and other data modeling techniques are constantly employed to automate the process of evaluating the risks of individuals and companies as they interact and work together. For example, a small business owner may need a loan from a bank to expand their business. The bank will collect data about this person and evaluate their supposed ability to pay back the loan based on models constructed using historical data. These models can sometimes yield unfair or biased results because they were constructed from data about people in the past that is no longer representative of the people today for whom they're being used. There is an old saying, "garbage in, garbage out," which, in this context, refers to the idea that if your analysis is based on incomplete or poor data, then your conclusions from that analysis can be no better. We need to be mindful of the sources of our data and the groups about whom we are trying to draw conclusions, as well as the possible differences between them, so as not to propagate inequity or unfairness.

Chapter 1: Key Terms

data
data science
population
sample
data science lifecycle
data preparation
data analysis
data storytelling
cloud (the cloud)
observational unit
variable
observation
tabular form
dataset
value
structured data
unstructured data
quantitative variable
discrete variable
continuous variable
categorical (qualitative) variable
text variable
integer variable
float variable
Boolean variable
data structuring
tidy data
metadata
raw data
data collection
data wrangling (preprocessing)
data management
cleaning data
transforming data
integrating data
descriptive analysis
diagnostics analysis
predictive analysis
prescriptive analysis
visualization
storytelling narrative

CHAPTER 1: CHAPTER REVIEW QUESTIONS

Section 1.1: Introduction to Data Science

1. What is the interdisciplinary field that uses scientific methods to extract knowledge from data?
2. What is the primary purpose of data analysis in data science?
3. Define a *population* in the context of data science.
4. Which other field, often confused with data science, applies data science principles to domain-specific problems?
5. What are some of the modern challenges that have shaped data science?
6. What constitutes a sample in data science?
7. In the context of data science, what is the significance of understanding samples and populations?
8. What combination of fields does data science encompass?
9. When was the term "data science" first proposed as an alternative for "computer science"?
10. What is the role of a data scientist in the context of the data science lifecycle?
11. What is the most significant reason for the importance of data and their analysis, according to the text?

Section 1.2: Data in Tables

12. In data science, what are the entities about which data are recorded called?
13. What are the recorded characteristics of observational units called?
14. What term is used to refer to the data for a single observational unit?
15. From the following set of numbers: 0.0, 2.71828, -3, 5, which one is most likely to be stored as a float variable?
16. Which type of variable is used to designate categories?
17. In the context of data science, what does it mean for data to be tidy?
18. What do we call the form when data are represented as a table with rows and columns?
19. Integers and floats are examples of what kind of variables?
20. What kind of variables only take on the values of TRUE and FALSE, sharing aspects of both categorical and quantitative variables?
21. What form of quantitative variables consists only of numbers with no decimal points?
22. Given the numbers 5000, -2.01, 1.3, and $1/2$, which one is most likely to be stored as an integer variable?
23. What form of quantitative variables consists of numbers that may have decimal points?
24. Given the set of numbers 0, 2, 5, and 1.5, which is most likely to be converted to a Boolean value in a data analysis context?
25. What form of categorical variables contains words or characters without order or numerical utility?
26. Given the numbers -1.0, 0.0, 3, and 4, which one could be stored as a float or converted to a Boolean variable?
27. Given the numbers 2.71828, -5, 3.14, and 2.01, which one is most likely to be stored as an integer in a data structure?
28. What is metadata?
29. When are data considered messy?
30. Which of the following numbers is likely to be stored as a float variable in a dataset: 1, 4.44, 144, or 0?
31. When data are in tabular form, what is stored in the columns? What is stored in the rows?
32. When the rows and columns of tabular data are fully populated, and each cell of the table corresponds to a single value, we consider the dataset to be clean and organized. What is the term we use to describe this?
33. Given the numbers 1243424, 1.2, -1.01, and 2.03, which is most likely to be stored as an integer variable?
34. What is the primary role of a data key or codebook?
35. In a tidy dataset, what does each column represent?

Section 1.3: Data Preparation

36. In the context of data science, what does it mean to prepare data?
37. What is the first step in data wrangling?
38. For what purpose is data transformation essential?
39. What is the goal of integrating datasets?
40. In the context of data preparation, what does data wrangling refer to?
41. Describe the process of data collection.
42. What is the importance of data quality and relevance in data collection?
43. What does the term "unstructured data" mean?
44. What is the process of organizing data (often into tables) called?

Section 1.4: Data Analysis and Storytelling

45. What is descriptive analysis in data science primarily concerned with?
46. What is the goal of diagnostic analytics in data science?
47. What is the goal of predictive analysis in data science?
48. What is the component of data storytelling that allows us to communicate statistics and analysis results in form of graphs and charts?
49. Describe prescriptive analysis in the context of data science.
50. What is the purpose of communication in the context of data science?
51. How is data visualization useful in data science?
52. What is the ultimate goal of data science storytelling?
53. What does effective storytelling in data science require?

Section 1.5: Data Science in Society and Industry

54. What is the primary driver for the growth of data science in society and industry?
55. How does data science help businesses engage with customers?
56. In the context of data science, what does the term "data deluge" refer to?
57. According to the text, for whom are data science skills essential?
58. What is the benefit of integrating data science into traditionally quantitative jobs?
59. What does the message "data science for all" emphasize?
60. According to the text, how can an organization's management use data science for decision-making?
61. According to the chapter, what benefits does data science offer to consumers, researchers, and companies?

Chapter 1: Explaining the Concepts

1

When you hear the word "data," what do you think of first? (What did you think of before reading this chapter?) How is this similar to or different from *tidy data*? What are the ways you see or work with data on a day-to-day basis?

2

Now that we've defined *data science*, have you done any data science in your life thus far? Can you envision putting data science to use at your current job or a future job? If so, how?

3

When it comes to data, we have covered two very special terms:

1. Observations
2. Variables

We covered these terms, mostly in the context of data in tables. Take a moment to remind yourself what an *observation* is and what a *variable* is. Because we want to do analysis and visualization with any kind of data we have, the ideas of *observations* and *variables* are important to keep in mind, even beyond tabular data.

- Imagine we want to conduct an analysis of a large collection of TikTok videos to which we've been given access.
  - Give some examples of characteristics, or *variables*, that we might be interested in investigating with this collection of videos.
  - What would you call the *observations* in this situation, and why?
  - Do your best to envision organizing your answers into a table format. Sketch this table out.
- Recall the Amazon Reviews example from earlier in the chapter, and imagine we want to analyze a large collection of Amazon Reviews to which we've been given access.
  - Give some examples of characteristics, or *variables*, that we might be interested in investigating with this collection of reviews.
  - What would you call the *observations* in this situation, and why?
  - Do your best to envision organizing your variables and observations into a table format. Sketch this table out.

Explain the importance of being able to visualize and identify the *observations* and *variables* in a collection of data for the purposes of preparing data for analysis and storytelling.

4

How is understanding data science important for consuming news and interacting with social media?

5

What are some examples of how you provide data to other people, companies, or organizations? To what extent can you control how much data you provide to each of these entities? Is this a good thing or a bad thing?

Chapter 1: Applying the Concepts

Activity: Identify Variable Uses and Types

Part I

Discussing data can seem daunting without the proper terminology to communicate ideas. To help practice the vocabulary discussed in this chapter, answer the questions.

Suppose your professor is interested in discovering if there is any relationship between students' sleeping habits and their final exam scores. To explore this relationship, your professor asks their 30 students on their final day to estimate how many hours they slept the previous night. The professor then records their final exam scores and begins the investigation.

1. In this scenario, what are the observational units? Why?
2. What are the variables?
3. What are the variable uses and types?

Part II

Discussing data can seem daunting without the proper terminology to communicate ideas. To help practice the vocabulary discussed in this chapter, answer the questions.

A doctor is studying a patient with a new, unidentified blood disease. The patient has recurring visits with the doctor to study the disease. At each visit, with the patient's consent (ethics are always a priority!), the doctor records the date, measures the patient's blood sugar (in mg/dL), weighs them (to the nearest pound), asks them whether it is true or false that they have eaten in the past hour, and records the patient's description of their symptoms each time they visit.

The doctor is considering how to store the data. She eventually decides on a tabular format.

1. In this format, what is stored in the rows?
2. Knowing your answer to question #1, what data would the doctor use to populate the rows?
3. What is stored in the columns?
4. Knowing your answer to question #3, what data would the doctor use to populate the columns?

The hospital's data science team is requesting information about how the doctors would like to store the data they collect. They are asking the doctors to indicate the variable type for each of the fields, as they are unsure about some of them. Please enter the appropriate variable type for each item.

1. Blood sugar:
2. Weight:
3. Eaten in the last hour:
4. Symptom description:

Activity: Tidy Messy Data

A team of doctors is studying patient symptoms. They recorded patient data, with their permission, into notes and asked an intern to convert the data into a table. Unfortunately, the intern, a student who hasn't read "Data Science for All," is unfamiliar with the principles of tidy data and created the following table:

TABLE 1.11
Messy data.

			eaten_in_the_last_hour	**symptom_description**
	date			Dizziness, migraines
	2022/08/02		TRUE	Dizziness, excessive hunger
	2022/08/16		FALSE	Low energy, lethargic
	2022/08/30		TRUE	Migraines, lethargic
	2022/09/13		FALSE	
		blood_sugar		**weight**
	75.3 mg/dL	2022/08/02		154.3 lbs
	77.2 mg/dL	2022/08/16		155.2 lbs
	74.5 mg/dL	2022/08/30		155.1 lbs
	73.2 mg/dL	2022/09/13		153.2 lbs

Help fix Table 1.11 by transferring elements from the disorganized table above into a new, tidy table.

Activity: Tidy Data for Better Analysis

Last year, Chef Emily, a cooking class instructor, rated the cooking skills of four students over three days. Each day, the students prepared a different dish (pizza, pasta, and cake), and Chef Emily rated the students' cooking skills from 0 (poor) to 100 (excellent) for each dish. Additionally, she recorded the time it took each student to prepare the dish (in minutes), as a measure of their efficiency.

In the original table (Table 1.12), each row corresponds to a single student, combining multiple observations for different dishes. However, Chef Emily wants each line to display just one student making one dish, along with that dish's cooking skill score and the time it took them to prepare it. Please create a new table for Chef Emily. Since there are three dishes, the new table will consist of 12 rows (three dishes for four people). The new table will have fewer columns, making it easier to understand and allowing for quick comparisons of how students performed with different dishes.

TABLE 1.12
Original table format.

student_id	pizza_skill	pizza_time	pasta_skill	pasta_time	cake_skill	cake_time
1	78	40	82	35	75	60
2	85	37	88	30	90	55
3	65	45	70	40	80	50
4	90	35	95	25	85	45

Activity: Investigate Metadata

Consider the metadata of a dataset in Table 1.13, which is about the survivors of the Titanic's historic shipwreck on April 15, 1912.

TABLE 1.13
Titanic survivors data by M. Yasser H., Kaggle.

variable_name	description	data_type	coding_scheme	unit_of_measurement	missing_values
passenger_id	Unique identifier for each passenger	integer	-	-	-
survived	Survival status	boolean	0 = Did not survive, 1 = Survived	-	-
p_class	Ticket class	integer	1 = 1st class, 2 = 2nd class, 3 = 3rd class	-	-
name	Name of the passenger	text	-	-	-
sex	Sex of the passenger	text	-	-	-
age	Age of the passenger	float	-	Years	177
sib_sp	Number of siblings/spouses aboard	integer	-	Count	-
par_ch	Number of parents/children aboard	integer	-	Count	-
ticket	Ticket number	text	-	-	-
fare	Passenger fare	float	-	Currency (likely Great British Pound or GBP)	-
cabin	Cabin number	text	-	-	687
embarked	Port of embarkation	text	C = Cherbourg, Q = Queenstown, S = Southampton	-	2

1. Why is the data type for the `survived` variable essential for data analysis?
2. What is the primary use of `passenger_id` as a unique identifier in data analysis?
3. What aspect of the Titanic's voyage does the `p_class` variable primarily represent?
4. Why does understanding the data type of each variable affect data analysis?
5. If we wanted to explore the hypothesis that higher-class passengers had a higher survival rate, which combination of variables would be most crucial to analyze together?

Activity: Illustrate Unstructured Data Using Structured Tables

Figure 1.11 is an example of Pearson's Instagram page. Figure 1.12 represents the data behind the page, using three tables: Users, Posts, and Followers.

For example, Figure 1.12's Users table contains one record for the PearsonOfficial user and billions of other records for other Instagram users. Likewise, the Posts table has the 587 posts we see in Figure 1.11, plus billions of records for all other users' Instagram posts. The Followers table, on the other hand, has over 28,700+ records from Figure 1.11, among billions of others associating users with their followers.

Figure 1.12 shows how to structure Pearson's Instagram page data into tables, much like what we did for Amazon reviews in Figures 1.4 and 1.5. Table 1.14 is a data key explaining the variables in Figure 1.12.

FIGURE 1.11
Pearson's Instagram page.

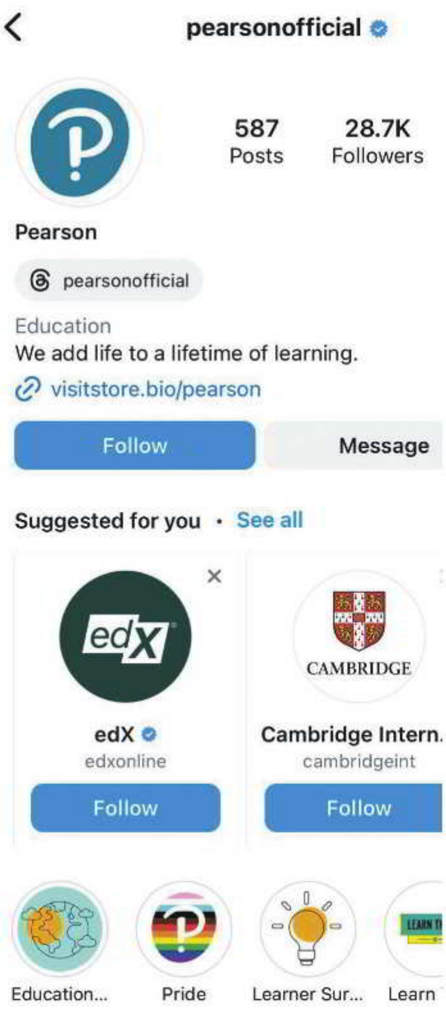
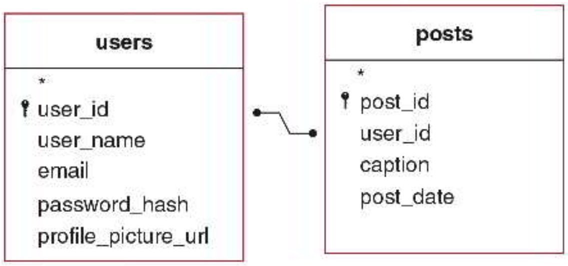


FIGURE 1.12
Transforming unstructured Pearson's Instagram page into structured tables connected by identification variable `user_id`.



Answer the following questions:

1. Which field in the `users` table is a unique numeric identifier for each user?
2. In which field and table is "Pearson" likely stored?
3. In which field and table is the link to the below image is stored?

4. To identify the user who created a specific post, which field and table should be used to link back to the Users table?

TABLE 1.14
Data key for the tables in Figure 1.12.

table	variables	explanation
users	user_id	Unique numeric identifier for each user.
users	user_name	The user's chosen name, used for logging in and display.
users	email	Email address associated with the account.
users	password_hash	Encrypted password for account security.
users	profile_picture_url	URL to the user's profile picture.
posts	post_id	Unique identifier for each post.
posts	user_id	Links to the Users table to identify the poster.
posts	caption	Text caption accompanying the post.
posts	post_date	Date and time when the post was created.

Activity: Identify Analysis Types

Part I

The hospital's data science team asks you to review the following analysis paragraphs and determine whether they are descriptive, diagnostic, predictive, or prescriptive.

1. We reviewed patient admission data over the past year, focusing on the high readmission rates within a month among cardiovascular patients, which stood at around 30%. Upon further investigation, we found that a significant portion of these readmissions, approximately 50%, were among patients with co-existing diabetes. Additionally, patients enrolled in our post-discharge follow-up program were less likely to be readmitted within a month.

   What type of analysis is this?

2. Our analysis suggests implementing a comprehensive post-discharge care program for cardiovascular patients with co-existing conditions such as diabetes. This could include regular follow-ups, medication management, and tailored health education. Furthermore, the analysis recommends scaling up our resources during this period to handle the predicted 15% increase in cardiovascular-related admissions next winter. This could involve adjusting staff schedules, ensuring sufficient inventory of necessary medical supplies, and coordinating with nearby healthcare facilities for overflow support. By adopting these strategies, we could reduce the readmission rate for cardiovascular patients from the predicted 30% to around 20%, thereby improving patient outcomes and optimizing resource utilization.

   What type of analysis is this?

3. We've observed significant trends in analyzing patient admission data over the past year at our hospital. Most admissions, approximately 40%, were related to cardiovascular conditions, with a notable spike in admissions during the winter months. The average length of stay for these patients was 4.5 days, although this extended to 6 days for patients aged 65 and over. It's also worth noting that around 30% of our cardiovascular patients were readmitted within a month of their initial discharge.

   What type of analysis is this?

4. According to our analysis, we expect a 15% increase in cardiovascular-related admissions over the winter season. Furthermore, without intervention, the readmission rate for cardiovascular patients within a month is likely to persist at around 30%.

   What type of analysis is this?

Part II
This activity builds on Part I to include elements of the data science lifecycle.

Dataset: Titanic-Dataset

A group of historians is studying the effect of classism in disaster scenarios. One disaster scenario of particular interest is the sinking of the RMS Titanic in 1912. In this event, approximately 1500 deaths occurred among those onboard. The historians are concerned that classism may have led to a disproportionately high number of casualties in the lower ticket classes. They also have secondary interests in how other passenger demographics relate to survival rates once their initial inquiry is answered. The historians published a full report for some new team members who are just learning the data science lifecycle, spelling out the data science components described in this chapter. Answer the questions to recognize the components and subcomponents of the data science lifecycle, and categories of data analysis.

We investigate the differences in the survival rate among the three passenger classes, using historically accurate data kept on the passengers of the Titanic. The ease of access to this data is a result of good data management.

1. We see that someone physically recorded the information about the passengers. What is this first step?
2. Someone digitized the physical records for the passengers and created a tidy table for our use. What is this second step?
3. Once the data is in a clean, easily accessible, and complete format, we can move on to the next step. In this step, we extract meaning from the data–in this case, through basic summary statistics. What is this third step?
4. Looking at Table 1.15, where summary statistics are provided for a past event, what type of analysis is this?

TABLE 1.15
Survival statistics of Titanic passengers separated by passenger class. Data from M. Yasser H., Kaggle.

passenger_class	survival_percentage	number_of_survivors	total_number_of_passengers
1st class	63.0	136	216
2nd class	47.3	87	184
3rd class	24.2	119	491

5. In another type of analysis we could perform on these data, we would address the underlying explanation for why these statistics are what they are. What type of analysis would that be?
6. If the historian placed these data into a historical context and used the numbers to support a cohesive narrative, what would he be doing?
7. The Titanic dataset has also been used to test data models to see if it is possible to predict a passenger's survival based on their characteristics such as age, ticket price, and sex. What kinds of analyses are these?
8. If we were to recommend taking some action in light of analytical insights, what kind of analysis would we be using?
9. However, the historian's work does not end here. Although we have demonstrated a difference in survival rates across passenger classes, passenger class is not the only factor in a passenger's survival rate. Other factors such as age and sex are also relevant in investigating differential survival rates. Data science improves iteration by iteration, a repeating process in which we build upon our previous work to enhance it. When we use the insights we gain from our analysis to fuel future analyses in new directions, what are we engaging in?

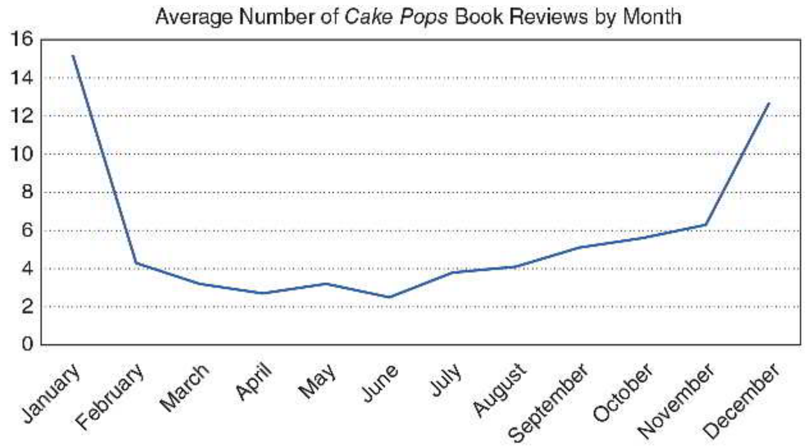
Activity: Optimize the Narrative

Consider the following visualization that graphs the average number of *Cake Pops* book ratings for each month of the year. Answer questions about the visualization based on Figure 1.13.

1. Looking at Figure 1.13, describe the relationship between the average number of *Cake Pops* book reviews and the time of year.
2. Looking at Figure 1.13, what ideas do you have about why the *Cake Pops* book shows certain review trends?
3. Open-Response Question: Use the STAR Framework to tell a story about Figure 1.13.

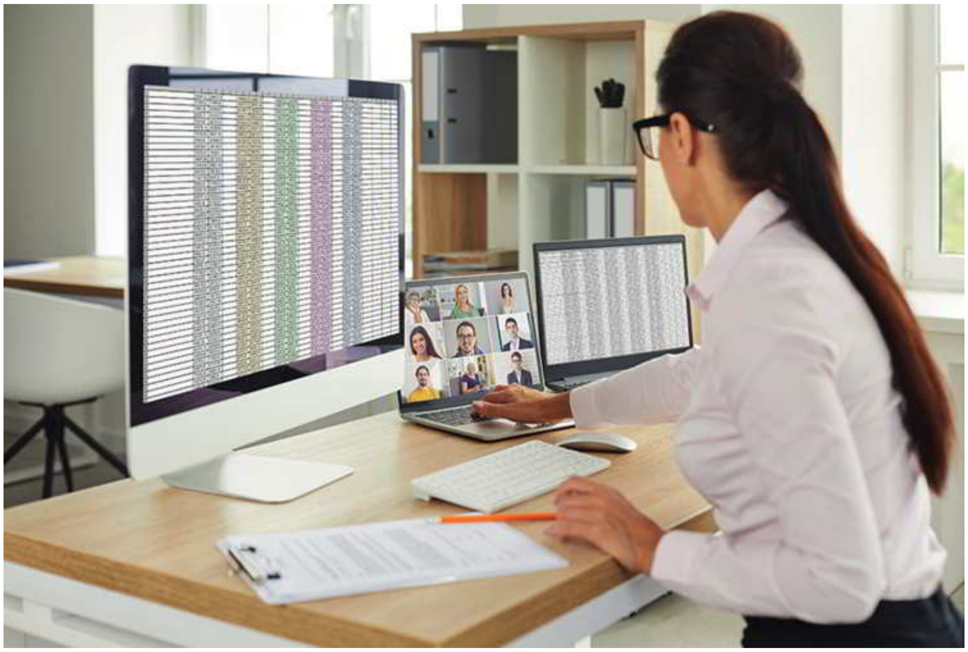
FIGURE 1.13

Average number of times the *Cake Pops* book was reviewed on Amazon by month over the past three years.



CHAPTER 2

DATA WRANGLING: PREPROCESSING



Studio Romantic/Shutterstock

LEARNING OBJECTIVES

- Outline the processes of data wrangling, focusing on their role in preparing data for analysis and storytelling.
- Apply strategies for addressing missing data within a dataset.
- Find and address anomalous values that may distort data analysis.
- Transform quantitative and categorical variables to enhance data structure and analysis.
- Reshape data into a tidy format, ensuring that each column represents a variable and each row an observation.
- Describe approaches to combining multiple datasets for comprehensive data analysis.

Introduction

Mirabel wants her political science senior project on student voting patterns to make policy recommendations about how her college can increase student participation in the next election. Her teacher gives her access to a unique dataset from last year's class on student pet preferences, grades, and demographics combined with data on those same students' voting patterns. She wonders whether younger students participate more in elections than older eligible voters do because they are excited about their new voting rights, or if they vote less because they are unsure about how to vote. She also ponders whether students with better grades might be more likely to vote. Based on a late-night debate with her roommate who is a cat lover, Mirabel is also curious whether "dog people" versus "cat people" vote differently.

Krakenimages.com/Shutterstock

Mirabel immediately dives into the dataset by viewing one record at a time. She is soon overwhelmed by the sheer amount of data from over five thousand students. She sees that it is impossible to absorb so much information by reading the responses one at a time. Then, Mirabel is relieved to see an option to download the results as a spreadsheet. She moves confidently back in front of her laptop and opens the file. She soon realizes she must clean the data. For example, some students wrote out their ages (e.g., "nineteen" instead of 19), others had impossible data (e.g., a GPA of 27.3), and others misspelled words (e.g., "catt" instead of "cat"). After analyzing her cleaned results, she discovers that age and GPA do not appear to have a relationship with voting behavior, and that dog lovers were somewhat more likely than cat lovers to vote in the most recent election. Based on her project's recommendations, voting advocates design "Rock the Vote" pizza parties on the next election day that walk first-year students to on-campus polling places, increasing the student vote at her college by 12%. They promote the event using cats, of course, to elevate the voting frequencies of "cat people."

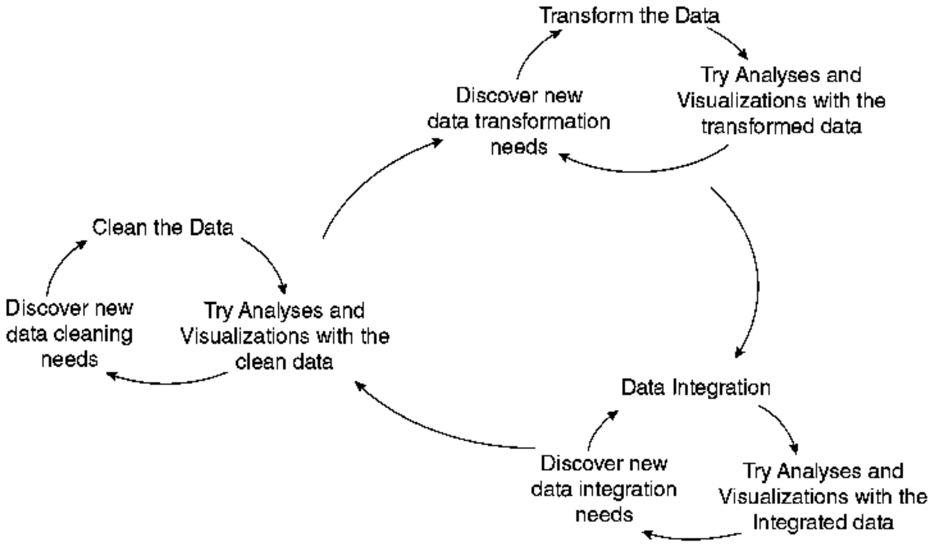
Sheila Fitzgerald/Shutterstock

SECTION 2.1: WHAT IS DATA WRANGLING?

Data wrangling, also known as data preprocessing, is the general term given to the process of preparing collected data for analysis, storytelling, or both. Data can come from a variety of sources in all shapes and sizes. Not all datasets are organized and formatted well into the tabular form known as *tidy data*. Even when datasets come in a tabular form, like Mirabel's dataset, the variables and their values are not always immediately ready for analysis and storytelling. Data wrangling can be broken down into three key components that all help transition data into a form more suitable for visualization and analysis. People usually wrangle data in this order, though they cycle within and between each step (Figure 2.1).

1. *Cleaning* the data.
2. *Transforming* the data.
3. *Integrating* the data.

FIGURE 2.1
The continuous cycles of data wrangling.



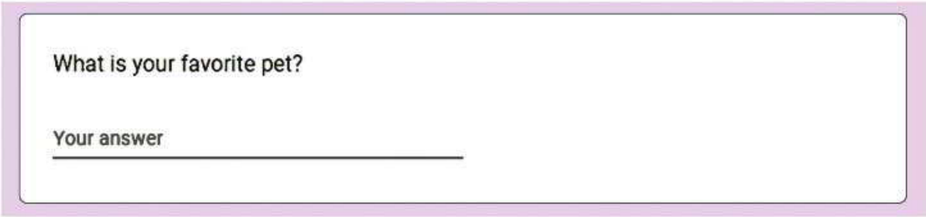
DEFINITION

Data wrangling, also known as **data preprocessing**, is the process of converting raw data into a more usable format by *cleaning*, *transforming*, and *integrating* data to facilitate effective analysis and storytelling.

Cleaning the Data

Recall that *cleaning* data involves fixing or removing incomplete, abnormal, or inconsistent data to enhance their quality for accurate analysis and storytelling. Making the data clean as a part of data wrangling might make it seem like data can initially be "dirty", but that is not quite accurate. Software tools for visualization and analysis require data to be in an organized and consistent format. Different visualization techniques and analysis methods will have their own requirements for this organization and format as well.

How might variables have inaccurate or inconsistent values? Let's revisit Mirabel's dataset, which included responses to an open-ended question:



In this format, students could have written their answers in the blank space, in whatever format they chose. Mirabel might see many different responses, including the following:

- dog
- Rover
- Birds
- parrot
- Dog
- cat
- cats
- Cat
- horses
- Felix
- DOG
- None
- I don't like animals

If Mirabel wants to explore dog versus cat preferences as part of her analysis, she probably only wants the responses here to represent three different possible values: dog, cat, unknown. Additional pet preferences, such as "Birds" and "horses," will not be useful to her dog vs. cat analysis. The first takeaway here is that Mirabel does not care about capitalization in the response; so, for example, "dog" and "Dog" should both become "dog".

The value of the last category, *unknown*, is a choice that is often up to the person working with the data, and this choice is very important. For instance, both the "None" answer and the "I don't like animals" answer would likely be converted to *unknown*. However, a response of "None" could imply that the person is an avid animal lover who can't choose a favorite, while "I don't like animals" would suggest something quite different. By cleaning this variable's values into *dog*, *cat*, and *unknown*, Mirabel is making the data more consistent, easier to visualize and analyze. However, she may also be losing a bit of information in the creation of the *unknown* category.

How might the observations in a dataset not accurately reflect the items from which the data came? If Mirabel's dataset includes people who submitted more than one response, then her dataset could end up with multiple rows for a single individual. It might also be the case that one of the respondents completed the form too quickly and entered nonsensical answers to multiple questions on the form (e.g., a value of 118 for age). Mirabel will want to address these duplicate records and implausible values before performing any analysis or storytelling, to ensure that her work accurately represents the individuals from which her data came.

We cover many of these cleaning approaches in this chapter. However, visualization (Chapter 3) and summarization (Chapters 4 and 5) are useful for discovering the ways in which data need to be cleaned, especially if the dataset is very large.

Transforming the Data

Recall that *transforming* data involves organizing and formatting data into a structure that is useful for efficient analysis. It may seem like the data we end up using for our visualization and analysis is completely different from the data we started with. However, data scientists are not magicians. The dataset contains information in the form it does, and it's up to the data scientist to figure out the best way to organize and extract that information to deliver insight. The process of transforming the data can involve many different steps, and there can be many different reasons to do so.

> **DEFINITION**
>
> **Data transformation** is the process used to convert a clean dataset into the form or structure needed for analysis and visualization.

For example, a dataset might contain a variable with the full date (e.g., January 10, 2023), but we might only be interested in the month and year. One transformation technique would create two new variables in the dataset from this original date variable, one for the month (e.g., January) and one for the year (e.g., 2023).

In general, transforming data can take many different forms. However, it is usually best practice to only expand our dataset to include any newly created variables and never replace our existing variables if we can avoid it. In the previous example, we would retain the original full date variable and simply add the month and year variables to our dataset. Similarly, Mirabel might keep the original responses to the pet preference question and create a new variable for dog-versus-cat preference.

We will cover some of the most common ways to transform our data into forms that can be more useful for visualization and analysis.

Integrating the Data

Recall that *integrating* the data is the process of combining multiple datasets from one or more sources to maximize the amount of information from which to draw insight. If we have multiple datasets, then once they have been cleaned and transformed, we should explore the possibility of combining them.

Usually this is useful because analyzing a single dataset alone would not be as informative as analyzing the combined dataset. When faced with two related datasets, there are a few primary ways to think about how they relate to each other:

- They could have the same set of variables, but different observations.
- They could have the same observations, but different variables.
- They could have some similar observations and some similar variables, but not identical sets of either.

After relating the two datasets, it is also important to decide *how* we want to combine them. It may be the case that we only want the observations from the first dataset—but with associated variables from the second dataset, or only observations from the second. It could be the case that we only want the variables in the first dataset for all the observations in the first *and* all the observations in the second dataset. There are numerous other ways to combine two datasets, and we will cover multiple approaches throughout this text.

All these data wrangling decisions and steps can help prepare data in a variety of ways for both visualization and analysis. For this reason, it is extremely common for data scientists to go through multiple iterations of data wrangling in the data science workflow, as they determine the most appropriate visualizations and analyses for the data. We will introduce advanced data management techniques for data integration in Chapter 5. For the sections that follow, we will assume that the dataset of interest is already in a tabular form.

Dataframes

For the purposes of this chapter, we will use the term *dataframe* to refer to a tabular dataset that can be stored, viewed, and processed with a variety of software tools. You may have seen dataframes before as tables (e.g., Tables 1.1–1.7 were all technically dataframes) or spreadsheets (e.g., in Microsoft Excel or Google Sheets).

DEFINITION

A **dataframe** is a tabular data structure with named columns.

Dataframes are usually more than just tables with columns and rows in spreadsheets and other tools, such as coding environments for R and Python, or statistical tools like StatCrunch. Dataframes often come with or lend themselves to functionality that can help compute and produce more information about the data. Spreadsheet, coding and statistical tools offer foundational data science functionality for working with data in many forms, but especially for dataframes. We can complete many tasks—that is, instruct the computer to fulfill complex or calculation-heavy functions—with these tools. For example, we can clean structured data by opening it in a spreadsheet tool and use the spreadsheet's built-in functions to recognize and address issues. Coding and statistical tools are similarly designed to provide functionality that interacts with dataframes. We clean, transform, and integrate data through the structure of a dataframe.

Big Picture Break

Wrangling data into a consistent and well-defined structure may not sound as glamorous as creating an interesting data visualization or performing an exciting analysis. In fact, the result of data wrangling is a new and improved dataset, *not* some fancy graphic or prediction model. However, the importance of data wrangling in the data science workflow cannot be overstated. Without thorough data wrangling, our data may never even be ready to be effectively visualized and analyzed. To this end, all of the steps undertaken throughout data wrangling must *also* be well documented and explained, just like any visualization or analysis. Data wrangling documentation is often the most important for fellow data scientists to be able to reproduce each other's work reliably.

SECTION 2.2: CLEANING MISSING DATA


Billion Photos/Shutterstock

Data wrangling is the process of cleaning data (converting from messy to tidy data), transforming it (making it ready for analysis), and integrating it (connecting it with more than one data source). Software tools aid these steps. We will explore examples that focus on cleaning data through functions such as search and replace, sorting and filtering, and other functions available in spreadsheet software. Although many different tools can accomplish the following data-cleaning steps, we will demonstrate how it could work with Excel or Google Sheets:

1. Open a dataset in Excel or Google Sheets.
2. Sort the columns to observe the range of values for each.
3. Use functions to calculate the average, sum, count, maximum, or minimum of a column of numbers.

4. Assess whether the values make sense. For example, if the *max* function returns a value of 928 for age, we know that the data have errors.
5. Correct errors. For example, use the *sort* function to isolate and delete all impossible values for age.

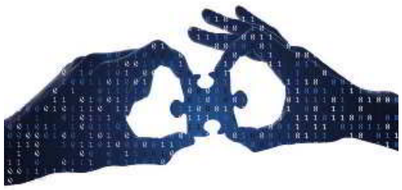
When we clean data, we often identify the following issues:

- Missing data
- Extreme data values
- Implausible values
- Incorrect formats
- Duplicate records

Exploring a dataset to identify and address these issues is essential for the remainder of the data science process to be of value.

Quick Tip

Remember, cleaning data is the first step in data wrangling. This foundational step not only ensures that our data analysis is built on accurate and reliable information, but also saves significant time. By doing it right the first time, we avoid the costly process of having to revisit and redo analyses built on flawed data.



Ink Drop/Shutterstock

Missing Data

When a dataframe has *missing data*, it means that at least some cells do not have values. In Table 2.1, a sample from Mirabel's dataset shows an example of observations with missing values for some variables. The observational units are the students represented in her dataset.

TABLE 2.1
Data with missing cells for some columns and rows.

first_name	age	gpa	favorite_pet	voted
Selma	18	3.1	Dog	0
Juan		3.2	Cat	1
Rozzlin	19	3.9		0

The table displays five variables about each individual (`first_name`, `age`, `gpa`, `favorite_pet`, and `voted`). Each row records one person's responses. The value in a specific cell, a single box in the dataframe, represents an individual's answer to a particular question. The second column of the dataframe represents *age*, and the first row contains Selma's responses. Selma's response has the cell value of 18 for *age*. However, it is noteworthy that Juan's *age* response,which should be in the second row, is missing.

Why would these missing data matter? Missing information might frustrate us in our analysis. Mirabel would not be able to identify Juan's responses about voting as belonging to groups of older versus younger respondents, which may have implications for strategic ways to get the vote out differently for each age group. Remember that a row in a dataframe represents each person's response across many columns (i.e., many variables).

What do we do with missing data? The best solution is to rerun the study with tighter response controls to avoid missing values (e.g., adding a dropdown menu of pet options instead of an open-ended response). However, this is not always possible, especially if the data come from a third party, as with Mirabel's dataset. Thus, we discuss a few other ways of addressing missing values in our data:

1. Removing observations with missing values.
2. Imputation from internal data.
3. Imputation from external data.
4. Other options.

DEFINITION

Imputation is the process of replacing missing data with substituted values derived from patterns in available data.

Try It Yourself: Identify Missing Data

Dataset: vote

GPA data is a way to understand people's educational performance and its potential relationship with voting behavior. A significant number of missing values can impact our analysis, potentially skewing results and obscuring key insights. By identifying the extent of missing `gpa` records, we aim to evaluate the dataset's reliability, inform subsequent data cleaning or imputation methods, and uncover any underlying patterns associated with these data gaps. This step is essential in ensuring that our analysis is both accurate and meaningful.

Answer the following questions (round float (i.e., non-integer) values to two decimal places):

1. How many missing values are there in the `gpa` column?
2. What is the total number of rows in the dataset?
3. What percentage of the dataset has missing `gpa` values?
4. How would you describe this using the STAR framework?

Tool Time–Available in MyLab Statistics

- Excel
- Python
- R
- StatCrunch

Removing Observations with Missing Values

One way to address missing values is to remove the entire observation if there are any missing values. This would mean deleting the person's response entirely, even if they answered some questions. For example, Mirabel would remove all of Juan's responses by deleting his row from the dataframe because the value for `age` was missing in the example above. The downside to this option is that Juan's other responses may have offered us essential insights. Mirabel should use caution before deleting data. For instance, imagine that a third of the respondents failed to indicate their age, but all the respondents did indicate their voting behavior in the most recent election. What would Mirabel do? She could decide not to group people by `age` rather than deleting observations. In summary, she can respond to missing data by ignoring the missing value's observation (e.g., Juan's entire record) or variable (e.g., all responses for `age`).

On the other hand, she might retain rows with missing values, presenting results with all available information. The downside to this approach is that some results may represent perspectives from those with missing values, while other analyses may not, sometimes resulting in divergent sample sizes. Consider the example in which Juan's `age` value is missing. Mirabel might report the average `gpa` including Juan's response, since his `gpa` is not missing. However, an accompanying report on `gpa` grouped by `age` would not include Juan as his age is missing. If there are many missing values, this can lead to confusion and even misrepresentation. Regardless of Mirabel's decisions, she must report them when publishing the information.

Big Picture Break

Retaining observations with missing data values often leads to biased results, as analyses based on the remaining data may not be representative of the whole. On the other hand, simply removing observations with missing values might reduce the size of the dataset, potentially losing valuable information. Each approach to handling missing data has its trade-offs, and the choice often depends on the context of the dataset and the nature of the missing data. In some cases, advanced techniques like machine learning, covered in Chapters 8–10, can be used for imputation, offering more nuanced handling of missing data. The key is to understand the implications of each method and to choose the one that aligns best with the goals and constraints of our specific analysis. In essence, handling missing data is not just a technical challenge but a strategic decision that can shape the direction and credibility of our data analysis. As data scientists, it's crucial to develop a keen understanding of these methods and their impact on the integrity of our analysis, transparently indicating our choices when we present results.

Imputation from Internal Data

Another option is to use *imputation*. As defined earlier in this chapter, imputation means estimating the missing value based on other similar observations. In other words, we would replace the missing value with a value that we hope is likely to be reasonably accurate. For example, suppose Juan indicates in response to additional questions that he graduated from high school four months ago, has a part-time job, and lives off campus. Mirabel might find hundreds of other respondents with the same profile and then average their ages (e.g., 18.2 years old) and replace Juan's missing `age`value with the newly calculated one. She can impute using averages or more advanced methods not covered in this chapter (e.g., machine learning methods). In using the existing dataset, Mirabel is *imputing from internal data*.

DEFINITION

The **mean** of a variable is the average of all the data values in a dataset for that variable. It is calculated by adding up all the values and then dividing this sum by the number of values.

The **median** of a variable is the middle value in a dataset for that variable when the values are arranged in ascending or descending order. If there is an even number of values, the median is the average of the two middle values.

Table 2.2 presents a historical dataset from an archaeological dig site. The data represent objects unearthed at different depths, which can provide us with an idea of the time period each object originates from. Please refer to the variables described in the list below.

1. `object_ID`: an identification number we assign to each archaeological object found
2. `type`: a classification of the object
3. `material`: what the object is primarily made of
4. `depth`: how deep the object was unearthed, measured in meters. (Note that objects found deeper underground are generally older.)
5. `age`: estimated age in years

It's common in archaeological excavations to find objects whose age is difficult to estimate directly. This could be due to lack of context, decay, or other factors.

In this example, we are looking to find the *mean* age of the objects listed in this dataset. However, several values in the `age` column are missing. We can deal with missing values in many ways. We will demonstrate two methods to deal with the missing `age` values: (1) *removing* observations with missing values, and (2) *imputing* them.

TABLE 2.2
Archeological dataset with missing values.

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	
O10	Coin	Silver	1.2	1900
O11	Jewelry	Gold	2.7	
O12	Pottery	Clay	2.2	2400

Method 1: First, we remove all observations with missing values for `age`. Then, we observe how this removal affects the data, for example, by calculating its mean. To calculate the mean age, we add up the known ages and divide by the number of objects with known ages. The sum of the known ages is

$$2000 + 3000 + 1500 + 2500 + 2000 + 1900 + 2400 = 15{,}300 \text{ years}$$

and there are seven objects with known ages.

So the mean age is

$$15{,}300/7 = 2185.71 \text{ years.}$$

For this calculation, note that the observations with missing age values were ignored or excluded. Although we originally described this process as *removing observations with missing values*, the actual dataset does not need to be reduced to only complete observations. Most, if not all, software tools will simply exclude observations with missing values when performing tasks that require those values.

Method 2: This time we impute missing values for `age` rather than remove them, again following up with a mean calculation to compare with that of Method 1. To impute the missing ages, let's use the mean age of the complete (or "known") observations, which, as calculated in Method 1, is 2185.71 years. This assumes that the age of the objects found at this archaeological site can generally be described well by their average value. Table 2.3 is our completed dataset:

After imputation, the sum of the ages is

$$2000 + 2185.71 + 3000 + 1500 + 2185.71 + 2500 + 2185.71$$
$$+ 2000 + 2185.71 + 1900 + 2185.71 + 2400 = 26{,}228.55$$

And we have 12 objects, so the mean age after imputation is still 2185.71 years.

So, we see that imputing missing values with the mean age will leave the mean age unchanged. We could also use the median instead of the mean to impute missing values.

TABLE 2.3
Archeological dataset with imputed values.

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	**2185.71**
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	**2185.71**
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	**2185.71**
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	**2185.71**
O10	Coin	Silver	1.2	1900
O11	Jewelry	Gold	2.7	**2185.71**
O12	Pottery	Clay	2.2	2400

Try It Yourself: Impute Missing Values from Internal Data

Dataset: vote

In our ongoing analysis of the voter survey dataset, we now tackle the issue of missing GPA values by employing the mean imputation method. Our objective is to create a new column in the dataset, which will replicate the original GPA data but with a key modification: missing values will be filled in with the mean GPA of the available data.

Questions

Answer the following questions (use two decimal places):

Before replacing missing data:

1. What is the mean GPA?
2. What is the median GPA?

After replacing the missing data using mean imputation:

3. What is the mean GPA?
4. What is the median GPA?
5. Describe the results using the STAR Framework.

Tool Time–Available in MyLab Statistics

Excel

Python

R

StatCrunch

Imputation from External Data

We have so far described *imputation from internal data* based on values within the same dataset. However, *imputation from external data* based on values from a different dataset is also possible. For example, in the case of Mirabel's dataset, she could access the college's roster that includes the year of birth for each student. Alternatively, if the college cannot provide data about birth dates, she could use public data about the average age of students in Juan's class. Even better, if she already has information about Juan from another dataset, and she is sure it is the same Juan, she should replace Juan's missing age value with the known age. However, she would not refer to this as an imputation because the values would be known, not estimated. When imputing data from external datasets, we should trust that the other sources of data are reputable and relevant to our original dataset.

The downside to imputation, whether from internal or external data, is that it makes assumptions and modifies the raw values, thereby influencing the analysis. These decisions should always be documented and reported.

Case Study: Data Wrangling in Criminal Justice Research

Dzelat/Shutterstock

The Criminal Justice Research Center is one of many organizations that use dataframes to analyze crime data. They have participated in one project, the Historical Violence Database (HVD), which is a collaborative research project that gathers data on the history of violent crime. Social scientists, the primary audience for this database, test theories of violent crime under different historical circumstances. When deciding how to code variables, the Criminal Justice Research Center carefully considered the needs of future users of the database. This included considering whether free-response entry might lead to erroneous or missing values.

Most dataframes from public agencies use numbers (i.e., codes) to refer to different crimes or weapons. For example, the FBI's Supplemental Homicide Reports reference murders committed with knives or cutting instruments using the number "20." However, the HVD opted to add a new text variable, spelling out the name of the weapon (e.g., "club"). This is because most of the social scientists and historians using this database are not trained data analysts. Using descriptive text data makes the database more accessible to its users. Considering how domain experts intend to use the data is critical when deciding how to record or transform variables.

Missing data pose many serious problems in criminology. They can obscure the types of crimes that have occurred, contribute to a lack of evidence or faulty evidence, and confuse criminal justice researchers. Data are rarely collected without missing values. Sometimes, researchers choose to ignore cases with missing values and only analyze cases where all cells contain data. This can be problematic, as it introduces bias if the cases with complete data are not representative of the cases that are missing data, especially as the number of cases with missing data increases. Strategies such as imputation have proven effective in criminological research when managing missing data.

The FBI's Uniform Crime Reporting (UCR) Program is a primary source of data on crime. The UCR has incorporated multiple forms of imputation into its data cleaning process. With *imputation from external data*, data from prior responses of the same respondent are used to impute the missing values. An advantage of this method is that it introduces less bias through its use of all available data. Imputation has allowed the UCR to produce more accurate and complete crime estimates at the state, regional, and national levels.

Other Options for Missing Data

We could also decide not to impute, but to replace missing values with an entirely new value specific to that missingness. For example, missing values for quantitative variables could be filled with "NA" (which is often a special value in software that uniquely denotes a missing value), or with a completely nonsensical value like -999. For categorical variables, missing values could literally be replaced with a value of "Missing" or "Unknown". By doing this, we can continue to include incomplete observations in the dataset, and our analysis can distinguish them from other observations where appropriate.

Although it may be cumbersome, we could also simply pursue two separate analyses with our data: one with the incomplete observations included, and one with the incomplete observations removed. Regardless, we should always think carefully about how we modify raw data, clearly noting the choices made to deal with missing values when writing a report.

An even more tedious strategy for addressing missing data is called *multiple imputation*. It involves imputing values for missing data many times, generating many different imputed datasets. From there, the data scientist analyzes each one of these imputed datasets and combines the results in some way.

In the end, handling missing data can be very complex. A data scientist must think carefully about the following things:

- What are the possible sources of missing data in my dataset?
- Is the missingness random, or do any patterns explain why values are missing?

  (In fact, the randomness of the missing data can affect the usefulness of the different strategies discussed here. Unfortunately, these details are beyond the scope of this text.)
- What strategies can I pursue to adjust my work for the missing values in my dataset?
- How can I document and communicate these missing data characteristics in my work most effectively?

Africa Studio/Shutterstock

SECTION 2.3: CLEANING ANOMALOUS VALUES

Cleaning data involves many types of anomalous, or "unusual," values. We focus on implausible values, extreme values, incorrect formats, and duplicate records in this section.

DEFINITION

Implausible values are data points that fall outside the realm of possible values, clearly indicating errors.

Extreme values are data points that are noticeably different from others, potentially affecting the overall analysis in noticeable ways.

Incorrect formats refer to data entries in an unexpected format, requiring correction to match the anticipated data type.

Duplicate records are repeated entries (e.g., observations) in a dataset, often identified and resolved using unique identifiers.

Implausible Values

Implausible values are quantitative data that are not within a range of anticipated values, to the point of being impossible. For example, a customer's credit score is typically between 300 and 850. If we come across a credit score value of $22,995$, or -53, we know the data must be a

Alexey Konkov/Shutterstock

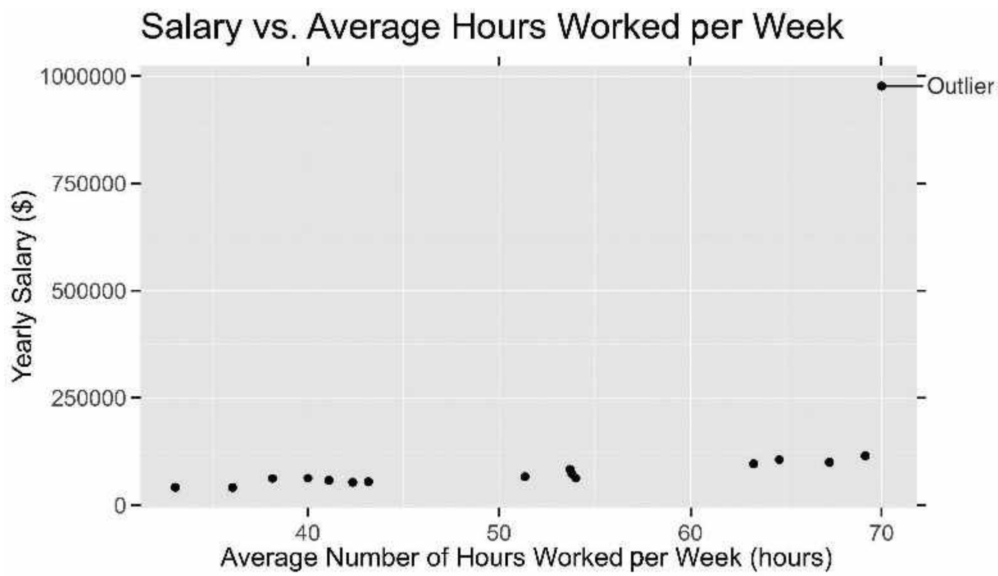
mistake of some sort because credit scores range from 300 to 850. Common sense tells us these values are incorrect. Depending on our judgment, we could replace these cells with missing values and proceed with any of the methods previously described in our discussion of missing data. Although it may require more time and effort, we could also return to the source of the data and correct these values instead of replacing them with missing values.

Extreme Data Values

Extreme data values are exceptionally higher or lower than the other values of a variable. Consider the results from Figure 2.2, which indicate that most of the respondents' annual earnings are between $33,000 and $200,000. What would we think if someone said their annual income was $977,000? This would be considered an extreme data value. Why does this matter? The $977,000 value could significantly alter our analyses. For example, if we include the $977,000 value and calculate the mean from those reported values, we find an average annual income of $103,375. If we do not include the $977,000 data point, our average annual income is $45,367, a noticeable difference. Reporting the $103,375 number may give the impression that a typical respondent earned over $100,000 when, in reality, they earned less than $100,000.

FIGURE 2.2

Depiction of an extra data value that is likely to be an outlier.



Though often used to mean the same thing as *extreme value*, an *outlier* is a value that we consider so extreme that we wish to fix or remove it. How do we know for sure whether a value is too extreme, i.e., likely to be an outlier? It requires some judgment, depending on the context. In the case of the income dataset in Figure 2.2, if all the respondents work for the same private company, it could be that the high-earning respondent is the owner. In such a case, there may be a way to verify the unusual data. In Chapter 4, we will discuss formal ways of identifying outlier observations.

DEFINITION

An **outlier** is an extreme data point so different from the majority that it may warrant correction or removal. Its extremity is determined by contextual judgment and verification.

Try It Yourself: Identify Extreme Data Values

Dataset: vote

We need to check our voter data for any unusually high or low ages. These extreme ages could be mistakes and might throw off our results. By finding the ages of the oldest and youngest 5% of observations in our data, we can decide if these numbers make sense or if we should exclude them to maintain the integrity and trustworthiness of our data.

Questions

Answer the following questions:

1. What ages encompass the oldest 5% of respondents?
2. What ages encompass the youngest 5% of respondents?
3. How many respondents have `age` values above 100?
4. How many respondents have `age` values below 18?
5. How extreme are the `age` values (use the STAR Framework)?

Tool Time–Available in MyLab Statistics

- Excel
- Python
- R
- StatCrunch

What should we do with extreme values? The answer is complex. In some cases, it depends on whether they are due to error. If so, we can try to fix or remove them. In other words, data scientists may treat extreme values as implausible and ignore the values or the entire rows from the analysis (reporting this, of course). In other cases, the values may be valid, but we need to think about whether they represent the population well. Thus, an analyst may run analyses both with and without the extreme value(s), and report on findings both ways. In the yearly salary example presented in Figure 2.2, we are working with a variable whose values are mostly concentrated together, with only one value separated from the bulk of them. This behavior is common with population- and age-related variables as well. Mathematically transforming such variables can help address the extremeness of some values.

Case Study: Dewey Defeats Truman and the Role of Data Wrangling

Byron H. Rollins

In 1948, the United States was in the grips of a presidential election, and the *Chicago Daily Tribune* made a historic mistake after election day. It published the headline "Dewey Defeats Truman," declaring Thomas E. Dewey the winner. The pollsters had relied heavily on telephone surveys, a cutting-edge methodology for that era. However, this approach was fundamentally flawed: telephones were a luxury that many couldn't afford, and as a result, the data gathered were biased toward a specific demographic—those who were wealthier and more likely to vote for Dewey. When the final votes were counted,

(*Continued*) ▶

(*Continued*)

Harry S. Truman had won, making the newspaper headline an infamous example of data gone awry.

Data wrangling might have helped. In the world of data science, wrangling serves as the critical bridge between raw data and meaningful analysis. If the pollsters in 1948 had taken this crucial step, they would have first identified the bias inherent in their telephone surveys. Recognizing this, they could have supplemented their data with other sources representing a broader swath of the population. This would have helped to balance out the bias introduced by the original telephone surveys.

Wrangling doesn't stop at merely cleaning up data; it also involves transforming it into a more analyzable form. If the pollsters had applied statistical techniques to weigh their data, they could have accounted for underrepresented groups, making their sample more reflective of the general voting population. This could have included techniques to account for uncertainties about their respondents.

The *Tribune's* headline serves as a potent reminder of how critical wrangling is to any data science endeavor. It shows that the quality of our data analysis can only ever be as good as the quality of our data. As this case study reveals, poor wrangling can lead to errors that reverberate through history, affecting not just election reporting but also the way we perceive the capabilities and limitations of data in decision-making across diverse fields.

Incorrect Formats

When we expect one data type, but a data value appears to be a different type, we are looking at an *incorrect format*. We may expect text but see a number, or vice versa. For example, in Mirabel's dataset, suppose that a student reported "seventeen" as their age, or "17+", instead of 17. Failing to correct this would mean that the value of that student's response would not be considered a number.

Additionally, Mirabel might find that someone has responded "Goldfisg" instead of "Goldfish" as their favorite pet. She may want to make the judgment call to edit a cell with an incorrect format if the data *almost* makes sense (e.g., she might replace "Goldfisg" with "Goldfish"). On the other hand, sometimes the value just doesn't make any sense (e.g., an income value might be nonsensical, like "slippery," or ambiguous, like "forty-ten"). In these cases, we have the option to remove the observation by deleting the entire row. Alternatively, we might just delete the value from the cell, causing the value with an incorrect format to be missing.

It is usually best practice to delete as little data as possible, so deleting the value in the cell would be the preferred method unless many cells in the row contain faulty data. However, in Mirabel's dataset of over 5000 responses, doing this by hand for each row would be nearly impossible! Therefore, she might need some help from tools to automate the process.

Irochka/123RF

Try It Yourself: Clean Incorrect Formats

Dataset: vote

Clean the `favorite_pet` column of the vote dataset. Answer the first question, then perform a basic cleaning of the data in this order:

- **A.** "dogs" should be changed to "dog"
- **B.** "cats" should be changed to "cat"
- **C.** any word that begins with "d" and ends with "g" should be changed to "dog"
- **D.** any word that begins with "c" and ends with "t" should be changed to "cat"

Once the dataset is cleaned, you will be ready to answer the remaining questions.

Questions

1. Before basic data cleaning for dogs and cats, how many people in this dataset indicated their favorite pet as:
   a. "dog"
   b. "cat"
2. After basic data cleaning for dog and cat, how many people in this dataset indicated their favorite pet as:
   a. "dog"
   b. "cat"
3. Describe the results using the STAR Framework.

Tool Time–Available in MyLab Statistics

Excel

Python

R

StatCrunch

Duplicate Records

The same record (observation) may appear more than once in a dataframe. For example, a Google Forms respondent may have submitted the same form multiple times, but we want to count each person's data only once when analyzing the data. We usually address this problem carefully by considering identifiers like a respondent's student ID number or email address. Such variables are expected to be unique for each observational unit. If two observations have the same value for a unique identifier, such as an email address, this may indicate a mistaken duplicate record. We often delete the duplicate records.

SECTION 2.4: TRANSFORMING QUANTITATIVE VARIABLES

In addition to dealing with problems like missing data, implausible values, extreme data values, incorrect formats, and duplicate records, data wrangling can include ways to enhance or further structure the data. We will discuss how to create new categorical variables from quantitative ones, change the scale of quantitative variables, and combine two or more quantitative variables.

Recall that *quantitative variables* are numerical in nature, allowing for mathematical computations, and can be either discrete, with countable values, or continuous. Also recall that *categorical variables* are often non-numeric and represent group or category memberships.

Creating a Categorical Variable from a Quantitative Variable

In some areas of the world, discounts are based on the age of the patron. A restaurant may charge less for a child or an older adult. This is an example of an age *category*. Who qualifies as a child, and who qualifies as a senior? The establishment can decide this. However, you may have noticed that an individual is frequently considered a child if they are 11 years old or younger, and an individual is frequently considered a senior if they are 60 years old or older. The conversion of someone's age to a label like "child" or "senior" is an example of *categorization*. In this case, we've created a categorical variable, `age_category`, from a quantitative variable, `age`.

Quick Tip

When categorizing data, ensure the categories are mutually exclusive (i.e., one row belongs to only one category) and collectively exhaustive (i.e., no row is left uncategorized). This clarity in categorization helps in creating more effective visualizations and analyses.

Generally, the process of *categorization* describes the way in which a new categorical variable is created from an existing quantitative variable. This is done by assigning a category or label to intervals of values of the quantitative variable. In the age example, it was

- Ages 0–11 years: child
- Ages 12–59 years: adult
- Ages 60+ years: senior

In this case, the age categories are more useful to a restaurant than the actual age value. This can be the case for a number of visualization- and analysis-related reasons, which we will discuss more later in this text. One of the most important steps in categorization is ensuring that all possible values of the quantitative variable are each associated with a single category. For example, someone who is 11 years old should NOT be considered both a child and an adult. Creating a categorical variable from a quantitative variable always, necessarily, results in the loss of information. This might be acceptable for our goals, but we should do it purposefully and thoughtfully.

Try It Yourself: Create a Categorical Variable from a Quantitative Variable

Dataset: vote

Create a new variable called `gpa_category` which contains a categorized version of the `gpa` variable according to the following:

- 2.00–2.99: low
- 3.00–4.00: high

Questions

1. How many high versus low GPA students are there?
2. Use STAR to describe the results.

Tool Time–Available in MyLab Statistics

- Excel
- Python
- R
- StatCrunch

Changing the Scale of a Quantitative Variable

Think about the last time you saw graphs or data visualizations that contained any sort of numbers. How many digits were there in each of the values displayed on the graph? We often have to make this choice as the person working with the data. A dataset may contain an income variable whose values are as low as $20,000 and as high as $800,000. Reading such large numbers on a data visualization can be challenging, so we might consider scaling these numbers down in a way that reduces the number of digits but still allows us to convey similar information. For example, we could create a new variable that represents income in thousands of US dollars, which is computed by dividing the original income variable by 1000.

$$\$20{,}000/1000 = \$20 \text{ (in thousands)}$$
$$\$800{,}000/1000 = \$800 \text{ (in thousands)}$$

Then, we could use this new variable in our data visualization and simply note that values are in thousands of dollars. In this way, readers will only see values from around $20 to around $800, which are considerably easier to digest.

Generally, the process of *scaling* or *transforming* describes the method of creating a new quantitative variable from an existing one by performing some mathematical operation on it. Specifically, *scaling* usually involves multiplying or dividing the original variable by a certain value. *Transforming*, in this restricted sense, refers to applying a more complex function to the original variable, such as taking the logarithm or square root of the values. (Throughout this text, the terms *data transformation* or *transforming* refer broadly to various types of alterations that make data easier to work with.) In the income example, we divided the original variable by 1000. Different situations with different variables may benefit from different types of mathematical operations. As long as we create a new variable without disrupting the raw, original data, we should feel free to explore how different transformations affect the insights the data can provide. Of course, we must ensure to document all our steps and work so that it can be explained and replicated if needed.

Case Study: GlobalGiving Teaches Nonprofits about Transforming Variables

RDNE Stock project/Pexels

GlobalGiving is a nonprofit organization that provides a platform for other nonprofits to raise funds from corporate donors such as Airbnb, 3M, and Twitch.tv. It aims to bring about positive change across various sectors including education, healthcare, and disaster relief. GlobalGiving helps nonprofits become more effective, empowering both donors and nonprofits to drive meaningful impact.

GlobalGiving says that many nonprofits have ample data but struggle to derive actionable insights. The problem? Their data are unorganized and disparate, making it difficult to measure the impact of their community programs. To address this, GlobalGiving teaches nonprofits how to centralize and clean their data, thereby transforming both quantitative and qualitative data. For instance, they helped an animal welfare program in reporting the ratio of rescued animals to attending volunteers by converting text-based animal categories into numeric counts. They also helped them transform quantitative data, such as pounds of food, into qualitative information like "too much food" or "too little food," to determine whether food was distributed evenly across all animals. GlobalGiving effectively teaches nonprofits to manage both types of data—quantitative and qualitative—for a comprehensive view of their programs' effectiveness, thereby facilitating better decision-making and reporting.

Combining Two or More Quantitative Variables

Oftentimes experts combine variables or characteristics in a way that enriches the insights we can gain from the data. For example, consider the computation of a "per capita" variable. "Per capita" means "per person," and it is a term used to understand statistics like the gross domestic product (GDP). We compute it by dividing a country's overall GDP by the country's population, thus finding the GDP per capita.

In this situation, our dataset might contain information on countries of the world, including their gross domestic products (GDPs) and their populations. To find a given country's GDP per capita, we would use the following equation.

```
gdp_per_capita = gdp / population
```

In the above scenario, we would be combining two variables—`gdp` and `population`—into a new variable, `gdp_per_capita`. We could use this new variable to learn more about goods and services produced *per person*, rather than for the population as a whole.

These kinds of relative measures can be much more useful for visualization and analysis because they help account for countries with different populations, or more generally, entities of different sizes. For example, the United States may have a much higher GDP than France, but this may not be surprising because of the drastically different sizes and populations of these two countries. In fact, their GDP per capita values are not as different as their national GDP values.

As with *data cleaning*, it is not uncommon for a data scientist to undergo multiple rounds of *data transformation* as they discover what their desired visualizations and analyses need with respect to the forms of the data. It is important to note that data transformation usually happens after data cleaning, so it is safe to assume that variable values are consistent and accurate. The cyclical nature of each data wrangling phase is illustrated in Figure 2.1.

Quantitative variables can represent numerous different types of variables, such as age, income, height, and weight. Once variables like these are clean, it will be useful to think about what kind of insights they can offer and how best to communicate those to a broader audience.

SECTION 2.5: TRANSFORMING CATEGORICAL VARIABLES

Categorical variables can represent numerous different types of variables, such as pet preference and date. Once variables like these are cleaned, it will be useful to think about what kind of insights they can offer and how best to communicate those to a broader audience.

Creating a Quantitative Variable from a Categorical Variable


Eric Isselee/Shutterstock

There are many analysis methods that we will learn about later, which can only work with data in numeric form. Some visualization and summarization techniques are only appropriate for numeric data. Therefore, we sometimes want to translate text data into numbers. For example, Mirabel's vote data variable `pet_ preference` from Table 2.1 has many possible text values including "Dog" and "Cat". We may want to associate `pet_preference` values with a tendency to vote. Do people with dogs vote more than people with cats? We can create Boolean variables (also known as dummy, indicator, dichotomous or binary variables) for each value. If `pet_preference`is “Dog,” we could create a new data column in the dataframe called `dog` and assign observations a value of 1 if `pet_preference` is a dog and 0 otherwise.

A common way to transform categorical variables into quantitative ones in data science is to create Boolean variables that represent different levels of a categorical variable.

Consider data in Table 2.4:

1. `student_id`—an ID we assign to each student who participated
2. `age`—self-reported age
3. `gpa`—the self-reported grade point average
4. `pet_preference`—the stated pet preference (e.g., Dog, Cat)
5. `voted`—whether they voted in the last election (1 = Yes, 0 = No)

We will create a new quantitative variable from the `pet_preference` categorical variable. Specifically, we will create new Boolean columns for each category in the original column (e.g., `prefers_dogs` and `prefers_cats`). If a student prefers dogs, they will have a 1 in `prefers_dogs` and `prefers_cats` and a 0 in the other new preference columns (e.g., a 0 in `prefers_cats` and `prefers_none`).

Here’s how this process would look:

TABLE 2.4
Original data.

student_id	age	gpa	pet_preference	voted
S1	19	3.5	Dog	Yes
S2	21	3.7	Cat	No
S3	20	3.2	None	Yes
S4	22	3.8	Dog	No
S5	19	3.1	Cat	Yes

TABLE 2.5
Data after creating Boolean variables from the categorical variable `pet_preference`.

student_id	age	gpa	pet_preference	prefers_dogs	prefers_cats	prefers_none	voted
S1	19	3.5	Dog	1	0	0	Yes
S2	21	3.7	Cat	0	1	0	No
S3	20	3.2	None	0	0	1	Yes
S4	22	3.8	Dog	1	0	0	No
S5	19	3.1	Cat	0	1	0	Yes

In Table 2.5, the `pet_preference` categorical variable has been transformed into three new binary variables: `prefers_dogs`, `prefers_cats`, and `prefers_none`. This quantitative transformation allows for numerical analysis and modeling that could not be accomplished with the original categorical data. For example, we can sum the values of `prefers_dogs` using the following equation:

$$1 + 0 + 0 + 1 + 0 = 2$$

Since there are five records, we know that $2/5 = 0.4 = 40\%$ of respondents prefer dogs. Creating quantitative variables from categorical variables is even more important in future chapters on machine learning.

Try It Yourself: Create a Boolean Variable That Represents a Categorical Variable

Dataset: vote

Create a Boolean variable called `drinking_age` where `drinking_age` = 1 if age is greater than or equal to 21, and 0 otherwise.

Questions

1. How many `drinking_age` values are 1? How many `drinking_age` values are 0?
2. How many people of drinking age voted? What percentage voted among people of drinking age? How many people not of drinking age voted? What percentage voted among people not of drinking age? Round float (i.e., non-integer) values to one decimal place.
3. Describe the results using the STAR Framework.

Tool Time–Available in MyLab Statistics

- Excel
- Python
- R
- StatCrunch

The process of creating a quantitative variable from a categorical variable is often called *encoding* because we are creating numeric codes associated with each category, or value, of the categorical variable. Boolean encoding is a special case of variable encoding where we assign a 1 or 0 during encoding. In transforming `pet_preference` to `prefers_dogs`, we were implementing *Boolean encoding* with the value of “Dog” as 1 and any other value as 0.

DEFINITION

Encoding is the process of converting a categorical variable into a quantitative one by assigning numeric codes to the values of the categorical variable.

Boolean encoding is the process of converting a categorical variable into a quantitative one by assigning a 1 or 0 to the values of the categorical variable.

In general, we can use whatever numeric values we want, but we must be extremely careful when doing so. The numeric values we use should reflect the categories as best as we can. In particular, the differences between the categories should align with the differences in the numeric values as much as possible. For example, suppose we have data on people's feelings about a particular movie with the following possible responses:

- Dislike
- Neutral
- Like

Table 2.6 presents four possible encoding schemes for these categories, each differing in how they represent these values.

TABLE 2.6
A table presenting four encoding schemes for Dislike, Neutral, and Like.

Response	Encoding Scheme 1	Encoding Scheme 2	Encoding Scheme 3	Encoding Scheme 4
Dislike	1	−1	0	0
Neutral	2	0	0	1
Like	3	1	1	1000

In Table 2.6, Encoding Scheme 1 is often the first type of encoding that people consider because it simply enumerates the responses to a question. There are three possible responses, so we number them 1, 2, and 3. Encoding Scheme 2 represents the responses in a way that is more explicitly oriented around the "Neutral" response: 0 represents the "Neutral" response, +1 represents the "Like" response, and −1 represents the "Dislike" response. Encoding Scheme 3 is a Boolean encoding where we treat "Dislike" and "Neutral" the same, both distinct from "Like" (e.g., we only want to consider those who like something, not those who are neutral or dislike it). Finally, Encoding Scheme 4 is an atypical example that appears to emphasize the "Like" response quite a bit. It's challenging to determine the best encoding scheme without knowing what types of visualization and analysis will be conducted. However, it's crucial to understand that each of these encoding schemes could result in different visualizations and analyses. Without knowing the type of visualization and analysis we want to conduct, it is difficult to determine which encoding is best.

Like our other data wrangling steps, encoding a quantitative variable may be an iterative process. Each time, it is informed by the visualizations and analysis performed.

Condense the Categories of a Categorical Variable

Sometimes free response results contain many different answers, some of which may not occur very often. When Mirabel's dataset involved students' pet preferences, she might have seen responses about dogs, cats, fish, birds, turtles, lizards, and more. Upon further inspection, she might discover that 90% of the responses were either "dog" or "cat". To simplify the rest of the work, she decides to create a new variable, `pet_preference_condensed`, that contains people's pet preference as "dog," "cat," "none," or "other." In doing this, Mirabel has merged all the values other than "dog," "cat," and "none" into a single "other" category, which is much easier to display and read in visualizations and summaries than the numerous other responses that occurred so rarely.

DEFINITION

Condensing the categories is the process of reducing the number of distinct values in a categorical variable by merging categories into broader ones. This simplifies the variable for analysis while reducing its informational detail.

Typically, condensing a categorical variable means decreasing the variety of its possible values. This process does not need to preserve any of the original variable's values. Of all our transformation methods thus far, this is one of the most important methods with which to create a new variable and not replace the existing variable. This is because the combining of categories, although it may end up being informative, is a reduction in the information contained in the variable. Note that Encoding Scheme 3 in Table 2.6 was actually an example of *condensing the categories*.

Splitting a Variable into Multiple Variables

Data can come in all shapes and sizes, and this includes variables. It's not uncommon for free-response questions to have answers in many different formats. Consider entering information for your birthdate. If you were given an unformatted prompt for your birthdate, you might enter it as follows:

July 7, 2007

And this might be exactly how your value for the birthdate variable appears in the dataset. However, the visualization and analysis of these data may involve the months and years in which people were born. This would require the extraction of the month and year from values of birthdate like the one above, ignoring the day. All software tools will have methods for extracting information like this.

In general, the *splitting* of a categorical variable describes the process of extracting pieces of information from its values and storing them in at least one new categorical variable (e.g., `month_name`) or quantitative variable (e.g., `year`). Figure 2.3 provides an example of such a splitting operation.

FIGURE 2.3
Example dataset with date variable split into `month` and `year` variables.

date	month	year
3/4/2003	March	2003
11/2/1998	November	1998
3/30/2013	March	2013
5/1/2009	May	2009
9/20/2012	September	2012

Try It Yourself: Splitting a Variable into Multiple Variables

Dataset: vote

We revisit the voter survey dataset with a focus on identifying variations in voting patterns across different birth months. Later, we aim to determine if there is a discernible relationship between voters' birth months and their voting behaviors. To prepare for a future analysis, we will modify the dataset by isolating the birth month from the existing birthdate information. This involves generating a new column dedicated solely to the birth month. This modification prepares the data, to explore in a future activity, how voting preferences may vary according to the birth month of the respondents.

Questions

1. How many people have a birthday in each month of the year?
2. Describe the results using the STAR Framework.

Tool Time–Available in MyLab Statistics

- Excel
- Python
- R
- StatCrunch

SECTION 2.6: RESHAPING A DATASET

The last type of data transformation we consider here is *reshaping* a dataset. Recall that a *tidy* dataset is one in which each column represents a variable, each row represents an observation, and each cell has a value. Often a raw dataset presented as a table will not be in a tidy format because of the shape that it's in.

Figure 2.4 provides an example of a Gapminder dataset that is not tidy. In this example, we have a row for each country in the world and a column for each year starting from 1800. Each cell contains the population value for a particular country and year combination. The three variables in this dataset are `country`, `year`, and `population`. However, the years only appear as column names and not as values in a column. A tidy version of this dataset would have three columns, one for each of these variables, and a row for each combination of country and year. Each combination of country and year (e.g., Afghanistan in 1800) can be considered as an observational unit.

The dataset in Figure 2.4 is an example of a *wide-format* dataset because it contains more columns than we might need. To tidy this dataset we aim to convert it into a *long format* with only three columns: `country`, `year`, and `population`. Figure 2.5 provides a preview of this tidy, long-format dataset.

The process of *reshaping a dataset* usually refers to the transformation of a dataset from wide format to long format, or from long format to wide format. The wide format version can be better for humans to read (e.g., to see as labels on a graph). Although the long format version seems less readable to a human, it is a better form for computing systems that can quickly use these variables (columns) for visualization and analysis. In either case, the goal is to end with a form that lends itself to visualization and analysis.

FIGURE 2.4
Wide format of the Gapminder dataset of population information for countries of the world each year starting from 1800.

country	1800	1801	1802	1803	1804	1805	1806	1807	1808	1809	1810
Afghanistan	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M
Angola	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M
Albania	400k	402k	404k	405k	407k	409k	411k	413k	414k	416k	418k
Andorra	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650
UAE	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k
Argentina	534k	520k	506k	492k	479k	466k	453k	441k	429k	417k	420k
Armenia	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k
Antigua and Barbuda	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k
Australia	200k	205k	211k	216k	222k	227k	233k	239k	246k	252k	259k
Austria	3M	3.02M	3.04M	3.05M	3.07M	3.09M	3.11M	3.12M	3.14M	3.16M	3.18M
Azerbaijan	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k
Burundi	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k
Belgium	3.25M	3.26M	3.27M	3.28M	3.29M	3.3M	3.3M	3.31M	3.32M	3.33M	3.34M

Gapminder

FIGURE 2.5
Tidy, long format version of Gapminder dataset, which contains population information for countries of the world each year starting from 1800.

country	year	population
Afghanistan	1800	3.28M
Afghanistan	1801	3.28M
Afghanistan	1802	3.28M
Afghanistan	1803	3.28M
Afghanistan	1804	3.28M
Afghanistan	1805	3.28M
Afghanistan	1806	3.28M
Afghanistan	1807	3.28M
Afghanistan	1808	3.28M
Afghanistan	1809	3.28M
Afghanistan	1810	3.28M
Afghanistan	1811	3.28M
Afghanistan	1812	3.28M
Afghanistan	1813	3.28M
Afghanistan	1814	3.28M
Afghanistan	1815	3.28M
Afghanistan	1816	3.28M
Afghanistan	1817	3.28M
Afghanistan	1818	3.28M
Afghanistan	1819	3.28M
Afghanistan	1820	3.29M
Afghanistan	1821	3.3M
Afghanistan	1822	3.31M
Afghanistan	1823	3.32M
Afghanistan	1824	3.34M

Gapminder

DEFINITION

Reshaping a dataset involves transforming its structure from a wide format, with extensive columns, to a long format, with fewer columns but more rows, or vice versa, to facilitate more effective visualization and analysis.

SECTION 2.7: COMBINING DATASETS

A vast number of data science projects and situations involve a single, often large, dataset from which to work and draw insight. Part of that work can involve seeking other sources of data that might be *integrated* to help shed light on patterns or relationships in the original data. Other times, a data science project will involve multiple datasets from the outset, but the *integration* of these datasets is a challenge. No matter the starting point, it is often the case that more data are better than less data. As part of data wrangling, we should be prepared to work with multiple datasets whenever possible.

Here, we will cover the most basic ways to combine two datasets. In Chapter 5 on Data Management, we will discuss more sophisticated methods. It's important to understand that additional cleaning and transforming may be necessary with the newly formed dataset after integration has occurred.

Stacking Datasets

Stacking two datasets is exactly what it sounds like. We have two datasets that have completely different observations, but identical variables. In this case, to combine these two datasets we can stack them vertically, one on top of the other. The order in which this is done does not matter while the combining is happening. For example, we might have average annual rainfall and temperature for four countries in one dataset (Table 2.7) and the same variables for three additional countries in another dataset (Table 2.8). Since both datasets have identical variables (`country`, `average_rainfall`, and `average_temperature`), we can stack them (Table 2.9).

TABLE 2.7
The average annual rainfall (inches) and temperature (Fahrenheit) in four countries

country	average_rainfall	average_temperature
Kenya	24.8	71.6
Tanzania	42.2	78.8
Uganda	47.2	71.6
Ethiopia	33.4	70.2

TABLE 2.8
The average annual rainfall (inches) and temperature (Fahrenheit) in three countries

country	average_rainfall	average_temperature
China	25.4	57.2
Japan	66.0	60.8
India	42.6	75.2

TABLE 2.9
Resulting dataframe after stacking Tables 2.7 and 2.8.

country	average_rainfall	average_temperature
Kenya	24.8	71.6
Tanzania	42.2	78.8
Uganda	47.2	71.6
Ethiopia	33.4	70.2
China	25.4	57.2
Japan	66.0	60.8
India	42.6	75.2

Augmenting a Dataset

Augmenting two datasets is, in some sense, the opposite of stacking two datasets. We have two datasets that have completely different variables but identical observations. In this case, to combine these two datasets we can put them together horizontally, one to the right of the other. The order in which this is done does not matter while the combining is happening.

To augment two datasets, the observations in one table must match the observations and sequence of those in the other. For example, one dataset might have Japan's average rainfall for four seasons (Table 2.10), and another might have Japan's average temperature for the same seasons (Table 2.11). If both datasets identify all four seasons in alphabetical order, we can augment the two datasets (Table 2.12).[1]

TABLE 2.10
Japan's average rainfall (inches) for four seasons.

season	average_rainfall
Fall	16.5
Winter	16.7
Spring	9.7
Summer	23.1

TABLE 2.11
Japan's average temperature (Fahrenheit) for four seasons.

season	average_temperature
Fall	59.4
Winter	43.2
Spring	58.6
Summer	81.9

TABLE 2.12
Resulting dataframe after augmenting Tables 2.10 and 2.11.

season	average_rainfall	average_temperature
Fall	16.5	59.4
Winter	16.7	43.2
Spring	9.7	58.6
Summer	23.1	81.9

[1] Augmenting dataset is distinct from data augmentation in machine learning, which involves creating new, synthetic data based on an original dataset to improve performance.

Merging Datasets

Merging two datasets is the process by which these sets are combined in a manner beyond mere stacking and augmenting. There are various ways this can occur. We may have two datasets with differing amounts of observations and variables. However, there must be at least one common variable between the two datasets. *Merging* these two datasets will happen by matching observations between them based on at least one of the common variables. Due to the complexity of this process, we will defer the rest of this discussion to Chapter 5.

Suppose a university's Student Life office has two datasets:

1. *Club Membership Data*: The dataset in Table 2.13 includes the student ID (`student_id`) and the name of the club (`club_name`) to which each student belongs.
2. *Academic Performance Data*: The dataset in Table 2.14 includes the student ID (`student_id`), their study major (`major`), and their grade point average (`gpa`).

TABLE 2.13
Club membership data.

student_id	club_name
S1	Drama Club
S1	Math Club
S2	Debate Team
S3	Math Club
S3	Science Society
S4	Drama Club
S5	Debate Team
S5	Science Society

TABLE 2.14
Academic performance data.

student_id	major	gpa
S1	Math	3.8
S2	English	3.5
S3	Science	3.9
S4	Drama	3.2
S5	Politics	3.7

Quick Tip

When integrating data from different sources, ensure consistency in units, scales, and format. We can clean one dataset to help it match the other better. For example, one might have two decimal places and the other only one, or one might use abbreviations while the other does not.

The Student Life office wants to investigate whether participation in certain clubs is associated with better academic performance. They hope this information will not only highlight the benefits of extracurricular involvement but also help them identify clubs that might be providing students with valuable skills or support.

We can merge these datasets on the `student_id` column to create a comprehensive dataset in Table 2.15.

The merged dataset in Table 2.15 allows us to investigate potential relationships between club participation and academic performance. This could potentially lead to exciting findings that can help promote club involvement and improve student success. We discuss merging datasets more thoroughly in Chapter 5: Data Management.

TABLE 2.15
Merged data.

student_id	club_name	major	gpa
S1	Drama Club	Math	3.8
S1	Math Club	Math	3.8
S2	Debate Team	English	3.5
S3	Math Club	Science	3.9
S3	Science Society	Science	3.9
S4	Drama Club	Drama	3.2
S5	Debate Team	Politics	3.7
S5	Science Society	Politics	3.7

Chapter 2: Putting It Together

Data wrangling is a crucial first step in any data analysis project. It prepares the data for exploration or storytelling, setting the stage for meaningful insights. In practical terms, this means converting raw data into a more digestible format, thereby saving time and effort in the subsequent stages of analysis and storytelling.

Understand the Significance of Data Wrangling

- Defined data wrangling as a precursor to data analysis and storytelling
- Emphasized the practical implications of converting raw data into a manageable format

Administer Strategies for Addressing Missing Data

- Covered various strategies to handle missing data, including removing observations and imputation
- Discussed how these methods can enhance the reliability of the dataset
- Highlighted the importance of a well-planned approach to dealing with missing data

Find and Address Anomalous Values

- Discussed the appearance of anomalous and outlier values in data
- Explained the impact of such values on the integrity of data analysis
- Stressed the importance of outlier detection methods for robust analysis

Transform Quantitative and Categorical Variables

- Examined ways to transform both quantitative and categorical variables
- Explained the importance of proper transformation in enhancing data structure
- Illustrated the benefits of these transformations through practical examples

Reshape Data into a Tidy Format

- Reviewed the concept of tidy data, where each column represents a variable, each row represents an observation, and each cell has a value
- Showed how reshaping data simplifies their structure for particular purposes

Combine Multiple Datasets

- Explained methods for integrating multiple datasets, such as stacking and merging
- Outlined the benefits of data integration in achieving a comprehensive analysis
- Highlighted how such integration can provide new avenues for insights

We can now navigate the complexities of data wrangling. Understanding how to handle missing data, detect anomalies, and adequately format datasets provides a solid foundation for data visualization and exploring the data through analysis.

Chapter 2: Ethics in Practice: Othering

Recall the favorite pet question in Mirabel's dataset. If most respondents answered "dog" or "cat" with a large variety of different responses for the remainder of the respondents, she could add an "other" category to represent them in her `favorite_pet`variable (e.g., "goldfish" and "I don't have a favorite"). While condensing the categories in this way might be useful for Mirabel's purposes, now we will discuss ethical considerations for variables that measure attributes attached to people's identities. For example, a survey could ask respondents about their religion, political identity, race, gender identity, sexual identity, or any number of other characteristics. Data scientists may be tempted to put less represented identities into an "other" category for convenience. However, people do not want to be marginalized as "others" and this condensing reduces the representativeness of the dataset to these individuals.

This process of grouping responses into an "unknown" or "other" category is a version of *othering*, a phenomenon in which some individuals or groups are defined and labeled as not fitting within the perceived norms of a social group. Othering should be avoided whenever possible and extremely well documented otherwise. For example, reducing respondents' political identity to "conservative," "liberal," and "other" is an oversimplification. Othering does not provide proper representation to the people surveyed. The historical use of this process and the requirement of certain procedures for minimum sample sizes underscore the need for more data that represent everyone well. We need to avoid othering. Until othering has been completely eradicated, we must remain as vigilant and cognizant as possible about who an "other" label is representing in a dataset, and whether that representation is tolerable or not.

Chapter 2: Key Terms

data wrangling/data preprocessing
data transformation
dataframe
imputation
mean
median
implausible values
extreme value/outlier
incorrect formats
duplicate records
encoding
Boolean encoding
condensing the categories
reshaping a dataset

CHAPTER 2: CHAPTER REVIEW QUESTIONS

Section 2.1: What Is Data Wrangling?

1. What term refers to the process of converting raw data into a tidy, organized format suitable for analysis?
2. What are the three key components of data wrangling?
3. Explain the data wrangling process.
4. What is the first step in data wrangling?
5. In the context of data wrangling, what does "integrating the data" refer to?
6. What is the purpose of cleaning data in data wrangling?
7. What does the term "dataframe" mean in the context of data wrangling?

Section 2.2: Cleaning Missing Data

8. What are three ways of dealing with missing data values?
9. What is the advantage of imputing missing values instead of leaving them missing?
10. What issues might arise when dropping rows with any missing values from a dataset?
11. When cleaning data, under which condition would it make the most sense to remove an entire row of data?
12. What term refers to estimating missing values based on other available information?
13. Describe a potential issue that arises from imputing missing data values.

For questions 14–17, please refer to the following scenario. Below is a table with data on 10 students. There are missing values scattered throughout:

student	state	age	gpa	favorite_color
1	NY	18	3.3	Blue
2	CA		2.9	Red
3	TX	20	3.4	Green
4	FL	19		Yellow
5	IL	21	3.3	Orange
6	CA	22	3.7	Purple
7	NJ		3.3	Blue
8	GA	23	2.8	Red
9	CA	18	3.0	Green
10	CA	20	3.3	

14. What is the mean GPA, ignoring only the missing GPA value?
15. What is the mean GPA, ignoring any row with a missing value?
16. Impute the CA student's missing age by taking the mean age of other CA students. What is their imputed age?
17. Impute the missing favorite color by the most frequent favorite color of students with the same GPA. What is the imputed color?

Section 2.3: Cleaning Anomalous Values

18. What term refers to data values that are conspicuously different from other values of the same variable?
19. If a dataset contains a person's age as 2995, how might this be addressed in data cleaning?
20. In data cleaning, which term refers to values that are clearly invalid or impossible?
21. Describe an example of incorrect formatting in data.
22. What is the best way to handle an extreme data value that has been verified to be correct?

Section 2.4: Transforming Quantitative Variables

23. What term refers to the creation of a new categorical variable by assigning categories to intervals of an existing quantitative variable?
24. How might a quantitative variable, such as annual income measured in dollars, be transformed for easier visualization?
25. Describe an example of creating a new quantitative variable by combining two existing quantitative variables.
26. How does scaling a quantitative variable by dividing its values help create better visualizations?
27. In general, what is the best practice when transforming variables?
28. What can be done to make a quantitative variable with a wide range easier to visualize and interpret without distorting the data?
29. How does creating a quantitative variable from a categorical one enable more types of analysis?
30. The following table displays the data usage of 10 students on campus, measured in megabytes (MB). For a campus-wide study, convert these data usage values to gigabytes (GB), with 1 GB = 1024 MB. Then calculate the mean and median data usage in GB, rounding to two decimal places.

student	data_usage	major	data_usage
Jackson	10587	Art	
Oneal	15789	CS	
Ortega	12678	Bio	
White	14567	Math	
Martinez	16987	Eng	
Dixon	12356	Psych	
Garcia	13489	Hist	
Harris	15782	Econ	
Brown	14256	Socio	
Baker	13567	Music	

For questions 31–32, please refer to the following scenario. The following table shows the heights of 10 athletes on campus, measured in inches (in).

31. Convert these heights to centimeters (cm) using the following conversion rate:

$$1 \text{ inch} = 2.54 \text{ centimeters}$$

student	height	major	height
Washington	71	Football	
Jefferson	74	Basketball	
Patel	68	Soccer	
Gupta	69	Baseball	
Li	74	Tennis	
Kim	75	Swimming	
Rodriguez	73	Track	
Garcia	72	Hockey	
Jackson	66	Wrestling	
Sharma	70	Golf	

and find the mean height in centimeters. Round to two decimal places.

32. In data cleaning, what might be done with text values in a quantitative variable like height?

For questions 33–34, please refer to the following scenario.
For a study on student health, the following table provides the weight (in lb), height (in inches), and majors of 10 students.

student	weight	height	major	bmi
A	120	62	Math	
B	135	64	Bio	
C	150	66	CS	
D	165	68	Eng	
E	180	70	Psych	
F	155	64	Hist	
G	160	66	Art	
H	175	68	Econ	
I	190	70	Socio	
J	205	72	Music	

33. Create a new variable, bmi by filling in the values of the final column's cells using the formula (rounded to one decimal place):

$$\text{bmi} = \text{Weight (lb)} / [\text{Height (in)}]^2 \times 703$$

34. What is the mean bmi for this dataset, rounded to one decimal place?

For questions 35–36, please refer to the following scenario. Use two decimal places for all answers.
The following table provides the distance (in miles), fuel consumption (in gallons), and cost per gallon (in $) for trips made by 10 students.

35. Calculate the cost per mile for each trip using the following formula:

$$\text{Cost per Mile (\$)} = (\text{Fuel Consumption (gallons)} * \text{Cost per Gallon (\$)}) / \text{Distance (miles)}$$

student_id	distance	fuel_consumption	cost_per_gallon	major	cost_per_mile
345634	50	2	3	Art	
534534	100	4	2.5	CS	
832344	150	5	3.5	Bio	
242345	200	6	3	Math	
859285	250	7	2.5	Eng	
894993	100	3	3	Psych	
388539	150	4	2.5	Hist	
393559	200	5	3.5	Econ	
395222	250	6	3	Socio	
825234	300	7	2.5	Mic	

36. What is the average cost per mile?

For questions 37–38, please refer to the following scenario.
In a physics experiment, the following table provides the mass (in kilogram), and acceleration (in meter/second^2).

object	mass	acceleration	force
1	5	0.5	
2	10	1	
3	15	1.5	
4	20	2	
5	25	2.5	
6	30	3	
7	35	3.5	
8	40	4	
9	45	4.5	
10	50	5	

37. Calculate the force (in Newton) applied to each cell in the final column using the formula (to one decimal place):

$$\text{force} = \text{mass} \times \text{acceleration}$$

38. What is the mean force applied to the objects (one decimal place)?

For questions 39–42, please refer to the following scenario.
Below are data on participant, age (years), height (inches), and weight (lbs).

participant	age	height	weight
1	32	5'11	175
2	55	5'5	132
3	62	5'9	195
4	18	5'3	120
5	23	6'1	180
6	43	5'7	150
7	51	5'8	170
8	28	5'4	135
9	47	6'0	200
10	61	5'2	142

39. Create a categorical `age_group` variable using the following categories:

- 18–35: Young Adult
- 36–55: Middle Aged
- 56+: Senior

40. How many participants would be in the Young Adult category?
41. How many would be Middle Aged?
42. How many would be Senior?

Section 2.5: Transforming Categorical Variables

43. How might the uncommon (low frequency) categories of a categorical variable be simplified, given that there are no concerns about othering?
44. What term refers to extracting multiple variables from a single categorical variable?
45. Why is it important to assign every possible value to a category when creating categories from a quantitative variable?
46. What is a Boolean variable? When is it used?
47. What are some other names for Boolean variables?
48. When encoding a categorical variable numerically, why is the choice of values important?
49. What term refers to reducing the number of distinct values a categorical variable can take?
50. Describe the process of condensing a categorical variable.
51. When would splitting a categorical variable into multiple new variables be useful?
52. Provide three examples of categorical variables.
53. What term is used to refer to the creation of quantitative variables from a categorical variable?
54. What is an example of creating a categorical variable from a quantitative one?
55. What is a potential mistake that can arise when encoding categorical data numerically?

For questions 56–58, please refer to the scenario below.
Below is a table with birthdate data for 10 people. Split the birthdate into Month, Day, and Year columns.

person	birth_date	month	day	year
1	01/22/1991			
2	04/17/1988			
3	09/05/1993			
4	11/27/1985			
5	03/14/1980			
6	05/29/1992			
7	02/03/1994			
8	10/11/1989			
9	06/25/1986			
10	08/19/1990			

56. How many people have January as their birth month?
57. How many were born in the last 15 days of a month?
58. What is the mean birth year, rounded to the nearest year?

For questions 59–61, please refer to the scenario below.
Below are responses from 10 participants who either agreed or disagreed with the statement:

"I am satisfied with the meal I just had at this restaurant."

participant	response	age
1	Strongly Disagree	35
2	Agree	29
3	Neutral	42
4	Strongly Agree	27
5	Agree	38
6	Disagree	31
7	Neutral	44
8	Agree	24
9	Strongly Agree	36
10	Disagree	39

59. Recode the responses as follows:

    Strongly Disagree = 1 | Disagree = 2 | Neutral = 3 | Agree = 4 | Strongly Agree = 5

60. What is the mean of the recorded responses rounded to one decimal place?
61. What is the median rounded to no decimal places?

For questions 62–63, please refer to the scenario below.
A researcher collected data on the college majors of 10 students. The major data are in the following format:
Department—Major
For example:
Business—Marketing

62. Split the `department_major` into department and major columns.

student	department_major	department	major
1	Computer Science—Software Engineering		
2	Physics—Applied Physics		
3	Business—Accounting		
4	History—American History		
5	Engineering—Electrical Engineering		
6	English—Creative Writing		
7	Chemistry—Biochemistry		
8	Education—Elementary Education		
9	Math—Applied Math		
10	Nursing—Nursing		

63. How many students have Business majors?

Section 2.6: Reshaping a Dataset

64. Describe the process of converting a dataset from wide format to long format.
65. Why might a data scientist prefer a long-format dataset?

Section 2.7: Combining Datasets

66. What is an advantage of stacking two datasets with the same variables but different observations?
67. Which data integration technique combines datasets with all the same variables but additional observations?
68. When would it be appropriate to augment two datasets by binding them horizontally?
69. What does "merging datasets" typically involve?
70. For which type of data integration must datasets have at least one common variable?
71. When merging two datasets, what must they have in common?
72. What term refers to binding two datasets together horizontally when they have the same observations but completely different variables?
73. What is the primary purpose of combining datasets?

For question 74, please consider the following two tables.

Table 1:

name	age	gpa
Alfonso	32	3.4
Kai	41	3.7
Sam	28	3.4

Table 2:

nickname	height	weight
Al	5'11"	165 lbs
KK	5'5"	132 lbs
Sam the Man	6'0"	180 lbs

74. If these tables were augmented by binding them horizontally, how many rows and how many columns would the resulting table have? (Don't count the column headings as rows.)

For questions 75–76, please consider the scenario below.
A graduate student has collected data on student grades in two sections of her math class.

The grades for Math in Section 1 are as follows:

name	math_test_1	math_test_2	section
Amanda	82	90	1
Diego	71	85	1
Sara	94	88	1

The grades for Math in Section 2 are as follows:

name	math_test_1	math_test_2	section
Tia	75	81	2
Anna	89	77	2
Sarah	93	88	2

75. If the graduate student stacks these data tables, how many rows would the new stacked table contain?
76. How many columns would the new stacked table have?

For questions 77–78, please refer to the scenario below.
A professor collected data on his students' performance across two semesters and wants to combine them into one large table.

Semester 1 Exam Scores:

name	exam_1	exam_2	final_exam	semester
Alex	82	90	93	1
Diego	77	88	81	1

Semester 2 Exam Scores:

name	exam_1	exam_2	final_exam	semester
Alex	86	92	95	2
Sam	80	85	79	2

77. If the professor stacks these two data tables, how many rows would the stacked table have?
78. How many columns would the stacked table have?

For questions 79–80, please consider the scenario below.
A teacher wants to analyze data on her students' performance. She has collected the following data:
Math test scores:

name	test_1	test_2	test_3
Anna	82	90	75
Diego	71	85	83
Sara	94	88	79
Tia	62	77	68

Reading test scores:

name	test_1	test_2
Anna	85	79
Diego	83	77
Sara	91	94

79. If the teacher merges these tables using Name, how many rows would the new table have?
80. How many columns would it have?

For questions 81–82, please consider the scenario below.
A company has two datasets:
Customer data:

customer_id	name	email
1	John	john@email.com
2	Sarah	sarah@email.com

Purchase data:

customer_id	date	amount
1	1/5/2023	$50
2	1/7/2023	$25
3	1/10/2023	$100

To better understand their customers' purchases, they want to merge these datasets using the `customer_id` field.

81. If they merge the datasets, how many unique `CustomerID` values would exist in the merged dataset?
82. How many rows would the merged dataset contain?

For questions 83–86, please consider the scenario below.
Below are data on the majors of 20 university students:

student	major	gpa	clubs
1	Biology	3.2	1
2	English	3.0	2
3	Math	3.5	0
4	History	3.2	1
5	Computer Science	3.8	0
6	Physics	3.4	2
7	English	3.1	3
8	Math	3.7	1
9	History	3.0	0
10	Biology	3.5	2
11	Computer Science	3.6	1
12	Physics	3.2	0
13	Arts	3.9	2
14	Math	3.3	1
15	English	2.9	0
16	Biology	3.8	2
17	History	3.0	3
18	Computer Science	3.5	1
19	Engineering	3.7	0
20	Arts	3.4	2

83. Merge the categories of the categorical variable, `major` into three categories:

- STEM: Biology, Chemistry, Physics, Math, Computer Science, Engineering
- Humanities: English, History, Arts
- Other: Any other majors

84. How many students have a STEM major?
85. How many have a Humanities major?
86. How many have an Other major?

For questions 87–89, please consider the following scenario.
A researcher collected data on crops grown by 15 farms:

farm	crop	acres	yield
1	Corn	80	8500
2	Soybeans	160	9200
3	Wheat	240	11,000
4	Corn	100	9000
5	Soybeans	200	10,000
6	Cotton	150	8000
7	Wheat	300	12,000
8	Corn	90	8000
9	Cotton	140	7500
10	Soybeans	180	9500
11	Wheat	220	10,500
12	Cotton	160	8500
13	Corn	110	9500
14	Soybeans	170	9000
15	Wheat	260	11,500

87. Merge the categories of the categorical variable, `crop`, into two categories:

- Row crops: Corn, Soybeans, Cotton
- Small grains: Wheat

88. How many farms grow row crops?
89. How many farms grow small grains?

Chapter 2: Explaining the Concepts

1

Think about all the forms or surveys, online or paper, that you've ever filled out in your life. This could even include creating a user profile and login credentials for the school in which you're currently enrolled, or the site through which you're accessing this text. List four or five of the fields (pieces of information) you provided on some of these forms. Then, list two or three ways in which a person could fill out these forms differently than you did, which would make the data *inconsistent*. Would you suggest changing the prompts on the form to help make the data more *consistent*, or do you think these issues are better handled by the data scientist after the data are collected?

2

Wrangling data to make it more ready for visualization can include a variety of tasks. For instance, we mentioned that Mirabel might have converted all of her `favorite_pet` variable values to one of *dog*, *cat*, or *unknown*. However, when these are displayed as values on a graph, it may be more appealing for readers to see *Dog*, *Cat*, and *Other*.

Alternatively, we could have a quantitative variable representing the total revenue of companies in a particular industry sector, which takes values in the billions of U.S. dollars. Should a visualization of this variable include values that look like "$10,000,000,000" or would this large number of digits take up too much space and be more easily digestible as something like "$10bn"?

Explain, with examples, what kind of cleaning and transformation steps would be especially useful for the visualization of different types of variables.

3

In our explanation of *transforming data*, we mentioned that it is very often the case that new variables are created, and existing variables are retained, not replaced. Explain why this is a good practice. Give examples, using some of our described *data transformation methods*, of how **replacing** an existing variable with the transformed version results in a loss of information in the dataset and hence a reduction in the quality of the dataset.

4

In our explanation of *integrating data*, we mentioned that the ways in which two datasets are related, and hence combined, can depend on the variables, the observations, or both. Explain why it is more useful to *clean* and *transform* the data of each dataset separately first, and then combine them rather than doing it the other way around. *Note: More cleaning or transforming may still be necessary after combining, but let's focus on the initial steps first.*

5

Documenting the data science workflow as we progress through it is extremely important. This is partly because data can come from the same source and in the same format at multiple points in time. For instance, another student, Miles, may want to use Mirabel's dataset responses as a model to collect data from a new group of students next year. All of the prompts will be the same, and the answers could involve many of the same *inconsistency* issues that Mirabel faced.

- Explain why having detailed instructions and explanations of Mirabel's data wrangling steps would be useful to Miles when he analyzes his new data.
- Generally, explain why it's good practice for Mirabel to ensure that the work she has done with the data can be replicated on her data and applied accurately to new data.

Chapter 2: Applying the Concepts

Activity: Identify Missing Data

Dataset: titanic_dataset

Now we step aboard the Titanic dataset. Identify the missing values in the `age` variable. This voyage goes beyond just finding missing data. You must deliberate on the implications of these missing values and strategize the best way to handle them. Are you going to steer around them, leaving them as they are, or are you going to anchor down and impute them? Your challenge here is not only to identify missing values, but also to weigh the pros and cons of different handling strategies and apply your chosen method.

Assess the presence of missing `age` values in the dataset as the first step in your analysis.

1. Overall, how many passengers were recorded in this dataset?
2. Of the total recorded passengers, how many were missing a value for the `age` variable?
3. What is the percentage of total recorded passengers who did not have their ages recorded?

Missing data can bias or distort your analysis, leading to incorrect conclusions. In the context of the Titanic dataset, if the ages of younger or older passengers are disproportionately missing, analysis around age may lead to misleading results.

4. If child passengers were less likely to have their ages recorded in this dataset than adult passengers, how would this affect the average age we calculate?

Understanding why data are missing can provide valuable insights. Were ages not recorded for certain passengers? Did passengers refuse to disclose their ages? The reasons can have implications on how we interpret the rest of the data.

5. Different strategies for handling missing data have different effects on the analysis. In one such strategy for handling missing data, the missing values are filled with educated guesses. What is this strategy called?

In the Titanic dataset, correctly handling missing age data could significantly impact any conclusions drawn about survival rates among different age groups.

Activity: Impute Missing Values from Internal Data

Dataset: world_happiness_report_2005_2021

We will be examining the missing data within the World Happiness Report for healthy_life_expectancy_at_birth. This variable has 58 missing values. We will investigate how different imputation methods influence calculations of average life expectancy, specifically using the mean and median imputation methods.

Use two decimal places for numeric responses.

Answer the following questions:

1. What average life expectancy does the mean imputation method produce? Round to two decimal places.
2. What average life expectancy does the median imputation method produce? Round to two decimal places.

Activity: Identify Extreme Data Values

Dataset: bird_watcher

Consider the birdwatcher's data. Are any of the ages extreme values?

Answer the following questions:

1. What age is the lowest value for the oldest 5% of birdwatchers?
2. What age is the highest value for the youngest 5% of birdwatchers?
3. How many birdwatchers are above 100?
4. How many birdwatchers are below 18?
5. Which values are impossible?
6. Explain this analysis using the STAR Framework.

Activity: Clean Incorrect Formats

Dataset: bird_watcher

This dataset originates from an Airbnb birdwatching experience, where the guide, a seasoned ornithologist, specializes in showcasing rare and captivating bird species such as sparrows, woodpeckers, and hummingbirds. The experience is tailored for nature enthusiasts and photographers, ranging from beginners to experts. After each tour, participants are asked to complete a short survey. The primary aim is to gather feedback on their experience, focusing on their age, favorite bird spotted during the tour, the lens size of their binoculars, and whether they would consider repeating the experience.

The survey was conducted over several years, accumulating responses from a diverse group of 23,893 birdwatchers. The dataset includes a unique identifier for each participant (`birdwatcher_id`), their age, their stated favorite bird from the tour, the lens size of their camera, a Boolean indicator of their interest in a repeat experience, and the date of their original tour. The guide is now interested in cleaning this data to analyze the relationship between participants' favorite birds and their likelihood of signing up for a second tour.

1. Clean the `favorite_bird` column of the birdwatcher dataset. Perform a basic cleaning of the data in this order:
   a. "sparrows" should be changed to "sparrow"
   b. "woodpeckers" should be changed to "woodpecker"
   c. "hummingbirds" should be changed to "hummingbird"
   d. any word that begins with "s" and ends with "w" should be changed to "sparrow"

   e. any word that begins with “w” and ends with “r” should be changed to “woodpecker”
   f. any word that begins with “h” and ends with “d” should be changed to “hummingbird”
2. Before basic data cleaning, how many people in this dataset indicated their favorite bird as:
   a. “sparrow”?
   b. “woodpecker”?
   c. “hummingbird”?
3. After basic data cleaning, how many people in this dataset indicated their favorite bird as:
   a. “sparrow”?
   b. “woodpecker”?
   c. “hummingbird”?
4. Explain the results using the STAR Framework.

Activity: Create a Categorical Variable from a Quantitative Variable

Dataset: ames_housing

We’re about to explore the suburban landscape of the housing market! We will use the Ames Housing dataset for this venture. Your mission? Give the `sale_price` (USD) variable a new spin. Convert this quantitative value into a new categorical variable with the following categories: "Cheap" for houses priced $0–$100,000, "Moderate" for houses priced $100,001–$200,000, and "Expensive" for houses priced $200,001 and above. Although these prices may not reflect the current state of the housing market, let’s extract some insight from our new categories.

Here are the steps we’re going to follow. First, create the new categorical `sale_price_category` variable as described above. Then, count how many houses fall into each of the categories. Finally, make a visualization to better convey the results. Answer the questions below.

1. For this dataset, most of the houses fall into the “Moderate” category. What does this tell us about the prices of most houses in the dataset?

   Suppose we are interested in breaking down the largest category into two new categories with smaller ranges. Recreate the counts and visualization after modifying the existing code to create these two new categories:

2. Split the “Moderate” category into two new categories:
   a. “Cheap Moderate”: $100,001–$150,000
   b. “Expensive Moderate”: $150,001–$200,000
3. How many houses are in the “Cheap Moderate” category?
4. Which category is the largest? Which category is the second largest?
5. How many houses are in the “Expensive” category?
6. How many houses are in the “Expensive Moderate” category?

Activity: Create a Boolean Variable That Represents a Categorical Variable

Dataset: zoo

Now we are going to the Zoo animal classification dataset. This dataset contains information about different animals, including a `class_type` variable that classifies animals as 1 = mammal, 2 = bird, 3 = reptile, 4 = fish, 5 = amphibian, 6 = insect, or 7 = other. Convert these class types into Boolean variables. This transformation is crucial in preparing the data for various animal analyses, which we could call “animalyses” to provoke a reaction.

Start by creating a new Boolean variable called `mammal`. This variable will take on a value of TRUE if the animal in question is a mammal, and FALSE otherwise.

1. After creating the `mammal` variable and counting the occurrences of TRUE and FALSE, how many mammals are in this dataset?

   Create Boolean variables for both birds and reptiles (Hint: `class_type = 2` for birds and `class_type = 3` for reptiles).

2. How many birds are in this dataset? How many reptiles are in this dataset?

In general, this Boolean encoding is useful for computers to encode categorical data, as it turns categories into TRUE/FALSE, or as the computer sees in binary, 1/0.

Creating Boolean variables from a categorical variable can simplify the data, making it more suitable for certain analyses. By converting the `class_type` variable into Boolean variables, we can perform a separate analysis for each class type. This could assist zoologists in understanding patterns unique to each animal class. For instance, this could be used in endangered species prediction, helping to determine conservation priorities.

Activity: Splitting a Variable into Multiple Variables

Dataset: bird_watcher

In this analysis, we revisit the birdwatcher dataset, this time with a focus on understanding seasonal patterns. Our goal is to examine if there's an association between the months of the year and the types of birds observed. To facilitate this, we first need to transform the dataset by isolating the month from the existing `date` variable's information. This involves creating a new column that exclusively represents the month. This step is crucial for enabling us to perform a more focused analysis on how birdwatching experiences vary by month, potentially providing insights into the seasonal preferences of different bird species.

1. How many entries were made in each month of the year?
2. Explain the results using the STAR Framework.

CHAPTER 2: TECH CENTER

PYTHON NOTEBOOK

Load Packages:

```
import pandas as pd
```

Import a Dataset:

Read a CSV file into Python:

```
dataset1 = pd.read_csv("Dataset1.csv")
```

Drop Rows with Missing Values:

Drop rows in a dataframe containing missing values

```
dataset1.dropna()
```

To return a dataframe with columns containing missing values removed

```
dataset1.dropna(axis=1)
```

To create a new dataframe (clean_columns_dataset) with all columns containing missing values removed from a dataset (e.g., dataset1).

```
clean_columns_dataset = dataset1.dropna(axis=1)
```

Fill Missing Values for Column q1 with the Mean of Column q1:

To assign the mean of column "q1" to "Mean_q1"

```
Mean_q1 = dataset1["q1"].mean()
```

To replace the missing values in the column with the mean of q1

```
dataset1["q1"] = dataset1["q1"].fillna(Mean_q1)
```

Replace a Value in Column XX and Row YY with a New Value:

Create user-assigned value

```
new_value = 999
```

Reassign row YY in column XX with 999

```
dataset1.at[2, "q2"] = new_value
```

Create a Categorical Variable from an Existing Quantitative Variable:

Create a new column called q1_groups, which will store group names based on the column q1 value:

```
dataset1['q1_groups']
```

Set cutoff values for each group in "q1_groups"

```
bins = [0, 50, 100, 300]
```

Set group names

```
labels = ['Low', 'Medium', 'High']
```

Create Low, Medium, and High groups based on data in column "q1", "right=" parameter set to True as default, determines whether each "bin" includes its rightmost edge or not

```
dataset1['q1_groups'] = pd.cut(dataset1['q1'], bins=bins,
labels=labels, right=False)
```

```
bins = [0, 50, 100, 300]
labels = ['Low', 'Medium', 'High']
dataset1['q1_groups'] = pd.cut(dataset1['q1'], bins = bins, labels = labels, right = False)
dataset1
```

	id	cat1	cat2	cat3	cat4	cat_binary	q1	q2	q3	q4	q1_groups
0	1	A	type_1	Y	300.0	1	52.199627	31.873114	9.649249	46.768104	Medium
1	2	B	type_3	X	100.0	1	77.308355	40.596928	7.193891	48.072335	Medium
2	3	A	type_3	X	100.0	0	95.874092	23.969227	3.499928	51.925680	Medium
3	4	B	type_1	X	300.0	0	11.732048	45.743154	2.543824	19.171565	Low
4	5	B	type_3	X	200.0	1	10.700414	2.467447	2.653033	5.990407	Low

Create a New Variable That Rescales an Existing Quantitative Variable:

Create a new column "q1_scaled" that contains the values in q1 divided by 1000

```
dataset1["q1_scaled"] = dataset1["q1"] / 1000
```

Create a New Variable from Existing Quantitative Variables:

Create a new column "q2_q3_sum" that contains the sum of the values in q2 and q3

```
dataset1["q2_q3_sum"] = dataset1["q2"] + dataset1["q3"]
```

Create a Quantitative Variable from an Existing Categorical Variable:

```
dataset1['cat2_numeric'] = dataset1['cat2'].astype('category').
cat.codes
```

Create a new column cat2_numeric that contains integer values encoding the categories in cat2: dataset1['cat2_numeric'] = dataset1['cat2'].astype('category').cat.codes

Merge the Categories of an Existing Categorical Variable:

To create a new column called "cat2_merged" with 2 categories (type_1_2 & type_3)

```
dataset1 ["cat2_merged"] = dataset1["cat2"].replace({"type_1":
"type_1_2", "type_2": "type_1_2"})
```

Split a Categorical Variable:

Split the cat2 variable by "_" into two separate columns

```
dataset1[['cat2_part1', 'cat2_part2']] =
dataset1['cat2'].str.split('_', n=1, expand=True)
```

Transformation into Wide Format Using pd.pivot()

Reshape to Long Format: Transform data from wide to long format, pivoting q1 and q2 columns into a longer format columns not included in the id_vars will be absent from the created table.

```
dataset1_melted = pd.melt(dataset1, id_vars=['id'], value_
vars=['q1', 'q2'], var_name='Quantity', value_name='Value')
```

Stack Datasets:

Stack datasets 1 & 2 on top of each other

```
stacked_dataset = pd.concat([dataset1, dataset2],
ignore_index=True)
```

Merge and Augment Datasets:

Merge the two datasets based on matching values within the id column

```
merged_dataset = pd.merge(dataset1, dataset2, on = "id")
```

Subset Using Text Comparisons in Categorical Variables:

Subset (filter) rows where the cat2 variable is equal to "type_1"

```
subset_cat = dataset1[dataset1["cat2"] == "type_1"]
```

Subset Using Numeric Comparison in Quantitative Variables:

Subset (filter) rows where the q1 variable is greater than 50

```
subset_numeric = dataset1[dataset1["q1"] > 50]
```

Subset Based on Missing Values:

Subset rows where the q1 variable is missing

```
subset_missing = dataset1[dataset1["q 1"].isnull()]
```

Subset rows where the q1 variable is not missing

```
subset_not_missing = dataset1[dataset1["q1"].notnull()]
```

Subset Based on Conditions Involving 2+ Variables:

Subset rows where q1 is greater than 50 and cat2 is equal to "type_1"

```
subset_conditions = dataset1[(dataset1["q1"] > 50) &
(dataset1["cat2"] == "type_1")]
```

R SCRIPT

Load Packages:

```
library(tidyverse)
```

Import a Dataset:

Read a CSV file into R:

```
dataset1 <- read_csv("Dataset1.csv")
```

Drop Rows with Missing Values:

Remove rows with any missing values

```
clean_dataset <- dataset1 %>% drop_na()
```

Drop Columns with Missing Values:

Remove columns containing any missing values

```
clean_columns_dataset <- dataset1 %>% select_if(~ !any(is.na(.)))
```

Fill Missing Values for Column q1 with the Mean of Column q1:

Calculate mean of q1 excluding NA values

```
mean_q1 <- mean(dataset1$q1, na.rm = TRUE)
```

Replace NA values in q1 with its mean

```
dataset1 <- dataset1 %>% mutate(q1 = case_when(is.na(q1) ~ mean_
q1, TRUE ~ q1))
```

Replace Value in Column XX and Row YY with a New Value:

Assign a new value

```
new_value <- 999
```

Assign a new value to a specific cell in Column XX and Row YY

```
dataset1$XX[YY] <- new_value
```

Create a Categorical Variable from an Existing Quantitative Variable:

Creating a new categorical variable from q1

```
dataset1 <- dataset1 %>%
mutate(q1_groups = case_when
(q1 >=0 & q1 <=50 ~ "Low",
q1 > 50 & q1 <= 100 ~ "Medium",
q1 > 100 & q1 <= 300 ~ "High"))
```

id	cat1	cat2	cat3	cat4	cat_binary	q1	q2	q3	q4	q1_groups
1	A	type_1	Y	300	1	52.19962736	31.8731139	9.649249408	46.7681040	Medium
2	B	type_3	X	100	1	77.30635541	40.5969281	7.193690620	48.0723350	Medium
3	A	type_3	X	100	0	95.87409231	23.9692275	3.499928436	51.9256798	Medium
4	B	type_1	X	300	0	11.73204804	45.7431544	2.543824011	19.1715654	Low
5	B	type_3	X	200	1	10.70041402	2.4674473	2.653033245	5.9904070	Low

Create a New Variable that Rescales an Existing Quantitative Variable:

Creating a scaled version of q1

```
dataset1 <- dataset1 %>% mutate(q1_scaled = q1 / 1000)
```

Create a New Variable from Existing Quantitative Variables:

Summing columns q2 and q3 to create a new variable

```
dataset1 <- dataset1 %>% mutate(q2_q3_sum = q2 + q3)
```

Create a Quantitative Variable from Existing Categorical Variable:

Converting a categorical variable cat2 to numeric

```
dataset1 <- dataset1 %>% mutate(cat2_numeric =
as.numeric(as.factor(cat2)))
```

Merge Categories of an Existing Categorical Variable:

Merging specific categories type_1 and type_2 into one category

```
dataset1 <- dataset1 %>%
mutate(cat2_merged = recode(cat2, type_1 = "type_1_2", type_2 =
"type_1_2"))
```

Split a Categorical Variable:

Splitting cat2 into multiple categories

```
dataset1 <- dataset1 %>%
separate(cat2, into = c("cat2_part1", "cat2_part2"), sep = "_")
```

Reshape to Long Format:

To pivot "q1" and "q2" columns from wide to long format

```
dataset1_melted <- dataset1 %>% pivot_longer(cols = c(q1, q2),
names_to = "Quantity", values_to = "Value")
```

Stack Datasets:

Stack datasets 1 & 2 on top of each other

```
stacked_dataset1 <- bind_rows(dataset1, dataset2)
```

Merge and Augment Datasets:

Merge the two datasets based on matching values within the id column

```
merged_dataset <- inner_join(dataset1, dataset2, by = "id")
```

Subset Using Text Comparisons in Categorical Variables:

Subset (filter) rows where the cat2 variable is equal to "type_1"

```
subset_cat2 <- dataset1 %>% filter(cat2 == "type_1")
```

Subset Using Numeric Comparison in Quantitative Variables:

Subset (filter) rows where the q1 variable is greater than 50

```
subset_numeric <- dataset1 %>% filter(q1 > 50)
```

Subset Based on Missing Values:

To subset rows based on missing values in a specific variable in R, use the filter() function with

```
is.na()
```

Subset rows where the q1 variable is missing (or not missing)

```
subset_missing <- dataset1 %>% filter(is.na(q1))subset_not_
missing <- dataset1 %>% filter(!is.na(q1))
```

Subset Based on Conditions Involving 2+ Variables:

To subset rows based on conditions involving two or more variables in R, use logical operators like

```
&
```

Subset rows where q1 is greater than 50 and cat2 is equal to "type_1"

```
subset_conditions <- dataset1 %>% filter(q1 > 50 & cat2 == "type_1")
```

EXCEL

Drop Rows with Missing Values:

1. Click on any cell in your dataset.
2. Go to the "Home" tab.
3. In the"Editing" group, click on "Find & Select."
4. Choose "Go To Special."
5. Select "Blanks" and click "OK". This will highlight all blank cells.
6. Right-click on any highlighted cell.
7. Select "Delete."
8. Choose "Entire row" in the delete dialog box.

Drop Columns with Missing Values:

1. Click on any cell in your dataset.
2. Go to the "Home" tab.
3. In the "Editing" group, click on "Find & Select".
4. Choose "Go To Special".
5. Select "Blanks" and click"OK". This highlights all blank cells.
6. Right-click on any highlighted cell.
7. Choose "Delete".
8. Select"Entire column" in the delete dialog box.

Fill Missing Values for Column XX with the Mean of Column XX:

1. Identify the column with missing values.
2. Click on an empty cell where you want to display the mean.
3. Enter the formula

   =AVERAGE(YourColumnRange)

   replacing "YourColumnRange" with the actual range of your column, for example,

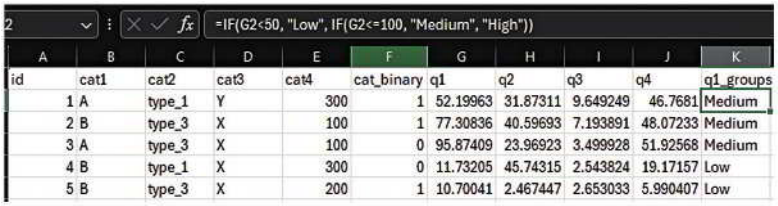
   =AVERAGE(F2:F100)
4. Press Enter to display the mean.
5. Select the cell with the mean value, right-click, and choose "Copy".
6. Select the cells with missing values in your column.
7. Right-click on the selected cells and choose "Paste Special".
8. In the Paste Special dialog, select "Values"and click"OK".

Replace Value in Column XX and Row YY:

1. Locate the cell you want to change.
2. Click on the cell.
3. Enter the new value.
4. Press Enter to confirm the change.

Create Categorical Variable from an Existing Quantitative Variable:

1. Insert a new column next to the column you want to categorize.
2. In the first cell of the new column, use an “IF” or“IFS” function to create categories based on the adjacent cell's value. For example, `=IF(F2<50,"Low",IF(F2<=100,"Medium", "High"))`
3. Copy this formula down the entire column.


2 =IF(G2<50, "Low", IF(G2<=100, "Medium", "High"))

	A	B	C	D	E	F	G	H	I	J	K
	id	cat1	cat2	cat3	cat4	cat_binary	q1	q2	q3	q4	q1_groups
	1	A	type_1	Y	300	1	52.19963	31.87311	9.649249	46.7681	Medium
	2	B	type_3	X	100	1	77.30836	40.59693	7.193891	48.07233	Medium
	3	A	type_3	X	100	0	95.87409	23.96923	3.499928	51.92568	Medium
	4	B	type_1	X	300	0	11.73205	45.74315	2.543824	19.17157	Low
	5	B	type_3	X	200	1	10.70041	2.467447	2.653033	5.990407	Low

Create New Quantitative Variable That Is a Scaling of an Existing Quantitative Variable:

1. Insert a new column next to the column you want to rescale.
2. In the first cell of the new column, enter a formula to rescale the value from the adjacent column. For example,

   =F2/1000

3. Copy this formula down the entire column.

Create New Quantitative Variable That Is a Combination of Existing Quantitative Variables:

To sum two quantitative variables:

1. Insert a new column where you want the sum.
2. In the first cell of the new column, enter a formula to add the two variables. For example,

   `=B2+C2`

3. Copy this formula down the entire column.

Create a Quantitative Variable from an Existing Categorical Variable:

1. Next to your categorical column, insert a blank column for the new quantitative data.
2. In the first cell of the new column, enter

   =IF(A2="Category1", 1, IF(A2="Category2", 2,…))

   adjusting for your specific categories and corresponding numeric values.
3. Copy this formula down the entire column.

Merge Categories of an Existing Categorical Variable:

1. Insert a new column next to the categorical column.
2. Use an “IF” function to merge categories. For example,

   `=IF(B2="type_1","type_1_2",IF(B2="type_2","type_1_2",B2))`

3. Copy this formula down the entire column.

Split Categorical Variable:

1. Insert new columns next to the categorical column.
2. Use the “LEFT”,“RIGHT”, or “MID” function to extract parts of the text. For example,

   `=LEFT(B2, FIND("_",B2)-1)” and “=MID(B2, FIND("_",B2)+1, LEN(B2))`

3. Copy these formulas down the respective columns.

Pivot Longer:

Excel has a built-in tool that allows us to extract all kind of information from our data without manually reshaping it.

1. Select your data and Go to "Insert" → "Pivot Table".
2. Choose placement for the Pivot Table.
3. Drag the identifier column (like ID) to "Rows".
4. Move columns to be pivoted to "Values".

5. Optional: The Filters field excludes data based on specified criteria, refining the displayed results.
6. Set "Values" field to "Count" or as needed.

Pivot Wider:

Excel has a built-in tool that allows us to extract all kind of information from our data without manually reshaping it.

1. Select your data and Go to "Insert" → "Pivot Table".
2. Choose placement for the Pivot Table.
3. Drag the column headers you want to pivot into the "Rows" area in the Pivot Table Field List. Drag any value field (e.g., the data corresponding to the headers) into the "Values" area.
4. Adjust the "Values" field settings if necessary (e.g., sum, count).

Stack Datasets:

1. Select the range of the second dataset.
2. Copy the range.
3. In the first dataset, go to the cell where you want the second dataset to begin.
4. Paste the copied range.

Merge Datasets Using XLOOKUP:

1. Ensure both datasets are in separate sheets.
2. In Dataset A, in a new column, enter

   =XLOOKUP(A2, Sheet2!$A$2:$A$100, Sheet2!$B$2:$B$100)

   where A2 is the ID in Dataset A and Sheet2 contains Dataset B. Dataset A's ID variable should be in column A2:A100 and match ID values in Dataset B, located in Sheet2's column B2:B100.
3. Drag the formula down to fill other rows in Dataset A.
4. Adjust XLOOKUP for other columns from Dataset B as needed.
5. Save the Excel file with the merged data.

Merge Datasets Using Power Query:

1. Open Excel, go to "Data" → "Get Data" → "From File" → "From Excel Workbook"

   Import both Dataset A and Dataset B by selecting and loading them.
2. In Excel, navigate to "Data" → "Get Data" → "Launch Power Query Editor".
3. In the Power Query Editor,
   a. Select Dataset A, go to "Home", then "Merge Queries".
   b. Choose Dataset B and select matching columns for merging (ID).
   c. Select the type of join (e.g., Inner join for matching records).
4. In the merged column, click the expand button and select columns from Dataset B to include.
5. Click"Close & Load" to load the merged data into a new worksheet.

Subset (Filter) the Rows/Observations of a Dataset by Criteria:

1. Click the top cell of the column you want to filter.
2. In the "Data" tab, click "Filter".
3. Click the dropdown menu in the column you selected in step 1.
4. Uncheck "Select All" and thenre-check the box next to the criteria you're interested in.

Subset using Text Comparisons in Categorical Variables:

1. Click the top cell of the column you want to filter.
2. In the "Data" tab, click "Filter".
3. Click the dropdown menu in the column you selected in step 1.
4. Check/uncheck the text values you wish to include or exclude.

Subset using Numeric Comparison in Quantitative Variables:

1. Click the top cell of the column you want to filter.
2. In the "Data" tab, click "Filter".
3. Click the dropdown menu in the column you selected in step 1, then select Number Filters.
4. Select a number filter (e.g., greater than, less than) and enter the value for comparison.

Subset Based on Missing Values:

1. Click the top cell of the column you want to filter.
2. In the "Data" tab, click the"Filter".
3. Click the dropdown menu in the column you selected in step 1.
4. Uncheck"(Blanks)" to exclude missing values or, check only"(Blanks)"to include them only.

Subset Used on Conditions Involving 2+ Variables:

1. Click the top cell of the columns you want to filter.
2. In the "Data" tab, click the"Filter".
3. Set conditions for each column involved by using the dropdown in each column header.

STATCRUNCH

Drop Rows with Missing Values:

1. Data → Row Selection → Detect → Empty
2. Ctrl+A to select all columns
3. Click Compute!
4. Edit → Rows → Delete

Drop Columns with Missing Values:

1. Data → Validate
2. Ctrl+A to select all columns
3. Select Display → Empty
4. Click Compute!
5. Edit → Columns → Delete
6. Choose the columns with found empty values.

Fill Missing Values for Column XX with the Mean of Column XX:

1. Stat → Summary Stats → Select Columns
2. Select the q1 column
3. Statistics → Mean
4. Click Compute!
5. Data → Compute → Expression
6. Enter a formula to replace missing values with the mean value
   a. For example, if the mean is 65.5 and the column is named 'q1', the formula would be:

      ```
      ifelse(isnull(q1), 65.5,q1)
      ```
7. Click Compute!

Create a Categorical Variable from an Existing Quantitative Variable:

1. At the top of a new, empty column, type "`q1_new`".
2. Data → Compute → Expression.
3. Enter an expression to categorize q1 in the new q1_new variable:

   ```
   ifelse(q1 <= 100, "Cheap", ifelse(q1 > 100 and q1 <= 150, "Low",
   ifelse(q1 > 150 and q1 <= 200, "Medium", "High")))
   ```
4. Click Compute!

Create a New Variable That Rescales an Existing Quantitative Variable:

1. Data → Compute → Expression.
2. Enter an expression to scale q2:
   ```
   q2/1000
   ```
3. Click Compute!

Create a New Variable from Existing Quantitative Variables:

1. Data → Compute → Expression
2. Enter an expression to add (combine) q2 and q3:

   q2+q3
3. Click Compute!

Create a Quantitative Variable from an Existing Categorical Variable:

1. Data → Compute → Expression
2. Enter an expression to assign numbers for each value of cat1

   ```
   ifelse(cat1="A",1,ifelse(cat1="B",2,3))
   ```
3. Click Compute!

Merge the Categories of an Existing Categorical Variable:

1. Data → Recode
2. Select categorical variable you want to recode (e.g., `cat3`)
3. Column label → type name of new column (e.g., `cat3_new`)
4. Options → Create new column(s)
5. Click Compute!
6. In the Recoded column, enter the values you want instead of the original values.
7. Click Compute!

Split a Categorical Variable:

1. Data → Arrange → Slice
2. Select column → Select categorical variable you want to split (e.g., `cat2`)
3. Delimiter → specify the delimiter you want to split on (e.g., _)
4. Click Compute!

Reshape in Long or Wide Format

- Not currently available in StatCrunch

Stack Datasets:

1. Select all of the cells corresponding to the first dataset
2. Copy this selection
3. Go to the cell where you want the second dataset to begin
4. Paste the copied selection

Merge Datasets

- Not currently available in StatCrunch

Subset Using Text Comparisons in Categorical Variables:

1. Data → Row Selection → Select Where
2. Enter expression:

   ```
   cat2="type_2"
   ```
3. Click Compute!

Subset using Numeric Comparison in Quantitative Variables:

1. Data → Row Selection → Select Where
2. Enter expression:

   ```
   q1>= 10
   ```
3. Click Compute!

Subset based on Missing Values:

1. Data → Row Selection → Select Where
2. Enter expression:

   ```
   !isnull(cat2)
   ```
3. Click Compute!

Subset based on Conditions Involving 2+ Variables:

1. Data → Row Selection → Select Where
2. Enter expression:

   ```
   "cat3"=100 and "q1">15
   ```
3. Click Compute!

CHAPTER 3

MAKING SENSE OF DATA THROUGH VISUALIZATION

Selinofoto/Shutterstock

LEARNING OBJECTIVES

- Use the fundamental ideas of the *grammar of graphics* to evaluate options for the construction of data visualizations.
- Employ techniques for visualizing a single quantitative variable, a single categorical variable, two variables, and three or more variables through various plots.
- Identify common pitfalls and dangers associated with visual misrepresentation in data visualization.
- Incorporate established guidelines to produce compelling and honest data visualizations.

BRENNAN DAVIS | HUNTER GLANZ

DATA SCIENCE FOR ALL

Brennan Davis
California Polytechnic State University

Hunter Glanz
California Polytechnic State University

DATA SCIENCE FOR ALL

Pearson

Content Development: David Dietz, Thomas Atwater, Amit Verma, Ravi Indiver
Content Management: Laura Briskman, Monique Bettencourt
Content Production: Mahalakshmi Usha, Nupur Anand, Audra Walsh, Robert Carroll, Dominick Frank, Sandra Rodriguez, Scott Disanno, Rachel Reeve, Guadalupe Edgar
Product Management: Karen Montgomery
Product Marketing: Demetrius Hall
Rights and Permissions: Ben Ferrini, Anjali Singh, Vignesh Murugamoorthy

Please contact www.AskPearsonSupport.com with any queries on this content.
Pearson puts care into the creation of our products. Our learning materials undergo content reviews to minimize bias and ensure the content is inclusive, accurate, and relevant. Please use this link to report instances of bias and inaccuracies: https://www.pearson.com/report-bias.html
You can learn more about Pearson's commitment to accessibility at https://www.pearson.com/us/accessibility.html

Cover images from Shutterstock, D.R.3D/Shutterstock, AAVAA / Shutterstock, and NasonovVasiliy/Shutterstock.

Acknowledgments of third-party content appear on the appropriate page within the text.

Library of Congress Cataloging-in-Publication Data

Data Science For All
Library of Congress Cataloging in Publication Control Number: 2024044768

$PrintCode

Pearson

Rental Edition:

ISBN 10: 0-13-832314-3
ISBN 13: 978-0-13-832314-1

Print Offer:

ISBN 10: 0-13-531141-1
ISBN 13: 978-0-13-531141-7

DEDICATION

To my beautiful wife, Anna, for her unwavering support and culinary delights; to my sons, Canyon, Levi, and Tobin, who were my pillars of encouragement; and to my son, Noe, and his wife, Dani, for their unwavering backing—this book is a testament to your love and support.

—Brennan

To my family, Maddie and Jake, for their unconditional support throughout all of this. To all of my friends and other family for always being there, no matter what.

—Hunter

ABOUT THE AUTHORS

Brennan Davis is the Richard and Julie Hood Professor and director of graduate analytics programs at the Orfalea College of Business at California Polytechnic State University (Cal Poly, San Luis Obispo). He received a BS in mathematics from the University of California, Los Angeles, an MBA from the Wharton School of Business at the University of Pennsylvania, and a PhD from the University of California, Irvine. Brennan currently teaches undergraduate and graduate analytics courses. In 2019, Brennan received the Emeritus Faculty Award for significant and meritorious achievement in contributing to student welfare.

Hunter Glanz is a Professor of Statistics and Data Science at California Polytechnic State University (Cal Poly, San Luis Obispo). He received a BS in mathematics and a BS in statistics from Cal Poly, followed by an MA and PhD in statistics from Boston University. He maintains a passion for data science, machine learning, and statistical computing and enjoys teaching courses in those areas. Hunter serves on numerous committees and organizations dedicated to delivering cutting-edge statistical and data science content to students and professionals alike, including being a founding board member of the California Alliance for Data Science Education. In 2019, Hunter received the Terrance Harris Excellence in Mentorship Award, and in 2020, he received the Outstanding Faculty Award in the Master's in Business Analytics program at Cal Poly.

CONTENTS

PREFACE

Data Science: Why Now?

In writing this book, we aim to democratize data literacy and empower every student with the skills to navigate our data-rich world. We recognize the need for this growing field of data science to be a part of everyone's life, both in their consumption of data day-to-day and in their working with data professionally or personally.

For decades, we have witnessed data's involvement in coursework and scholarly projects across campus. We have observed the necessity for data-related skills and comprehension in the diverse professions into which our students have graduated. The ubiquity of data has the potential to unite us in the process of ingesting information, thinking critically to draw conclusions, and communicating with those around us.

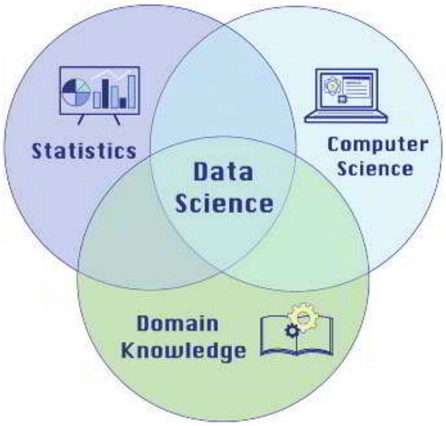
As an exciting and rich blend of statistics, computer science, and domain application, the field of data science has much to offer everyone in their pursuits. This book aims to take the most applicable ideas from these component fields and put them in the hands of students so they can apply them immediately to their everyday lives. By giving them hands-on experience with every concept, this book will provide a *lingua franca*, enabling them to fluently converse with data no matter their background, major, or professional goals.

The time is right for a textbook that *distinguishes* itself by offering unparalleled support through meticulously designed content that caters to true beginners, ensuring quality, clarity, and accessibility without sacrificing depth. With a commitment to fostering an inclusive learning environment, we provide a comprehensive, up-to-date curriculum, complemented by innovative, evolving tools for assessment and engagement. This holistic approach guarantees a learning experience that is both enriching and enduring, setting a new standard for academic resources in the data science domain.

Approach

As California Polytechnic State University instructors, we work daily to help our students "Learn by Doing," a motto that is also a core part of our professional identity. We, therefore, make it a core part of this text. The firm belief that data work requires technology is closely linked with this motto, but software should not be a barrier to engaging with data science. To this end, we have filled our text with opportunities to get students working with data early and often in the software package of the instructor's choice (R, Python, Excel, or StatCrunch).

A foundation in data science is not exclusive to those with a quantitative background. We have built our text to be accessible to students with no prior statistics or computer science knowledge and minimal mathematical experience. Instead, this text provides students of all backgrounds with an entry point to the world of data and any future coursework that involves data. It is not merely a modernization of a traditional first course in statistics or computer science. Instead, it distills these component fields into an overview of what data science means and what it involves in today's world.



Prepare, Analyze, and Communicate

Today's students already are consumers of data and will likely be directly engaged with data work in their future careers. We believe it is essential that they understand what the complete life cycle of a data science project can entail. Chapter 1 introduces the field of data science and our three overarching components of a data science project: data preparation, data analysis, and data storytelling.

One of the most exciting facets of data work is its iterative nature. Working with data is a labor of learning, analyzing, and communicating. We then revisit these steps as new pieces of data are introduced. Analysis can result in findings that necessitate revisiting or using a different analysis technique. Completing a data science project also requires communicating these analyses in different formats.

Data Preparation

Data preparation can include:

- Data collection
- Data organization
- Understanding the data
- Wrangling (cleaning, transforming, and integrating) the data.

Data often require preparation before they're suitable for visualization and analysis. It's essential to first grasp the data's structure and source. As we delve into data exploration, it becomes clear that further organization or manipulation might be necessary for practical analysis and storytelling. We'll explore several prevalent techniques for transforming data into formats ready for analytical and narrative purposes.

Data Analysis

Data analysis can include:

- Describing data with numerical summaries
- Testing hypotheses
- Predictive modeling
- Understand the various relationships in a dataset.

Data analysis spans a spectrum of methods, each serving distinct objectives and questions. Initially, we describe data using summary statistics and other essential metrics, condensing key characteristics into concise numerical or verbal forms. This is where data science starts to branch away from introductory statistics, which typically focuses on making inferences about various population parameters. While hypothesis testing remains vital for deriving conclusions, data science emphasizes examining proportions, means, and correlations using single samples, extending to two-sample inference in contexts like A/B testing. Furthermore, data science expands beyond traditional linear regression, encompassing a more comprehensive range of supervised and unsupervised modeling techniques. This includes, but is not limited to, methods like decision trees, random forests, *k*-nearest neighbors, logistic regression, and *k*-means clustering, providing a more comprehensive toolkit for analyzing complex datasets.

Data Storytelling

Data storytelling can include:

- Visualizations
- Written reports
- Oral presentations
- Recommendations to a client.

Students must master compelling data storytelling, enabling them to uncover and articulate the narratives hidden within data. This skill goes beyond merely presenting findings. It involves probing what the data reveal and identifying what information might still be missing.

A pivotal aspect of data storytelling is creating and interpreting compelling data visualizations with software tools. Students can effectively convey complex information by weaving a narrative that links data preparation, analysis, and visual storytelling, making it accessible and actionable for decision-makers. Our text integrates a data storytelling framework called STAR, encouraging students to apply this approach at each stage of their data-related tasks.

Topical Coverage

We designed this text and the accompanying MyLab Statistics course for a first-semester course in introductory data science. The structure of our Table of Contents mirrors this approach, allocating additional time for certain sections at the instructor's discretion. However, this text can be used flexibly. Chapters 1–4 introduce data preparation, storytelling, visualization, and foundational analysis. Chapters 5–7 delve into more advanced data analysis techniques, and Chapters 8–10 cover contemporary machine learning techniques. Recognizing the dynamic nature of data science, we've designed the curriculum with the flexibility to incorporate specific topics or projects that align with your unique emphasis.

Chapter 1 introduces data science as an academic discipline, an increasingly critical component of every business, and an inevitable piece of our everyday lives. It also introduces the anatomy of data as consisting of observations, variables, and values. We distill the data science workflow stages into three overarching categories: data preparation, data analysis, and data storytelling.

Students *apply* this knowledge, including:

- distinguishing between different data types,
- practicing storytelling, and
- recognizing analysis types.

Chapter 2 introduces data wrangling as the first stage in the data science workflow to preprocess data into a cleaner, more friendly form for analysis and storytelling. At the end of data wrangling efforts, data should be consistent and in the structure needed for analysis and storytelling.

Students *apply* this knowledge, including:

- applying strategies for addressing missing data,
- finding and addressing anomalous values, and
- transforming data into a more usable structure.

Chapter 3 introduces data visualization as a primary way to communicate data using graphical techniques. We introduce the grammar of graphics and continue the STAR data storytelling framework as ways to unite the creation of all data visualizations in insightful ways. By the end of the chapter, students will have seen standard techniques for visualizing one, two, three, and four variables simultaneously. We emphasize the dangers of misrepresentation and provide guidelines.

Students *apply* this knowledge, including creating and interpreting:

- histograms and density plots;
- box plots and violin plots;
- bar charts; and
- scatterplots, line charts, and bubble charts.

Chapter 4 introduces exploratory data analysis via numerical metrics that capture key characteristics in a dataset, such as central tendency and spread. Students are reminded of the different shapes distributions have, but then shown how these shapes and the resistance of specific metrics to extreme values can affect the values of summary metrics. We finish with the characterization and identification of outliers.

Students *apply* this knowledge, including:

- calculating and evaluating summary statistics,
- identifying distribution shapes,
- computing correlations, and
- determining outliers.

Chapter 5 introduces the process of formulating questions we can ask of our data and how to make those requests using a software tool. We show students how to perform standard data queries (in generic, tool-agnostic ways) by subsetting a dataset by variables or observations and sorting, aggregating, and combining datasets. For this chapter only, we introduce structured query language (SQL) as a tool option alongside Excel, Python, R, and StatCrunch.

Students *apply* this knowledge, including:

- selecting data,
- filtering data,

- ordering data,
- summarizing data, and
- merging data.

Chapter 6 revisits the concept of variability, first introduced in Chapter 4, and explores its relationship to uncertainty and predictability in our data and the target population. To clarify the role of uncertainty in data analysis, we introduce fundamental ideas in probability. Subsequently, we focus on diagnosing and assessing variability in data visualizations. Students also leave Chapter 6 with a better understanding of sampling methods.

Students *apply* this knowledge, including:

- assessing variability,
- estimating likelihoods of events using data, and
- performing simulations.

Chapter 7 broadly introduces statistical inference and then walks students through simulation-based inference for a population proportion and correlation. We cover inference for a population mean and the difference between two population means via t-tests. In particular, the two-sample t-test is introduced in the context of an A/B Test. Students leave Chapter 7 with a better understanding of statistical inference and the difference between experiments and observational studies.

Students *apply* the knowledge, including:

- choosing appropriate methods for statistical inference,
- making statistical inferences in observational studies and experiments, and
- understanding false positives and false negatives.

Chapter 8 provides an overview of machine learning and artificial intelligence, with particular attention to the choices and considerations required in a data science problem involving machine learning. In particular, students are introduced to differentiating model characteristics, with the goal of guiding the students on identifying which models to explore. Then, they see the metrics that inform final model evaluation and selection.

Students *apply* the knowledge, including:

- distinguishing between regression and classification methods,
- comparing methods with respect to interpretability and complexity, and
- evaluating machine learning methods using training and test data.

Chapter 9 dives deeper into supervised machine learning as the class of methods appropriate for a problem involving a particular response variable of interest, such as regression or classification. We build intuition and understanding about traditional linear regression models with different types of explanatory variables and introduce logistic regression for classification. We also present decision trees, random forests, and k-nearest neighbors models for regression and classification.

Students *apply* this knowledge, including:

- distinguishing different types of supervised learning methods,
- evaluating and comparing regression methods on test data, and
- evaluating and comparing classification methods on test data.

Chapter 10 expands on unsupervised machine learning as the class of methods appropriate for when our problem does not involve a response variable of interest. We outline three of the most common categories of unsupervised learning: dimension reduction, anomaly detection, and clustering. Students are introduced to k-means and hierarchical clustering, emphasizing the former in evaluating clustering results.

Students *apply* this knowledge, including:

- detecting anomalies,
- calculating similarity using distance, and
- comparing cluster analysis results.

Alternative Pathways through the Content

We have arranged the 10 chapters of this book to align with much of the data science work we are familiar with and our recommended order. However, data science work is iterative and can proceed in a different order than we present. For example, many professors feel more comfortable teaching data visualization earlier than it is currently placed.

Chapters 2–5 can be taught in any order. To ensure the richest possible experience for students, they should come before Chapters 6–10. Then, we recommend the following:

- Chapter 6 should come first after Chapters 1–5.
- Chapter 7 should come immediately after Chapter 6.
- Chapter 8 should come immediately after Chapter 7.
- Chapters 9 and 10 could be covered in either order after completing Chapter 8.

While the table of contents has 10 chapters, we recommend teaching at a slower pace for Chapters 8–10, especially Chapter 9. While students might have been exposed to graphing concepts, foundational statistics, and coding, machine learning will likely be new to most, and these chapters benefit from a slower pace. We promise that it is worthwhile!

Integrating Software

In this text, we have elected to support R, Python, Microsoft Excel, and StatCrunch (with support for SQL as topics allow). Screenshots from different tools are used throughout the text. While we recognize that instructors will likely select one of these, we believe exposure to multiple tools is helpful to students who will encounter different technologies in their careers.

Tool Time

Excel

Python

R

StatCrunch

The use of technology should not be a barrier to success in this course. We recommend cloud-based tools (e.g., Posit Cloud and Google Colab) for running the code in this course. However, we support any tool, cloud-based or device-installed, that runs R and Python code. We have created numerous resources (available in MyLab Statistics) to support students and instructors as they get started:

- Video guides in setting up cloud-based tools, Posit Cloud and Google Colab, for running R and Python code (recommended)
- Video guides in setting up R and Python on students' devices
- Video support for software functions in R, Python, Excel, and StatCrunch
- R and Python code for every "Try It Yourself" feature, where applicable
- Excel and StatCrunch click paths for every "Try It Yourself" feature, where available and applicable

All datasets are available in MyLab Statistics and the Pearson Math and Statistics Resource site (www.pearsonhighered.com/mathstatsresources).

Join our GitHub for code, data, and other digital assets: https://github.com/datasci4all

R

If you're using R for any portion of the journey through our text, please install the correct packages so that your output matches ours. Each chapter involves a different set of packages. Check the corresponding Tech Center resource for the packages needed in that chapter.

If the Tech Center includes a line such as

library(tidyverse)

then you need to make sure you have the tidyverse package installed. This only needs to be done once per machine by running the following line of code:

install.packages("tidyverse")

Python

If you're using Python for any portion of the journey through our text, please install the correct libraries so that your output matches ours. Each chapter involves a different set of libraries. Check the corresponding Tech Center resource for the libraries needed in that chapter.

If the Tech Center includes a line such as

import pandas

then you need to make sure you have the pandas library installed. This only needs to be done once per machine by running the following line of code:

pip install pandas

Excel and StatCrunch

If you're using Excel and/or StatCrunch for any portion of the journey through our text, make sure that you are working with a fresh copy of supplied data files for any exercise. It can be easy to overlook work from previous exercises or activities that will give you incorrect results. Note that not every activity is currently possible with standard Excel and StatCrunch functionality; we have noted these where applicable and recommend exploring the activities in R or Python if desired.

Video Support

As instructors, we know that students are always asking for video support. They want to watch videos if they miss lectures, for test preparation, for software support, or just to remind themselves how to do something that might have looked easy in class. We also know students don't want to watch 15-minute lecture videos. So, we have created a suite of video support for students, ensuring that their textbook authors are essentially "on call" to help them when needed. We have recorded over 500 videos:

- Lecture videos that introduce each section;
- Example videos that walk students through selected exercises in each chapter, and
- Software support videos.

Guiding Students

All chapters (except Chapter 1) open with a **story** centering on a young person encountering a data science investigation in their school, place of work, or personal life. Lessons presented in a given chapter tie back to the opening narrative, applying concepts to real life.

Case Studies are integrated throughout each chapter, exploring chapter-wise concepts in business, technology, environmental sciences, health, sports, and more.

In the margins, we supply **Quick Tips** as additional guidance on challenging material, warnings about common mistakes, and further details expounding on pedagogy.

We periodically answer the ubiquitous question, "Why does this all matter?" via **Big Picture Breaks**, which take a step back from minutiae to offer additional context.

Each chapter concludes with a **Putting It Together** feature, with broken-down bullets so that students can reinforce what they have learned.

Storytelling is a fundamental aspect of data science, and to emphasize its importance, we have developed a unique framework specific to this title. The **STAR Framework** invites students to communicate an engaging narrative that not only showcases data but also provides context and prompts action. We introduce this framework in Chapter 1 and extensively apply it throughout. Here is an overview of the STAR Framework's components:

1. *Scope the Data's Context*: We begin by detailing the data's background, including the observational units, variables, their types, and the data source.
2. *Track the Question Behindthe Exploration*: We identify the crucial questions and underlying tensions the data aim to resolve.
3. *Articulate the Results*: We focus on clearly presenting the findings derived from the data.
4. *Respond with Next Steps*: We conclude by suggesting further actions or investigations guided by the results.

We designed the STAR Framework as a teaching tool, guiding students through a structured process of thinking about and communicating with data.

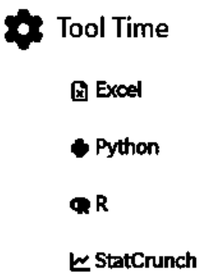
Guiding Instructors

Teaching a new course is always a challenge. We have included a robust instructor resource package, including the following:

- An instructor resource manual written by Hunter Glanz, with notes from his own introduction to data science course. This provides an overview of each chapter. It includes notes, on what to emphasize in each class and suggestions for presenting the material.
- PowerPoint lecture slides that include essential graphics and follow the sequence and philosophy of the text. They can be downloaded from MyLab Statistics or www.pearson.com.
- A complete instructor solutions manual containing worked-out solutions to all text exercises and case study answers. It can be downloaded from MyLab Statistics or www.pearson.com.
- TestGen® (www.pearson.com/testgen), which enables instructors to build, edit, print, and administer tests using a computerized bank of questions developed to cover all the text's objectives. TestGen is algorithmically based, allowing instructors to create multiple but equivalent versions of the same question or test with the click of a button. Instructors can also modify test bank questions or add new questions. The software and test bank are available for download from Pearson's online catalog, www.pearson.com. The questions are also assignable in MyLab Statistics.
- Accessibility: Pearson works continuously to ensure our products are as accessible as possible to all students. We are working toward achieving WCAG 2.0 Level AA and Section 508 standards, as expressed in the Pearson Guidelines for Accessible Educational Web Media, www.pearson.com/mylab/statistics/accessibility.

Active Learning

Because these lessons are buildable, we heavily incorporate student engagement as they learn. Concepts are interspersed with **Try It Yourself** (TiY) exercises, where students can practice what they have just learned. Where applicable, the TiY exercises are accompanied by our Tool Time feature, which offers technology-specific solutions and tips (e.g., Excel, Python, R, and StatCrunch). Answers to all TiYs are available to students at the end of the text.



For each TiY exercise, we also provide an accompanying **Applying the Concepts** (AtC) activity at the end of the chapter. The goal is a scaffolding approach, wherein students practice with their in-chapter TiYs and then complete AtCs as assigned with less guidance. Comprehensive answers are available to instructors in MyLab Statistics.

AtCs are also available in MyLab Statistics, which can be assigned as MediaShareproject-based assignments for group or individual work. They come with customizable rubrics designed to offer grading support to instructors.

Designed to provoke further thought and investigation, we have also included **Explaining the Concept** questions at the end of each chapter, and available in MyLab Statistics for assignment. These open-ended questions are great for formative or summative assessment or just as an in-class exercise.

Integrated Review

Data Science for All was created with no prerequisites necessary for success beyond the completion of high school math. However, we realize that students may need extra support with some mathematical or statistical topics. This content has been added to the MyLab Statistics course with just-in-time support via auto-graded exercises, worksheets, and videos. This allows for targeted support for students in their time of need. This also fits a corequisite model.

Putting It Together

Data Science for All is a comprehensive yet accessible journey into the world of data science designed for students of all majors and backgrounds. Our book demystifies data science, covering its entire lifecycle from preparation and analysis to storytelling, all through a hands-on approach. We emphasize "Learning By Doing," integrating the STAR framework and various software tools and making the experience engaging and practical.

We warmly invite you to explore *Data Science for All*. Whether you're a student, educator, or curious learner, this book is a gateway to the vital skills of data science, applicable in every facet of modern life. We invite you to connect, joining us on this enlightening journey to uncover the power of data.

Sincerely,
Brennan Davis and Hunter Glanz
bdavis39@calpoly.edu
hglanz@calpoly.edu

ACKNOWLEDGMENTS

We express our profound gratitude to those closest to us. Hunter thanks Maddie Schroth-Glanz and Jake for their invaluable support and encouragement. Brennan thanks Anna, Canyon, Levi, Tobin, Noe, and Dani Davis for their immeasurable advocacy and inspiration.

We also extend our heartfelt thanks to those we feel professionally and personally close through this process. Laura Briskman, Karen Montgomery, Demetrius Hall, and Monique Bettencourt believed in our idea to launch a new book from the very beginning. They gave us their time, introduced us to their networks, and skillfully guided us, knowing exactly when to rein us in and when to encourage us forward. Through this process, they have become more than colleagues to us; they are friends. We love our tradition of enjoying fun and unusual desserts together, a testament to the sweet journey we've shared.

We are also grateful to Melissa Melton, Sean Pepper Leader, and Brogan Pietrocini for their dedicated efforts in bringing this project to fruition.

We sincerely thank the content production team led by Nupur Anand, Sandra Rodriguez, and Scott Disanno for their meticulous work. We are also thankful for the exceptional media production overseen by Audra Walsh and Stephanie Woodward.

We acknowledge the valuable contributions of the MathXL development team, particularly Bob Carroll and Phil Oslin, for their technical expertise. Lastly, we express our gratitude to our developmental editor, David Dietz, whose insights and guidance have been instrumental in shaping this book.

REVIEWERS

We are indebted to the many reviewers, class testers, and students who provided us with guidance and feedback as we developed this product. We apologize for any omissions.

We are also indebted to our many corporate reviewers for their support and guidance through this process, including Abe Weston at Google, John Frith at Urban Science, Rex Briggs at Claritas, and Lucas Pierce at Character.AI.

Bagui, Sikha—University of West Florida

Ballion, Tatiana—Midland University

Beagley, Jonathan—Valparaiso University

Beckman, Matthew—Pennsylvania State University

Bergeler, Elmar—University of Texas at San Antonio

Brown, Austin—Kennesaw State University

Brown, Robert—Carroll Community College

Burgin, Joe—College of Southern Maryland

Burns, Jared W.—Seton Hill University

Carter, Perry—Texas Tech University

Cheng, Fuxia—Illinois State University

Coe, Jacquelyn—Central Oregon Community College

Combs, Adam—Robert Morris University

Dalkilic, Mehmet—Indiana University

Davis, Amanda—Forsyth Technical Community College

Ding, Junhua—University of North Texas

Do, Thuy—Luther College

Drummey, Kevin—Loyola University Maryland

Fernando, Harshini—Purdue University Northwest

Finotti, Luis—University of Tennessee, Knoxville

Firouzian, Shawn—MiraCosta College

Fitzpatrick, Kelly—County College of Morris

Ghriga, Mohammed—Long Island University

Gold, Kevin—Boston University

Habimana, Jean Remy—Southwestern University

Hardin, Jo—Pomona College

Hashemi, Navid—College of Charleston

Hendricks, Todd—GSU Perimeter College

Holay, Sandeep—Southeast Community College

Hong, Sungbum—Jackson State University

Houser, Shannon—University of Alabama at Birmingham

Kader, Idrissa—Northern Virginia Community College

Karunaratne, Susitha—Purdue University

Kinney, David—Bellevue University

Kirmani, Syed—University of Northern Iowa

Kitchings, N. Clayton—University of North Georgia

Kox, Erin—Fanshawe College

Kristin, Lilly—Columbus State University

Kurban, Hasan—Texas A&M University

Kuyper, Arend—Northwestern University

Lewis, Mark—Alfred University

Long, Michael—Polk State College

Lyne, Heidi—Joliet Junior College

Magnum, Amanda—Converse College

Malachowski, James—Northwest Arkansas Community College

McLean, Jeffrey—University of North Carolina at Chapel Hill

Mogull, Michael—Cuesta College

Montgomery, Aaron—Baldwin Wallace University

Neal, Matthew—Denison University

Nicolae, Dan—University of Chicago

Onunwor, Enyinda—Saint Paul College

Pearsall, Sam—Los Angeles Pierce College

Poliak, Adam—Bryn Mawr College

Rampure, Suraj—University of California at San Diego

Harsy Ramsey, Amanda—Lewis University

Rathor, Shekhar—University of Central Oklahoma

Ritter, Carrie—Randolph Community College
Roberts, David—University of Calgary
Rosson, Holly—Warren Wilson College
Schell, William—Montana State University
Scoville, Nicholas—Ursinus College
Seefield, Kim—Nashua Community College
Shamir, Lior—Kansas State University
Sharma, Sharad—University of North Texas
Singh, Vivek—Rutgers University
Smith, Alice E.—Auburn University
Sourile, Daniel—Joliet Junior College
Spence, Dianna—University of North Georgia
Starin, Karen—Columbus State Community College
Szczurek, Piotr—Lewis University
Thurman, Alexis—County College of Morris
Williams, Therese—University of Central Oklahoma
Xu, Jie—George Mason University
Xue, Yajiong (Lucky)—East Carolina University
Zhou, Yi—Columbus State University

INDEX OF ACTIVITIES

These activities are designed for students to learn how to interact with data in different ways. They take a scaffolded approach, where students start with software support and then are asked to apply what they've learned to the same objective.

Chapter 5: Data Management

Chapter 6: Understanding Uncertainty, Probability, and Variability

Chapter 7: Drawing Conclusions from Data

Chapter 8: Machine Learning

Chapter 9: Supervised Learning

Software Libraries to Install

For those using Python or R, the following libraries are required to complete activities in each chapter. In Chapter 7 only, an extra tool pack will be required for those using Excel. No libraries need to be installed for those using StatCrunch.

Python

Chapters 2–10
pandas

Chapter 3
plotnine
wordcloud
matplotlib
squarify

Chapter 4
numpy
scipy.stats
plotnine

Chapter 5
numpy

Chapter 6
numpy
plotnine
random

Chapter 7
numpy
scipy
sklearn.utils
plotnine
random

Chapter 8
sklearn.model_selection
sklearn.tree
sklearn.metrics
sklearn.pipeline
sklearn.compose
sklearn.preprocessing

Chapter 9
numpy
scipy
sklearn.linear_model
sklearn.preprocessing
plotnine
sklearn.model_selection
sklearn.tree
sklearn.ensemble
sklearn.neighbors
sklearn.linear_model
sklearn.metrics
sklearn.compose
sklearn.pipeline
statsmodels.api

Chapter 10
numpy
sklearn.cluster
plotnine
scipy.spatial

R

Chapters 2–10
tidyverse

Chapter 2
lubridate

Chapter 3
wordcloud
RColorBrewer

Chapter 7
broom

Chapter 8
rpart
caret
tidymodels

Chapter 9
kknn
vip
tidymodels
broom
caret
ranger
rsample
rpart
randomForest
Metrics

Excel

Activities in Microsoft Excel are all able to be completed in standard Excel except for Chapter 7, which requires the (free) Analysis ToolPak. To enable the Analysis ToolPak, follow these steps:

1. File → Options → Add-Ins.
2. In the Manage box.
   a. Select Excel Add-Ins.
   b. Click Go.
3. Check the Analysis ToolPak checkbox.
   a. Click OK.

CHAPTER 1

WHAT IS DATA SCIENCE?

GaudiLab/Shutterstock

LEARNING OBJECTIVES

- Interpret the definition and scope of data science, including its interdisciplinary nature and lifecycle.
- Recognize the benefits of presenting data as a tidy table.
- Distinguish between different data types.
- Describe a coherent data preparation pipeline, considering the various formats and types of data that may be encountered.
- Explore the various types of data analysis for compelling storytelling.
- Evaluate the growing impact of data science in society and industry, including ethical considerations.

Introduction

Data science impacts many aspects of our world. Streaming media services use viewers' past entertainment choices to customize recommended content. Doctors use human genome data to prescribe medicines tailored to patients' unique gene profiles. Farmers use soil and weather data to determine suitable watering and harvest times, thereby improving crop production and reducing waste. Historians mine thousands of declassified U.S. State Department memos to learn how diplomacy affected the Cold War in the 20th century. University administrators quickly filter data on tens of thousands of students to match them with donors' scholarship criteria. Coaches use real-time wearable technology data to help their athletes win championships. These are just a few examples of how data science revolutionizes our lives.


Shutterstock

A whale surfaces near offshore wind energy technologies in the Pacific Ocean. From Ocean News & Technology.

Data science also helps us do good on a more global scale. Consider the tensions between energy progress and marine life preservation. While energy providers aim to increase renewable energy through offshore wind technologies, marine scientists are concerned about how new offshore wind energy sites might affect nearby marine mammals. Offshore wind technologies have the potential to benefit civilization, but their noise might disrupt marine mammals' foraging, orientation, hunting, and communication, which are already threatened by human-made disturbances (Figure 1.1).

California marine scientists use data science to monitor the deep-water ocean ecosystem near the state's offshore wind energy technologies. Specifically, they combine computer science and statistics using multiple data sources: acoustic recordings, underwater camera images, and Monterey Bay Aquarium Research Institute classifications. They use computer algorithms and statistics to recognize marine mammals and track their behavior. This automated data science helps the California Energy Commission promote clean energy while simultaneously protecting marine life.

Using data sources as diverse as photographic images, acoustic recordings, animal tagging data, and animal classifications to make important planning decisions would have been far less feasible just a few years ago. New approaches provided by data science make such advances possible.

FIGURE 1.1
Sound waves overlap in the ocean soundscape. From Ocean News & Technology.

NOAA Fisheries

SECTION 1.1: INTRODUCTION TO DATA SCIENCE

Before defining *data science*, we should define *data*. For this textbook, we will use the following definition:

DEFINITION

Data are information from a source or sources that can be useful for insights and decision-making after being processed in various ways.

As we will see, data can be more than just numbers. We'll use the word as a plural noun (e.g., "these data are recorded…"). We can be less formal about the *science* part of data science; think of science as a way of obtaining knowledge.

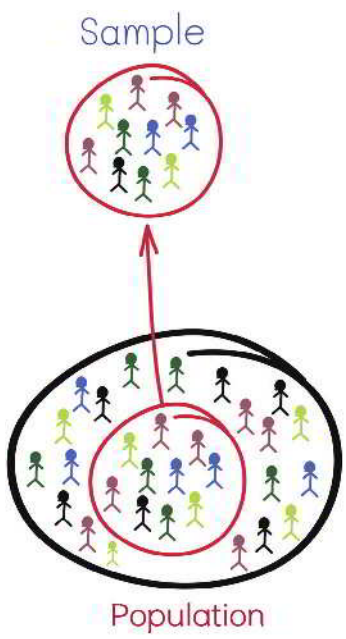
DEFINITION

Data science is an interdisciplinary field that uses scientific methods, processes, and systems to extract knowledge and insights from data across various domains.

The main reason data and their analyses are important is because we are constantly interested in describing or drawing conclusions about a *population* of interest based on a more feasibly accessed subset of that population, called a *sample*.

DEFINITION

A **population** is the set of every item or individual that is of interest for a particular question. For example, all used car owners in the United States.



lamnee/123RF

Because it's not practical to collect data on an entire population of interest, we are forced to take a sample from this population and work with data from this sample.

DEFINITION

A **sample** is a subset of the population of interest.

For example, if our population of interest were all used car owners in Europe, one possible sample would be the used car owners responding to a survey sent to a randomly generated list of 1000 people in Europe.

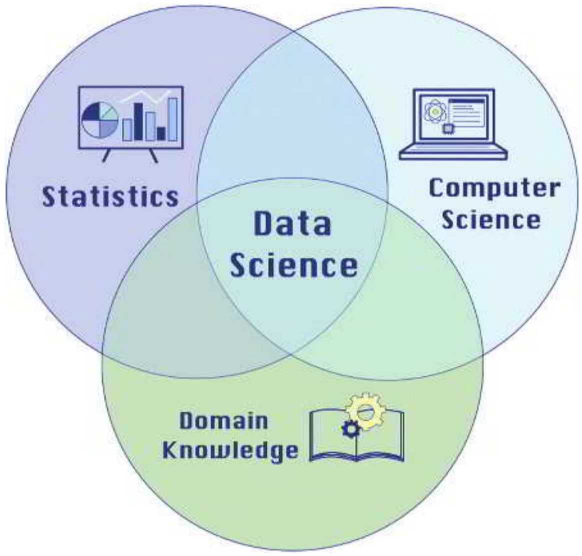
The ability to understand how people work with data, from a sample, to communicate insights about populations of interest is crucial for living an informed life in today's society of news and other media.

Combining Statistics, Computing, and Expert Knowledge

Data science combines statistics, computer science, and expertise from various domains to extract value from data (Figure 1.2). For example, when we ask Spotify to play our favorite songs, when Netflix recommends shows we enjoy binge-watching, or when we see products on Amazon's homepage tailored to our preferences, we experience the power of data science. These online suggestions are made using algorithms based on user-provided data, running on lightning-fast servers.

FIGURE 1.2

Data science overlaps with computer science, statistics, and domain knowledge.



Sometimes, the distinctions between data science and data analytics need to be clarified. *Data analytics* is the applicationof data science to a domain-specific problem. The term *data science* has been traced back to 1974 when Peter Naur proposed it as an alternative name for computer science. In these ways, it also resembles the field of statistics, which is the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data. More modern challenges, such as the largeness, complexity, and accessibility of datasets, and the increased complexity of many analysis methods, have necessitated the incorporation of ideas and tools from computer science and statistics to handle such data and extract insights successfully.

Modern data science recognizes that data are everywhere and can be incredibly useful through collaboration with statistics, computer science, and every imaginable domain. While the field of data science is still evolving, there is a consensus that it involves all three pieces of the Venn diagram in Figure 1.2.

It may be helpful to distinguish "data science" as a field and "data scientist" as a job title. Different companies and organizations will likely have distinct responsibilities for those working as data scientists. However, most data scientists are expected to be able to gain actionable insights from data.

As data science has evolved alongside technology, it has become one of the most in-demand professions. Data scientists and data science skills are necessary assets for almost all organizations.

Data Science Lifecycle

As data scientists, we generally follow a process that consists of *iterative* rounds of (1) data preparation, (2) data analysis, and (3) data storytelling for better insights and decision-making. By *iterative*, we mean that the steps often repeat and build on each other, improving over time. Figure 1.3 illustrates these components as useful exercises at multiple points in the workflow to help people make better decisions. We typically start with preliminary data preparation, which involves cleaning and organizing the data. After preliminary data preparation, data scientists may repeat the other components of the process to refine their understanding, which could include additional data preparation. For example, after preparing visualization data, we may dive into data analysis with our cleaned data to find patterns and relationships between variables. Alternatively, we may jump to data storytelling with a visualization that tells us how to prepare or analyze the data better.

FIGURE 1.3

The data science lifecycle comprises three iterative data science components that are repeated and improved as necessary: preparation, analysis, and storytelling.

We often revisit data preparation after data analysis and storytelling. For instance, analyzing or visualizing data can reveal new insights that prompt a return to preparing the data in a different way. At any point in the cycle, a discovery can spark further questions and analyses, feeding back into the cycle. Finally, we are ready for publicized data storytelling (e.g., posting it online, or distributing it privately to a decision maker), wherein we share a summary of our findings, often accompanied by compelling visual narratives.

Sometimes, this process also includes a data collection stage before the first data preparation step. Data collection may involve conducting surveys to gather customer feedback, using sensors to collect environmental data, or scraping websites for social media trends. Data scientists may collect more data as needs arise at any point in the cycle.

We call this the *data science lifecycle*, which occurs when all stages are completed at least once. Restarting the process based on findings from the previous cycle is essential because data science often introduces as many new questions as it answers. Figure 1.3 illustrates the data science lifecycle by showing data preparation, analysis, and storytelling as a rotation.

Quick Tip

Remember, before diving into data preparation, it's crucial to identify the question, problem or conjecture driving our data analysis. This initial step guides the purpose and direction of our data preparation, ensuring that the process is aligned with our objectives.

DEFINITION

The **data science lifecycle** is a system of three recurring steps: preparation, analysis, and storytelling. It starts with preliminary preparation and ends with publicized storytelling.

Data preparation is the process of collecting, cleaning, transforming, integrating, and managing data to structure it for effective analysis, storytelling, and decision-making.

Data analysis is the process of examining prepared data to uncover valuable insights and guide decision-making.

Data storytelling is the communication of data insights through summaries, visualizations, and narratives.

Data Preparation

Evgeny Tomeev/Shutterstock

Data preparation is the first component in which we collect (e.g., access) and wrangle (e.g., clean, transform, and integrate) data. As the name suggests, data preparation involves working with the data to get it into an increasingly structured format that is helpful for analysis and communication. We will spend the first few chapters of *Data Science for All* learning how to prepare data. Specifically, we will learn about representing data in tabular form (Chapter 1) and wrangling data to make it more useful (Chapter 2). Data preparation is often the most time-consuming part of a data scientist's job. It is like preparing ingredients to make a salad. In data preparation, much like meticulously washing, chopping, and mixing various ingredients to create a well-balanced salad, we methodically clean, transform, and integrate data to build a solid foundation for insightful analysis and storytelling.

Data Analysis

Christian Jung/Shutterstock

Data analysis is the extraction of meaning from prepared data to answer questions such as "What has happened?", "What might happen in the future?", and "What should we do?". We can perform data analysis by exploring (Chapter 4), querying (Chapter 5), understanding uncertainty (the study of how likely an event is to occur, covered in Chapter 6), learning how to draw conclusions from data (Chapter 7), and modeling data (e.g., supervised and unsupervised *machine learning* covered in Chapters 8, 9, and 10). We will learn how to conduct and interpret typical data science analyses. It is akin to creating your salad from the prepared ingredients.

Data Storytelling

Evgeny Karandaev/Shutterstock

Data storytelling is a component of data science aimed at maximizing decision-making utility. The goal is to ensure that findings from data analysis are suitable for making informed decisions. Success in communicating our discoveries depends on how well we prepare and analyze data. Chapter 3 covers the topic of *visualization in storytelling.* We use storytelling to assist data scientists in making decisions about data preparation and analysis. Storytelling can help both technical and non-technical stakeholders understand the results of data analysis.Consider this as presenting the salad in a way that makes people want to eat it.

Data preparation, analysis and storytelling are interrelated, so you will find them in all chapters. One reason these components can reoccur

Case Study: Netflix Uses Data Science for a Better Customer Experience

Urbazon/E+/Getty Images

Netflix uses data science to create a state-of-the-art customer service streaming site. It collects all kinds of customer data: when and how often subscribers watch a show, whether they binge-watch it or take some time to complete a series, and whether they pause an episode and resume it later. These data are used to create personalized account attributes for each subscriber, helping Netflix make content recommendations. Netflix's data scientists take advantage of visualization by structuring raw data into consumable dashboards, such as graphs and charts, for analysts to explore and translate into new features for the service.

Netflix uses the data science practices outlined in this textbook for preparing its data, analyzing its traffic and operations, maintaining its information security, and managing experimentation using A/B tests. These tests are trials in which, for example, the company sends one of two subscription offers to randomly selected customers to see which offer is accepted more often and by whom.

Overall, Netflix has wisely put all the data to good use. By incorporating concepts such as data analysis, machine learning, and statistics, data science has helped Netflix grow exponentially and become one of the top streaming sites in the world.

throughout the data science workflow is that the audience of the work can change. For example, the first iterations of data preparation and data storytelling are often most useful for the data scientist who is actually working with and analyzing the data. These rounds give them the understanding of the data needed to process, analyze, and communicate it in the best ways for downstream consumers like their professor, boss, colleagues, or customers. To this end, the final step in the data science lifecycle is to publicize the data storytelling component with visualizations and communication built for these consumers.

Data Science Tools

Data tools support the implementation of data science. These tools are software applications that facilitate preparation, analysis, and storytelling. They include coding tools (e.g., R and Python) as well as spreadsheet tools (e.g., Excel). Imagine trying to analyze online reviews without these technologies. Let's say you're an analyst at Amazon, aiming to understand if and how comments listing pros and cons influence customers' purchase decisions. If you or your team had to read and categorize every review for every product on Amazon, then compare those results to product sales histories, it could take a lifetime. However, tools like R, Python, or Excel could expedite the process, scanning through comments and searching for pros and cons, lists, and keywords in mere seconds.

We will include information and resources about data technologies in every chapter throughout this text. We will also cover scalable and robust tools for data preparation, analysis, and storytelling. *Data ethics*, also included in every chapter throughout this text, is the application of social justice and moral and ethical principles to data science.

DEFINITION

The **cloud** is a system of computer servers accessible over the internet, allowing us to deal with large amounts of data.

Case Study: NASA Uses Cloud Services to Stream Real-Time Mars Footage


Triff/Shutterstock

The National Aeronautics and Space Administration (NASA) is an agency of the United States government focused on the civil space program and space research. NASA's Curiosity Mission involved a meticulous and complex landing procedure for its interplanetary rover, Curiosity. NASA wanted to share real-time footage of the launch with people worldwide, so it turned to *cloud* services to achieve this. NASA's Jet Propulsion Laboratory uses cloud services to stream images and videos from Curiosity's Mars landing. The use of cloud services enables NASA to ensure the availability of its content around the globe. Thus, in real time, NASA can provide hundreds of gigabits per second of groundbreaking images of Mars for hundreds of thousands of viewers.

SECTION 1.2: DATA IN TABLES

As we will see, data come from a wide variety of sources in a wide variety of forms. Both the sources and the forms will help dictate the specific steps we need to take to extract meaningful insight from the data. Although the exact steps will almost never be the same, the general stages of this process will always include preparation, analysis, and storytelling. Before we delve into the ways in which we prepare data, we need to understand the anatomy of data.

Observations and Variables

Tables are a way to organize our data, structuring them in rows of *observational units* and columns of *variables*.

DEFINITION

An entity about which data are recorded is called an **observational unit**.

The recorded characteristic of an observational unit is the **variable** of interest.

The data for a single observational unit is sometimes called an **observation**.

Data in **tabular form** are organized into a table of rows and columns for clarity and ease of use.

A **dataset** is a collection of data.

A variable's **value** is the attribute or quantity given to a single observational unit within a dataset.

Many datasets exist in a *tabular form***,** meaning a table with rows and columns (Table 1.1). In the dataset below, the observational units in the rows are students, and the variables in the columns are first name, age, GPA, favorite pet species, and whether each person (observational unit) voted in a recent election.

For the tabular data we will discuss in this chapter, a dataset is a collection of variables with multiple observations. Therefore, a tabular dataset is like a table with many rows and columns. A dataset can also be represented in tabular form as a sheet in spreadsheets (e.g., Excel and Google Sheets), which we will discuss throughout this text.

TABLE 1.1
Example data from student election in tabular form.

variable

observational unit

first_name	age	gpa	favorite_pet	voted
Selma	18	3.7	Dog	0
Jerry	19	3.5	Cat	1
Sanjiv	17	3.1	Dog	1
Olivia	18	3.9	Other	0
Hector	18	3.6	Fish	0

Quick Tip

In practice, the data scientist collecting these data would use a student ID number or some other confidential identifier in place of names to protect subjects' personal information.

A variable's value is the information stored about each observational unit. In Table 1.1, each student is an observational unit in a particular row of our dataset. For example, Selma is someone in our dataset whose age is 18 and whose GPA is 3.7. Her age (18) and GPA (3.7) serve as values.

The values of the variables `age` and `gpa` would be stored on the computer as 18 and 3.7, respectively, for Selma. We would be able to find the correct age and GPA for any of our students, assuming the information stored in each student's `age` and `gpa` variables is accurate.

Pavel Bobrovskiy/Shutterstock

Variables and Data Types

Tables with defined variables (e.g., spreadsheets) are a very common form of structured data. Unstructured data, such as documents or images, do not directly fit into analysis tools like spreadsheets. Structured data versus unstructured data is like comparing organized belongings to unorganized ones. Structured data are often ordered in rows and columns.

DEFINITION

Structured data are information that is organized in a standardized format for efficient access, often in tables or databases.

A **database** is an organized collection of data, typically stored electronically, that allows for easy storage, retrieval, and manipulation of information.

Unstructured data are information that lacks predefined organization, such as text, images, or videos.

For data that can be represented as variables of observational units in something as neat as a spreadsheet, there are two critical ways to think about variables and their values:

1. The variable's use. Throughout the rest of this book, we will discuss numerous ways to use data, including creating summaries, visualizations, and analyses.
   a. *Quantitative* variables take numeric values on which mathematical operations make sense. In Table 1.1, `gpa` is an example of a quantitative variable.
      i. *Discrete variables* are quantitative variables that take on values that can be written down in a list. In Table 1.1, `age` (in whole numbers of years) is an example of a discrete variable.
      ii. *Continuous variables* are quantitative variables that can take on any value within an interval of values. The body temperature of the students is an example of a continuous variable.

b. *Categorical (qualitative) variables* take values that are category designations. In Table 1.1, `favorite_pet` is an example of a categorical variable.

2. The variable's type. The type of data the variable contains determines how the variable's values are stored in a computer. These variable types include:

    a. *Text*: The variable contains words or characters without order or numerical utility (e.g., a `first_name` value of "Hector" or `favorite_pet` value of "Dog"). A text variable is sometimes referred to as a string variable. Categorical variables are often stored as text variables.

    b. *Integer*: The variable contains numbers with no decimal points (e.g., an `age` value of 18).

    c. *Float*: The variable holds numbers with decimal points (e.g., a `gpa` value of 3.1). Because a number may have many decimal places, the number can be quite long. This variable is sometimes referred to as a decimal variable.

    d. *Boolean*: The variable contains only two possible values: TRUE or FALSE. Booleans are also referred to as *indicator variables*, *logical variables*, *binary variables*, *dichotomous variables*, or *dummy variables*. Sometimes, the values are stored as 1 or 0, representing TRUE or FALSE, respectively (e.g., the `voted` value of 1 or 0, meaning it is TRUE or FALSE that the student indicated they voted).

Try It Yourself: Identify Variable Uses and Types

Consider the following ten variables and a sample of their values:

`height`: 5.7, 6.0, 5.8, 5.9, 6.1
`age`: 21, 22, 23, 24, 25
`name`: 'Alex', 'Beth', 'Charlie', 'Dana', 'Eli'
`gpa`: 3.5, 3.6, 3.7, 3.8, 3.9
`graduated`: True, False, True, True, False
`pets`: 0, 1, 2, 0, 1
`birth_month`: 'January', 'February', 'March', 'April', 'May'
`siblings`: 2, 1, 0, 3, 2
`fulltime_student`: False, True, True, False, True
`shoe_size`: 8, 9, 10, 11, 12

Your task is to identify each variable's use and type. Remember, the possible uses are quantitative and categorical, and the possible types are text, integer, float, and Boolean. This exercise should help you better understand and identify different variable types, which is an essential skill when working with data.

Data scientists name their variables. Sometimes, the names are helpful to others, and sometimes they are not. Consider a variable named `var_1`. This variable name is unhelpful. Without a data key or codebook, we have no idea what information is stored in this variable. A *data key*, or *codebook,* is a reference document that defines the variables and the types of information they contain. Table 1.2 shows an example data key.

Creating meaningful names related to the contents of their variables is a thoughtful process. For example, suppose we create a Boolean variable where people with a cell value of 1 indicate that their favorite pet is a fish and 0 if they indicate some other animal. Table 1.3 has a poorly named variable, `var_1`, which does not describe the Boolean variable well. A better name might be `favorite_pet_is_fish`.

Fesus/123RF

It is best practice—often required by software—to name variables using a single word without symbols like %, ^, &, *, $, #, ~, or @ that often have other coding purposes. Spaces are also usually prohibited in variable names when coding. In Table 1.3, the variables are named to follow these conventions. We use the snake_case convention in this textbook because it is the most common naming method used in Python and R. However, other conventions exist like UpperCamelCase, which separates words without spaces using changes in case. To prepare you for the real world we may use UpperCamelCase occasionally in this textbook, too.

TABLE 1.2
Data key with both uninformative and meaningful variable names.

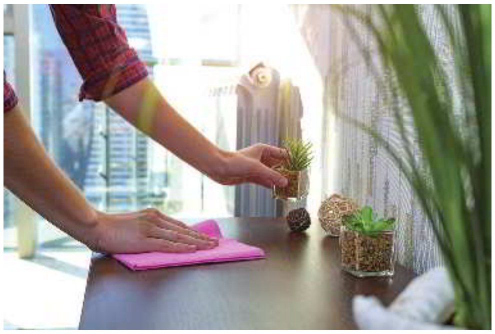
Uninformative Variable Name	Meaningful Variable Name	Description	Type of Data
`ar_n`	`artist_name`	Name of artist	Text
`al_t`	`album_title`	Album title	Text
`p_r`	`pitch_score`	Review score from online music magazine *Pitchfork*	Float
`g_n`	`grammy_nominated`	Was the album nominated for a Grammy Award?	Boolean
`b_p`	`peak_chart`	Peak charting position on the U.S. *Billboard* 200	Integer
`s`	`sales_us`	Number of certified sales in the U.S.	Integer

TABLE 1.3
Data with an uninformatively named variable, `var_1`.

first_name	favorite_pet	var_1
Selma	Dog	0
Jerry	Cat	0
Sanjiv	Dog	0
Olivia	Other	0
Hector	Fish	1

Tidy Data

Tabular datasets are often delivered to us in a format that is not yet ready to work with. While a tabular shape is considered structured, it's important to remember that data can be further structured in preparation for analysis. The creation of summaries, data visualizations, and analyses will require that our data be organized in a consistent, well-defined structure. If our dataset is not originally organized in such a structure, it's referred to as messy data. While unstructured data are unorganized, messy data have been poorly organized. Table 1.4 provides an example of messy data, where the first two rows appear to contain a mixture of column headers and data values, and the third column contains data on more than one variable.



Goffkein.pro/Shutterstock

TABLE 1.4
Messy data.

Class	Sporting	Hound	NonSporting	Herding	
	Age	Name_and_ID	Breed		Class
	Ten	Otis18654	BORDERCOLLIE		he
	Five	Tito20862	DACHSHUND		ho
	Fifteen	Chloe12368	POODLE		ns
	One	Olive56731	BULLDOG		ns
	One	Olive56731	BULLDOG		ns
	Six	Georgie90052	LABRADOR		sp

The process of converting messy data into a form ready for summarizing, visualizing, and analyzing is a large part of data preparation and usually involves both cleaning (discussed later in this chapter) and structuring. At the data analysis stage of a data science project, it's helpful to be working with *tidy data*. With tidy data:

1. Every column is a variable.
2. Every row is an observation.
3. Every cell is a single value.

DEFINITION

Data structuring is the process of organizing data, often into tables.

Tidy data is a data structure where each column represents a distinct variable, each row corresponds to a unique observation, and each cell contains only one value.

Table 1.4 is an example of data that are not tidy. Further preparation may need to be done for us to implement specific visualizations or analyses, but a tidy dataset is well understood and straightforward to work from. Table 1.5 displays a clean, tidy version of the data in Table 1.4, where we now have a column for every variable, a row for every observation, and a cell for every value. Whether data are considered tidy or messy may depend on how a unit is defined. For example, tidying data depends on knowing that `dog_id` and `age` are numerical variables, and `class`, `name`, and `breed` are text variables.

TABLE 1.5
The messy data are now tidy.

dog_id	class	name	age	breed
18654	Herding	Otis	10	Collie
20862	Hound	Tito	5	Dachshund
12368	NonSporting	Chloe	15	Poodle
56731	NonSporting	Olive	1	Bulldog
90052	Sporting	Georgie	6	Labrador

The previous messy data we reviewed obviously needed cleaning. However, some data are messy because they do not fit a purpose for analysis. For example, suppose we observed multiple student performances across different subjects: Math, English, and History. We want to analyze the data in a way that considers each subject a different observation. Table 1.7 is considered messy because it does not adhere to the second principle of tidy data: every row should be a single observation. In other words, each row in Table 1.7 represents multiple observations for the stated purpose: to analyze the data across multiple subjects. It certainly looks tidier than Table 1.6. However, we still want to tidy it more to adhere to the purpose.

Table 1.8 is a tidy version of Table 1.7, with separate columns for `subject`, `score`, and `hours_studied`. Each row in Table 1.8 represents a single observation, i.e., a student's score and hours studied for a specific subject. This structure might make the table easier to analyze and visualize.

Try It Yourself: Tidy Messy Data

Africa Studio/Shutterstock

Take a look at Table 1.6, which shows student performance across different subjects. You might notice that the table is a bit messy! It has displaced column headers and confused values in the rows. However, each observational unit is fortunately within one row, making your task easier. Please tidy this table!

TABLE 1.6
A messy table example.

student_id		Score
	Subject	
1	Math	85
1	90	Chemistry
2	Physics	88
2	Physics	88
88	Math	2
Match	3	88
3	Chemistry	85

TABLE 1.7
The data are messy because each row is a series of observations.

student_id	math_ score	math_hours_ studied	english_ score	english_hours_ studied	history_ score	history_hours_ studied
1	85	10	90	12	88	9
2	90	11	88	11	90	10
3	88	10	89	12	85	8
4	92	12	85	11	90	9

TABLE 1.8
The messy data are now tidy.

student_id	subject	score	hours_studied
1	Math	85	10
1	English	90	12
1	History	88	9
2	Math	90	11
2	English	88	11
2	History	90	10
3	Math	88	10
3	English	89	12
3	History	85	8
4	Math	92	12
4	English	85	11
4	History	90	9

Try It Yourself: Tidy Data for Better Analysis

Over three days, Coach John rated the athletic performance of camp participants from 0 (low performance) to 100 (high performance) in three different sports. He also recorded the number of minutes they took to run a mile in the morning of the observation, as a measure of their endurance. Specifically, he rated their performance in volleyball on Monday, basketball on Wednesday, and water polo on Friday, each time after recording their endurance. Coach John wants each observation to be a distinct sports performance. A tidier version of Table 1.9 should be created for Coach John's purpose.

TABLE 1.9
The original data are considered messy for Coach John's purpose.

student_jersey	volleyball_rating	volleyball_endurance	basketball_rating	basketball_endurance	water_polo_rating	water_polo_endurance
1	87	7.11	90	7.09	88	7.10
2	97	6.75	88	6.57	90	6.66
3	81	8.11	89	7.75	85	8.01
4	97	5.98	85	6.03	90	6.12

Big Picture Break

We just described the usefulness of data organized in a *tidy* way, and what characteristics make a dataset tidy. Many of the data preparation techniques discussed in the next section, and in future chapters, are especially useful for getting *messy* data into a *tidy* format. However, it should be noted that there are many formats that a structured dataset can be in that are consistent, well-defined, and *not technically tidy*. In the end, it is often the case that the intended uses of a dataset help drive the formats it needs to take. For example, one structure may be especially useful for a particular visualization, while another structure may be better suited for certain modeling or analysis methods. People doing data science work with a dataset should be familiar with multiple techniques for wrangling a dataset into the shapes needed for different types of visualization, analysis, and storytelling.

Metadata

It is extremely common that the data you are working with will require their own description and documentation. These descriptions are not about the values or observations within your dataset, but about the dataset itself. These data about your dataset are appropriately called *metadata*.

Quick Tip

Metadata are almost always the first things we want to know about a dataset before actually looking at any of the data itself. Think of metadata as the information you would want someone to tell you when they hand you a new dataset, so that you can start work quickly and easily.

DEFINITION

Metadata are information that describe and provide context about a dataset.

The codebook, or data key, mentioned in the previous section, is a common place to store important metadata. To this end, the following characteristics are important pieces of metadata:

- Variable names in the dataset.
- Description of the variables in the dataset.

- The type of data stored in each variable in the dataset.
- The coding scheme (if the variables are coded).
  - For example, in Table 1.3, we need to understand what a value of 0 signifies and what a value of 1 represents in the `var_1` variable.
- The units of measurement for each of the variables in the dataset.
- The date the dataset was created.
- The author and/or sources of the dataset.
- The number of missing values.

This information should be stored in a file accompanying the dataset, wherever it resides, and not in the dataset file itself.

Try It Yourself: Investigate Metadata

Dataset: Titanic-Dataset

Consider Table 1.10 with the metadata for a dataset about survivors of the Titanic's historic shipwreck on April 15, 1912.

TABLE 1.10
From a dataset provided by M. Yasser H., accessed at Kaggle.[1]

variable_name	description	data_type	coding_scheme	unit_of_ measurement	missing_values
passenger_id	Unique identifier for each passenger	integer	-	-	-
survived	Survival status	boolean	0 = Did not survive, 1 = Survived	-	-
p_class	Ticket class	integer	1 = 1st class, 2 = 2nd class, 3 = 3rd class	-	-
name	Name of the passenger	text	-	-	-
sex	Sex of the passenger	text	-	-	-
age	Age of the passenger	float	-	Years	177
sib_sp	Number of siblings/spouses aboard	integer	-	Count	-
par_ch	Number of parents/children aboard	integer	-	Count	-
ticket	Ticket number	text	-	-	-
fare	Passenger fare	float	-	Currency (likely Great British Pound or GBP)	-
cabin	Cabin number	text	-	-	687
embarked	Port of embarkation	text	C = Cherbourg, Q = Queenstown, S = Southampton	-	2

[1] Yasser H., M. (2022). Titanic Dataset. Kaggle. https://www.kaggle.com/datasets/yasserh/titanic-dataset/

(*Continued*) ▶

(*Continued*)

Question

Fill in the blank: Why is metadata important when analyzing a dataset, like the Titanic passenger list?

Metadata helps us understand what the data mean. For example, we understand that 'Survived' coded as '1' means ________________, which is crucial for analyzing survival patterns in the Titanic dataset.

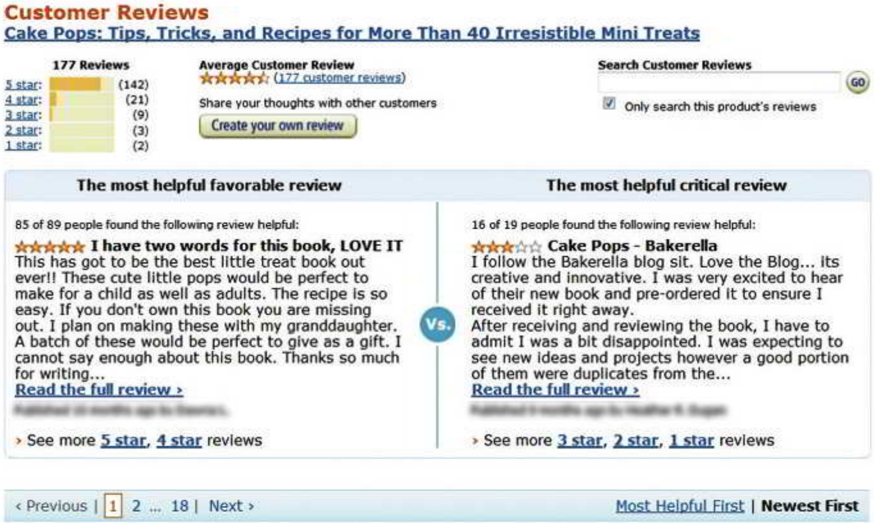
SECTION 1.3: DATA PREPARATION

Data come from many sources in different formats. As we just saw, they may be quantitative—using numbers—or qualitative—using words, photos, or graphs. They may be stored on a computer, such as in a database of demographic data organized by zip code, downloadable from a government server, or accessed "live," such as real-time data from a stock exchange. They may come ready for analysis, like a dataset of media consumption patterns purchased from a data provider like Nielsen, or they may need to be cleaned before use, as with the extraction of social media posts.

We will see in the discussion of data wrangling below that *data cleaning* is different from *data tidying*, as discussed earlier in the chapter. Data can be structured, such as those found in spreadsheets, or unstructured, like the text found in Amazon reviews (Figure 1.4).

FIGURE 1.4

A sample Amazon product review.



Amazon.com, Inc.

In any form, recall that unstructured datameans the data are not organized in predefined data formats and, therefore, are not yet useful for analysis. Note that structured dataare the opposite: they are organized in predefined formats and fit nicely into relatable frameworks like tables or spreadsheets, many of which are ready for analysis. Extracting meaning from unstructured data is more difficult. For example, grouping Amazon reviews by the same geographic region would be challenging before the data are structured. In addition to structuring data, a data scientist must prepare it by *cleaning* it to integrate the data with other data sources.

Data preparation has three parts:

- data collection
- data wrangling
- data management.

DEFINITION

Raw data are data in the form originally collected that have not been processed for use in analysis, storytelling or other purposes.

Data collection is the systematic gathering of relevant data from various sources.

Data wrangling, also known as **data preprocessing**, is the process of converting raw data into a more usable format. This is achieved by cleaning, transforming, and integrating the data to facilitate effective analysis and storytelling.

Data management is the storage of data to make it available for analysis.

Data Collection

Data collection is the systematic process of gathering relevant data from various sources such as databases, websites, surveys, or logs recorded in scientific experiments or field research. The quality and relevance of the data collected form the bedrock of our analyses and subsequent insights. Even the most advanced data analysis cannot compensate for low-quality or irrelevant data. Therefore, it's important that we contemplate the specifics of what data are needed, the sources from which they can be procured, and the method of collection to be employed.


Abhishekspadmanabhan/123RF

We can collect data from a variety of sources, and these sources often depend on the field of study. For example, suppose a wildlife conservation group wants to study the movement patterns and behaviors of birds in a particular wetland. They decide to collect data by:

- Tagging a sample of birds with small tracking devices to record GPS locations over time.
- Installing motion-sensor cameras throughout the wetland to capture images of the birds.
- Making visual observations from viewing blinds and documenting behaviors such as feeding, nesting, and interactions.
- Recording audio of bird calls and analyzing their characteristics and frequency.
- Collecting samples of plants and insects in the habitat, and noting the birds associating with each food source.
- Acquiring regional weather and seasonal data to associate environmental factors with activity.
- Partnering with other researchers to obtain existing observational datasets on similar species in nearby wetlands.

Creativity is often required to gather relevant data when not working with traditional transactions and metrics. We discuss data collection further in Chapters 6 and 7.

After data collection, our data don't just sit there—they undergo a makeover. In the data wrangling stage, we clean and transform the data, preparing it for its big moment: the analysis. We might need to clean the data, deal with any errors, or address missing values, to ensure our analysis can accurately reflect what the data are trying to tell us.

Data Wrangling: Clean, Transform, and Integrate

Data wrangling, also known as data preprocessing, is the process of converting raw data into forms necessary for analysis and storytelling. Specifically, wrangling data involves cleaning, transforming, and integrating data to make them useful for analysis and storytelling.

DEFINITION

Cleaning data involves fixing or removing incomplete, abnormal, or inconsistent data to enhance their quality for accurate analysis and storytelling.

Transforming data involves organizing and formatting data into a structure that is useful for efficient analysis.

Integrating data is the process of merging multiple data sources to provide comprehensive insights that surpass what each individual source could offer alone.

Quick Tip

Even though data cleaning is distinct from data tidying, they are both critical pieces of data preparation. Many forms of visualization and analysis require data to be in a tidy format.

Step 1: Clean

Cleaning deals with data that are incomplete, abnormal, or inconsistent. We clean data to ensure that they are of the highest quality possible before analyzing them and presenting our findings. Cleaning data is distinct from tidying data: cleaning improves data quality, while tidying optimizes their structure.

Cleaning data can be a complicated process that requires judgment calls. The best strategy for a given dataset that needs cleaning may depend on the industry or the situation. However, it's always important to report any adjustments that were made to your data during the cleaning process. We discuss some best practices of data cleaning in Chapter 2.

Step 2: Transform

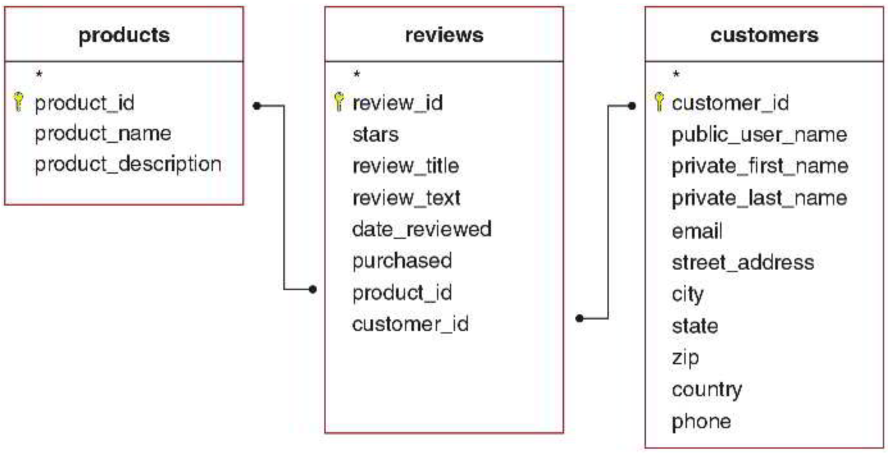
MK photograp55/Shutterstock

As mentioned, we often need to structure data when it comes to us in ways that would otherwise be difficult to analyze. *Data transformation* is the process of organizing and formatting data to facilitate efficient analysis. For example, consider a database of blog entries from the past ten years. How would we analyze those millions of unstructured blog entries? Would we only include text written by the blog owner? What about responses? How would we explore everything? It would be difficult because there needs to be a structure relating responses to original posts.

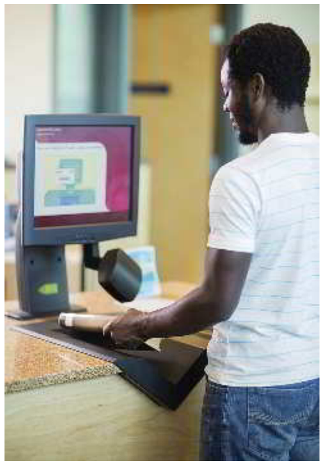
Unstructured data do not reside in traditional, organized databases (such as a collection of related spreadsheets). Instead, they have an internal structure (e.g., the name of a blog's author with her product review). Figure 1.5 is an example of transforming the unstructured data from Figure 1.4 into structured data. The *Cake Pops* book, the reviews of which are shown in Figure 1.4, would be one entry in the Products table in Figure 1.5, along with all the other millions of products sold by Amazon.

FIGURE 1.5

Transforming unstructured Amazon reviews into structured tables, connected by identification variables such as `product_id` and `customer_id`.



Similarly, the two reviews we see in Figure 1.4 would be two entries in the Reviews table in Figure 1.5, which could contain tens or hundreds of millions of entries. The two customers who wrote these reviews would be two entries in the Customers table in Figure 1.5, which could also contain millions of entries. We acknowledge that Amazon's data management methods are more sophisticated, handling billions of daily transactions in the cloud.



Tyler Olson/Shutterstock

Unstructured data may be generated by humans interacting with machines. A turnstile counter at an amusement park is an example of humans interacting with a machine. There can also be machine-to-machine communication, sometimes called M2M. For example, automated weather monitors inform an energy company's power plants about projected electricity demand and productivity.

Unstructured data are diverse. They may include personal messaging such as email, instant messages, and chats; business documents like reports, presentations, and survey responses; web content such as site pages, blogs, wikis, audio files, photos, and videos; and sensor output like scanner transactions.

Sometimes, there is a trade-off between efficiency and organization. In such cases, the data scientist may structure the data by organizing them in whatever convenient format works for the analysis needed. For example, large amounts of big data accessed in real time may need to be processed quickly. It may be impractical to force the data into a structure because of the time and storage space limitations. Instead, the data scientist would analyze the data *without* structuring it. On the other hand, structuring even large amounts of data may be worthwhile if the data are to be deeply analyzed and integrated with other data sources to discover patterns.

It's often the case that a structured dataset can be further structured, or prepared for analysis, by transforming some of the existing data. For example, a spreadsheet showing reported crimes with a single column for the date of each report is already well-structured. However, it may be of interest to separate the day, month, and year into their own columns for visualization and analysis. Such transformations fall under the umbrella of data wrangling.

Step 3: Integrate

Once a dataset has been cleaned and structured, it is ready for analysis. However, integrating it with other datasets often offers essential insights. *Data integration* involves combining two or more datasets to provide more insights than each dataset would yield individually. Integration is crucial for publicly available data, where analysts have likely already harvested insights. When data are easily obtained from a single, publicly available source, the competitive advantage of the information by itself is low.

Still AB/Shutterstock

We can typically find unique insights by integrating one data source with another, especially if the second data source is proprietary (i.e., not public). Suppose an umbrella company considers a list of keyword search frequencies for "umbrella" within each U.S. state over particular time periods, from a data source like Google Trends. A competitor could easily find these data too. The value of the keyword search frequencies for competitive insights is limited. However, after integrating this dataset with internal sales information and external weather

Try It Yourself: Illustrate Unstructured Data Using Structured Tables

Consider the related tables in Figure 1.5, based on the Amazon reviews in Figure 1.4. Please fill in the blanks.

1. "Cake Pops" from Figure 1.4 is likely stored in the _________ field of Figure 1.5's _________ table.
2. "Cake Pops - Bakerella" from Figure 1.4 is likely a record in Figure 1.5's _________ field in the _________ table.
3. To find all reviews for a specific product on Amazon, the _________ field in Figure 1.5's `reviews table` is used to link to the corresponding product in the `products table`.
4. To identify the customer who wrote a review on Amazon, the _________ field in Figure 1.5's `reviews table` is used to link to the customer's record in the `customers table`.

data, it would be possible to model the predictive value of keyword searches, and to use that data to estimate expected umbrella sales across states at different times of the year.

We will delve more into data wrangling in Chapter 2 and data management in Chapter 5. However, next, we will briefly discuss the place of data management within data preparation, alongside data collection and data wrangling.

Data Management

Data management is the process of storing data and facilitating its access. After collecting and wrangling data, we often want to store them in a tabular form within a secure database. When data are integrated from multiple sources, we may connect multiple tables with each other. We also want people to access our data efficiently, ensuring the right information is available when needed. When executed effectively, data management allows us to store raw data as a meaningful resource for in-depth analyses in various academic and industry contexts.

SECTION 1.4: DATA ANALYSIS AND STORYTELLING

Data Analysis

Data analysis is the process of examining datasets to uncover useful information, which can inform conclusions and support decision-making. It involves applying statistical, logical, and mathematical techniques to sift through data, identify patterns, trends, and relationships. This phase is critical in data science because it translates data into insights that can drive informed actions and strategies.

Exploratory Data Analysis (EDA) is a preliminary step in the data analysis process to explore and examine datasets. Data analysis can be broken into four main categories: descriptive, diagnostic, predictive, and prescriptive analysis. As the first step, exploratory data analysis often focuses on descriptive and diagnostic statistics but can sometimes adopt predictive and prescriptive analysis as well.

DEFINITION

Descriptive analysis systematically examines current and historical data, utilizing various benchmarks, to provide a detailed view of past and present outcomes.

Diagnostic analysis investigates data to determine the possible causes and reasons behind past events.

Predictive analysis uses statistical methods to predict future events based on historical data.

Prescriptive analysis provides recommendations on actions to take.


Ruth Black/Shutterstock

Descriptive analysis examines what has happened in the past or what is happening at the present moment, based on current data. For instance, we could use the data represented in Figure 1.4 to determine the average number of stars the *Cake Pops* book receives compared to the average ratings of products in the "dessert books" category. We might find that our *Cake Pops* book has an average rating of 4.68, while the average rating for all dessert books is higher, at 4.81.

Diagnostic analysis goes beyond asking *what* happened to *why* it happened. For example, we could ask whether our *Cake Pops* book's average ratings are lower than all dessert books because of the reviewers' geography. We might discover that half of the *Cake Pops* book ratings come primarily from a single region that is particularly tough on dessert books in general. Within this single region, the *Cake Pops* book's average rating is 4.58, while the average rating for all dessert books is also lower than the global average at 4.61. Outside that region, the *Cake Pops* book's ratings are much higher at 4.81, equal to the category's global average rating. These data would help the *Cake Pops* authors understand why their book has lower average ratings. In other words, diagnostic analysis reveals that they may be focusing too much on selling to customers from a region that does not appreciate dessert books, and they can adjust their marketing strategy based on that information.

Predictive analysis seeks to determine what is likely to happen in the future by estimating future behavior based on past patterns. For instance, we could predict next month's average customer review ratings based on past seasonal data. These data show that reviews for the product category of dessert books tend to worsen from January to July and improve from August to December. Data scientists could also create a mathematical formula, known as a "predictive model." This model uses machine learning to identify which regions are most likely to appreciate the *Cake Pops* book, based on other facts about the customers and product.

Prescriptive analysis asks what actions should be taken, building on predictive analysis. For example, marketers could use predictive models based on seasonality and geographical region to recommend the action of a promotional campaign in the more advantageous months of August to December, in the particular areas more favorable to dessert books during those months. In other words, prescriptive analysis would optimize seasonality and location data for good reviews.

Try It Yourself: Identify Analysis Types

Consider four simple analyses related to the *Cake Pops* book ratings, and identify each as either descriptive, diagnostic, predictive, or prescriptive:

- A targeted marketing campaign in regions with a higher appreciation for bakery books should increase *Cake Pops* book ratings by 4%.
- Customers who bought at least two other bakery books rated the *Cake Pops* book 4.71 on average, which is higher than the 4.55 average rating from customers who only bought the *Cake Pops* book. This indicates that a general interest in bakery played a role in its ratings.
- Based on current bakery book review trends, the average rating of the *Cake Pops* book is projected to decrease by 0.5 stars (half a star) in the next year.
- The *Cake Pops* book has an average rating of 4.68, whereas the overall average rating for general bakery books is 4.79.

Data Storytelling: Visualization, Narrative, and the STAR Framework

Data storytelling is the translation of complex data preparation and data analysis into a narrative that can be easily understood by an audience not necessarily trained in data science. Many data scientists focus on gaining data and analysis skill sets, but eventually, they must learn how best to communicate their findings. However, we approach the concept of storytelling cautiously. While it is certainly not the creation of a fictitious story, it is also not sufficient to tell a true story without presenting evidence based on the hard work of preparation and analysis. It is also insufficient to omit the complex assumptions and decisions that data scientists must make to obtain their results. It is crucial that any data scientist who hears the story can replicate it themselves based on the description of the process. Storytelling supports decision-making by succinctly communicating the data scientist's highly sophisticated mathematical, statistical, and computer science methods. We tell stories with data through visualization and narrative. We offer the STAR Framework as a helpful aid when communicating data results.

DEFINITION

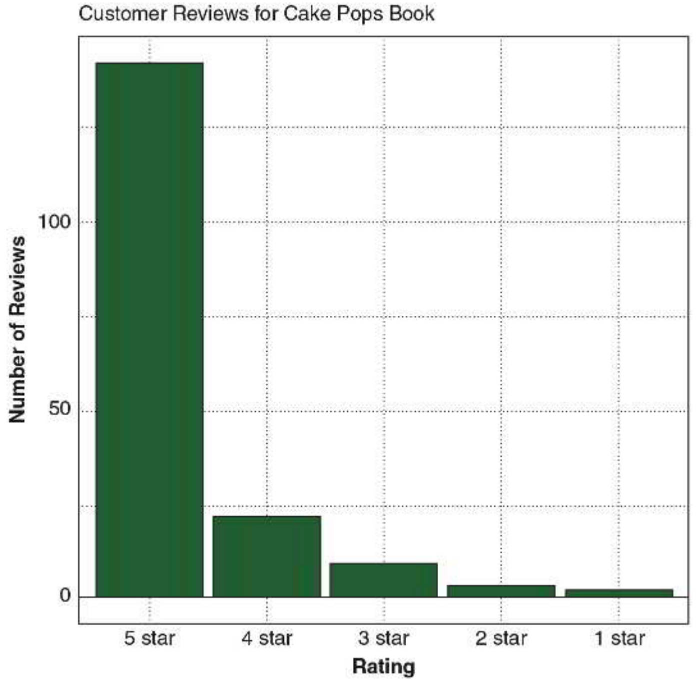
Visualization is the presentation of data in a graphical format.

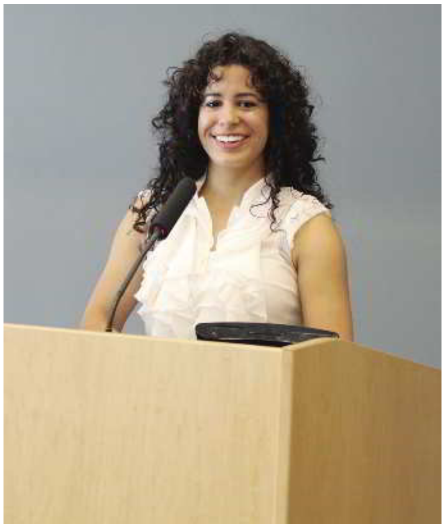
A **storytelling narrative** in data science is the art of presenting data in an accessible and engaging manner.

Visualization

Visualizationis a meaningful way to represent information concisely in a graphical format, facilitating storytelling. For example, Figure 1.6 shows the distribution of customer reviews for the *Cake Pops* book from Figure 1.4 in a visual format. This communicates that the book has received an overwhelming number of five-star reviews.

FIGURE 1.6
Visualization shows the number of customer reviews for the *Cake Pops* book.





Val lawless/Shutterstock

Learning data science requires practice in choosing chart types, making charts, and interpreting charts. In addition to the focus on data visualization in Chapter 3, we will see visualization examples throughout many chapters. This is because visualization is central to effective data preparation, analysis, and storytelling.

Narrative

The best data preparation and analysis are only helpful if communicated effectively to the right people. A well-crafted narrative makes information more accessible to a diverse audience, including scientists, managers, farmers, artists, scholars, and policymakers. Therefore, we also aim to enhance both spoken and written presentations. The ultimate goal of data science is to tell a story that aids people in making better decisions in any field. For instance, geneticists can make more informed decisions about medicine, historians can interpret digitized documents more accurately, farmers can identify which crops will maximize sustainable cultivation, and ecologists can better manage wildlife populations. The art of storytelling requires presenting multiple facets of data in a manner that most people can understand.

STAR Framework

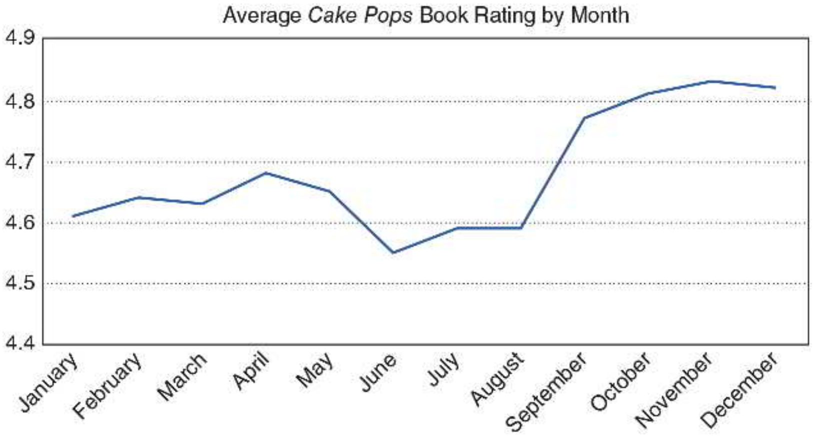
We introduce the STAR Framework as a structured approach to communication in data storytelling. Each component of STAR serves as a step towards crafting a compelling narrative that not only presents data but also contextualizes it and drives action. Here's how each part of the STAR Framework functions:

1. **Scope the Data's Context**: This initial step characterizes the data and their context, including the observational units, variables and their types, and data source.
2. **Track the Question Behind the Exploration**: Here, we identify the questions and core tensions we hope the data will address.
3. **Articulate the Results**: In this phase, we clearly explain the results.
4. **Respond with Next Steps**: The final component encourages potential actions or investigations.

Try It Yourself: Optimize the Narrative

Consider the following visualization that draws a line between the average *Cake Pops* book rating for each month of the year. Answer questions about the visualization based on Figure 1.7.

FIGURE 1.7
Average Amazon ratings of the *Cake Pops* book by month over the past three years.



1. Which statement best describes Figure 1.7 in terms of months and seasons (e.g., fall, winter, spring, and summer)?

   The average *Cake Pops* book rating . . .

   A. is unrelated to the time of year.
   B. changes with the seasons.
   C. increases each month.
   D. changes each month unrelated to the seasons.

2. Which statement about the *Cake Pops* book is most likely, given the data in Figure 1.7?

   The *Cake Pops* book features . . .

   A. month-themed cake pop recipes.
   B. a new cake pop recipe each week.
   C. cake pop recipes for each season of the year.

(*Continued*) ▶

(*Continued*)

D. cake pop recipes for each season emphasizing holidays in September, October, November, and December.
E. cake pop recipes for each season emphasizing December holidays.
F. cake pop recipes for each season emphasizing spring holidays.

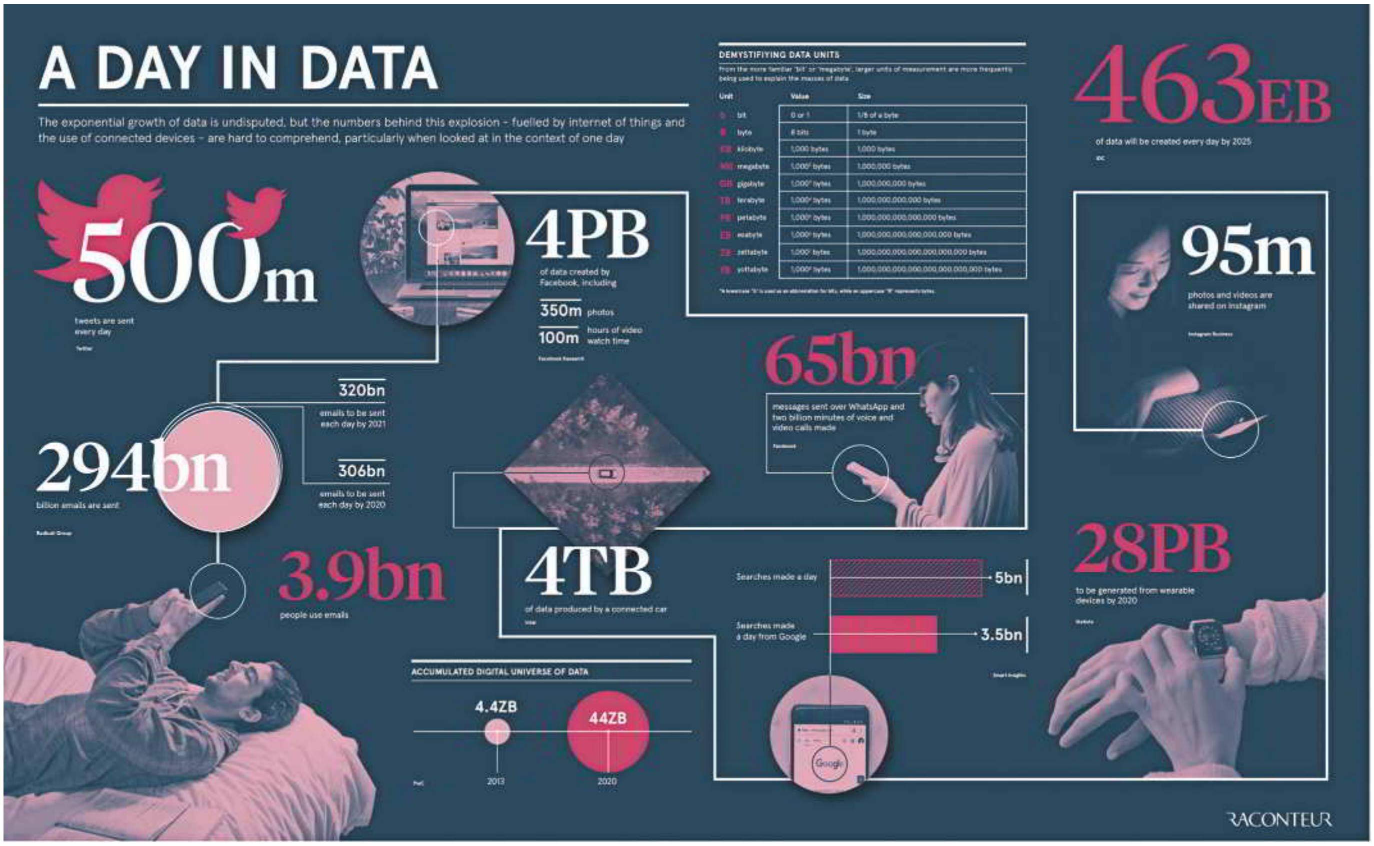
3. Open-Response Question: Use the STAR Framework to tell a story about Figure 1.7.

SECTION 1.5: DATA SCIENCE IN SOCIETY AND INDUSTRY

The Growth of Data Science

Data science is a growing field due to previously unimaginable amounts of available data. The number of people engaged in exploring and learning from these data is not keeping pace with this growth. The amount of global data is staggering (Figure 1.8). Living in the digital age, we can access new kinds of data that were rarely seen in previous decades.

FIGURE 1.8
Global data are increasing exponentially. From Jeff Desjardins, Visual Capitalist.



Jeff Desjardins, How Much Data Is Generated Each Day?, Visual Capitalist, April 15, 2019, https://www.visualcapitalist.com/how-much-data-is-generated-each-day/

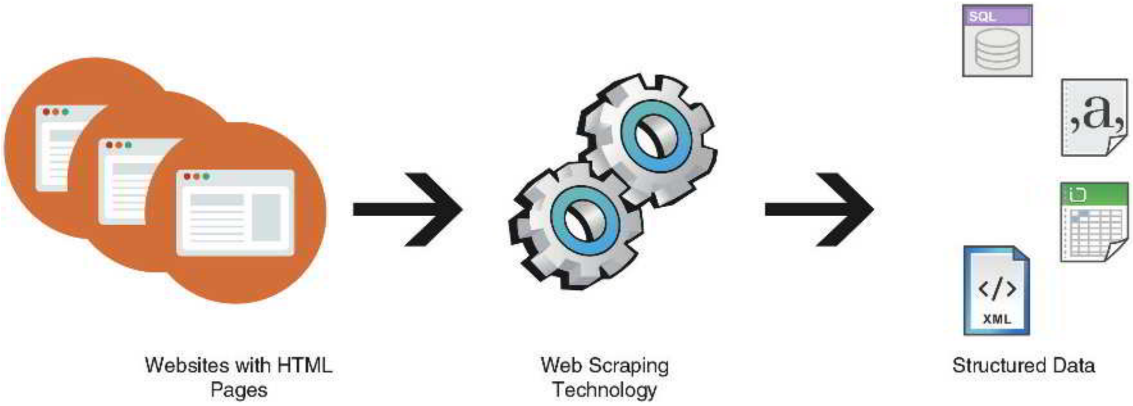
Consider how the following data sources could benefit people:

- Amazon reviews assist customers in making more informed purchase decisions. For instance, review data aids firms in improving their products and services for the benefit of customers.

- Social scientists monitor videos featuring cultural trends found on social media. For example, a consulting firm performs video analysis of people wearing yellow for World Mental Health Day in videos posted on TikTok.
- Impactful technological innovations from online forums provide business insights. For example, a start-up company searches online forums devoted to new uses of artificial intelligence in household products, like Amazon Alexa.
- Web-scraping, the automated extraction of information from public websites, supports the study of technological trends. For example, a professor monitors technology review websites to teach students about the latest product offerings (Figure 1.9).

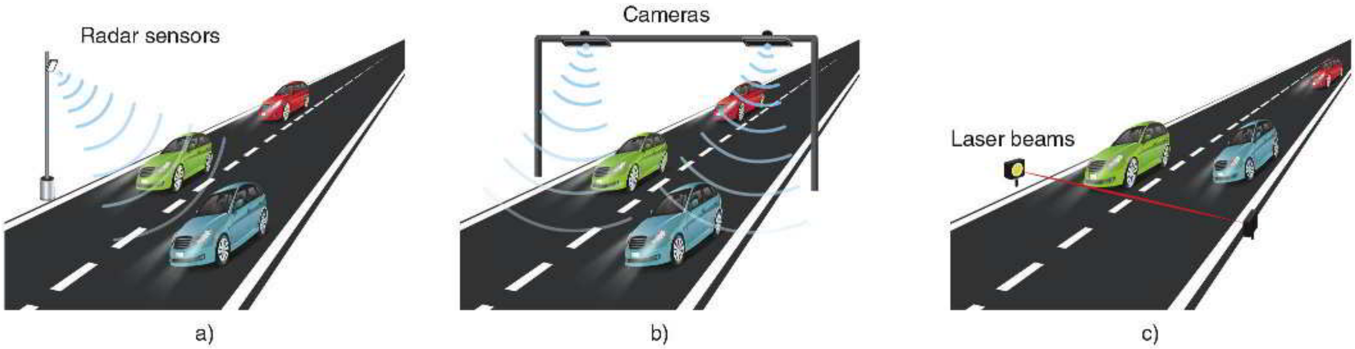
Olegdudko/123RF

FIGURE 1.9
Web-scraping translates HTML from thousands of websites into useful data tables.



- Machine-generated data influence public-sector administration. For example, a city planner analyzes changing traffic patterns from road sensors (Figure 1.10).

FIGURE 1.10
Radar sensors, cameras, and laser beams generate data to estimate traffic problems.



Data Science Everywhere

We don't have to be in the workplace to experience data science. Data science affects nearly every part of our lives.

Do you know anyone who looks at their Fitbit or smartwatch for anything other than discovering the number of steps they've taken that day? These wearable devices can generate vast amounts of data. Data scientists and other public health researchers are figuring out how to use these large but simple accelerometer data to predict a user's activity at any given moment. These health professionals may be able to apply this new source of data to improve health by monitoring the activity levels of Fitbit wearers. It turns out that people are not very good at self-reporting how much

Irabel8/Shutterstock

time they spend each day doing physical activities. Therefore, if medical professionals can take advantage of wearable monitor data volunteered by at-risk individuals, there is hope that health outcomes can be improved.

Have you ever been distracted by the sounds of cars or trains traveling near your home? Now, imagine this, but many times worse, and in the ocean! Sound travels faster and more easily in the sea than in the air. Therefore, locating noise-generating coastal roads and mapping shipping routes to minimize the negative impacts on ocean wildlife present challenging tasks. The ocean is vast, though. So, how do we know where creatures are in order to avoid them? Because various shapes and sizes of sound vibrations traveling through water are detectable at great distances, many tools can be used to collect and analyze acoustic data. Researchers use special microphones to listen for sounds made by wildlife, and other tools like sonar can be used to assess what's under the water in real time. Regardless of the technology, these acoustic data must be prepared, analyzed, and communicated to inform what we do in the ocean.

The *scientific method* involves posing a question, making a hypothesis, conducting a test with an experiment, and analyzing the resulting data to draw conclusions. Therefore, by definition, anyone conducting science or testing hypotheses will be doing some form of data science through their work with data to investigate their question of interest, as we saw in the previous two situations.

Today's unprecedented plethora of data has often been dubbed the "data deluge," and for a good reason. While the number of people doing data science grows daily, everyone is a consumer of data science through news, social media, pop culture, etc. To this end, *Data Science for All* aims to equip everyone with the literacy to be a competent consumer of data science and a few tools to start doing data science.

Data's Importance to Business and Industry


Drazen Zigic/Shutterstock

Data science is in high demand. In a recent global survey by Gartner, over 400 leaders revealed their highest-priority technology opportunities for growth. Data science is the primary tool for addressing these opportunities. In particular, data science identifies areas ripe for disruption.

For example, data show marketers which product categories have responded well to past disruptions and reveal which related categories still need to be disrupted. Similarly, data science helps firms understand how best to engage their customers. For instance, data may immediately show which strategies—such as contests, news, or giveaways—engage more customers for fewer dollars. In summary, data science applies technology to provide a deeper understanding of issues critical to organizations' missions.

Job expectations are increasingly integrated with data science. *Data Science for All* reveals the importance of data science for professional roles that may seem to have little use for quantitative analysis. For example, graphic designers must understand the power of A/B testing to validate their designs. Copywriters are interested in ways to use social listening to inform their content strategies. Journalists should understand how a machine learning model might help them design targeted narratives for various audiences. Customer service representatives need analytics to assess the success of the interactive strategies they practice. Professional sellers may focus on predicting sales leads. Public relations managers may analyze social and news media to participate in and exchange information with the public.

Likewise, traditionally, quantitative jobs are also more integrated across data science domains. Rather than hiring analysts with deep expertise in a narrow area, companies seek data professionals with a broad understanding of how data interact with fields such as agriculture, architecture, biology, business, history, journalism, etc. Today's data scientists need to know how their analyses and storytelling affect the qualitative aspects of the workplace. Equipped with these competencies, they can design enduring employee experiences, create engaging media content, craft effective public communication, produce fantastic new goods and services, and make the world a better place.

Case Study: Amazon Uses Data for Customers, Ads, and Fraud Prevention

Elpisterra/Shutterstock

Amazon uses data science to analyze customer purchase patterns, tailor its ads, and audit fraudulent activity. Sometimes, it feels like Amazon knows what we need even before we do. Amazon manages millions of purchases daily without direct personal contact. It does this with three specific strategies: dynamic pricing, product recommendations, and fraud protection.

Dynamic pricing is a strategy in which companies continually change the price of a product based on the data they collect from customers, estimating how much each customer is willing to pay. Amazon, for example, changes product prices approximately 2.5 million times daily. This means the cost of a typical product shifts every ten minutes. Thanks to the company's massive amounts of data, it can analyze factors ranging from competitor pricing to available inventory. It then uses sophisticated algorithms to make informed decisions about pricing.

Customers benefit when Amazon configures their product recommendations based on items already in their carts, products they've purchased before, and products they have browsed. Any action a customer takes on their website, any click—Amazon will use that data. They can learn what each customer wants and likes and can recommend appropriate products. Product recommendations are attributed to 35% of Amazon's annual sales.

Amazon protects itself and its customers from retail fraud. It collects more than 2000 historical and real-time data points on every order and then uses machine learning algorithms to identify transactions with an elevated likelihood of being fraudulent. Additionally, the company uses data science to tweak algorithms, enabling it to scrutinize suspicious return requests. Therefore, if the data shows that a person has returned too many items or reveals other unusual patterns, the company is notified to investigate further.

"Data Science for All" implies that everyone should understand how data science impacts their areas of interest. These skills can be applied to almost every job and academic field. Moreover, some readers of *Data Science for All* may discover that they are interested in pursuing a career in data science. Companies appreciate employees who can serve as intermediaries between the qualitative and quantitative aspects of business, fostering understanding between high-level business managers and information systems specialists. Data science values curiosity, creativity, strategy, and insight in all careers. While data science does not necessarily require the full expertise of a computer scientist or statistician, we aim to impart the ability to formulate meaningful data queries, interpret data science results, critique its methods, and make sound recommendations based on data science output.

Chapter 1: Putting It Together

Data science is a new world made possible by the digital age, where data are rich and plentiful. We explored data science's multidisciplinary applications and overall lifecycle. We also navigated through different data types, preparation pipelines, methods of analysis, and the societal impact of data science.

Interpret the Definition and Scope of Data Science

- Discussed the interdisciplinary nature of data science.
- Emphasized its applicability across various fields, such as healthcare, finance, and social sciences.
- Outlined the typical lifecycle of a data science project, from data collection to insights.

Recognize the Need for Data to Be Organized as a Tidy Table

- Explained how the source and form of data influence the subsequent analysis.
- Recognized tidy data resides in tables where every column is a variable, every row is an observation, and every cell is a single value.

Distinguish between Different Data Types

- Examined the different uses of data, such as quantitative and categorical.
- Highlighted the importance of understanding data types for effective data preparation and analysis.

Describe a Coherent Data Preparation Pipeline

- Explored various data preparation techniques and their sequence in a typical pipeline.
- Provided insights on assembling an effective pipeline tailored to specific data types and formats.

Evaluate Various Types of Data Analysis

- Covered the four major types of data analysis: descriptive, diagnostic, predictive, and prescriptive.
- Explained how these methods contribute to compelling storytelling and decision-making.
- Illustrated the use of these methods in a real-world scenario (e.g., *Cake Pops* customer reviews on Amazon), to emphasize their practical importance.

Evaluate the Impact and Growth of Data Science

- Discussed the transformative role of data science in society and industry.
- Highlighted its rapid growth and the increasing demand for data science skills.

Now that we appreciate the depth and breadth of data science, we understand the importance of careful data preparation, the diversity of analytical methods available, and the significant impact of data science in the modern world. We are ready to wrangle with data to get it ready for storytelling and analysis.

Chapter 1: Ethics in Practice: Some Risks in Data Science

While data science provides valuable insights, it also carries ethical risks around bias, transparency, and consent that should be carefully considered. A nuanced, conscientious approach to data ethics is just as important as the data themselves. If we can record or measure something, we can collect data about it. And if we can collect data about an entity or process, we can try to optimize it. By keeping ethics front and center, data collectors can gather meaningful insights while respecting human dignity.

Consider the following examples that have likely affected us all:

- Email spam filters are designed to divert unwanted emails from our inboxes.
- University planners predict the number of sections offered in a particular course based on historical and present data.
- Policymakers use aerial pictures of wildlife to estimate population sizes in state and national parks, and other protected areas.
- Medical researchers use the human genome and health-related data to determine factors that decrease disease risk.
- Demographers and political scientists use polls and surveys to collect demographic and public opinion data, then summarize and report on them.

Because it's nearly impossible for us as individuals to perceive the world around us at all possible scales, we rely on other individuals or organizations to summarize the information and data we lack. A fundamental understanding of data science is paramount to trusting what these other authorities provide. Each of us must ask the following critical questions:

- What are the sources of the data I have?
- What are the sources of the data summarized for me?
- Were the data collected ethically?
- Are the data being misrepresented?

As useful as data science can be for understanding the world around us and enhancing our lives, it is not without its shortcomings. Some of these are related to issues of ethics. Two examples are: (1) issues of privacy, and (2) issues of fairness/bias in decision-making algorithms.

A massive amount of the technology and software that we use on a daily basis collects a plethora of data about us; particularly our behavior and usage of these tools (e.g., web browsers, websites, and mobile phone apps). With enough of this information, people and companies can explicitly identify users and uncover their personal information. Some companies may want to use this information to customize a user's web browsing experience, while other entities may have more malicious goals. Even with the former, having your entire web browsing experience potentially controlled by advertisers may not be desirable. Your private life may become much less private.

Machine learning and other data modeling techniques are constantly employed to automate the process of evaluating the risks of individuals and companies as they interact and work together. For example, a small business owner may need a loan from a bank to expand their business. The bank will collect data about this person and evaluate their supposed ability to pay back the loan based on models constructed using historical data. These models can sometimes yield unfair or biased results because they were constructed from data about people in the past that is no longer representative of the people today for whom they're being used. There is an old saying, "garbage in, garbage out," which, in this context, refers to the idea that if your analysis is based on incomplete or poor data, then your conclusions from that analysis can be no better. We need to be mindful of the sources of our data and the groups about whom we are trying to draw conclusions, as well as the possible differences between them, so as not to propagate inequity or unfairness.

Chapter 1: Key Terms

data
data science
population
sample
data science lifecycle
data preparation
data analysis
data storytelling
cloud (the cloud)
observational unit
variable
observation
tabular form
dataset
value
structured data
unstructured data
quantitative variable
discrete variable
continuous variable
categorical (qualitative) variable
text variable
integer variable
float variable
Boolean variable
data structuring
tidy data
metadata
raw data
data collection
data wrangling (preprocessing)
data management
cleaning data
transforming data
integrating data
descriptive analysis
diagnostics analysis
predictive analysis
prescriptive analysis
visualization
storytelling narrative

CHAPTER 1: CHAPTER REVIEW QUESTIONS

Section 1.1: Introduction to Data Science

1. What is the interdisciplinary field that uses scientific methods to extract knowledge from data?
2. What is the primary purpose of data analysis in data science?
3. Define a *population* in the context of data science.
4. Which other field, often confused with data science, applies data science principles to domain-specific problems?
5. What are some of the modern challenges that have shaped data science?
6. What constitutes a sample in data science?
7. In the context of data science, what is the significance of understanding samples and populations?
8. What combination of fields does data science encompass?
9. When was the term "data science" first proposed as an alternative for "computer science"?
10. What is the role of a data scientist in the context of the data science lifecycle?
11. What is the most significant reason for the importance of data and their analysis, according to the text?

Section 1.2: Data in Tables

12. In data science, what are the entities about which data are recorded called?
13. What are the recorded characteristics of observational units called?
14. What term is used to refer to the data for a single observational unit?
15. From the following set of numbers: 0.0, 2.71828, -3, 5, which one is most likely to be stored as a float variable?
16. Which type of variable is used to designate categories?
17. In the context of data science, what does it mean for data to be tidy?
18. What do we call the form when data are represented as a table with rows and columns?
19. Integers and floats are examples of what kind of variables?
20. What kind of variables only take on the values of TRUE and FALSE, sharing aspects of both categorical and quantitative variables?
21. What form of quantitative variables consists only of numbers with no decimal points?
22. Given the numbers 5000, -2.01, 1.3, and $1/2$, which one is most likely to be stored as an integer variable?
23. What form of quantitative variables consists of numbers that may have decimal points?
24. Given the set of numbers 0, 2, 5, and 1.5, which is most likely to be converted to a Boolean value in a data analysis context?
25. What form of categorical variables contains words or characters without order or numerical utility?
26. Given the numbers -1.0, 0.0, 3, and 4, which one could be stored as a float or converted to a Boolean variable?
27. Given the numbers 2.71828, -5, 3.14, and 2.01, which one is most likely to be stored as an integer in a data structure?
28. What is metadata?
29. When are data considered messy?
30. Which of the following numbers is likely to be stored as a float variable in a dataset: 1, 4.44, 144, or 0?
31. When data are in tabular form, what is stored in the columns? What is stored in the rows?
32. When the rows and columns of tabular data are fully populated, and each cell of the table corresponds to a single value, we consider the dataset to be clean and organized. What is the term we use to describe this?
33. Given the numbers 1243424, 1.2, -1.01, and 2.03, which is most likely to be stored as an integer variable?
34. What is the primary role of a data key or codebook?
35. In a tidy dataset, what does each column represent?

Section 1.3: Data Preparation

36. In the context of data science, what does it mean to prepare data?
37. What is the first step in data wrangling?
38. For what purpose is data transformation essential?
39. What is the goal of integrating datasets?
40. In the context of data preparation, what does data wrangling refer to?
41. Describe the process of data collection.
42. What is the importance of data quality and relevance in data collection?
43. What does the term "unstructured data" mean?
44. What is the process of organizing data (often into tables) called?

Section 1.4: Data Analysis and Storytelling

45. What is descriptive analysis in data science primarily concerned with?
46. What is the goal of diagnostic analytics in data science?
47. What is the goal of predictive analysis in data science?
48. What is the component of data storytelling that allows us to communicate statistics and analysis results in form of graphs and charts?
49. Describe prescriptive analysis in the context of data science.
50. What is the purpose of communication in the context of data science?
51. How is data visualization useful in data science?
52. What is the ultimate goal of data science storytelling?
53. What does effective storytelling in data science require?

Section 1.5: Data Science in Society and Industry

54. What is the primary driver for the growth of data science in society and industry?
55. How does data science help businesses engage with customers?
56. In the context of data science, what does the term "data deluge" refer to?
57. According to the text, for whom are data science skills essential?
58. What is the benefit of integrating data science into traditionally quantitative jobs?
59. What does the message "data science for all" emphasize?
60. According to the text, how can an organization's management use data science for decision-making?
61. According to the chapter, what benefits does data science offer to consumers, researchers, and companies?

Chapter 1: Explaining the Concepts

1

When you hear the word "data," what do you think of first? (What did you think of before reading this chapter?) How is this similar to or different from *tidy data*? What are the ways you see or work with data on a day-to-day basis?

2

Now that we've defined *data science*, have you done any data science in your life thus far? Can you envision putting data science to use at your current job or a future job? If so, how?

3

When it comes to data, we have covered two very special terms:

1. Observations
2. Variables

We covered these terms, mostly in the context of data in tables. Take a moment to remind yourself what an *observation* is and what a *variable* is. Because we want to do analysis and visualization with any kind of data we have, the ideas of *observations* and *variables* are important to keep in mind, even beyond tabular data.

- Imagine we want to conduct an analysis of a large collection of TikTok videos to which we've been given access.
  - Give some examples of characteristics, or *variables*, that we might be interested in investigating with this collection of videos.
  - What would you call the *observations* in this situation, and why?
  - Do your best to envision organizing your answers into a table format. Sketch this table out.
- Recall the Amazon Reviews example from earlier in the chapter, and imagine we want to analyze a large collection of Amazon Reviews to which we've been given access.
  - Give some examples of characteristics, or *variables*, that we might be interested in investigating with this collection of reviews.
  - What would you call the *observations* in this situation, and why?
  - Do your best to envision organizing your variables and observations into a table format. Sketch this table out.

Explain the importance of being able to visualize and identify the *observations* and *variables* in a collection of data for the purposes of preparing data for analysis and storytelling.

4

How is understanding data science important for consuming news and interacting with social media?

5

What are some examples of how you provide data to other people, companies, or organizations? To what extent can you control how much data you provide to each of these entities? Is this a good thing or a bad thing?

Chapter 1: Applying the Concepts

Activity: Identify Variable Uses and Types

Part I

Discussing data can seem daunting without the proper terminology to communicate ideas. To help practice the vocabulary discussed in this chapter, answer the questions.

Suppose your professor is interested in discovering if there is any relationship between students' sleeping habits and their final exam scores. To explore this relationship, your professor asks their 30 students on their final day to estimate how many hours they slept the previous night. The professor then records their final exam scores and begins the investigation.

1. In this scenario, what are the observational units? Why?
2. What are the variables?
3. What are the variable uses and types?

Part II

Discussing data can seem daunting without the proper terminology to communicate ideas. To help practice the vocabulary discussed in this chapter, answer the questions.

A doctor is studying a patient with a new, unidentified blood disease. The patient has recurring visits with the doctor to study the disease. At each visit, with the patient's consent (ethics are always a priority!), the doctor records the date, measures the patient's blood sugar (in mg/dL), weighs them (to the nearest pound), asks them whether it is true or false that they have eaten in the past hour, and records the patient's description of their symptoms each time they visit.

The doctor is considering how to store the data. She eventually decides on a tabular format.

1. In this format, what is stored in the rows?
2. Knowing your answer to question #1, what data would the doctor use to populate the rows?
3. What is stored in the columns?
4. Knowing your answer to question #3, what data would the doctor use to populate the columns?

The hospital's data science team is requesting information about how the doctors would like to store the data they collect. They are asking the doctors to indicate the variable type for each of the fields, as they are unsure about some of them. Please enter the appropriate variable type for each item.

1. Blood sugar:
2. Weight:
3. Eaten in the last hour:
4. Symptom description:

Activity: Tidy Messy Data

A team of doctors is studying patient symptoms. They recorded patient data, with their permission, into notes and asked an intern to convert the data into a table. Unfortunately, the intern, a student who hasn't read "Data Science for All," is unfamiliar with the principles of tidy data and created the following table:

TABLE 1.11
Messy data.

			eaten_in_the_last_hour	symptom_description
	date			Dizziness, migraines
	2022/08/02		TRUE	Dizziness, excessive hunger
	2022/08/16		FALSE	Low energy, lethargic
	2022/08/30		TRUE	Migraines, lethargic
	2022/09/13		FALSE	
		blood_sugar		weight
	75.3 mg/dL	2022/08/02		154.3 lbs
	77.2 mg/dL	2022/08/16		155.2 lbs
	74.5 mg/dL	2022/08/30		155.1 lbs
	73.2 mg/dL	2022/09/13		153.2 lbs

Help fix Table 1.11 by transferring elements from the disorganized table above into a new, tidy table.

Activity: Tidy Data for Better Analysis

Last year, Chef Emily, a cooking class instructor, rated the cooking skills of four students over three days. Each day, the students prepared a different dish (pizza, pasta, and cake), and Chef Emily rated the students' cooking skills from 0 (poor) to 100 (excellent) for each dish. Additionally, she recorded the time it took each student to prepare the dish (in minutes), as a measure of their efficiency.

In the original table (Table 1.12), each row corresponds to a single student, combining multiple observations for different dishes. However, Chef Emily wants each line to display just one student making one dish, along with that dish's cooking skill score and the time it took them to prepare it. Please create a new table for Chef Emily. Since there are three dishes, the new table will consist of 12 rows (three dishes for four people). The new table will have fewer columns, making it easier to understand and allowing for quick comparisons of how students performed with different dishes.

TABLE 1.12
Original table format.

student_id	pizza_skill	pizza_time	pasta_skill	pasta_time	cake_skill	cake_time
1	78	40	82	35	75	60
2	85	37	88	30	90	55
3	65	45	70	40	80	50
4	90	35	95	25	85	45

Activity: Investigate Metadata

Consider the metadata of a dataset in Table 1.13, which is about the survivors of the Titanic's historic shipwreck on April 15, 1912.

TABLE 1.13
Titanic survivors data by M. Yasser H., Kaggle.

variable_name	description	data_type	coding_scheme	unit_of_measurement	missing_values
passenger_id	Unique identifier for each passenger	integer	-	-	-
survived	Survival status	boolean	0 = Did not survive, 1 = Survived	-	-
p_class	Ticket class	integer	1 = 1st class, 2 = 2nd class, 3 = 3rd class	-	-
name	Name of the passenger	text	-	-	-
sex	Sex of the passenger	text	-	-	-
age	Age of the passenger	float	-	Years	177
sib_sp	Number of siblings/spouses aboard	integer	-	Count	-
par_ch	Number of parents/children aboard	integer	-	Count	-
ticket	Ticket number	text	-	-	-
fare	Passenger fare	float	-	Currency (likely Great British Pound or GBP)	-
cabin	Cabin number	text	-	-	687
embarked	Port of embarkation	text	C = Cherbourg, Q = Queenstown, S = Southampton	-	2

1. Why is the data type for the `survived` variable essential for data analysis?
2. What is the primary use of `passenger_id` as a unique identifier in data analysis?
3. What aspect of the Titanic's voyage does the `p_class` variable primarily represent?
4. Why does understanding the data type of each variable affect data analysis?
5. If we wanted to explore the hypothesis that higher-class passengers had a higher survival rate, which combination of variables would be most crucial to analyze together?

Activity: Illustrate Unstructured Data Using Structured Tables

Figure 1.11 is an example of Pearson's Instagram page. Figure 1.12 represents the data behind the page, using three tables: Users, Posts, and Followers.

For example, Figure 1.12's Users table contains one record for the PearsonOfficial user and billions of other records for other Instagram users. Likewise, the Posts table has the 587 posts we see in Figure 1.11, plus billions of records for all other users' Instagram posts. The Followers table, on the other hand, has over 28,700+ records from Figure 1.11, among billions of others associating users with their followers.

Figure 1.12 shows how to structure Pearson's Instagram page data into tables, much like what we did for Amazon reviews in Figures 1.4 and 1.5. Table 1.14 is a data key explaining the variables in Figure 1.12.

FIGURE 1.11
Pearson's Instagram page.

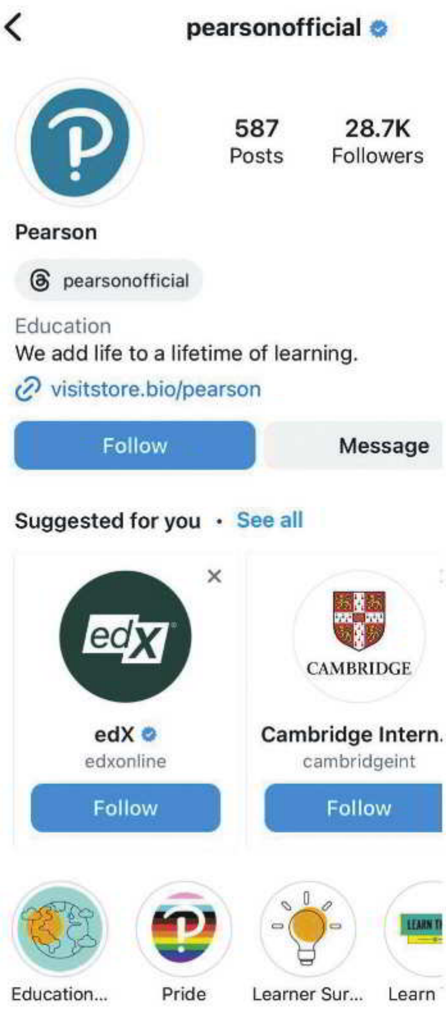
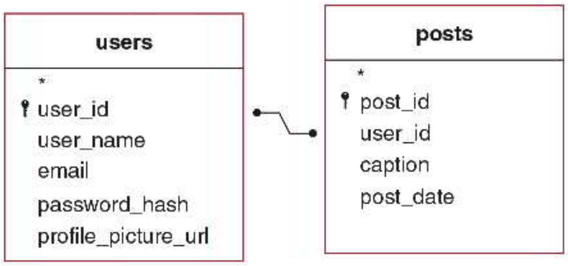


FIGURE 1.12
Transforming unstructured Pearson's Instagram page into structured tables connected by identification variable `user_id`.



Answer the following questions:

1. Which field in the `users` table is a unique numeric identifier for each user?
2. In which field and table is "Pearson" likely stored?
3. In which field and table is the link to the below image is stored?

4. To identify the user who created a specific post, which field and table should be used to link back to the Users table?

TABLE 1.14
Data key for the tables in Figure 1.12.

table	variables	explanation
users	user_id	Unique numeric identifier for each user.
users	user_name	The user's chosen name, used for logging in and display.
users	email	Email address associated with the account.
users	password_hash	Encrypted password for account security.
users	profile_picture_url	URL to the user's profile picture.
posts	post_id	Unique identifier for each post.
posts	user_id	Links to the Users table to identify the poster.
posts	caption	Text caption accompanying the post.
posts	post_date	Date and time when the post was created.

Activity: Identify Analysis Types

Part I

The hospital's data science team asks you to review the following analysis paragraphs and determine whether they are descriptive, diagnostic, predictive, or prescriptive.

1. We reviewed patient admission data over the past year, focusing on the high readmission rates within a month among cardiovascular patients, which stood at around 30%. Upon further investigation, we found that a significant portion of these readmissions, approximately 50%, were among patients with co-existing diabetes. Additionally, patients enrolled in our post-discharge follow-up program were less likely to be readmitted within a month.

   What type of analysis is this?
2. Our analysis suggests implementing a comprehensive post-discharge care program for cardiovascular patients with co-existing conditions such as diabetes. This could include regular follow-ups, medication management, and tailored health education. Furthermore, the analysis recommends scaling up our resources during this period to handle the predicted 15% increase in cardiovascular-related admissions next winter. This could involve adjusting staff schedules, ensuring sufficient inventory of necessary medical supplies, and coordinating with nearby healthcare facilities for overflow support. By adopting these strategies, we could reduce the readmission rate for cardiovascular patients from the predicted 30% to around 20%, thereby improving patient outcomes and optimizing resource utilization.

   What type of analysis is this?
3. We've observed significant trends in analyzing patient admission data over the past year at our hospital. Most admissions, approximately 40%, were related to cardiovascular conditions, with a notable spike in admissions during the winter months. The average length of stay for these patients was 4.5 days, although this extended to 6 days for patients aged 65 and over. It's also worth noting that around 30% of our cardiovascular patients were readmitted within a month of their initial discharge.

   What type of analysis is this?
4. According to our analysis, we expect a 15% increase in cardiovascular-related admissions over the winter season. Furthermore, without intervention, the readmission rate for cardiovascular patients within a month is likely to persist at around 30%.

   What type of analysis is this?

Part II
This activity builds on Part I to include elements of the data science lifecycle.

Dataset: Titanic-Dataset

A group of historians is studying the effect of classism in disaster scenarios. One disaster scenario of particular interest is the sinking of the RMS Titanic in 1912. In this event, approximately 1500 deaths occurred among those onboard. The historians are concerned that classism may have led to a disproportionately high number of casualties in the lower ticket classes. They also have secondary interests in how other passenger demographics relate to survival rates once their initial inquiry is answered. The historians published a full report for some new team members who are just learning the data science lifecycle, spelling out the data science components described in this chapter. Answer the questions to recognize the components and subcomponents of the data science lifecycle, and categories of data analysis.

We investigate the differences in the survival rate among the three passenger classes, using historically accurate data kept on the passengers of the Titanic. The ease of access to this data is a result of good data management.

1. We see that someone physically recorded the information about the passengers. What is this first step?
2. Someone digitized the physical records for the passengers and created a tidy table for our use. What is this second step?
3. Once the data is in a clean, easily accessible, and complete format, we can move on to the next step. In this step, we extract meaning from the data–in this case, through basic summary statistics. What is this third step?
4. Looking at Table 1.15, where summary statistics are provided for a past event, what type of analysis is this?

TABLE 1.15
Survival statistics of Titanic passengers separated by passenger class. Data from M. Yasser H., Kaggle.

passenger_class	survival_percentage	number_of_survivors	total_number_of_passengers
1st class	63.0	136	216
2nd class	47.3	87	184
3rd class	24.2	119	491

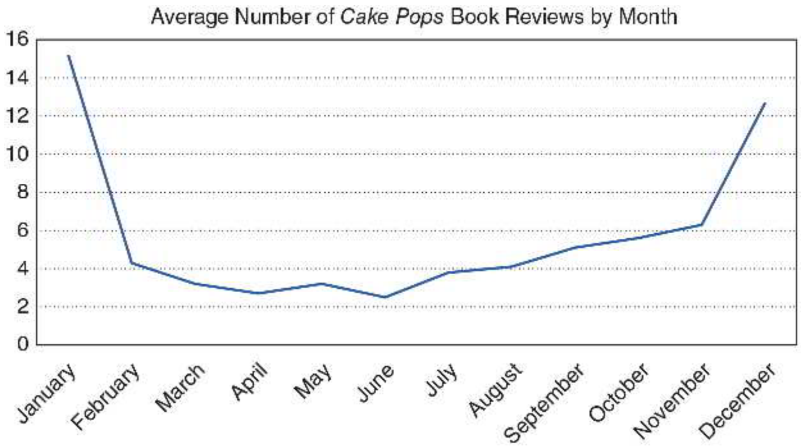
5. In another type of analysis we could perform on these data, we would address the underlying explanation for why these statistics are what they are. What type of analysis would that be?
6. If the historian placed these data into a historical context and used the numbers to support a cohesive narrative, what would he be doing?
7. The Titanic dataset has also been used to test data models to see if it is possible to predict a passenger's survival based on their characteristics such as age, ticket price, and sex. What kinds of analyses are these?
8. If we were to recommend taking some action in light of analytical insights, what kind of analysis would we be using?
9. However, the historian's work does not end here. Although we have demonstrated a difference in survival rates across passenger classes, passenger class is not the only factor in a passenger's survival rate. Other factors such as age and sex are also relevant in investigating differential survival rates. Data science improves iteration by iteration, a repeating process in which we build upon our previous work to enhance it. When we use the insights we gain from our analysis to fuel future analyses in new directions, what are we engaging in?

Activity: Optimize the Narrative

Consider the following visualization that graphs the average number of *Cake Pops* book ratings for each month of the year. Answer questions about the visualization based on Figure 1.13.

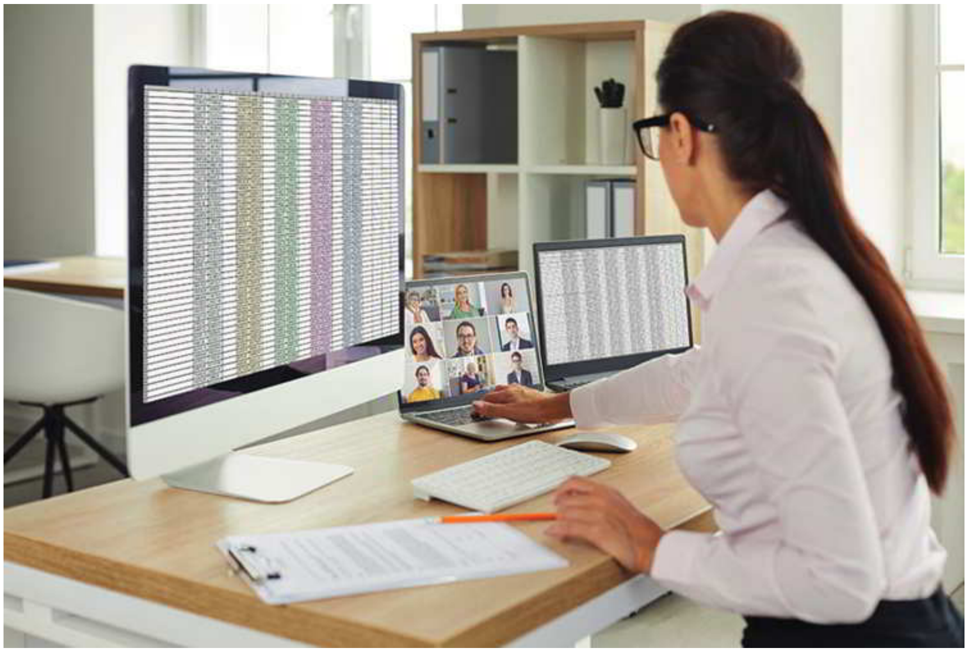
1. Looking at Figure 1.13, describe the relationship between the average number of *Cake Pops* book reviews and the time of year.
2. Looking at Figure 1.13, what ideas do you have about why the *Cake Pops* book shows certain review trends?
3. Open-Response Question: Use the STAR Framework to tell a story about Figure 1.13.

FIGURE 1.13
Average number of times the *Cake Pops* book was reviewed on Amazon by month over the past three years.



CHAPTER 2

DATA WRANGLING: PREPROCESSING



Studio Romantic/Shutterstock

LEARNING OBJECTIVES

- Outline the processes of data wrangling, focusing on their role in preparing data for analysis and storytelling.
- Apply strategies for addressing missing data within a dataset.
- Find and address anomalous values that may distort data analysis.
- Transform quantitative and categorical variables to enhance data structure and analysis.
- Reshape data into a tidy format, ensuring that each column represents a variable and each row an observation.
- Describe approaches to combining multiple datasets for comprehensive data analysis.

Introduction

Mirabel wants her political science senior project on student voting patterns to make policy recommendations about how her college can increase student participation in the next election. Her teacher gives her access to a unique dataset from last year's class on student pet preferences, grades, and demographics combined with data on those same students' voting patterns. She wonders whether younger students participate more in elections than older eligible voters do because they are excited about their new voting rights, or if they vote less because they are unsure about how to vote. She also ponders whether students with better grades might be more likely to vote. Based on a late-night debate with her roommate who is a cat lover, Mirabel is also curious whether "dog people" versus "cat people" vote differently.

Krakenimages.com/Shutterstock

Mirabel immediately dives into the dataset by viewing one record at a time. She is soon overwhelmed by the sheer amount of data from over five thousand students. She sees that it is impossible to absorb so much information by reading the responses one at a time. Then, Mirabel is relieved to see an option to download the results as a spreadsheet. She moves confidently back in front of her laptop and opens the file. She soon realizes she must clean the data. For example, some students wrote out their ages (e.g., "nineteen" instead of 19), others had impossible data (e.g., a GPA of 27.3), and others misspelled words (e.g., "catt" instead of "cat"). After analyzing her cleaned results, she discovers that age and GPA do not appear to have a relationship with voting behavior, and that dog lovers were somewhat more likely than cat lovers to vote in the most recent election. Based on her project's recommendations, voting advocates design "Rock the Vote" pizza parties on the next election day that walk first-year students to on-campus polling places, increasing the student vote at her college by 12%. They promote the event using cats, of course, to elevate the voting frequencies of "cat people."

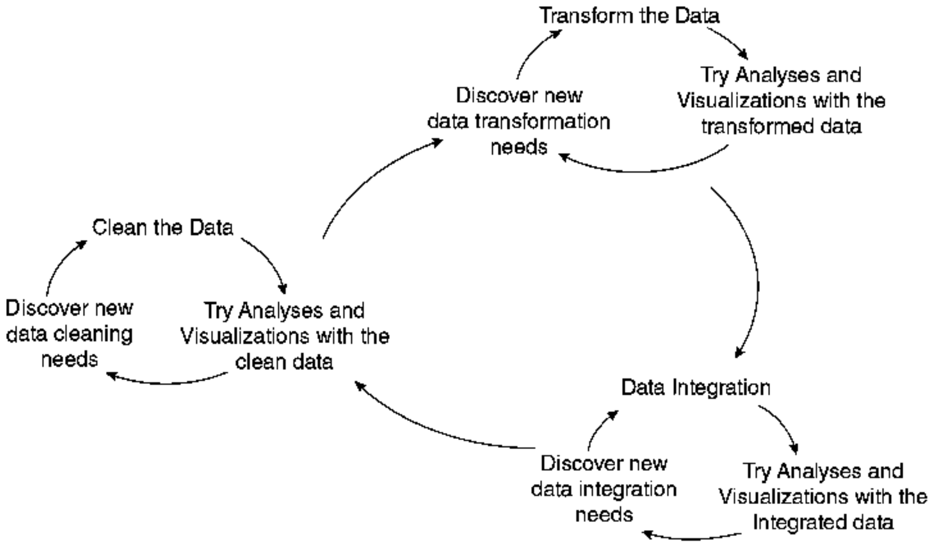
Sheila Fitzgerald/Shutterstock

SECTION 2.1: WHAT IS DATA WRANGLING?

Data wrangling, also known as data preprocessing, is the general term given to the process of preparing collected data for analysis, storytelling, or both. Data can come from a variety of sources in all shapes and sizes. Not all datasets are organized and formatted well into the tabular form known as *tidy data*. Even when datasets come in a tabular form, like Mirabel's dataset, the variables and their values are not always immediately ready for analysis and storytelling. Data wrangling can be broken down into three key components that all help transition data into a form more suitable for visualization and analysis. People usually wrangle data in this order, though they cycle within and between each step (Figure 2.1).

1. *Cleaning* the data.
2. *Transforming* the data.
3. *Integrating* the data.

FIGURE 2.1
The continuous cycles of data wrangling.



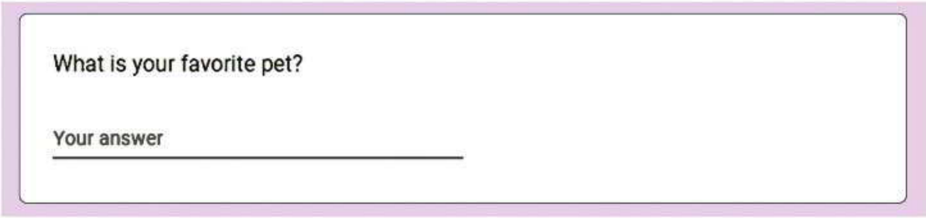
DEFINITION

Data wrangling, also known as **data preprocessing**, is the process of converting raw data into a more usable format by *cleaning*, *transforming*, and *integrating* data to facilitate effective analysis and storytelling.

Cleaning the Data

Recall that *cleaning* data involves fixing or removing incomplete, abnormal, or inconsistent data to enhance their quality for accurate analysis and storytelling. Making the data clean as a part of data wrangling might make it seem like data can initially be "dirty", but that is not quite accurate. Software tools for visualization and analysis require data to be in an organized and consistent format. Different visualization techniques and analysis methods will have their own requirements for this organization and format as well.

How might variables have inaccurate or inconsistent values? Let's revisit Mirabel's dataset, which included responses to an open-ended question:



In this format, students could have written their answers in the blank space, in whatever format they chose. Mirabel might see many different responses, including the following:

- dog
- Rover
- Birds
- parrot
- Dog
- cat
- cats
- Cat
- horses
- Felix
- DOG
- None
- I don't like animals

If Mirabel wants to explore dog versus cat preferences as part of her analysis, she probably only wants the responses here to represent three different possible values: dog, cat, unknown. Additional pet preferences, such as "Birds" and "horses," will not be useful to her dog vs. cat analysis. The first takeaway here is that Mirabel does not care about capitalization in the response; so, for example, "dog" and "Dog" should both become "dog".

The value of the last category, *unknown*, is a choice that is often up to the person working with the data, and this choice is very important. For instance, both the "None" answer and the "I don't like animals" answer would likely be converted to *unknown*. However, a response of "None" could imply that the person is an avid animal lover who can't choose a favorite, while "I don't like animals" would suggest something quite different. By cleaning this variable's values into *dog*, *cat*, and *unknown*, Mirabel is making the data more consistent, easier to visualize and analyze. However, she may also be losing a bit of information in the creation of the *unknown* category.

How might the observations in a dataset not accurately reflect the items from which the data came? If Mirabel's dataset includes people who submitted more than one response, then her dataset could end up with multiple rows for a single individual. It might also be the case that one of the respondents completed the form too quickly and entered nonsensical answers to multiple questions on the form (e.g., a value of 118 for age). Mirabel will want to address these duplicate records and implausible values before performing any analysis or storytelling, to ensure that her work accurately represents the individuals from which her data came.

We cover many of these cleaning approaches in this chapter. However, visualization (Chapter 3) and summarization (Chapters 4 and 5) are useful for discovering the ways in which data need to be cleaned, especially if the dataset is very large.

Transforming the Data

Recall that *transforming* data involves organizing and formatting data into a structure that is useful for efficient analysis. It may seem like the data we end up using for our visualization and analysis is completely different from the data we started with. However, data scientists are not magicians. The dataset contains information in the form it does, and it's up to the data scientist to figure out the best way to organize and extract that information to deliver insight. The process of transforming the data can involve many different steps, and there can be many different reasons to do so.

DEFINITION

Data transformation is the process used to convert a clean dataset into the form or structure needed for analysis and visualization.

For example, a dataset might contain a variable with the full date (e.g., January 10, 2023), but we might only be interested in the month and year. One transformation technique would create two new variables in the dataset from this original date variable, one for the month (e.g., January) and one for the year (e.g., 2023).

In general, transforming data can take many different forms. However, it is usually best practice to only expand our dataset to include any newly created variables and never replace our existing variables if we can avoid it. In the previous example, we would retain the original full date variable and simply add the month and year variables to our dataset. Similarly, Mirabel might keep the original responses to the pet preference question and create a new variable for dog-versus-cat preference.

We will cover some of the most common ways to transform our data into forms that can be more useful for visualization and analysis.

Integrating the Data

Recall that *integrating* the data is the process of combining multiple datasets from one or more sources to maximize the amount of information from which to draw insight. If we have multiple datasets, then once they have been cleaned and transformed, we should explore the possibility of combining them.

Usually this is useful because analyzing a single dataset alone would not be as informative as analyzing the combined dataset. When faced with two related datasets, there are a few primary ways to think about how they relate to each other:

- They could have the same set of variables, but different observations.
- They could have the same observations, but different variables.
- They could have some similar observations and some similar variables, but not identical sets of either.

After relating the two datasets, it is also important to decide *how* we want to combine them. It may be the case that we only want the observations from the first dataset—but with associated variables from the second dataset, or only observations from the second. It could be the case that we only want the variables in the first dataset for all the observations in the first *and* all the observations in the second dataset. There are numerous other ways to combine two datasets, and we will cover multiple approaches throughout this text.

All these data wrangling decisions and steps can help prepare data in a variety of ways for both visualization and analysis. For this reason, it is extremely common for data scientists to go through multiple iterations of data wrangling in the data science workflow, as they determine the most appropriate visualizations and analyses for the data. We will introduce advanced data management techniques for data integration in Chapter 5. For the sections that follow, we will assume that the dataset of interest is already in a tabular form.

Dataframes

For the purposes of this chapter, we will use the term *dataframe* to refer to a tabular dataset that can be stored, viewed, and processed with a variety of software tools. You may have seen dataframes before as tables (e.g., Tables 1.1–1.7 were all technically dataframes) or spreadsheets (e.g., in Microsoft Excel or Google Sheets).

DEFINITION

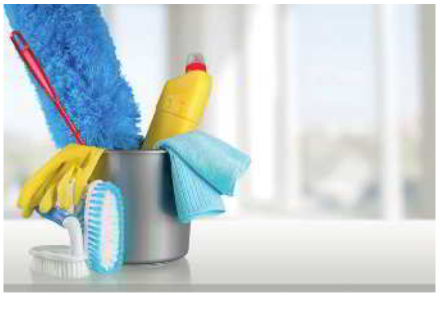
A **dataframe** is a tabular data structure with named columns.

Dataframes are usually more than just tables with columns and rows in spreadsheets and other tools, such as coding environments for R and Python, or statistical tools like StatCrunch. Dataframes often come with or lend themselves to functionality that can help compute and produce more information about the data. Spreadsheet, coding and statistical tools offer foundational data science functionality for working with data in many forms, but especially for dataframes. We can complete many tasks—that is, instruct the computer to fulfill complex or calculation-heavy functions—with these tools. For example, we can clean structured data by opening it in a spreadsheet tool and use the spreadsheet's built-in functions to recognize and address issues. Coding and statistical tools are similarly designed to provide functionality that interacts with dataframes. We clean, transform, and integrate data through the structure of a dataframe.

Big Picture Break

Wrangling data into a consistent and well-defined structure may not sound as glamorous as creating an interesting data visualization or performing an exciting analysis. In fact, the result of data wrangling is a new and improved dataset, *not* some fancy graphic or prediction model. However, the importance of data wrangling in the data science workflow cannot be overstated. Without thorough data wrangling, our data may never even be ready to be effectively visualized and analyzed. To this end, all of the steps undertaken throughout data wrangling must *also* be well documented and explained, just like any visualization or analysis. Data wrangling documentation is often the most important for fellow data scientists to be able to reproduce each other's work reliably.

SECTION 2.2: CLEANING MISSING DATA



Billion Photos/Shutterstock

Data wrangling is the process of cleaning data (converting from messy to tidy data), transforming it (making it ready for analysis), and integrating it (connecting it with more than one data source). Software tools aid these steps. We will explore examples that focus on cleaning data through functions such as search and replace, sorting and filtering, and other functions available in spreadsheet software. Although many different tools can accomplish the following data-cleaning steps, we will demonstrate how it could work with Excel or Google Sheets:

1. Open a dataset in Excel or Google Sheets.
2. Sort the columns to observe the range of values for each.
3. Use functions to calculate the average, sum, count, maximum, or minimum of a column of numbers.

4. Assess whether the values make sense. For example, if the *max* function returns a value of 928 for age, we know that the data have errors.
5. Correct errors. For example, use the *sort* function to isolate and delete all impossible values for age.

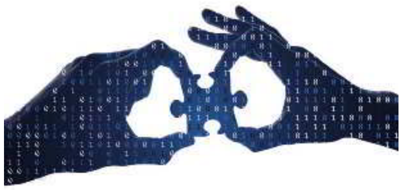
When we clean data, we often identify the following issues:

- Missing data
- Extreme data values
- Implausible values
- Incorrect formats
- Duplicate records

Exploring a dataset to identify and address these issues is essential for the remainder of the data science process to be of value.

Quick Tip

Remember, cleaning data is the first step in data wrangling. This foundational step not only ensures that our data analysis is built on accurate and reliable information, but also saves significant time. By doing it right the first time, we avoid the costly process of having to revisit and redo analyses built on flawed data.



Ink Drop/Shutterstock

Missing Data

When a dataframe has *missing data*, it means that at least some cells do not have values. In Table 2.1, a sample from Mirabel's dataset shows an example of observations with missing values for some variables. The observational units are the students represented in her dataset.

TABLE 2.1
Data with missing cells for some columns and rows.

first_name	age	gpa	favorite_pet	voted
Selma	18	3.1	Dog	0
Juan		3.2	Cat	1
Rozzlin	19	3.9		0

The table displays five variables about each individual (`first_name`, `age`, `gpa`, `favorite_pet`, and `voted`). Each row records one person's responses. The value in a specific cell, a single box in the dataframe, represents an individual's answer to a particular question. The second column of the dataframe represents *age*, and the first row contains Selma's responses. Selma's response has the cell value of 18 for *age*. However, it is noteworthy that Juan's *age* response,which should be in the second row, is missing.

Why would these missing data matter? Missing information might frustrate us in our analysis. Mirabel would not be able to identify Juan's responses about voting as belonging to groups of older versus younger respondents, which may have implications for strategic ways to get the vote out differently for each age group. Remember that a row in a dataframe represents each person's response across many columns (i.e., many variables).

What do we do with missing data? The best solution is to rerun the study with tighter response controls to avoid missing values (e.g., adding a dropdown menu of pet options instead of an open-ended response). However, this is not always possible, especially if the data come from a third party, as with Mirabel's dataset. Thus, we discuss a few other ways of addressing missing values in our data:

1. Removing observations with missing values.
2. Imputation from internal data.
3. Imputation from external data.
4. Other options.

DEFINITION

Imputation is the process of replacing missing data with substituted values derived from patterns in available data.

Try It Yourself: Identify Missing Data

Dataset: vote

GPA data is a way to understand people's educational performance and its potential relationship with voting behavior. A significant number of missing values can impact our analysis, potentially skewing results and obscuring key insights. By identifying the extent of missing `gpa` records, we aim to evaluate the dataset's reliability, inform subsequent data cleaning or imputation methods, and uncover any underlying patterns associated with these data gaps. This step is essential in ensuring that our analysis is both accurate and meaningful.

Answer the following questions (round float (i.e., non-integer) values to two decimal places):

1. How many missing values are there in the `gpa` column?
2. What is the total number of rows in the dataset?
3. What percentage of the dataset has missing `gpa` values?
4. How would you describe this using the STAR framework?

Tool Time–Available in MyLab Statistics

Excel

Python

R

StatCrunch

Removing Observations with Missing Values

One way to address missing values is to remove the entire observation if there are any missing values. This would mean deleting the person's response entirely, even if they answered some questions. For example, Mirabel would remove all of Juan's responses by deleting his row from the dataframe because the value for `age` was missing in the example above. The downside to this option is that Juan's other responses may have offered us essential insights. Mirabel should use caution before deleting data. For instance, imagine that a third of the respondents failed to indicate their age, but all the respondents did indicate their voting behavior in the most recent election. What would Mirabel do? She could decide not to group people by `age` rather than deleting observations. In summary, she can respond to missing data by ignoring the missing value's observation (e.g., Juan's entire record) or variable (e.g., all responses for `age`).

On the other hand, she might retain rows with missing values, presenting results with all available information. The downside to this approach is that some results may represent perspectives from those with missing values, while other analyses may not, sometimes resulting in divergent sample sizes. Consider the example in which Juan's `age` value is missing. Mirabel might report the average `gpa` including Juan's response, since his `gpa` is not missing. However, an accompanying report on `gpa` grouped by `age` would not include Juan as his age is missing. If there are many missing values, this can lead to confusion and even misrepresentation. Regardless of Mirabel's decisions, she must report them when publishing the information.

Big Picture Break

Retaining observations with missing data values often leads to biased results, as analyses based on the remaining data may not be representative of the whole. On the other hand, simply removing observations with missing values might reduce the size of the dataset, potentially losing valuable information. Each approach to handling missing data has its trade-offs, and the choice often depends on the context of the dataset and the nature of the missing data. In some cases, advanced techniques like machine learning, covered in Chapters 8–10, can be used for imputation, offering more nuanced handling of missing data. The key is to understand the implications of each method and to choose the one that aligns best with the goals and constraints of our specific analysis. In essence, handling missing data is not just a technical challenge but a strategic decision that can shape the direction and credibility of our data analysis. As data scientists, it's crucial to develop a keen understanding of these methods and their impact on the integrity of our analysis, transparently indicating our choices when we present results.

Imputation from Internal Data

Another option is to use *imputation*. As defined earlier in this chapter, imputation means estimating the missing value based on other similar observations. In other words, we would replace the missing value with a value that we hope is likely to be reasonably accurate. For example, suppose Juan indicates in response to additional questions that he graduated from high school four months ago, has a part-time job, and lives off campus. Mirabel might find hundreds of other respondents with the same profile and then average their ages (e.g., 18.2 years old) and replace Juan's missing `age`value with the newly calculated one. She can impute using averages or more advanced methods not covered in this chapter (e.g., machine learning methods). In using the existing dataset, Mirabel is *imputing from internal data*.

DEFINITION

The **mean** of a variable is the average of all the data values in a dataset for that variable. It is calculated by adding up all the values and then dividing this sum by the number of values.

The **median** of a variable is the middle value in a dataset for that variable when the values are arranged in ascending or descending order. If there is an even number of values, the median is the average of the two middle values.

Table 2.2 presents a historical dataset from an archaeological dig site. The data represent objects unearthed at different depths, which can provide us with an idea of the time period each object originates from. Please refer to the variables described in the list below.

1. `object_ID`: an identification number we assign to each archaeological object found
2. `type`: a classification of the object
3. `material`: what the object is primarily made of
4. `depth`: how deep the object was unearthed, measured in meters. (Note that objects found deeper underground are generally older.)
5. `age`: estimated age in years

It's common in archaeological excavations to find objects whose age is difficult to estimate directly. This could be due to lack of context, decay, or other factors.

In this example, we are looking to find the *mean* age of the objects listed in this dataset. However, several values in the `age` column are missing. We can deal with missing values in many ways. We will demonstrate two methods to deal with the missing `age` values: (1) *removing* observations with missing values, and (2) *imputing* them.

TABLE 2.2
Archeological dataset with missing values.

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	
O10	Coin	Silver	1.2	1900
O11	Jewelry	Gold	2.7	
O12	Pottery	Clay	2.2	2400

Method 1: First, we remove all observations with missing values for `age`. Then, we observe how this removal affects the data, for example, by calculating its mean. To calculate the mean age, we add up the known ages and divide by the number of objects with known ages. The sum of the known ages is

$$2000 + 3000 + 1500 + 2500 + 2000 + 1900 + 2400 = 15{,}300 \text{ years}$$

and there are seven objects with known ages.

So the mean age is

$$15{,}300/7 = 2185.71 \text{ years.}$$

For this calculation, note that the observations with missing age values were ignored or excluded. Although we originally described this process as *removing observations with missing values*, the actual dataset does not need to be reduced to only complete observations. Most, if not all, software tools will simply exclude observations with missing values when performing tasks that require those values.

Method 2: This time we impute missing values for `age` rather than remove them, again following up with a mean calculation to compare with that of Method 1. To impute the missing ages, let's use the mean age of the complete (or "known") observations, which, as calculated in Method 1, is 2185.71 years. This assumes that the age of the objects found at this archaeological site can generally be described well by their average value. Table 2.3 is our completed dataset:

After imputation, the sum of the ages is

$$2000 + 2185.71 + 3000 + 1500 + 2185.71 + 2500 + 2185.71$$
$$+ 2000 + 2185.71 + 1900 + 2185.71 + 2400 = 26{,}228.55$$

And we have 12 objects, so the mean age after imputation is still 2185.71 years.

So, we see that imputing missing values with the mean age will leave the mean age unchanged. We could also use the median instead of the mean to impute missing values.

TABLE 2.3
Archeological dataset with imputed values.

object_id	type	material	depth	age
O1	Pottery	Clay	1	2000
O2	Jewelry	Gold	2	**2185.71**
O3	Tool	Bronze	1.5	3000
O4	Weapon	Iron	3.5	1500
O5	Coin	Silver	1.5	**2185.71**
O6	Pottery	Clay	2.5	2500
O7	Jewelry	Gold	1	**2185.71**
O8	Weapon	Bronze	3	2000
O9	Tool	Iron	4	**2185.71**
O10	Coin	Silver	1.2	1900
O11	Jewelry	Gold	2.7	**2185.71**
O12	Pottery	Clay	2.2	2400

Try It Yourself: Impute Missing Values from Internal Data

Dataset: vote

In our ongoing analysis of the voter survey dataset, we now tackle the issue of missing GPA values by employing the mean imputation method. Our objective is to create a new column in the dataset, which will replicate the original GPA data but with a key modification: missing values will be filled in with the mean GPA of the available data.

Questions

Answer the following questions (use two decimal places):

Before replacing missing data:

1. What is the mean GPA?
2. What is the median GPA?

After replacing the missing data using mean imputation:

3. What is the mean GPA?
4. What is the median GPA?
5. Describe the results using the STAR Framework.

Tool Time–Available in MyLab Statistics

- Excel
- Python
- R
- StatCrunch

Imputation from External Data

We have so far described *imputation from internal data* based on values within the same dataset. However, *imputation from external data* based on values from a different dataset is also possible. For example, in the case of Mirabel's dataset, she could access the college's roster that includes the year of birth for each student. Alternatively, if the college cannot provide data about birth dates, she could use public data about the average age of students in Juan's class. Even better, if she already has information about Juan from another dataset, and she is sure it is the same Juan, she should replace Juan's missing age value with the known age. However, she would not refer to this as an imputation because the values would be known, not estimated. When imputing data from external datasets, we should trust that the other sources of data are reputable and relevant to our original dataset.

The downside to imputation, whether from internal or external data, is that it makes assumptions and modifies the raw values, thereby influencing the analysis. These decisions should always be documented and reported.

Case Study: Data Wrangling in Criminal Justice Research

Dzelat/Shutterstock

The Criminal Justice Research Center is one of many organizations that use dataframes to analyze crime data. They have participated in one project, the Historical Violence Database (HVD), which is a collaborative research project that gathers data on the history of violent crime. Social scientists, the primary audience for this database, test theories of violent crime under different historical circumstances. When deciding how to code variables, the Criminal Justice Research Center carefully considered the needs of future users of the database. This included considering whether free-response entry might lead to erroneous or missing values.

Most dataframes from public agencies use numbers (i.e., codes) to refer to different crimes or weapons. For example, the FBI's Supplemental Homicide Reports reference murders committed with knives or cutting instruments using the number "20." However, the HVD opted to add a new text variable, spelling out the name of the weapon (e.g., "club"). This is because most of the social scientists and historians using this database are not trained data analysts. Using descriptive text data makes the database more accessible to its users. Considering how domain experts intend to use the data is critical when deciding how to record or transform variables.

Missing data pose many serious problems in criminology. They can obscure the types of crimes that have occurred, contribute to a lack of evidence or faulty evidence, and confuse criminal justice researchers. Data are rarely collected without missing values. Sometimes, researchers choose to ignore cases with missing values and only analyze cases where all cells contain data. This can be problematic, as it introduces bias if the cases with complete data are not representative of the cases that are missing data, especially as the number of cases with missing data increases. Strategies such as imputation have proven effective in criminological research when managing missing data.

The FBI's Uniform Crime Reporting (UCR) Program is a primary source of data on crime. The UCR has incorporated multiple forms of imputation into its data cleaning process. With *imputation from external data*, data from prior responses of the same respondent are used to impute the missing values. An advantage of this method is that it introduces less bias through its use of all available data. Imputation has allowed the UCR to produce more accurate and complete crime estimates at the state, regional, and national levels.

Other Options for Missing Data

We could also decide not to impute, but to replace missing values with an entirely new value specific to that missingness. For example, missing values for quantitative variables could be filled with "NA" (which is often a special value in software that uniquely denotes a missing value), or with a completely nonsensical value like -999. For categorical variables, missing values could literally be replaced with a value of "Missing" or "Unknown". By doing this, we can continue to include incomplete observations in the dataset, and our analysis can distinguish them from other observations where appropriate.

Although it may be cumbersome, we could also simply pursue two separate analyses with our data: one with the incomplete observations included, and one with the incomplete observations removed. Regardless, we should always think carefully about how we modify raw data, clearly noting the choices made to deal with missing values when writing a report.

An even more tedious strategy for addressing missing data is called *multiple imputation*. It involves imputing values for missing data many times, generating many different imputed datasets. From there, the data scientist analyzes each one of these imputed datasets and combines the results in some way.

In the end, handling missing data can be very complex. A data scientist must think carefully about the following things:

- What are the possible sources of missing data in my dataset?
- Is the missingness random, or do any patterns explain why values are missing?

  (In fact, the randomness of the missing data can affect the usefulness of the different strategies discussed here. Unfortunately, these details are beyond the scope of this text.)
- What strategies can I pursue to adjust my work for the missing values in my dataset?
- How can I document and communicate these missing data characteristics in my work most effectively?


Africa Studio/Shutterstock

SECTION 2.3: CLEANING ANOMALOUS VALUES

Cleaning data involves many types of anomalous, or "unusual," values. We focus on implausible values, extreme values, incorrect formats, and duplicate records in this section.

DEFINITION

Implausible values are data points that fall outside the realm of possible values, clearly indicating errors.

Extreme values are data points that are noticeably different from others, potentially affecting the overall analysis in noticeable ways.

Incorrect formats refer to data entries in an unexpected format, requiring correction to match the anticipated data type.

Duplicate records are repeated entries (e.g., observations) in a dataset, often identified and resolved using unique identifiers.

Implausible Values

Implausible values are quantitative data that are not within a range of anticipated values, to the point of being impossible. For example, a customer's credit score is typically between 300 and 850. If we come across a credit score value of 22,995, or -53, we know the data must be a

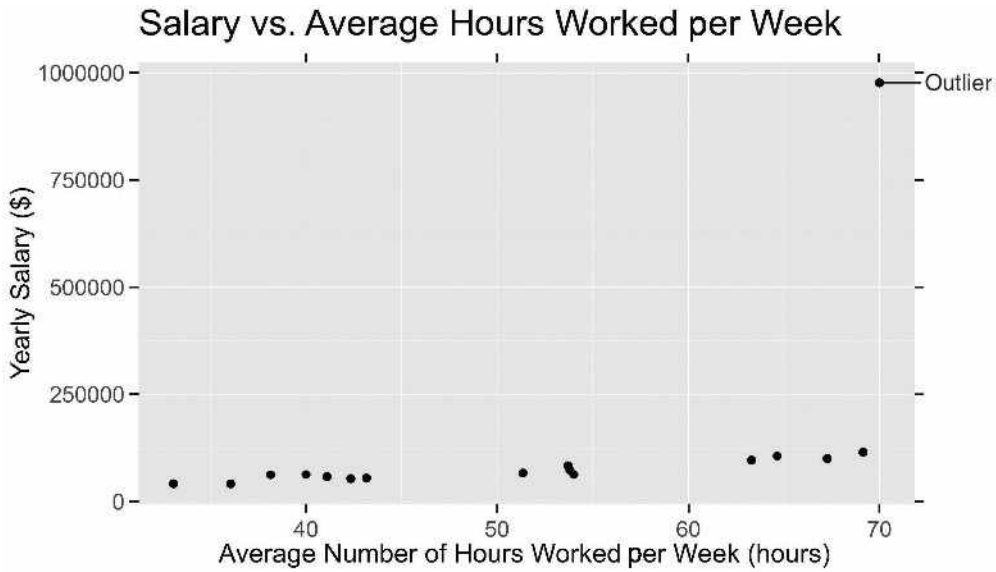

Alexey Konkov/Shutterstock

mistake of some sort because credit scores range from 300 to 850. Common sense tells us these values are incorrect. Depending on our judgment, we could replace these cells with missing values and proceed with any of the methods previously described in our discussion of missing data. Although it may require more time and effort, we could also return to the source of the data and correct these values instead of replacing them with missing values.

Extreme Data Values

Extreme data values are exceptionally higher or lower than the other values of a variable. Consider the results from Figure 2.2, which indicate that most of the respondents' annual earnings are between $33,000 and $200,000. What would we think if someone said their annual income was $977,000? This would be considered an extreme data value. Why does this matter? The $977,000 value could significantly alter our analyses. For example, if we include the $977,000 value and calculate the mean from those reported values, we find an average annual income of $103,375. If we do not include the $977,000 data point, our average annual income is $45,367, a noticeable difference. Reporting the $103,375 number may give the impression that a typical respondent earned over $100,000 when, in reality, they earned less than $100,000.

FIGURE 2.2
Depiction of an extra data value that is likely to be an outlier.



Though often used to mean the same thing as *extreme value*, an *outlier* is a value that we consider so extreme that we wish to fix or remove it. How do we know for sure whether a value is too extreme, i.e., likely to be an outlier? It requires some judgment, depending on the context. In the case of the income dataset in Figure 2.2, if all the respondents work for the same private company, it could be that the high-earning respondent is the owner. In such a case, there may be a way to verify the unusual data. In Chapter 4, we will discuss formal ways of identifying outlier observations.

DEFINITION

An **outlier** is an extreme data point so different from the majority that it may warrant correction or removal. Its extremity is determined by contextual judgment and verification.

Try It Yourself: Identify Extreme Data Values

Dataset: vote

We need to check our voter data for any unusually high or low ages. These extreme ages could be mistakes and might throw off our results. By finding the ages of the oldest and youngest 5% of observations in our data, we can decide if these numbers make sense or if we should exclude them to maintain the integrity and trustworthiness of our data.

Questions

Answer the following questions:

1. What ages encompass the oldest 5% of respondents?
2. What ages encompass the youngest 5% of respondents?
3. How many respondents have `age` values above 100?
4. How many respondents have `age` values below 18?
5. How extreme are the `age` values (use the STAR Framework)?

Tool Time–Available in MyLab Statistics

Excel

Python

R

StatCrunch

What should we do with extreme values? The answer is complex. In some cases, it depends on whether they are due to error. If so, we can try to fix or remove them. In other words, data scientists may treat extreme values as implausible and ignore the values or the entire rows from the analysis (reporting this, of course). In other cases, the values may be valid, but we need to think about whether they represent the population well. Thus, an analyst may run analyses both with and without the extreme value(s), and report on findings both ways. In the yearly salary example presented in Figure 2.2, we are working with a variable whose values are mostly concentrated together, with only one value separated from the bulk of them. This behavior is common with population- and age-related variables as well. Mathematically transforming such variables can help address the extremeness of some values.

Case Study: Dewey Defeats Truman and the Role of Data Wrangling

Byron H. Rollins

In 1948, the United States was in the grips of a presidential election, and the *Chicago Daily Tribune* made a historic mistake after election day. It published the headline “Dewey Defeats Truman,” declaring Thomas E. Dewey the winner. The pollsters had relied heavily on telephone surveys, a cutting-edge methodology for that era. However, this approach was fundamentally flawed: telephones were a luxury that many couldn’t afford, and as a result, the data gathered were biased toward a specific demographic—those who were wealthier and more likely to vote for Dewey. When the final votes were counted,

(*Continued*) ▶

(*Continued*)

Harry S. Truman had won, making the newspaper headline an infamous example of data gone awry.

Data wrangling might have helped. In the world of data science, wrangling serves as the critical bridge between raw data and meaningful analysis. If the pollsters in 1948 had taken this crucial step, they would have first identified the bias inherent in their telephone surveys. Recognizing this, they could have supplemented their data with other sources representing a broader swath of the population. This would have helped to balance out the bias introduced by the original telephone surveys.

Wrangling doesn't stop at merely cleaning up data; it also involves transforming it into a more analyzable form. If the pollsters had applied statistical techniques to weigh their data, they could have accounted for underrepresented groups, making their sample more reflective of the general voting population. This could have included techniques to account for uncertainties about their respondents.

The *Tribune's* headline serves as a potent reminder of how critical wrangling is to any data science endeavor. It shows that the quality of our data analysis can only ever be as good as the quality of our data. As this case study reveals, poor wrangling can lead to errors that reverberate through history, affecting not just election reporting but also the way we perceive the capabilities and limitations of data in decision-making across diverse fields.

Incorrect Formats

When we expect one data type, but a data value appears to be a different type, we are looking at an *incorrect format*. We may expect text but see a number, or vice versa. For example, in Mirabel's dataset, suppose that a student reported "seventeen" as their age, or "17+", instead of 17. Failing to correct this would mean that the value of that student's response would not be considered a number.

Additionally, Mirabel might find that someone has responded "Goldfisg" instead of "Goldfish" as their favorite pet. She may want to make the judgment call to edit a cell with an incorrect format if the data *almost* makes sense (e.g., she might replace "Goldfisg" with "Goldfish"). On the other hand, sometimes the value just doesn't make any sense (e.g., an income value might be nonsensical, like "slippery," or ambiguous, like "forty-ten"). In these cases, we have the option to remove the observation by deleting the entire row. Alternatively, we might just delete the value from the cell, causing the value with an incorrect format to be missing.

Irochka/123RF

It is usually best practice to delete as little data as possible, so deleting the value in the cell would be the preferred method unless many cells in the row contain faulty data. However, in Mirabel's dataset of over 5000 responses, doing this by hand for each row would be nearly impossible! Therefore, she might need some help from tools to automate the process.

Try It Yourself: Clean Incorrect Formats

Dataset: vote

Clean the `favorite_pet` column of the vote dataset. Answer the first question, then perform a basic cleaning of the data in this order:

A. "dogs" should be changed to "dog"
B. "cats" should be changed to "cat"
C. any word that begins with "d" and ends with "g" should be changed to "dog"
D. any word that begins with "c" and ends with "t" should be changed to "cat"

Once the dataset is cleaned, you will be ready to answer the remaining questions.

Questions

1. Before basic data cleaning for dogs and cats, how many people in this dataset indicated their favorite pet as:
   a. "dog"
   b. "cat"
2. After basic data cleaning for dog and cat, how many people in this dataset indicated their favorite pet as:
   a. "dog"
   b. "cat"
3. Describe the results using the STAR Framework.

Tool Time–Available in MyLab Statistics

Excel

Python

R

StatCrunch

Duplicate Records

The same record (observation) may appear more than once in a dataframe. For example, a Google Forms respondent may have submitted the same form multiple times, but we want to count each person's data only once when analyzing the data. We usually address this problem carefully by considering identifiers like a respondent's student ID number or email address. Such variables are expected to be unique for each observational unit. If two observations have the same value for a unique identifier, such as an email address, this may indicate a mistaken duplicate record. We often delete the duplicate records.

SECTION 2.4: TRANSFORMING QUANTITATIVE VARIABLES

In addition to dealing with problems like missing data, implausible values, extreme data values, incorrect formats, and duplicate records, data wrangling can include ways to enhance or further structure the data. We will discuss how to create new categorical variables from quantitative ones, change the scale of quantitative variables, and combine two or more quantitative variables.

Recall that *quantitative variables* are numerical in nature, allowing for mathematical computations, and can be either discrete, with countable values, or continuous. Also recall that *categorical variables* are often non-numeric and represent group or category memberships.

Creating a Categorical Variable from a Quantitative Variable

In some areas of the world, discounts are based on the age of the patron. A restaurant may charge less for a child or an older adult. This is an example of an age *category*. Who qualifies as a child, and who qualifies as a senior? The establishment can decide this. However, you may have noticed that an individual is frequently considered a child if they are 11 years old or younger, and an individual is frequently considered a senior if they are 60 years old or older. The conversion of someone's age to a label like "child" or "senior" is an example of *categorization*. In this case, we've created a categorical variable, `age_category`, from a quantitative variable, `age`.

Quick Tip

When categorizing data, ensure the categories are mutually exclusive (i.e., one row belongs to only one category) and collectively exhaustive (i.e., no row is left uncategorized). This clarity in categorization helps in creating more effective visualizations and analyses.

Generally, the process of *categorization* describes the way in which a new categorical variable is created from an existing quantitative variable. This is done by assigning a category or label to intervals of values of the quantitative variable. In the age example, it was

- Ages $0-11$ years: child
- Ages $12-59$ years: adult
- Ages $60+$ years: senior

In this case, the age categories are more useful to a restaurant than the actual age value. This can be the case for a number of visualization- and analysis-related reasons, which we will discuss more later in this text. One of the most important steps in categorization is ensuring that all possible values of the quantitative variable are each associated with a single category. For example, someone who is 11 years old should NOT be considered both a child and an adult. Creating a categorical variable from a quantitative variable always, necessarily, results in the loss of information. This might be acceptable for our goals, but we should do it purposefully and thoughtfully.

Try It Yourself: Create a Categorical Variable from a Quantitative Variable

Dataset: vote

Create a new variable called `gpa_category` which contains a categorized version of the `gpa` variable according to the following:

- 2.00–2.99: low
- 3.00–4.00: high

Questions

1. How many high versus low GPA students are there?
2. Use STAR to describe the results.

Tool Time–Available in MyLab Statistics

Excel

Python

R

StatCrunch

Changing the Scale of a Quantitative Variable

Think about the last time you saw graphs or data visualizations that contained any sort of numbers. How many digits were there in each of the values displayed on the graph? We often have to make this choice as the person working with the data. A dataset may contain an income variable whose values are as low as $\$20,000$ and as high as $\$800,000$. Reading such large numbers on a data visualization can be challenging, so we might consider scaling these numbers down in a way that reduces the number of digits but still allows us to convey similar information. For example, we could create a new variable that represents income in thousands of US dollars, which is computed by dividing the original income variable by 1000.

$$\$20,000/1000 = \$20 \text{ (in thousands)}$$
$$\$800,000/1000 = \$800 \text{ (in thousands)}$$

Then, we could use this new variable in our data visualization and simply note that values are in thousands of dollars. In this way, readers will only see values from around \$20 to around $\$800$, which are considerably easier to digest.

Generally, the process of *scaling* or *transforming* describes the method of creating a new quantitative variable from an existing one by performing some mathematical operation on it. Specifically, *scaling* usually involves multiplying or dividing the original variable by a certain value. *Transforming*, in this restricted sense, refers to applying a more complex function to the original variable, such as taking the logarithm or square root of the values. (Throughout this text, the terms *data transformation* or *transforming* refer broadly to various types of alterations that make data easier to work with.) In the income example, we divided the original variable by 1000. Different situations with different variables may benefit from different types of mathematical operations. As long as we create a new variable without disrupting the raw, original data, we should feel free to explore how different transformations affect the insights the data can provide. Of course, we must ensure to document all our steps and work so that it can be explained and replicated if needed.

Case Study: GlobalGiving Teaches Nonprofits about Transforming Variables

RDNE Stock project/Pexels

GlobalGiving is a nonprofit organization that provides a platform for other nonprofits to raise funds from corporate donors such as Airbnb, 3M, and Twitch.tv. It aims to bring about positive change across various sectors including education, healthcare, and disaster relief. GlobalGiving helps nonprofits become more effective, empowering both donors and nonprofits to drive meaningful impact.

GlobalGiving says that many nonprofits have ample data but struggle to derive actionable insights. The problem? Their data are unorganized and disparate, making it difficult to measure the impact of their community programs. To address this, GlobalGiving teaches nonprofits how to centralize and clean their data, thereby transforming both quantitative and qualitative data. For instance, they helped an animal welfare program in reporting the ratio of rescued animals to attending volunteers by converting text-based animal categories into numeric counts. They also helped them transform quantitative data, such as pounds of food, into qualitative information like “too much food” or “too little food,” to determine whether food was distributed evenly across all animals. GlobalGiving effectively teaches nonprofits to manage both types of data—quantitative and qualitative—for a comprehensive view of their programs’ effectiveness, thereby facilitating better decision-making and reporting.

Combining Two or More Quantitative Variables

Oftentimes experts combine variables or characteristics in a way that enriches the insights we can gain from the data. For example, consider the computation of a “per capita” variable. “Per capita” means “per person,” and it is a term used to understand statistics like the gross domestic product (GDP). We compute it by dividing a country’s overall GDP by the country’s population, thus finding the GDP per capita.

In this situation, our dataset might contain information on countries of the world, including their gross domestic products (GDPs) and their populations. To find a given country’s GDP per capita, we would use the following equation.

```
gdp_per_capita = gdp / population
```

In the above scenario, we would be combining two variables—`gdp` and `population`—into a new variable, `gdp_per_capita`. We could use this new variable to learn more about goods and services produced *per person*, rather than for the population as a whole.

These kinds of relative measures can be much more useful for visualization and analysis because they help account for countries with different populations, or more generally, entities of different sizes. For example, the United States may have a much higher GDP than France, but this may not be surprising because of the drastically different sizes and populations of these two countries. In fact, their GDP per capita values are not as different as their national GDP values.

As with *data cleaning*, it is not uncommon for a data scientist to undergo multiple rounds of *data transformation* as they discover what their desired visualizations and analyses need with respect to the forms of the data. It is important to note that data transformation usually happens after data cleaning, so it is safe to assume that variable values are consistent and accurate. The cyclical nature of each data wrangling phase is illustrated in Figure 2.1.

Quantitative variables can represent numerous different types of variables, such as age, income, height, and weight. Once variables like these are clean, it will be useful to think about what kind of insights they can offer and how best to communicate those to a broader audience.

SECTION 2.5: TRANSFORMING CATEGORICAL VARIABLES

Categorical variables can represent numerous different types of variables, such as pet preference and date. Once variables like these are cleaned, it will be useful to think about what kind of insights they can offer and how best to communicate those to a broader audience.

Creating a Quantitative Variable from a Categorical Variable


Eric Isselee/Shutterstock

There are many analysis methods that we will learn about later, which can only work with data in numeric form. Some visualization and summarization techniques are only appropriate for numeric data. Therefore, we sometimes want to translate text data into numbers. For example, Mirabel's vote data variable `pet_ preference` from Table 2.1 has many possible text values including "Dog" and "Cat". We may want to associate `pet_preference` values with a tendency to vote. Do people with dogs vote more than people with cats? We can create Boolean variables (also known as dummy, indicator, dichotomous or binary variables) for each value. If `pet_preference`is "Dog," we could create a new data column in the dataframe called `dog` and assign observations a value of 1 if `pet_preference` is a dog and 0 otherwise.

A common way to transform categorical variables into quantitative ones in data science is to create Boolean variables that represent different levels of a categorical variable.

Consider data in Table 2.4:

1. `student_id`—an ID we assign to each student who participated
2. `age`—self-reported age
3. `gpa`—the self-reported grade point average
4. `pet_preference`—the stated pet preference (e.g., Dog, Cat)
5. `voted`—whether they voted in the last election (1 = Yes, 0 = No)

We will create a new quantitative variable from the `pet_preference` categorical variable. Specifically, we will create new Boolean columns for each category in the original column (e.g., `prefers_dogs` and `prefers_cats`). If a student prefers dogs, they will have a 1 in `prefers_dogs` and `prefers_cats` and a 0 in the other new preference columns (e.g., a 0 in `prefers_cats` and `prefers_none`).

Here's how this process would look:

TABLE 2.4
Original data.

student_id	age	gpa	pet_preference	voted
S1	19	3.5	Dog	Yes
S2	21	3.7	Cat	No
S3	20	3.2	None	Yes
S4	22	3.8	Dog	No
S5	19	3.1	Cat	Yes

TABLE 2.5
Data after creating Boolean variables from the categorical variable `pet_preference`.

student_id	age	gpa	pet_preference	prefers_dogs	prefers_cats	prefers_none	voted
S1	19	3.5	Dog	1	0	0	Yes
S2	21	3.7	Cat	0	1	0	No
S3	20	3.2	None	0	0	1	Yes
S4	22	3.8	Dog	1	0	0	No
S5	19	3.1	Cat	0	1	0	Yes

In Table 2.5, the `pet_preference` categorical variable has been transformed into three new binary variables: `prefers_dogs`, `prefers_cats`, and `prefers_none`. This quantitative transformation allows for numerical analysis and modeling that could not be accomplished with the original categorical data. For example, we can sum the values of `prefers_dogs` using the following equation:

$$1 + 0 + 0 + 1 + 0 = 2$$

Since there are five records, we know that $2/5 = 0.4 = 40\%$ of respondents prefer dogs. Creating quantitative variables from categorical variables is even more important in future chapters on machine learning.

Try It Yourself: Create a Boolean Variable That Represents a Categorical Variable

Dataset: vote

Create a Boolean variable called `drinking_age` where `drinking_age` = 1 if age is greater than or equal to 21, and 0 otherwise.

Questions

1. How many `drinking_age` values are 1? How many `drinking_age` values are 0?
2. How many people of drinking age voted? What percentage voted among people of drinking age? How many people not of drinking age voted? What percentage voted among people not of drinking age? Round float (i.e., non-integer) values to one decimal place.
3. Describe the results using the STAR Framework.

Tool Time–Available in MyLab Statistics

- Excel
- Python
- R
- StatCrunch

The process of creating a quantitative variable from a categorical variable is often called *encoding* because we are creating numeric codes associated with each category, or value, of the categorical variable. Boolean encoding is a special case of variable encoding where we assign a 1 or 0 during encoding. In transforming `pet_preference` to `prefers_dogs`, we were implementing *Boolean encoding* with the value of "Dog" as 1 and any other value as 0.

DEFINITION

Encoding is the process of converting a categorical variable into a quantitative one by assigning numeric codes to the values of the categorical variable.

Boolean encoding is the process of converting a categorical variable into a quantitative one by assigning a 1 or 0 to the values of the categorical variable.

In general, we can use whatever numeric values we want, but we must be extremely careful when doing so. The numeric values we use should reflect the categories as best as we can. In particular, the differences between the categories should align with the differences in the numeric values as much as possible. For example, suppose we have data on people's feelings about a particular movie with the following possible responses:

- Dislike
- Neutral
- Like

Table 2.6 presents four possible encoding schemes for these categories, each differing in how they represent these values.

TABLE 2.6
A table presenting four encoding schemes for Dislike, Neutral, and Like.

Response	Encoding Scheme 1	Encoding Scheme 2	Encoding Scheme 3	Encoding Scheme 4
Dislike	1	−1	0	0
Neutral	2	0	0	1
Like	3	1	1	1000

In Table 2.6, Encoding Scheme 1 is often the first type of encoding that people consider because it simply enumerates the responses to a question. There are three possible responses, so we number them 1, 2, and 3. Encoding Scheme 2 represents the responses in a way that is more explicitly oriented around the "Neutral" response: 0 represents the "Neutral" response, +1 represents the "Like" response, and −1 represents the "Dislike" response. Encoding Scheme 3 is a Boolean encoding where we treat "Dislike" and "Neutral" the same, both distinct from "Like" (e.g., we only want to consider those who like something, not those who are neutral or dislike it). Finally, Encoding Scheme 4 is an atypical example that appears to emphasize the "Like" response quite a bit. It's challenging to determine the best encoding scheme without knowing what types of visualization and analysis will be conducted. However, it's crucial to understand that each of these encoding schemes could result in different visualizations and analyses. Without knowing the type of visualization and analysis we want to conduct, it is difficult to determine which encoding is best.

Like our other data wrangling steps, encoding a quantitative variable may be an iterative process. Each time, it is informed by the visualizations and analysis performed.

Condense the Categories of a Categorical Variable

Sometimes free response results contain many different answers, some of which may not occur very often. When Mirabel's dataset involved students' pet preferences, she might have seen responses about dogs, cats, fish, birds, turtles, lizards, and more. Upon further inspection, she might discover that 90% of the responses were either "dog" or "cat". To simplify the rest of the work, she decides to create a new variable, `pet_preference_condensed`, that contains people's pet preference as "dog," "cat," "none," or "other." In doing this, Mirabel has merged all the values other than "dog," "cat," and "none" into a single "other" category, which is much easier to display and read in visualizations and summaries than the numerous other responses that occurred so rarely.

DEFINITION

Condensing the categories is the process of reducing the number of distinct values in a categorical variable by merging categories into broader ones. This simplifies the variable for analysis while reducing its informational detail.

Typically, condensing a categorical variable means decreasing the variety of its possible values. This process does not need to preserve any of the original variable's values. Of all our transformation methods thus far, this is one of the most important methods with which to create a new variable and not replace the existing variable. This is because the combining of categories, although it may end up being informative, is a reduction in the information contained in the variable. Note that Encoding Scheme 3 in Table 2.6 was actually an example of *condensing the categories*.

Splitting a Variable into Multiple Variables

Data can come in all shapes and sizes, and this includes variables. It's not uncommon for free-response questions to have answers in many different formats. Consider entering information for your birthdate. If you were given an unformatted prompt for your birthdate, you might enter it as follows:

July 7, 2007

And this might be exactly how your value for the birthdate variable appears in the dataset. However, the visualization and analysis of these data may involve the months and years in which people were born. This would require the extraction of the month and year from values of birthdate like the one above, ignoring the day. All software tools will have methods for extracting information like this.

In general, the *splitting* of a categorical variable describes the process of extracting pieces of information from its values and storing them in at least one new categorical variable (e.g., `month_name`) or quantitative variable (e.g., `year`). Figure 2.3 provides an example of such a splitting operation.

FIGURE 2.3
Example dataset with date variable split into `month` and `year` variables.

date	month	year
3/4/2003	March	2003
11/2/1998	November	1998
3/30/2013	March	2013
5/1/2009	May	2009
9/20/2012	September	2012

Try It Yourself: Splitting a Variable into Multiple Variables

Dataset: vote

We revisit the voter survey dataset with a focus on identifying variations in voting patterns across different birth months. Later, we aim to determine if there is a discernible relationship between voters' birth months and their voting behaviors. To prepare for a future analysis, we will modify the dataset by isolating the birth month from the existing birthdate information. This involves generating a new column dedicated solely to the birth month. This modification prepares the data, to explore in a future activity, how voting preferences may vary according to the birth month of the respondents.

Questions

1. How many people have a birthday in each month of the year?
2. Describe the results using the STAR Framework.

Tool Time–Available in MyLab Statistics

- Excel
- Python
- R
- StatCrunch

SECTION 2.6: RESHAPING A DATASET

The last type of data transformation we consider here is *reshaping* a dataset. Recall that a *tidy* dataset is one in which each column represents a variable, each row represents an observation, and each cell has a value. Often a raw dataset presented as a table will not be in a tidy format because of the shape that it's in.

Figure 2.4 provides an example of a Gapminder dataset that is not tidy. In this example, we have a row for each country in the world and a column for each year starting from 1800. Each cell contains the population value for a particular country and year combination. The three variables in this dataset are `country`, `year`, and `population`. However, the years only appear as column names and not as values in a column. A tidy version of this dataset would have three columns, one for each of these variables, and a row for each combination of country and year. Each combination of country and year (e.g., Afghanistan in 1800) can be considered as an observational unit.

The dataset in Figure 2.4 is an example of a *wide-format* dataset because it contains more columns than we might need. To tidy this dataset we aim to convert it into a *long format* with only three columns: `country`, `year`, and `population`. Figure 2.5 provides a preview of this tidy, long-format dataset.

The process of *reshaping a dataset* usually refers to the transformation of a dataset from wide format to long format, or from long format to wide format. The wide format version can be better for humans to read (e.g., to see as labels on a graph). Although the long format version seems less readable to a human, it is a better form for computing systems that can quickly use these variables (columns) for visualization and analysis. In either case, the goal is to end with a form that lends itself to visualization and analysis.

FIGURE 2.4
Wide format of the Gapminder dataset of population information for countries of the world each year starting from 1800.

country	1800	1801	1802	1803	1804	1805	1806	1807	1808	1809	1810
Afghanistan	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M	3.28M
Angola	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M	1.57M
Albania	400k	402k	404k	405k	407k	409k	411k	413k	414k	416k	418k
Andorra	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650	2650
UAE	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k	40.2k
Argentina	534k	520k	506k	492k	479k	466k	453k	441k	429k	417k	420k
Armenia	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k	413k
Antigua and Barbuda	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k	37k
Australia	200k	205k	211k	216k	222k	227k	233k	239k	246k	252k	259k
Austria	3M	3.02M	3.04M	3.05M	3.07M	3.09M	3.11M	3.12M	3.14M	3.16M	3.18M
Azerbaijan	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k	880k
Burundi	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k	899k
Belgium	3.25M	3.26M	3.27M	3.28M	3.29M	3.3M	3.3M	3.31M	3.32M	3.33M	3.34M

Gapminder

FIGURE 2.5
Tidy, long format version of Gapminder dataset, which contains population information for countries of the world each year starting from 1800.

country	year	population
Afghanistan	1800	3.28M
Afghanistan	1801	3.28M
Afghanistan	1802	3.28M
Afghanistan	1803	3.28M
Afghanistan	1804	3.28M
Afghanistan	1805	3.28M
Afghanistan	1806	3.28M
Afghanistan	1807	3.28M
Afghanistan	1808	3.28M
Afghanistan	1809	3.28M
Afghanistan	1810	3.28M
Afghanistan	1811	3.28M
Afghanistan	1812	3.28M
Afghanistan	1813	3.28M
Afghanistan	1814	3.28M
Afghanistan	1815	3.28M
Afghanistan	1816	3.28M
Afghanistan	1817	3.28M
Afghanistan	1818	3.28M
Afghanistan	1819	3.28M
Afghanistan	1820	3.29M
Afghanistan	1821	3.3M
Afghanistan	1822	3.31M
Afghanistan	1823	3.32M
Afghanistan	1824	3.34M

Gapminder

DEFINITION

Reshaping a dataset involves transforming its structure from a wide format, with extensive columns, to a long format, with fewer columns but more rows, or vice versa, to facilitate more effective visualization and analysis.

SECTION 2.7: COMBINING DATASETS

A vast number of data science projects and situations involve a single, often large, dataset from which to work and draw insight. Part of that work can involve seeking other sources of data that might be *integrated* to help shed light on patterns or relationships in the original data. Other times, a data science project will involve multiple datasets from the outset, but the *integration* of these datasets is a challenge. No matter the starting point, it is often the case that more data are better than less data. As part of data wrangling, we should be prepared to work with multiple datasets whenever possible.

Here, we will cover the most basic ways to combine two datasets. In Chapter 5 on Data Management, we will discuss more sophisticated methods. It's important to understand that additional cleaning and transforming may be necessary with the newly formed dataset after integration has occurred.

Stacking Datasets

Stacking two datasets is exactly what it sounds like. We have two datasets that have completely different observations, but identical variables. In this case, to combine these two datasets we can stack them vertically, one on top of the other. The order in which this is done does not matter while the combining is happening. For example, we might have average annual rainfall and temperature for four countries in one dataset (Table 2.7) and the same variables for three additional countries in another dataset (Table 2.8). Since both datasets have identical variables (`country`, `average_rainfall`, and `average_temperature`), we can stack them (Table 2.9).

TABLE 2.7
The average annual rainfall (inches) and temperature (Fahrenheit) in four countries

country	average_rainfall	average_temperature
Kenya	24.8	71.6
Tanzania	42.2	78.8
Uganda	47.2	71.6
Ethiopia	33.4	70.2

TABLE 2.8
The average annual rainfall (inches) and temperature (Fahrenheit) in three countries

country	average_rainfall	average_temperature
China	25.4	57.2
Japan	66.0	60.8
India	42.6	75.2

TABLE 2.9
Resulting dataframe after stacking Tables 2.7 and 2.8.

country	average_rainfall	average_temperature
Kenya	24.8	71.6
Tanzania	42.2	78.8
Uganda	47.2	71.6
Ethiopia	33.4	70.2
China	25.4	57.2
Japan	66.0	60.8
India	42.6	75.2

Augmenting a Dataset

Augmenting two datasets is, in some sense, the opposite of stacking two datasets. We have two datasets that have completely different variables but identical observations. In this case, to combine these two datasets we can put them together horizontally, one to the right of the other. The order in which this is done does not matter while the combining is happening.

To augment two datasets, the observations in one table must match the observations and sequence of those in the other. For example, one dataset might have Japan's average rainfall for four seasons (Table 2.10), and another might have Japan's average temperature for the same seasons (Table 2.11). If both datasets identify all four seasons in alphabetical order, we can augment the two datasets (Table 2.12).[1]

TABLE 2.10
Japan's average rainfall (inches) for four seasons.

season	average_rainfall
Fall	16.5
Winter	16.7
Spring	9.7
Summer	23.1

TABLE 2.11
Japan's average temperature (Fahrenheit) for four seasons.

season	average_temperature
Fall	59.4
Winter	43.2
Spring	58.6
Summer	81.9

TABLE 2.12
Resulting dataframe after augmenting Tables 2.10 and 2.11.

season	average_rainfall	average_temperature
Fall	16.5	59.4
Winter	16.7	43.2
Spring	9.7	58.6
Summer	23.1	81.9

[1] Augmenting dataset is distinct from data augmentation in machine learning, which involves creating new, synthetic data based on an original dataset to improve performance.

Merging Datasets

Merging two datasets is the process by which these sets are combined in a manner beyond mere stacking and augmenting. There are various ways this can occur. We may have two datasets with differing amounts of observations and variables. However, there must be at least one common variable between the two datasets. *Merging* these two datasets will happen by matching observations between them based on at least one of the common variables. Due to the complexity of this process, we will defer the rest of this discussion to Chapter 5.

Suppose a university's Student Life office has two datasets:

1. *Club Membership Data*: The dataset in Table 2.13 includes the student ID (`student_id`) and the name of the club (`club_name`) to which each student belongs.
2. *Academic Performance Data*: The dataset in Table 2.14 includes the student ID (`student_id`), their study major (`major`), and their grade point average (`gpa`).

TABLE 2.13
Club membership data.

student_id	club_name
S1	Drama Club
S1	Math Club
S2	Debate Team
S3	Math Club
S3	Science Society
S4	Drama Club
S5	Debate Team
S5	Science Society

TABLE 2.14
Academic performance data.

student_id	major	gpa
S1	Math	3.8
S2	English	3.5
S3	Science	3.9
S4	Drama	3.2
S5	Politics	3.7

Quick Tip

When integrating data from different sources, ensure consistency in units, scales, and format. We can clean one dataset to help it match the other better. For example, one might have two decimal places and the other only one, or one might use abbreviations while the other does not.

The Student Life office wants to investigate whether participation in certain clubs is associated with better academic performance. They hope this information will not only highlight the benefits of extracurricular involvement but also help them identify clubs that might be providing students with valuable skills or support.

We can merge these datasets on the `student_id` column to create a comprehensive dataset in Table 2.15.

The merged dataset in Table 2.15 allows us to investigate potential relationships between club participation and academic performance. This could potentially lead to exciting findings that can help promote club involvement and improve student success. We discuss merging datasets more thoroughly in Chapter 5: Data Management.

TABLE 2.15
Merged data.

student_id	club_name	major	gpa
S1	Drama Club	Math	3.8
S1	Math Club	Math	3.8
S2	Debate Team	English	3.5
S3	Math Club	Science	3.9
S3	Science Society	Science	3.9
S4	Drama Club	Drama	3.2
S5	Debate Team	Politics	3.7
S5	Science Society	Politics	3.7

Chapter 2: Putting It Together

Data wrangling is a crucial first step in any data analysis project. It prepares the data for exploration or storytelling, setting the stage for meaningful insights. In practical terms, this means converting raw data into a more digestible format, thereby saving time and effort in the subsequent stages of analysis and storytelling.

Understand the Significance of Data Wrangling

- Defined data wrangling as a precursor to data analysis and storytelling
- Emphasized the practical implications of converting raw data into a manageable format

Administer Strategies for Addressing Missing Data

- Covered various strategies to handle missing data, including removing observations and imputation
- Discussed how these methods can enhance the reliability of the dataset
- Highlighted the importance of a well-planned approach to dealing with missing data

Find and Address Anomalous Values

- Discussed the appearance of anomalous and outlier values in data
- Explained the impact of such values on the integrity of data analysis
- Stressed the importance of outlier detection methods for robust analysis

Transform Quantitative and Categorical Variables

- Examined ways to transform both quantitative and categorical variables
- Explained the importance of proper transformation in enhancing data structure
- Illustrated the benefits of these transformations through practical examples

Reshape Data into a Tidy Format

- Reviewed the concept of tidy data, where each column represents a variable, each row represents an observation, and each cell has a value
- Showed how reshaping data simplifies their structure for particular purposes

Combine Multiple Datasets

- Explained methods for integrating multiple datasets, such as stacking and merging
- Outlined the benefits of data integration in achieving a comprehensive analysis
- Highlighted how such integration can provide new avenues for insights

We can now navigate the complexities of data wrangling. Understanding how to handle missing data, detect anomalies, and adequately format datasets provides a solid foundation for data visualization and exploring the data through analysis.

Chapter 2: Ethics in Practice: Othering

Recall the favorite pet question in Mirabel's dataset. If most respondents answered "dog" or "cat" with a large variety of different responses for the remainder of the respondents, she could add an "other" category to represent them in her `favorite_pet`variable (e.g., "goldfish" and "I don't have a favorite"). While condensing the categories in this way might be useful for Mirabel's purposes, now we will discuss ethical considerations for variables that measure attributes attached to people's identities. For example, a survey could ask respondents about their religion, political identity, race, gender identity, sexual identity, or any number of other characteristics. Data scientists may be tempted to put less represented identities into an "other" category for convenience. However, people do not want to be marginalized as "others" and this condensing reduces the representativeness of the dataset to these individuals.

This process of grouping responses into an "unknown" or "other" category is a version of *othering*, a phenomenon in which some individuals or groups are defined and labeled as not fitting within the perceived norms of a social group. Othering should be avoided whenever possible and extremely well documented otherwise. For example, reducing respondents' political identity to "conservative," "liberal," and "other" is an oversimplification. Othering does not provide proper representation to the people surveyed. The historical use of this process and the requirement of certain procedures for minimum sample sizes underscore the need for more data that represent everyone well. We need to avoid othering. Until othering has been completely eradicated, we must remain as vigilant and cognizant as possible about who an "other" label is representing in a dataset, and whether that representation is tolerable or not.

Chapter 2: Key Terms

data wrangling/data preprocessing
data transformation
dataframe
imputation
mean
median
implausible values
extreme value/outlier
incorrect formats
duplicate records
encoding
Boolean encoding
condensing the categories
reshaping a dataset

CHAPTER 2: CHAPTER REVIEW QUESTIONS

Section 2.1: What Is Data Wrangling?

1. What term refers to the process of converting raw data into a tidy, organized format suitable for analysis?
2. What are the three key components of data wrangling?
3. Explain the data wrangling process.
4. What is the first step in data wrangling?
5. In the context of data wrangling, what does "integrating the data" refer to?
6. What is the purpose of cleaning data in data wrangling?
7. What does the term "dataframe" mean in the context of data wrangling?

Section 2.2: Cleaning Missing Data

8. What are three ways of dealing with missing data values?
9. What is the advantage of imputing missing values instead of leaving them missing?
10. What issues might arise when dropping rows with any missing values from a dataset?
11. When cleaning data, under which condition would it make the most sense to remove an entire row of data?
12. What term refers to estimating missing values based on other available information?
13. Describe a potential issue that arises from imputing missing data values.

For questions 14–17, please refer to the following scenario. Below is a table with data on 10 students. There are missing values scattered throughout:

student	state	age	gpa	favorite_color
1	NY	18	3.3	Blue
2	CA		2.9	Red
3	TX	20	3.4	Green
4	FL	19		Yellow
5	IL	21	3.3	Orange
6	CA	22	3.7	Purple
7	NJ		3.3	Blue
8	GA	23	2.8	Red
9	CA	18	3.0	Green
10	CA	20	3.3	

14. What is the mean GPA, ignoring only the missing GPA value?
15. What is the mean GPA, ignoring any row with a missing value?
16. Impute the CA student's missing age by taking the mean age of other CA students. What is their imputed age?
17. Impute the missing favorite color by the most frequent favorite color of students with the same GPA. What is the imputed color?

Section 2.3: Cleaning Anomalous Values

18. What term refers to data values that are conspicuously different from other values of the same variable?
19. If a dataset contains a person's age as 2995, how might this be addressed in data cleaning?
20. In data cleaning, which term refers to values that are clearly invalid or impossible?
21. Describe an example of incorrect formatting in data.
22. What is the best way to handle an extreme data value that has been verified to be correct?

Section 2.4: Transforming Quantitative Variables

23. What term refers to the creation of a new categorical variable by assigning categories to intervals of an existing quantitative variable?
24. How might a quantitative variable, such as annual income measured in dollars, be transformed for easier visualization?
25. Describe an example of creating a new quantitative variable by combining two existing quantitative variables.
26. How does scaling a quantitative variable by dividing its values help create better visualizations?
27. In general, what is the best practice when transforming variables?
28. What can be done to make a quantitative variable with a wide range easier to visualize and interpret without distorting the data?
29. How does creating a quantitative variable from a categorical one enable more types of analysis?
30. The following table displays the data usage of 10 students on campus, measured in megabytes (MB). For a campus-wide study, convert these data usage values to gigabytes (GB), with $1 \text{ GB} = 1024 \text{ MB}$. Then calculate the mean and median data usage in GB, rounding to two decimal places.

student	data_usage	major	data_usage
Jackson	10587	Art	
Oneal	15789	CS	
Ortega	12678	Bio	
White	14567	Math	
Martinez	16987	Eng	
Dixon	12356	Psych	
Garcia	13489	Hist	
Harris	15782	Econ	
Brown	14256	Socio	
Baker	13567	Music	

For questions 31–32, please refer to the following scenario. The following table shows the heights of 10 athletes on campus, measured in inches (in).

31. Convert these heights to centimeters (cm) using the following conversion rate:

$$1 \text{ inch} = 2.54 \text{ centimeters}$$

student	height	major	height
Washington	71	Football	
Jefferson	74	Basketball	
Patel	68	Soccer	
Gupta	69	Baseball	
Li	74	Tennis	
Kim	75	Swimming	
Rodriguez	73	Track	
Garcia	72	Hockey	
Jackson	66	Wrestling	
Sharma	70	Golf	

and find the mean height in centimeters. Round to two decimal places.

32. In data cleaning, what might be done with text values in a quantitative variable like height?

For questions 33–34, please refer to the following scenario.
For a study on student health, the following table provides the weight (in lb), height (in inches), and majors of 10 students.

student	weight	height	major	bmi
A	120	62	Math	
B	135	64	Bio	
C	150	66	CS	
D	165	68	Eng	
E	180	70	Psych	
F	155	64	Hist	
G	160	66	Art	
H	175	68	Econ	
I	190	70	Socio	
J	205	72	Music	

33. Create a new variable, bmi by filling in the values of the final column's cells using the formula (rounded to one decimal place):

$$\text{bmi} = \text{Weight (lb)} / [\text{Height (in)}]^2 \times 703$$

34. What is the mean bmi for this dataset, rounded to one decimal place?

For questions 35–36, please refer to the following scenario. Use two decimal places for all answers.
The following table provides the distance (in miles), fuel consumption (in gallons), and cost per gallon (in \$) for trips made by 10 students.

35. Calculate the cost per mile for each trip using the following formula:

$$\text{Cost per Mile } (\$) = (\text{Fuel Consumption } (\text{gallons}) * \text{Cost per Gallon } (\$)) \;/\; \text{Distance } (\text{miles})$$

student_id	distance	fuel_consumption	cost_per_gallon	major	cost_per_mile
345634	50	2	3	Art	
534534	100	4	2.5	CS	
832344	150	5	3.5	Bio	
242345	200	6	3	Math	
859285	250	7	2.5	Eng	
894993	100	3	3	Psych	
388539	150	4	2.5	Hist	
393559	200	5	3.5	Econ	
395222	250	6	3	Socio	
825234	300	7	2.5	Mic	

36. What is the average cost per mile?

For questions 37–38, please refer to the following scenario.
In a physics experiment, the following table provides the mass (in kilogram), and acceleration (in meter/second^2).

object	mass	acceleration	force
1	5	0.5	
2	10	1	
3	15	1.5	
4	20	2	
5	25	2.5	
6	30	3	
7	35	3.5	
8	40	4	
9	45	4.5	
10	50	5	

37. Calculate the force (in Newton) applied to each cell in the final column using the formula (to one decimal place):

$$\text{force} = \text{mass} \times \text{acceleration}$$

38. What is the mean force applied to the objects (one decimal place)?

For questions 39–42, please refer to the following scenario.
Below are data on participant, age (years), height (inches), and weight (lbs).

participant	age	height	weight
1	32	5'11	175
2	55	5'5	132
3	62	5'9	195
4	18	5'3	120
5	23	6'1	180
6	43	5'7	150
7	51	5'8	170
8	28	5'4	135
9	47	6'0	200
10	61	5'2	142

39. Create a categorical `age_group` variable using the following categories:

- 18–35: Young Adult
- 36–55: Middle Aged
- 56+: Senior

40. How many participants would be in the Young Adult category?
41. How many would be Middle Aged?
42. How many would be Senior?

Section 2.5: Transforming Categorical Variables

43. How might the uncommon (low frequency) categories of a categorical variable be simplified, given that there are no concerns about othering?
44. What term refers to extracting multiple variables from a single categorical variable?
45. Why is it important to assign every possible value to a category when creating categories from a quantitative variable?
46. What is a Boolean variable? When is it used?
47. What are some other names for Boolean variables?
48. When encoding a categorical variable numerically, why is the choice of values important?
49. What term refers to reducing the number of distinct values a categorical variable can take?
50. Describe the process of condensing a categorical variable.
51. When would splitting a categorical variable into multiple new variables be useful?
52. Provide three examples of categorical variables.
53. What term is used to refer to the creation of quantitative variables from a categorical variable?
54. What is an example of creating a categorical variable from a quantitative one?
55. What is a potential mistake that can arise when encoding categorical data numerically?

For questions 56–58, please refer to the scenario below.
Below is a table with birthdate data for 10 people. Split the birthdate into Month, Day, and Year columns.

person	birth_date	month	day	year
1	01/22/1991			
2	04/17/1988			
3	09/05/1993			
4	11/27/1985			
5	03/14/1980			
6	05/29/1992			
7	02/03/1994			
8	10/11/1989			
9	06/25/1986			
10	08/19/1990			

56. How many people have January as their birth month?
57. How many were born in the last 15 days of a month?
58. What is the mean birth year, rounded to the nearest year?

For questions 59–61, please refer to the scenario below.
Below are responses from 10 participants who either agreed or disagreed with the statement:

"I am satisfied with the meal I just had at this restaurant."

participant	response	age
1	Strongly Disagree	35
2	Agree	29
3	Neutral	42
4	Strongly Agree	27
5	Agree	38
6	Disagree	31
7	Neutral	44
8	Agree	24
9	Strongly Agree	36
10	Disagree	39

59. Recode the responses as follows:

    Strongly Disagree = 1 | Disagree = 2 | Neutral = 3 | Agree = 4 | Strongly Agree = 5

60. What is the mean of the recorded responses rounded to one decimal place?
61. What is the median rounded to no decimal places?

For questions 62–63, please refer to the scenario below.
A researcher collected data on the college majors of 10 students. The major data are in the following format:
Department—Major
For example:
Business—Marketing

62. Split the `department_major` into department and major columns.

student	department_major	department	major
1	Computer Science—Software Engineering		
2	Physics—Applied Physics		
3	Business—Accounting		
4	History—American History		
5	Engineering—Electrical Engineering		
6	English—Creative Writing		
7	Chemistry—Biochemistry		
8	Education—Elementary Education		
9	Math—Applied Math		
10	Nursing—Nursing		

63. How many students have Business majors?

Section 2.6: Reshaping a Dataset

64. Describe the process of converting a dataset from wide format to long format.
65. Why might a data scientist prefer a long-format dataset?

Section 2.7: Combining Datasets

66. What is an advantage of stacking two datasets with the same variables but different observations?
67. Which data integration technique combines datasets with all the same variables but additional observations?
68. When would it be appropriate to augment two datasets by binding them horizontally?
69. What does "merging datasets" typically involve?
70. For which type of data integration must datasets have at least one common variable?
71. When merging two datasets, what must they have in common?
72. What term refers to binding two datasets together horizontally when they have the same observations but completely different variables?
73. What is the primary purpose of combining datasets?

For question 74, please consider the following two tables.

Table 1:

name	age	gpa
Alfonso	32	3.4
Kai	41	3.7
Sam	28	3.4

Table 2:

nickname	height	weight
Al	5'11"	165 lbs
KK	5'5"	132 lbs
Sam the Man	6'0"	180 lbs

74. If these tables were augmented by binding them horizontally, how many rows and how many columns would the resulting table have? (Don't count the column headings as rows.)

For questions 75–76, please consider the scenario below.
A graduate student has collected data on student grades in two sections of her math class.

The grades for Math in Section 1 are as follows:

name	math_test_1	math_test_2	section
Amanda	82	90	1
Diego	71	85	1
Sara	94	88	1

The grades for Math in Section 2 are as follows:

name	math_test_1	math_test_2	section
Tia	75	81	2
Anna	89	77	2
Sarah	93	88	2

75. If the graduate student stacks these data tables, how many rows would the new stacked table contain?
76. How many columns would the new stacked table have?

For questions 77–78, please refer to the scenario below.
A professor collected data on his students' performance across two semesters and wants to combine them into one large table.

Semester 1 Exam Scores:

name	exam_1	exam_2	final_exam	semester
Alex	82	90	93	1
Diego	77	88	81	1

Semester 2 Exam Scores:

name	exam_1	exam_2	final_exam	semester
Alex	86	92	95	2
Sam	80	85	79	2

77. If the professor stacks these two data tables, how many rows would the stacked table have?
78. How many columns would the stacked table have?

For questions 79–80, please consider the scenario below.
A teacher wants to analyze data on her students' performance. She has collected the following data:
Math test scores:

name	test_1	test_2	test_3
Anna	82	90	75
Diego	71	85	83
Sara	94	88	79
Tia	62	77	68

Reading test scores:

name	test_1	test_2
Anna	85	79
Diego	83	77
Sara	91	94

79. If the teacher merges these tables using Name, how many rows would the new table have?
80. How many columns would it have?

For questions 81–82, please consider the scenario below.
A company has two datasets:
Customer data:

customer_id	name	email
1	John	john@email.com
2	Sarah	sarah@email.com

Purchase data:

customer_id	date	amount
1	1/5/2023	$50
2	1/7/2023	$25
3	1/10/2023	$100

To better understand their customers' purchases, they want to merge these datasets using the `customer_id` field.

81. If they merge the datasets, how many unique `CustomerID` values would exist in the merged dataset?
82. How many rows would the merged dataset contain?

For questions 83–86, please consider the scenario below.
Below are data on the majors of 20 university students:

student	major	gpa	clubs
1	Biology	3.2	1
2	English	3.0	2
3	Math	3.5	0
4	History	3.2	1
5	Computer Science	3.8	0
6	Physics	3.4	2
7	English	3.1	3
8	Math	3.7	1
9	History	3.0	0
10	Biology	3.5	2
11	Computer Science	3.6	1
12	Physics	3.2	0
13	Arts	3.9	2
14	Math	3.3	1
15	English	2.9	0
16	Biology	3.8	2
17	History	3.0	3
18	Computer Science	3.5	1
19	Engineering	3.7	0
20	Arts	3.4	2

83. Merge the categories of the categorical variable, `major` into three categories:

- STEM: Biology, Chemistry, Physics, Math, Computer Science, Engineering
- Humanities: English, History, Arts
- Other: Any other majors

84. How many students have a STEM major?
85. How many have a Humanities major?
86. How many have an Other major?

For questions 87–89, please consider the following scenario.
A researcher collected data on crops grown by 15 farms:

farm	crop	acres	yield
1	Corn	80	8500
2	Soybeans	160	9200
3	Wheat	240	11,000
4	Corn	100	9000
5	Soybeans	200	10,000
6	Cotton	150	8000
7	Wheat	300	12,000
8	Corn	90	8000
9	Cotton	140	7500
10	Soybeans	180	9500
11	Wheat	220	10,500
12	Cotton	160	8500
13	Corn	110	9500
14	Soybeans	170	9000
15	Wheat	260	11,500

87. Merge the categories of the categorical variable, `crop`, into two categories:

- Row crops: Corn, Soybeans, Cotton
- Small grains: Wheat

88. How many farms grow row crops?
89. How many farms grow small grains?

Chapter 2: Explaining the Concepts

1

Think about all the forms or surveys, online or paper, that you've ever filled out in your life. This could even include creating a user profile and login credentials for the school in which you're currently enrolled, or the site through which you're accessing this text. List four or five of the fields (pieces of information) you provided on some of these forms. Then, list two or three ways in which a person could fill out these forms differently than you did, which would make the data *inconsistent*. Would you suggest changing the prompts on the form to help make the data more *consistent*, or do you think these issues are better handled by the data scientist after the data are collected?

2

Wrangling data to make it more ready for visualization can include a variety of tasks. For instance, we mentioned that Mirabel might have converted all of her `favorite_pet` variable values to one of *dog*, *cat*, or *unknown*. However, when these are displayed as values on a graph, it may be more appealing for readers to see *Dog*, *Cat*, and *Other*.

Alternatively, we could have a quantitative variable representing the total revenue of companies in a particular industry sector, which takes values in the billions of U.S. dollars. Should a visualization of this variable include values that look like "$10, 000, 000, 000" or would this large number of digits take up too much space and be more easily digestible as something like "$10bn"?

Explain, with examples, what kind of cleaning and transformation steps would be especially useful for the visualization of different types of variables.

3

In our explanation of *transforming data*, we mentioned that it is very often the case that new variables are created, and existing variables are retained, not replaced. Explain why this is a good practice. Give examples, using some of our described *data transformation methods*, of how **replacing** an existing variable with the transformed version results in a loss of information in the dataset and hence a reduction in the quality of the dataset.

4

In our explanation of *integrating data*, we mentioned that the ways in which two datasets are related, and hence combined, can depend on the variables, the observations, or both. Explain why it is more useful to *clean* and *transform* the data of each dataset separately first, and then combine them rather than doing it the other way around. *Note: More cleaning or transforming may still be necessary after combining, but let's focus on the initial steps first.*

5

Documenting the data science workflow as we progress through it is extremely important. This is partly because data can come from the same source and in the same format at multiple points in time. For instance, another student, Miles, may want to use Mirabel's dataset responses as a model to collect data from a new group of students next year. All of the prompts will be the same, and the answers could involve many of the same *inconsistency* issues that Mirabel faced.

- Explain why having detailed instructions and explanations of Mirabel's data wrangling steps would be useful to Miles when he analyzes his new data.
- Generally, explain why it's good practice for Mirabel to ensure that the work she has done with the data can be replicated on her data and applied accurately to new data.

Chapter 2: Applying the Concepts

Activity: Identify Missing Data

Dataset: titanic_dataset

Now we step aboard the Titanic dataset. Identify the missing values in the `age` variable. This voyage goes beyond just finding missing data. You must deliberate on the implications of these missing values and strategize the best way to handle them. Are you going to steer around them, leaving them as they are, or are you going to anchor down and impute them? Your challenge here is not only to identify missing values, but also to weigh the pros and cons of different handling strategies and apply your chosen method.

Assess the presence of missing `age` values in the dataset as the first step in your analysis.

1. Overall, how many passengers were recorded in this dataset?
2. Of the total recorded passengers, how many were missing a value for the `age` variable?
3. What is the percentage of total recorded passengers who did not have their ages recorded?

Missing data can bias or distort your analysis, leading to incorrect conclusions. In the context of the Titanic dataset, if the ages of younger or older passengers are disproportionately missing, analysis around age may lead to misleading results.

4. If child passengers were less likely to have their ages recorded in this dataset than adult passengers, how would this affect the average age we calculate?

Understanding why data are missing can provide valuable insights. Were ages not recorded for certain passengers? Did passengers refuse to disclose their ages? The reasons can have implications on how we interpret the rest of the data.

5. Different strategies for handling missing data have different effects on the analysis. In one such strategy for handling missing data, the missing values are filled with educated guesses. What is this strategy called?

In the Titanic dataset, correctly handling missing age data could significantly impact any conclusions drawn about survival rates among different age groups.

Activity: Impute Missing Values from Internal Data

Dataset: world_happiness_report_2005_2021

We will be examining the missing data within the World Happiness Report for healthy_life_expectancy_at_birth. This variable has 58 missing values. We will investigate how different imputation methods influence calculations of average life expectancy, specifically using the mean and median imputation methods.

Use two decimal places for numeric responses.

Answer the following questions:

1. What average life expectancy does the mean imputation method produce? Round to two decimal places.
2. What average life expectancy does the median imputation method produce? Round to two decimal places.

Activity: Identify Extreme Data Values

Dataset: bird_watcher

Consider the birdwatcher's data. Are any of the ages extreme values?

Answer the following questions:

1. What age is the lowest value for the oldest 5% of birdwatchers?
2. What age is the highest value for the youngest 5% of birdwatchers?
3. How many birdwatchers are above 100?
4. How many birdwatchers are below 18?
5. Which values are impossible?
6. Explain this analysis using the STAR Framework.

Activity: Clean Incorrect Formats

Dataset: bird_watcher

This dataset originates from an Airbnb birdwatching experience, where the guide, a seasoned ornithologist, specializes in showcasing rare and captivating bird species such as sparrows, woodpeckers, and hummingbirds. The experience is tailored for nature enthusiasts and photographers, ranging from beginners to experts. After each tour, participants are asked to complete a short survey. The primary aim is to gather feedback on their experience, focusing on their age, favorite bird spotted during the tour, the lens size of their binoculars, and whether they would consider repeating the experience.

The survey was conducted over several years, accumulating responses from a diverse group of 23,893 birdwatchers. The dataset includes a unique identifier for each participant (`birdwatcher_id`), their age, their stated favorite bird from the tour, the lens size of their camera, a Boolean indicator of their interest in a repeat experience, and the date of their original tour. The guide is now interested in cleaning this data to analyze the relationship between participants' favorite birds and their likelihood of signing up for a second tour.

1. Clean the `favorite_bird` column of the birdwatcher dataset. Perform a basic cleaning of the data in this order:
   a. "sparrows" should be changed to "sparrow"
   b. "woodpeckers" should be changed to "woodpecker"
   c. "hummingbirds" should be changed to "hummingbird"
   d. any word that begins with "s" and ends with "w" should be changed to "sparrow"

   e. any word that begins with "w" and ends with "r" should be changed to "woodpecker"
   f. any word that begins with "h" and ends with "d" should be changed to "hummingbird"
2. Before basic data cleaning, how many people in this dataset indicated their favorite bird as:
   a. "sparrow"?
   b. "woodpecker"?
   c. "hummingbird"?
3. After basic data cleaning, how many people in this dataset indicated their favorite bird as:
   a. "sparrow"?
   b. "woodpecker"?
   c. "hummingbird"?
4. Explain the results using the STAR Framework.

Activity: Create a Categorical Variable from a Quantitative Variable

Dataset: ames_housing

We're about to explore the suburban landscape of the housing market! We will use the Ames Housing dataset for this venture. Your mission? Give the `sale_price` (USD) variable a new spin. Convert this quantitative value into a new categorical variable with the following categories: "Cheap" for houses priced $0–$100,000, "Moderate" for houses priced $100,001–$200,000, and "Expensive" for houses priced $200,001 and above. Although these prices may not reflect the current state of the housing market, let's extract some insight from our new categories.

Here are the steps we're going to follow. First, create the new categorical `sale_price_category` variable as described above. Then, count how many houses fall into each of the categories. Finally, make a visualization to better convey the results. Answer the questions below.

1. For this dataset, most of the houses fall into the "Moderate" category. What does this tell us about the prices of most houses in the dataset?

   Suppose we are interested in breaking down the largest category into two new categories with smaller ranges. Recreate the counts and visualization after modifying the existing code to create these two new categories:

2. Split the "Moderate" category into two new categories:
   a. "Cheap Moderate": $100,001–$150,000
   b. "Expensive Moderate": $150,001–$200,000

3. How many houses are in the "Cheap Moderate" category?
4. Which category is the largest? Which category is the second largest?
5. How many houses are in the "Expensive" category?
6. How many houses are in the "Expensive Moderate" category?

Activity: Create a Boolean Variable That Represents a Categorical Variable

Dataset: zoo

Now we are going to the Zoo animal classification dataset. This dataset contains information about different animals, including a `class_type` variable that classifies animals as 1 = mammal, 2 = bird, 3 = reptile, 4 = fish, 5 = amphibian, 6 = insect, or 7 = other. Convert these class types into Boolean variables. This transformation is crucial in preparing the data for various animal analyses, which we could call "animalyses" to provoke a reaction.

Start by creating a new Boolean variable called `mammal`. This variable will take on a value of TRUE if the animal in question is a mammal, and FALSE otherwise.

1. After creating the `mammal` variable and counting the occurrences of TRUE and FALSE, how many mammals are in this dataset?

   Create Boolean variables for both birds and reptiles (Hint: `class_type = 2` for birds and `class_type = 3` for reptiles).

2. How many birds are in this dataset? How many reptiles are in this dataset?

In general, this Boolean encoding is useful for computers to encode categorical data, as it turns categories into TRUE/FALSE, or as the computer sees in binary, 1/0.

Creating Boolean variables from a categorical variable can simplify the data, making it more suitable for certain analyses. By converting the `class_type` variable into Boolean variables, we can perform a separate analysis for each class type. This could assist zoologists in understanding patterns unique to each animal class. For instance, this could be used in endangered species prediction, helping to determine conservation priorities.

Activity: Splitting a Variable into Multiple Variables

Dataset: bird_watcher

In this analysis, we revisit the birdwatcher dataset, this time with a focus on understanding seasonal patterns. Our goal is to examine if there's an association between the months of the year and the types of birds observed. To facilitate this, we first need to transform the dataset by isolating the month from the existing `date` variable's information. This involves creating a new column that exclusively represents the month. This step is crucial for enabling us to perform a more focused analysis on how birdwatching experiences vary by month, potentially providing insights into the seasonal preferences of different bird species.

1. How many entries were made in each month of the year?
2. Explain the results using the STAR Framework.

CHAPTER 2: TECH CENTER

PYTHON NOTEBOOK

Load Packages:

```
import pandas as pd
```

Import a Dataset:

Read a CSV file into Python:

```
dataset1 = pd.read_csv("Dataset1.csv")
```

Drop Rows with Missing Values:

Drop rows in a dataframe containing missing values

```
dataset1.dropna()
```

To return a dataframe with columns containing missing values removed

```
dataset1.dropna(axis=1)
```

To create a new dataframe (clean_columns_dataset) with all columns containing missing values removed from a dataset (e.g., dataset1).

```
clean_columns_dataset = dataset1.dropna(axis=1)
```

Fill Missing Values for Column q1 with the Mean of Column q1:

To assign the mean of column "q1" to "Mean_q1"

```
Mean_q1 = dataset1["q1"].mean()
```

To replace the missing values in the column with the mean of q1

```
dataset1["q1"] = dataset1["q1"].fillna(Mean_q1)
```

Replace a Value in Column XX and Row YY with a New Value:

Create user-assigned value

```
new_value = 999
```

Reassign row YY in column XX with 999

```
dataset1.at[2, "q2"] = new_value
```

Create a Categorical Variable from an Existing Quantitative Variable:

Create a new column called q1_groups, which will store group names based on the column q1 value:

```
dataset1['q1_groups']
```

Set cutoff values for each group in "q1_groups"

```
bins = [0, 50, 100, 300]
```

Set group names

```
labels = ['Low', 'Medium', 'High']
```

Create Low, Medium, and High groups based on data in column "q1", "right=" parameter set to True as default, determines whether each "bin" includes its rightmost edge or not

```
dataset1['q1_groups'] = pd.cut(dataset1['q1'], bins=bins,
labels=labels, right=False)
```

```
bins = [0, 50, 100, 300]
labels = ['Low', 'Medium', 'High']
dataset1['q1_groups'] = pd.cut(dataset1['q1'], bins = bins, labels = labels, right = False)
dataset1
```

	id	cat1	cat2	cat3	cat4	cat_binary	q1	q2	q3	q4	q1_groups
0	1	A	type_1	Y	300.0	1	52.199627	31.873114	9.649249	46.768104	Medium
1	2	B	type_3	X	100.0	1	77.308355	40.596928	7.193891	48.072335	Medium
2	3	A	type_3	X	100.0	0	95.874092	23.969227	3.499928	51.925680	Medium
3	4	B	type_1	X	300.0	0	11.732048	45.743154	2.543824	19.171565	Low
4	5	B	type_3	X	200.0	1	10.700414	2.467447	2.653033	5.990407	Low

Create a New Variable That Rescales an Existing Quantitative Variable:

Create a new column "q1_scaled" that contains the values in q1 divided by 1000

```
dataset1["q1_scaled"] = dataset1["q1"] / 1000
```

Create a New Variable from Existing Quantitative Variables:

Create a new column "q2_q3_sum" that contains the sum of the values in q2 and q3

```
dataset1["q2_q3_sum"] = dataset1["q2"] + dataset1["q3"]
```

Create a Quantitative Variable from an Existing Categorical Variable:

```
dataset1['cat2_numeric'] = dataset1['cat2'].astype('category').
cat.codes
```

Create a new column cat2_numeric that contains integer values encoding the categories in cat2: dataset1['cat2_numeric'] = dataset1['cat2'].astype('category').cat.codes

Merge the Categories of an Existing Categorical Variable:

To create a new column called "cat2_merged" with 2 categories (type_1_2 & type_3)

```
dataset1 ["cat2_merged"] = dataset1["cat2"].replace({"type_1":
"type_1_2", "type_2": "type_1_2"})
```

Split a Categorical Variable:

Split the cat2 variable by "_" into two separate columns

```
dataset1[['cat2_part1', 'cat2_part2']] =
dataset1['cat2'].str.split('_', n=1, expand=True)
```

Transformation into Wide Format Using pd.pivot()

Reshape to Long Format: Transform data from wide to long format, pivoting q1 and q2 columns into a longer format columns not included in the id_vars will be absent from the created table.

```
dataset1_melted = pd.melt(dataset1, id_vars=['id'], value_
vars=['q1', 'q2'], var_name='Quantity', value_name='Value')
```

Stack Datasets:

Stack datasets 1 & 2 on top of each other

```
stacked_dataset = pd.concat([dataset1, dataset2],
ignore_index=True)
```

Merge and Augment Datasets:

Merge the two datasets based on matching values within the id column

```
merged_dataset = pd.merge(dataset1, dataset2, on = "id")
```

Subset Using Text Comparisons in Categorical Variables:

Subset (filter) rows where the cat2 variable is equal to "type_1"

```
subset_cat = dataset1[dataset1["cat2"] == "type_1"]
```

Subset Using Numeric Comparison in Quantitative Variables:

Subset (filter) rows where the q1 variable is greater than 50

```
subset_numeric = dataset1[dataset1["q1"] > 50]
```

Subset Based on Missing Values:

Subset rows where the q1 variable is missing

```
subset_missing = dataset1[dataset1["q 1"].isnull()]
```

Subset rows where the q1 variable is not missing

```
subset_not_missing = dataset1[dataset1["q1"].notnull()]
```

Subset Based on Conditions Involving 2+ Variables:

Subset rows where q1 is greater than 50 and cat2 is equal to "type_1"

```
subset_conditions = dataset1[(dataset1["q1"] > 50) &
(dataset1["cat2"] == "type_1")]
```

R SCRIPT

Load Packages:

```
library(tidyverse)
```

Import a Dataset:

Read a CSV file into R:

```
dataset1 <- read_csv("Dataset1.csv")
```

Drop Rows with Missing Values:

Remove rows with any missing values

```
clean_dataset <- dataset1 %>% drop_na()
```

Drop Columns with Missing Values:

Remove columns containing any missing values

```
clean_columns_dataset <- dataset1 %>% select_if(~ !any(is.na(.)))
```

Fill Missing Values for Column q1 with the Mean of Column q1:

Calculate mean of q1 excluding NA values

```
mean_q1 <- mean(dataset1$q1, na.rm = TRUE)
```

Replace NA values in q1 with its mean

```
dataset1 <- dataset1 %>% mutate(q1 = case_when(is.na(q1) ~ mean_
q1, TRUE ~ q1))
```

Replace Value in Column XX and Row YY with a New Value:

Assign a new value

```
new_value <- 999
```

Assign a new value to a specific cell in Column XX and Row YY

```
dataset1$XX[YY] <- new_value
```

Create a Categorical Variable from an Existing Quantitative Variable:

Creating a new categorical variable from q1

```
dataset1 <- dataset1 %>%
mutate(q1_groups = case_when
(q1 >=0 & q1 <=50 ~ "Low",
q1 > 50 & q1 <= 100 ~ "Medium",
q1 > 100 & q1 <= 300 ~ "High"))
```

id	cat1	cat2	cat3	cat4	cat_binary	q1	q2	q3	q4	q1_groups
1	A	type_1	Y	300	1	52.19962736	31.8731139	9.649249408	46.7681040	Medium
2	B	type_3	X	100	1	77.30835541	40.5969281	7.193890620	48.0723350	Medium
3	A	type_3	X	100	0	95.87409231	23.9692275	3.499928436	51.9256798	Medium
4	B	type_1	X	300	0	11.73204804	45.7431544	2.543824011	19.1715654	Low
5	B	type_3	X	200	1	10.70041402	2.4674473	2.653033245	5.9904070	Low

Create a New Variable that Rescales an Existing Quantitative Variable:

Creating a scaled version of q1

```
dataset1 <- dataset1 %>% mutate(q1_scaled = q1 / 1000)
```

Create a New Variable from Existing Quantitative Variables:

Summing columns q2 and q3 to create a new variable

```
dataset1 <- dataset1 %>% mutate(q2_q3_sum = q2 + q3)
```

Create a Quantitative Variable from Existing Categorical Variable:

Converting a categorical variable cat2 to numeric

```
dataset1 <- dataset1 %>% mutate(cat2_numeric =
as.numeric(as.factor(cat2)))
```

Merge Categories of an Existing Categorical Variable:

Merging specific categories type_1 and type_2 into one category

```
dataset1 <- dataset1 %>%
mutate(cat2_merged = recode(cat2, type_1 = "type_1_2", type_2 =
"type_1_2"))
```

Split a Categorical Variable:

Splitting cat2 into multiple categories

```
dataset1 <- dataset1 %>%
separate(cat2, into = c("cat2_part1", "cat2_part2"), sep = "_")
```

Reshape to Long Format:

To pivot "q1" and "q2" columns from wide to long format

```
dataset1_melted <- dataset1 %>% pivot_longer(cols = c(q1, q2),
names_to = "Quantity", values_to = "Value")
```

Stack Datasets:

Stack datasets 1 & 2 on top of each other

```
stacked_dataset1 <- bind_rows(dataset1, dataset2)
```

Merge and Augment Datasets:

Merge the two datasets based on matching values within the id column

```
merged_dataset <- inner_join(dataset1, dataset2, by = "id")
```

Subset Using Text Comparisons in Categorical Variables:

Subset (filter) rows where the cat2 variable is equal to "type_1"

```
subset_cat2 <- dataset1 %>% filter(cat2 == "type_1")
```

Subset Using Numeric Comparison in Quantitative Variables:

Subset (filter) rows where the q1 variable is greater than 50

```
subset_numeric <- dataset1 %>% filter(q1 > 50)
```

Subset Based on Missing Values:

To subset rows based on missing values in a specific variable in R, use the filter() function with

```
is.na()
```

Subset rows where the q1 variable is missing (or not missing)

```
subset_missing <- dataset1 %>% filter(is.na(q1))subset_not_
missing <- dataset1 %>% filter(!is.na(q1))
```

Subset Based on Conditions Involving 2+ Variables:

To subset rows based on conditions involving two or more variables in R, use logical operators like

```
&
```

Subset rows where q1 is greater than 50 and cat2 is equal to "type_1"

```
subset_conditions <- dataset1 %>% filter(q1 > 50 & cat2 == "type_1")
```

EXCEL

Drop Rows with Missing Values:

1. Click on any cell in your dataset.
2. Go to the "Home" tab.
3. In the"Editing" group, click on "Find & Select."
4. Choose "Go To Special."
5. Select "Blanks" and click "OK". This will highlight all blank cells.
6. Right-click on any highlighted cell.
7. Select "Delete."
8. Choose "Entire row" in the delete dialog box.

Drop Columns with Missing Values:

1. Click on any cell in your dataset.
2. Go to the "Home" tab.
3. In the "Editing" group, click on "Find & Select".
4. Choose "Go To Special".
5. Select "Blanks" and click"OK". This highlights all blank cells.
6. Right-click on any highlighted cell.
7. Choose "Delete".
8. Select"Entire column" in the delete dialog box.

Fill Missing Values for Column XX with the Mean of Column XX:

1. Identify the column with missing values.
2. Click on an empty cell where you want to display the mean.
3. Enter the formula

   =AVERAGE(YourColumnRange)

   replacing "YourColumnRange" with the actual range of your column, for example,

   =AVERAGE(F2:F100)
4. Press Enter to display the mean.
5. Select the cell with the mean value, right-click, and choose "Copy".
6. Select the cells with missing values in your column.
7. Right-click on the selected cells and choose "Paste Special".
8. In the Paste Special dialog, select "Values"and click"OK".

Replace Value in Column XX and Row YY:

1. Locate the cell you want to change.
2. Click on the cell.
3. Enter the new value.
4. Press Enter to confirm the change.

Create Categorical Variable from an Existing Quantitative Variable:

1. Insert a new column next to the column you want to categorize.
2. In the first cell of the new column, use an "IF" or"IFS" function to create categories based on the adjacent cell's value. For example, `=IF(F2<50,"Low",IF(F2<=100,"Medium","High"))`
3. Copy this formula down the entire column.

2 =IF(G2<50, "Low", IF(G2<=100, "Medium", "High"))

A	B	C	D	E	F	G	H	I	J	K
id	cat1	cat2	cat3	cat4	cat_binary	q1	q2	q3	q4	q1_groups
1	A	type_1	Y	300	1	52.19963	31.87311	9.649249	46.7681	Medium
2	B	type_3	X	100	1	77.30836	40.59693	7.193891	48.07233	Medium
3	A	type_3	X	100	0	95.87409	23.96923	3.499928	51.92568	Medium
4	B	type_1	X	300	0	11.73205	45.74315	2.543824	19.17157	Low
5	B	type_3	X	200	1	10.70041	2.467447	2.653033	5.990407	Low

Create New Quantitative Variable That Is a Scaling of an Existing Quantitative Variable:

1. Insert a new column next to the column you want to rescale.
2. In the first cell of the new column, enter a formula to rescale the value from the adjacent column. For example,

   =F2/1000
3. Copy this formula down the entire column.

Create New Quantitative Variable That Is a Combination of Existing Quantitative Variables:

To sum two quantitative variables:

1. Insert a new column where you want the sum.
2. In the first cell of the new column, enter a formula to add the two variables. For example,

   `=B2+C2`
3. Copy this formula down the entire column.

Create a Quantitative Variable from an Existing Categorical Variable:

1. Next to your categorical column, insert a blank column for the new quantitative data.
2. In the first cell of the new column, enter

   =IF(A2="Category1", 1, IF(A2="Category2", 2,…))

   adjusting for your specific categories and corresponding numeric values.
3. Copy this formula down the entire column.

Merge Categories of an Existing Categorical Variable:

1. Insert a new column next to the categorical column.
2. Use an "IF" function to merge categories. For example,

   `=IF(B2="type_1","type_1_2",IF(B2="type_2","type_1_2",B2))`
3. Copy this formula down the entire column.

Split Categorical Variable:

1. Insert new columns next to the categorical column.
2. Use the "LEFT","RIGHT", or "MID" function to extract parts of the text. For example,

   `=LEFT(B2, FIND("_",B2)-1)" and "=MID(B2, FIND("_",B2)+1, LEN(B2))`
3. Copy these formulas down the respective columns.

Pivot Longer:

Excel has a built-in tool that allows us to extract all kind of information from our data without manually reshaping it.

1. Select your data and Go to "Insert" → "Pivot Table".
2. Choose placement for the Pivot Table.
3. Drag the identifier column (like ID) to "Rows".
4. Move columns to be pivoted to "Values".

5. Optional: The Filters field excludes data based on specified criteria, refining the displayed results.
6. Set "Values" field to "Count" or as needed.

Pivot Wider:

Excel has a built-in tool that allows us to extract all kind of information from our data without manually reshaping it.

1. Select your data and Go to "Insert" → "Pivot Table".
2. Choose placement for the Pivot Table.
3. Drag the column headers you want to pivot into the "Rows" area in the Pivot Table Field List. Drag any value field (e.g., the data corresponding to the headers) into the "Values" area.
4. Adjust the "Values" field settings if necessary (e.g., sum, count).

Stack Datasets:

1. Select the range of the second dataset.
2. Copy the range.
3. In the first dataset, go to the cell where you want the second dataset to begin.
4. Paste the copied range.

Merge Datasets Using XLOOKUP:

1. Ensure both datasets are in separate sheets.
2. In Dataset A, in a new column, enter

   =XLOOKUP(A2, Sheet2!$A$2:$A$100, Sheet2!$B$2:$B$100)

   where A2 is the ID in Dataset A and Sheet2 contains Dataset B. Dataset A's ID variable should be in column A2:A100 and match ID values in Dataset B, located in Sheet2's column B2:B100.
3. Drag the formula down to fill other rows in Dataset A.
4. Adjust XLOOKUP for other columns from Dataset B as needed.
5. Save the Excel file with the merged data.

Merge Datasets Using Power Query:

1. Open Excel, go to "Data" → "Get Data" → "From File" → "From Excel Workbook"

   Import both Dataset A and Dataset B by selecting and loading them.
2. In Excel, navigate to "Data" → "Get Data" → "Launch Power Query Editor".
3. In the Power Query Editor,
   a. Select Dataset A, go to "Home", then "Merge Queries".
   b. Choose Dataset B and select matching columns for merging (ID).
   c. Select the type of join (e.g., Inner join for matching records).
4. In the merged column, click the expand button and select columns from Dataset B to include.
5. Click"Close & Load" to load the merged data into a new worksheet.

Subset (Filter) the Rows/Observations of a Dataset by Criteria:

1. Click the top cell of the column you want to filter.
2. In the "Data" tab, click "Filter".
3. Click the dropdown menu in the column you selected in step 1.
4. Uncheck "Select All" and thenre-check the box next to the criteria you're interested in.

Subset using Text Comparisons in Categorical Variables:

1. Click the top cell of the column you want to filter.
2. In the "Data" tab, click "Filter".
3. Click the dropdown menu in the column you selected in step 1.
4. Check/uncheck the text values you wish to include or exclude.

Subset using Numeric Comparison in Quantitative Variables:

1. Click the top cell of the column you want to filter.
2. In the "Data" tab, click "Filter".
3. Click the dropdown menu in the column you selected in step 1, then select Number Filters.
4. Select a number filter (e.g., greater than, less than) and enter the value for comparison.

Subset Based on Missing Values:

1. Click the top cell of the column you want to filter.
2. In the “Data” tab, click the“Filter”.
3. Click the dropdown menu in the column you selected in step 1.
4. Uncheck"(Blanks)" to exclude missing values or, check only"(Blanks)"to include them only.

Subset Used on Conditions Involving 2+ Variables:

1. Click the top cell of the columns you want to filter.
2. In the “Data” tab, click the“Filter”.
3. Set conditions for each column involved by using the dropdown in each column header.

STATCRUNCH

Drop Rows with Missing Values:

1. Data → Row Selection → Detect → Empty
2. Ctrl+A to select all columns
3. Click Compute!
4. Edit → Rows → Delete

Drop Columns with Missing Values:

1. Data → Validate
2. Ctrl+A to select all columns
3. Select Display → Empty
4. Click Compute!
5. Edit → Columns → Delete
6. Choose the columns with found empty values.

Fill Missing Values for Column XX with the Mean of Column XX:

1. Stat → Summary Stats → Select Columns
2. Select the q1 column
3. Statistics → Mean
4. Click Compute!
5. Data → Compute → Expression
6. Enter a formula to replace missing values with the mean value
   a. For example, if the mean is 65.5 and the column is named 'q1', the formula would be:

      ```
      ifelse(isnull(q1), 65.5,q1)
      ```
7. Click Compute!

Create a Categorical Variable from an Existing Quantitative Variable:

1. At the top of a new, empty column, type “`q1_new`”.
2. Data → Compute → Expression.
3. Enter an expression to categorize q1 in the new q1_new variable:

   ```
   ifelse(q1 <= 100, "Cheap", ifelse(q1 > 100 and q1 <= 150, "Low",
   ifelse(q1 > 150 and q1 <= 200, "Medium", "High")))
   ```
4. Click Compute!

Create a New Variable That Rescales an Existing Quantitative Variable:

1. Data → Compute → Expression.
2. Enter an expression to scale q2:

   ```
   q2/1000
   ```
3. Click Compute!

Create a New Variable from Existing Quantitative Variables:

1. Data → Compute → Expression
2. Enter an expression to add (combine) q2 and q3:

   q2+q3
3. Click Compute!

Create a Quantitative Variable from an Existing Categorical Variable:

1. Data → Compute → Expression
2. Enter an expression to assign numbers for each value of cat1

   `ifelse(cat1="A",1,ifelse(cat1="B",2,3))`

3. Click Compute!

Merge the Categories of an Existing Categorical Variable:

1. Data → Recode
2. Select categorical variable you want to recode (e.g., `cat3`)
3. Column label → type name of new column (e.g., `cat3_new`)
4. Options → Create new column(s)
5. Click Compute!
6. In the Recoded column, enter the values you want instead of the original values.
7. Click Compute!

Split a Categorical Variable:

1. Data → Arrange → Slice
2. Select column → Select categorical variable you want to split (e.g., `cat2`)
3. Delimiter → specify the delimiter you want to split on (e.g., _)
4. Click Compute!

Reshape in Long or Wide Format

- Not currently available in StatCrunch

Stack Datasets:

1. Select all of the cells corresponding to the first dataset
2. Copy this selection
3. Go to the cell where you want the second dataset to begin
4. Paste the copied selection

Merge Datasets

- Not currently available in StatCrunch

Subset Using Text Comparisons in Categorical Variables:

1. Data → Row Selection → Select Where
2. Enter expression:

   `cat2="type_2"`

3. Click Compute!

Subset using Numeric Comparison in Quantitative Variables:

1. Data → Row Selection → Select Where
2. Enter expression:

   `q1>= 10`

3. Click Compute!

Subset based on Missing Values:

1. Data → Row Selection → Select Where
2. Enter expression:

   `!isnull(cat2)`

3. Click Compute!

Subset based on Conditions Involving 2+ Variables:

1. Data → Row Selection → Select Where
2. Enter expression:

   `“cat3”=100 and “q1”>15`

3. Click Compute!

CHAPTER 3

MAKING SENSE OF DATA THROUGH VISUALIZATION

Selinofoto/Shutterstock

LEARNING OBJECTIVES

- Use the fundamental ideas of the *grammar of graphics* to evaluate options for the construction of data visualizations.
- Employ techniques for visualizing a single quantitative variable, a single categorical variable, two variables, and three or more variables through various plots.
- Identify common pitfalls and dangers associated with visual misrepresentation in data visualization.
- Incorporate established guidelines to produce compelling and honest data visualizations.